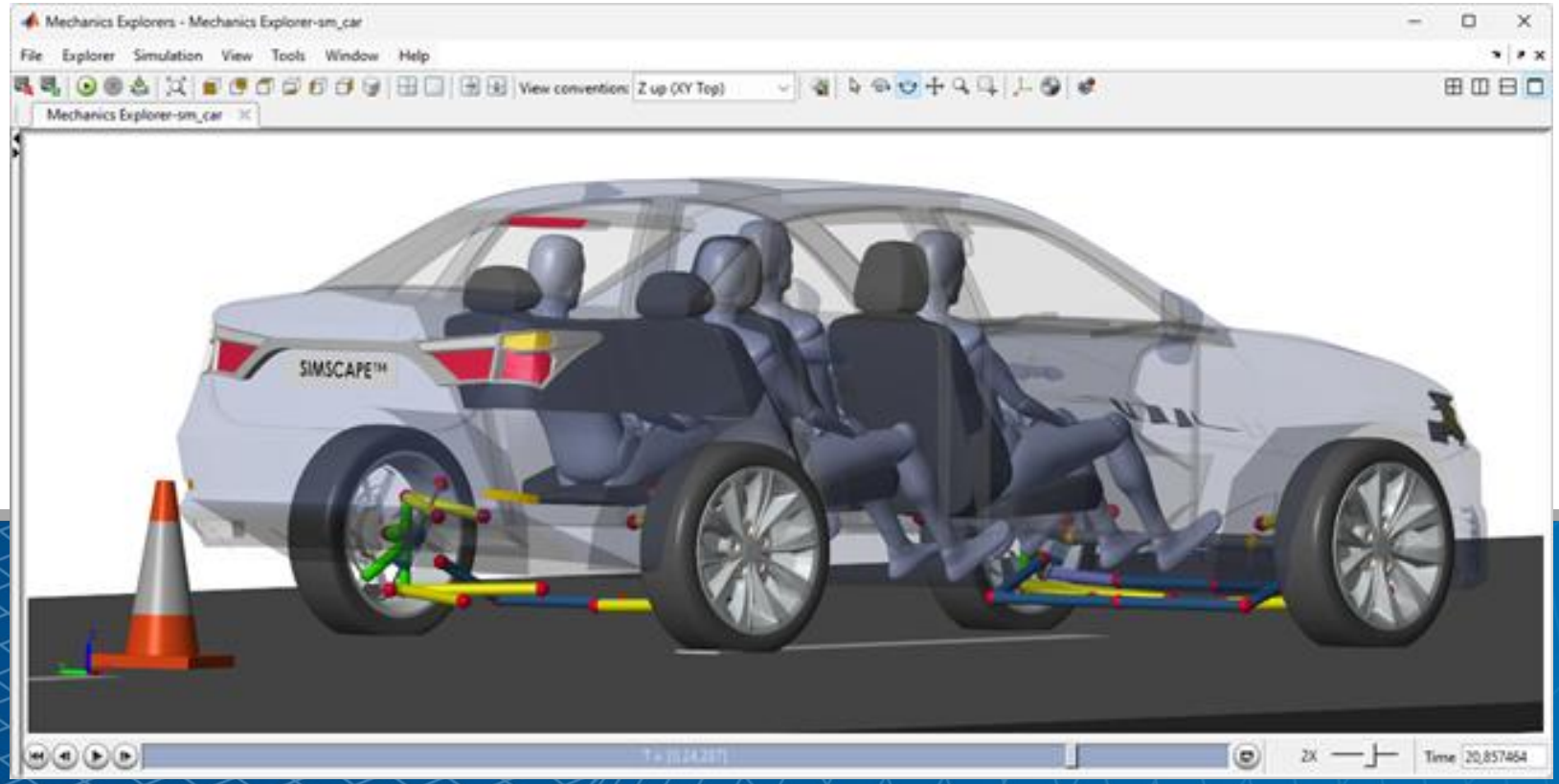
Simscape Vehicle Templates Overview



1. Getting Started
2. Overview
3. Subsystem Models
4. Parameters
5. Variants
6. Maneuvers
7. Suspension (Linkages)
8. Bushings
9. Steering
10. Anti-Roll Bar
11. Suspension (Axle)
12. Springs and Dampers
13. Tires
14. Scene and Terrain
15. Brakes
16. Powertrain
17. Closed-Loop Driver Model
18. Trailers
19. Workflows
20. Managing Customizations

Updated July 2026 for version 4.12

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17. Closed-Loop Driver Model
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Simscape Vehicle Templates Getting Started

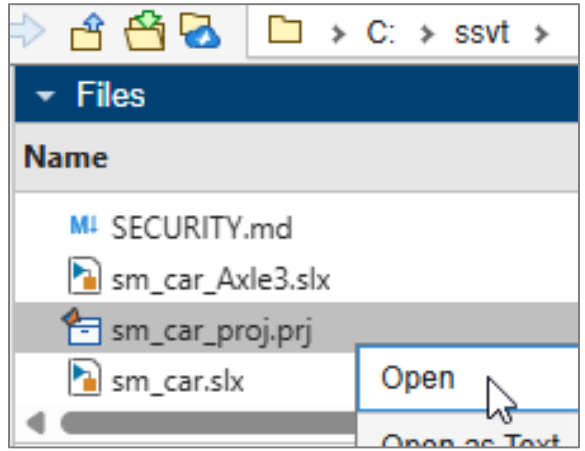


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Getting Started

- Getting Files
- Open Project
- Configure Test using the App
- Evaluate Results using the App





Simscape™ Vehicle Templates Available on File Exchange

- Download Simscape Vehicle Templates directly from the File Exchange
  - Get files saved for use with MATLAB® releases R2018b and higher
    - **Recommend R2022a and higher**
  - Files hosted on GitHub so we can react quickly to requests



https://www.mathworks.com/solutions/physical-modeling/simscape-vehicle-templates.html

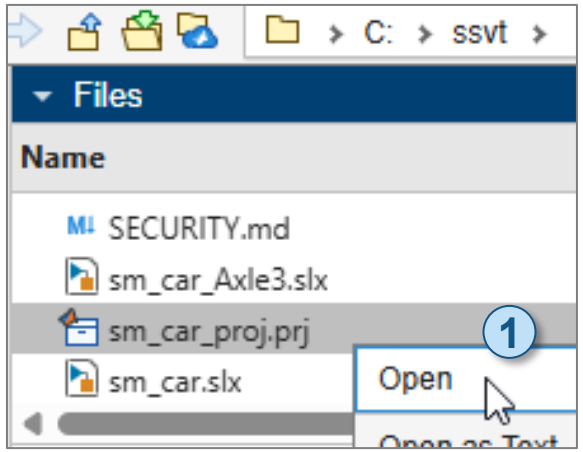
https://www.mathworks.com/matlabcentral/fileexchange/79484-simscape-vehicle-templates

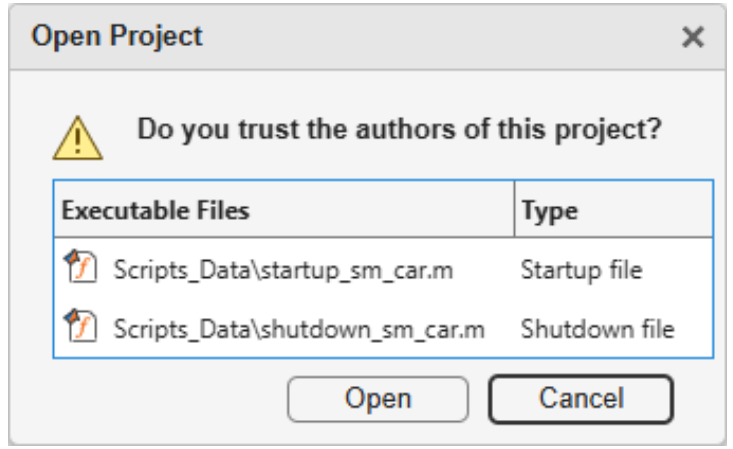
Open Project

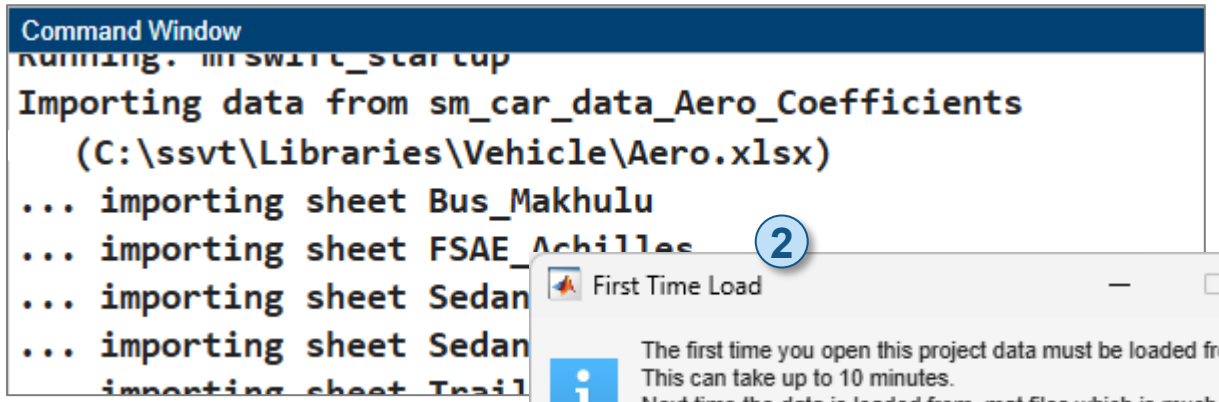
1. After downloading or cloning, open `sm_car_proj.prj`
   - If asked, click the “Open” button

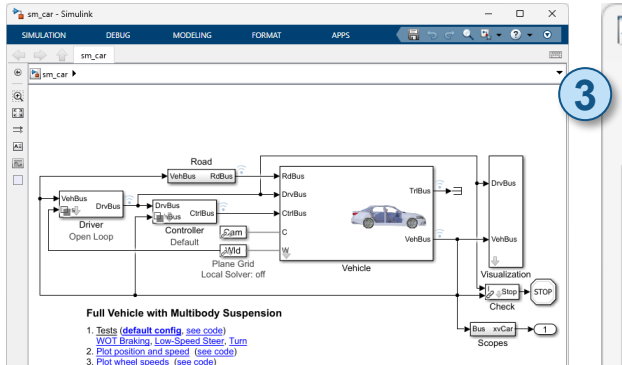
2. First time project opens, all default parameters are loaded from Excel
   - After that, loads faster from .mat

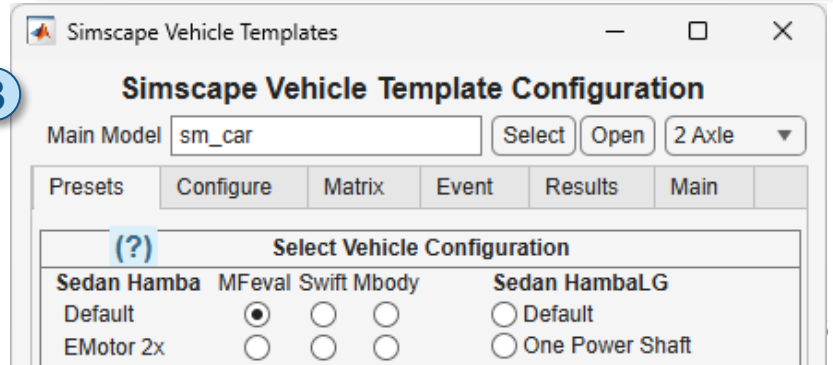
3. Eventually, app and model open





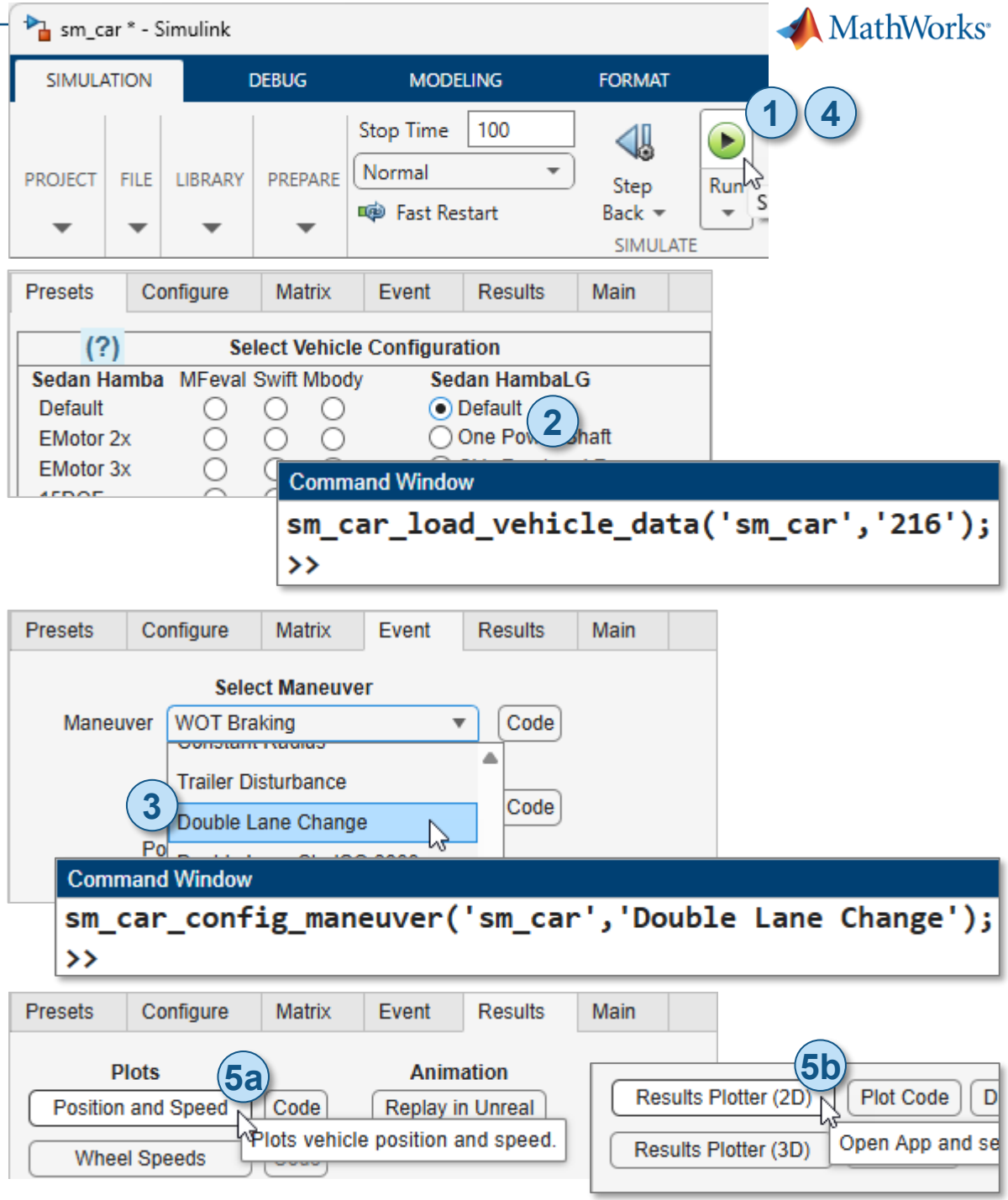




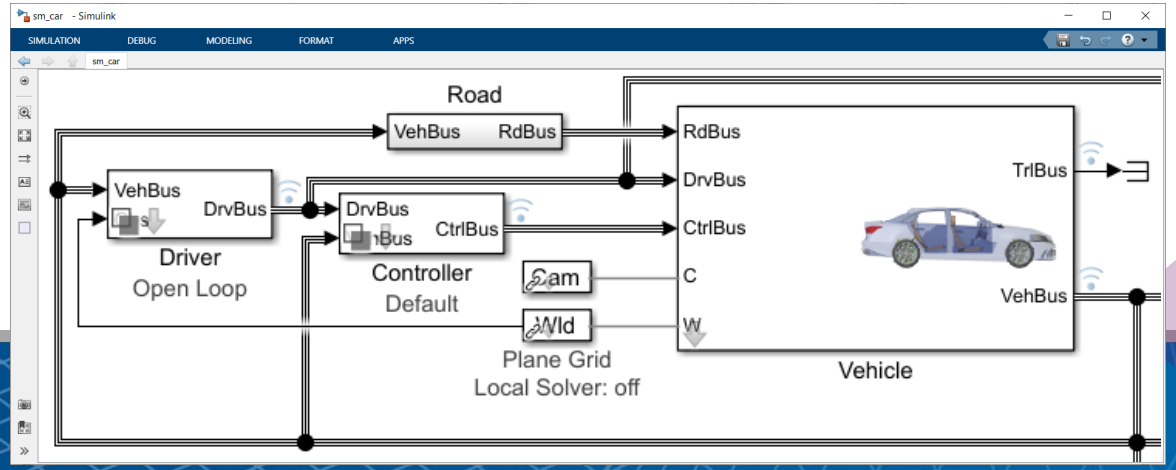


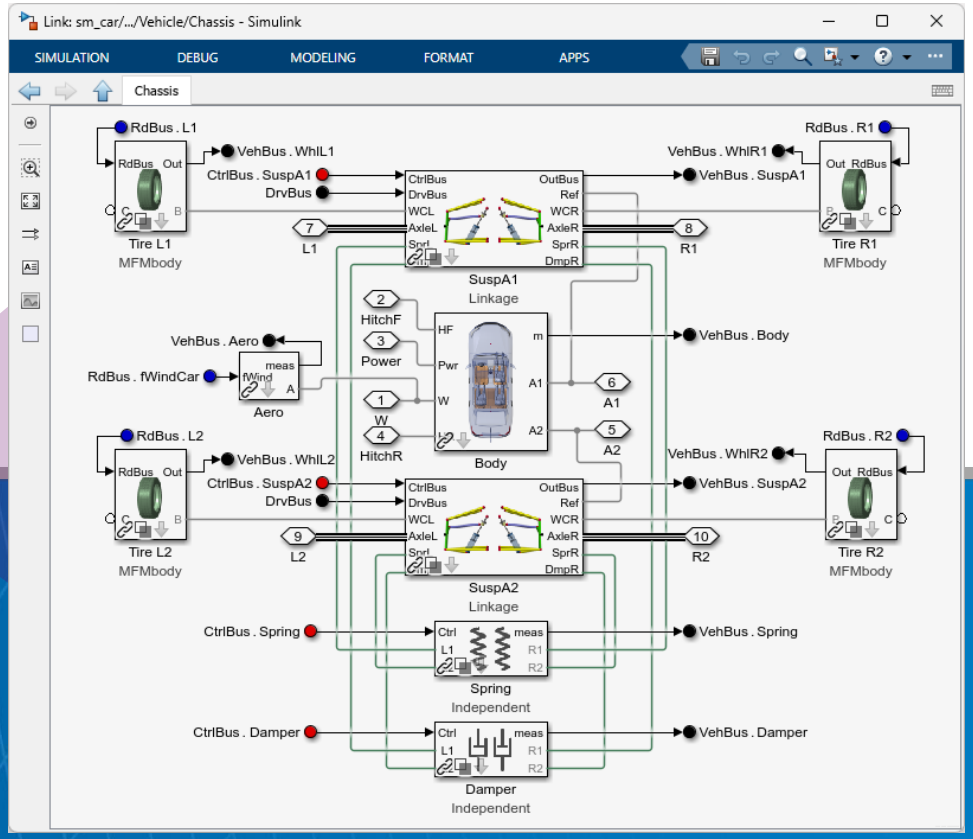
First Test: Double Lane Change

1. Run model – simple accel/decel test
2. In app on tab “Preset”, select “Default” under Sedan HambaLG
   - Command that the app executed is echoed to Command Window
3. In app on tab “Event”, select “Double Lane Change” for Maneuver
   - Command echoed to Command Window
4. Run model again
5. In app on tab “Results”
   a. Press the “Position and Speed” button
   b. Press the “Results Plotter (2D)” button



Simscape Vehicle Templates Overview

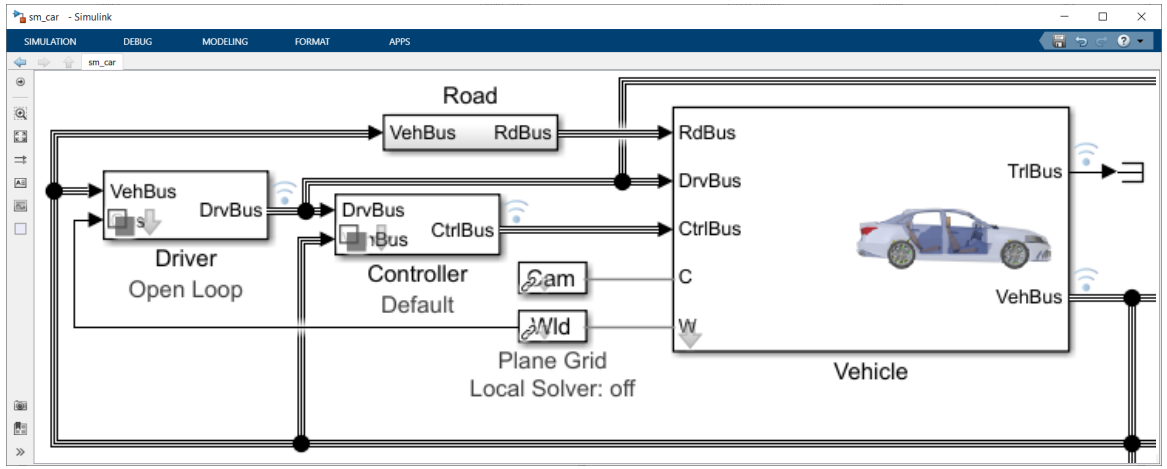


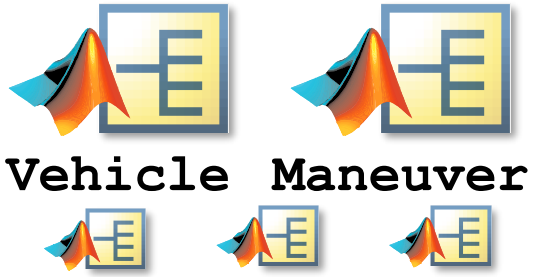


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Overview

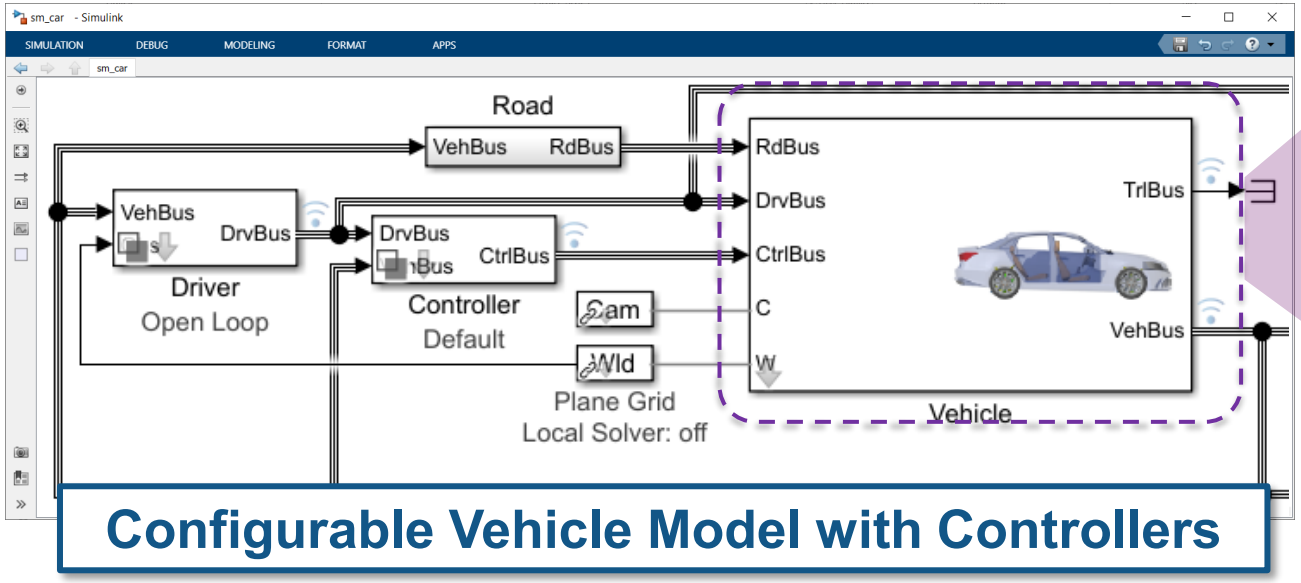
- Model Structure
- Parameterization
- Model Inputs
- Model Outputs
- Configuration App



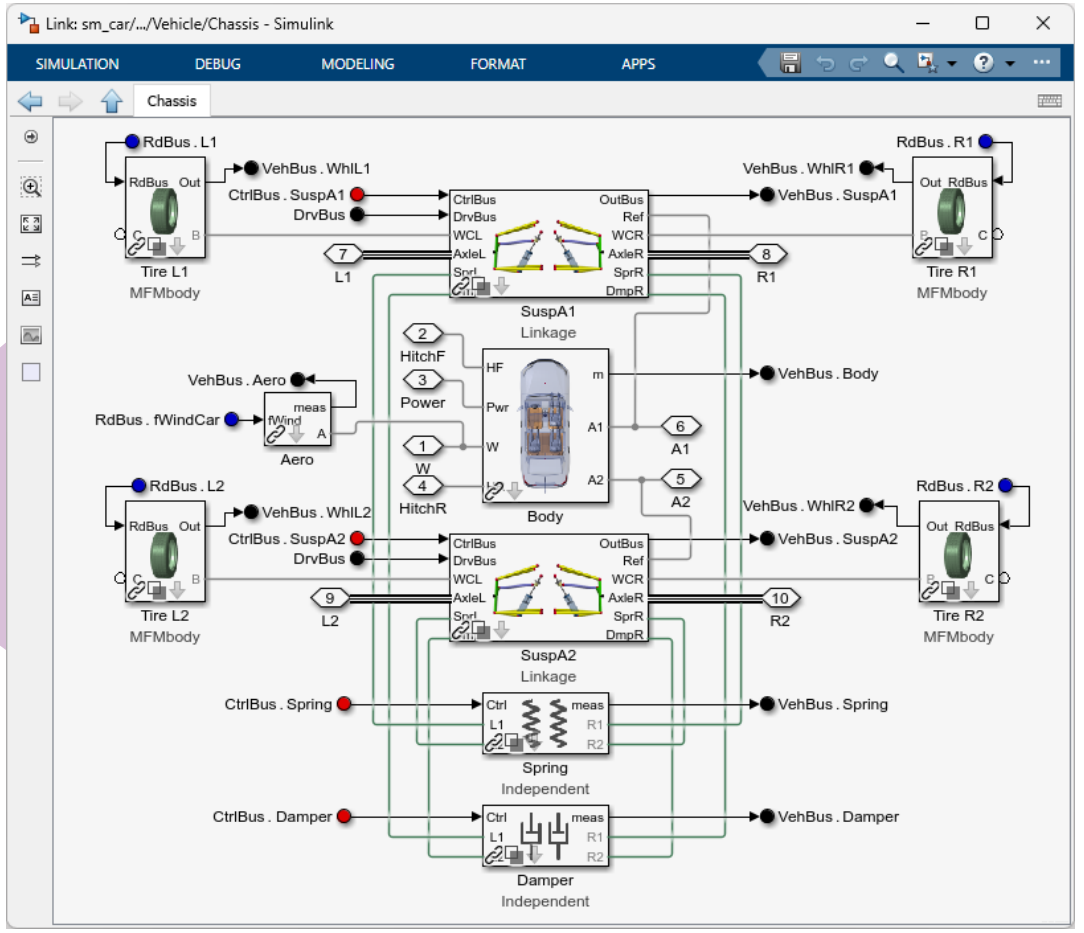


Model Structure

- Vehicle model with physical connections enables easy customization

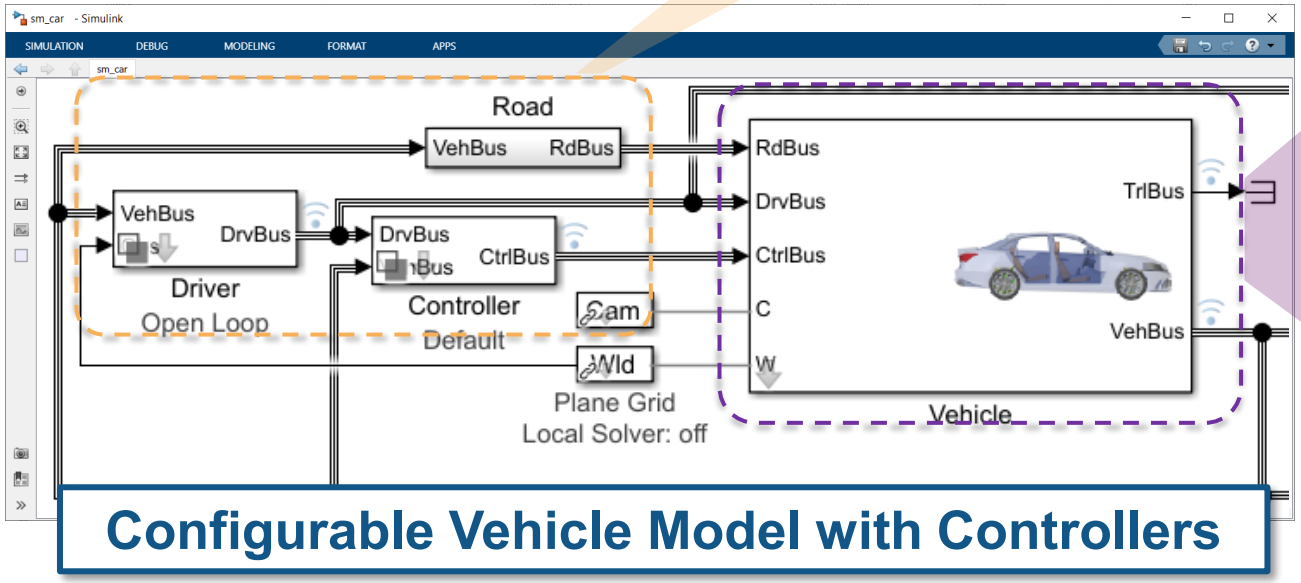


Configurable Vehicle Model with Controllers

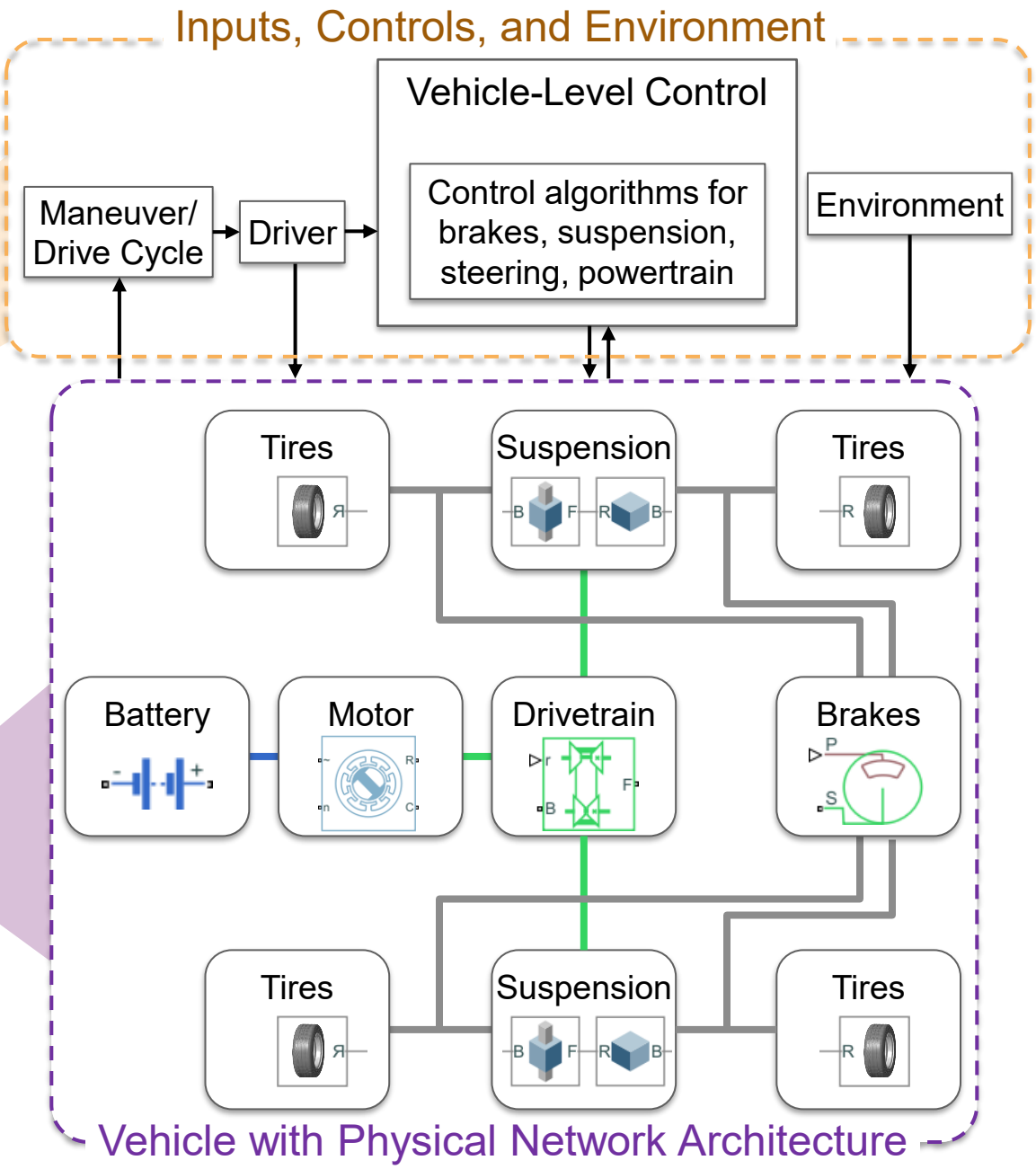


Model Structure

- Vehicle model with physical connections enables easy customization
- Integrated control algorithms and vehicle model in a single environment



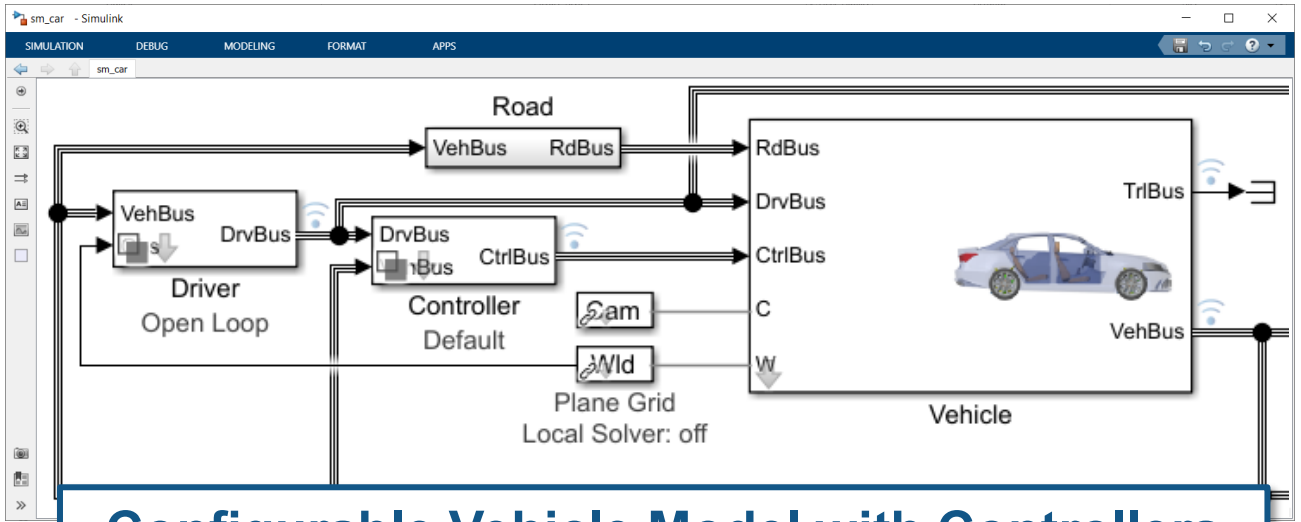
Configurable Vehicle Model with Controllers



Model Structure

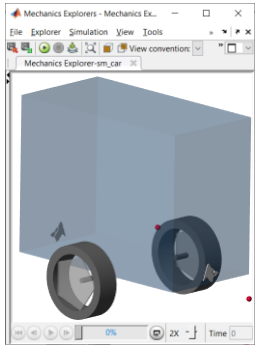
1 Axle
2 Axle

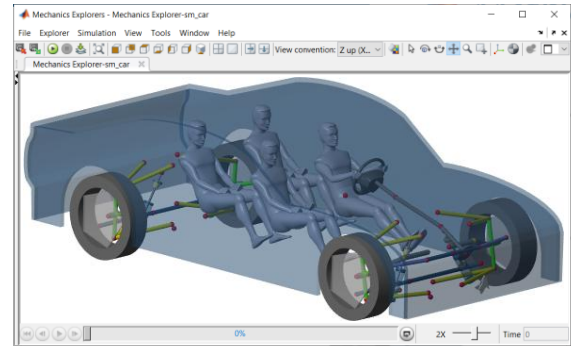
- Vehicle model with physical connections enables easy customization
- Integrated control algorithms and vehicle model in a single environment
- Parameterized and specific designs



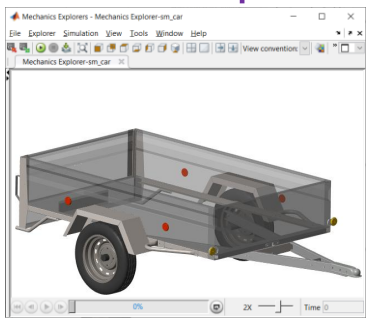
Configurable Vehicle Model with Controllers

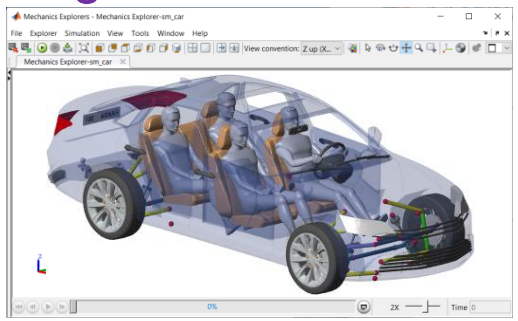
Parameterized Models

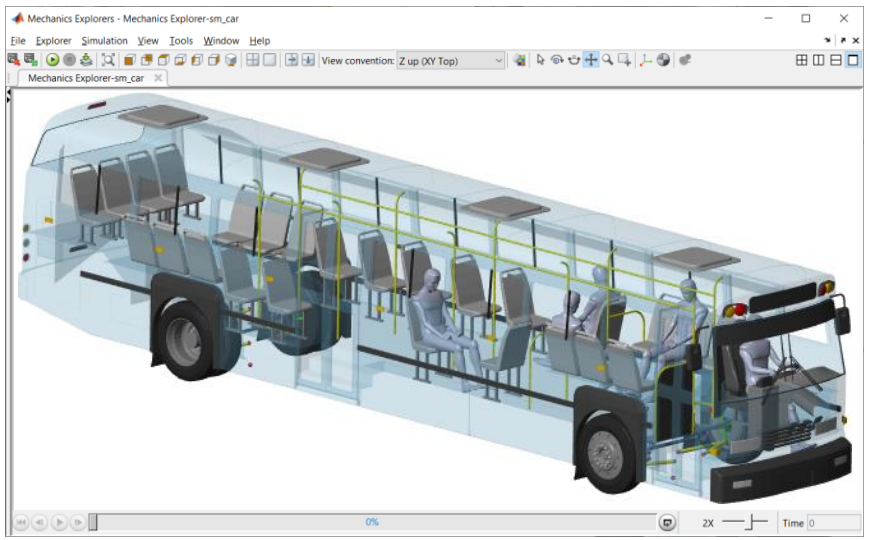




Specific Designs

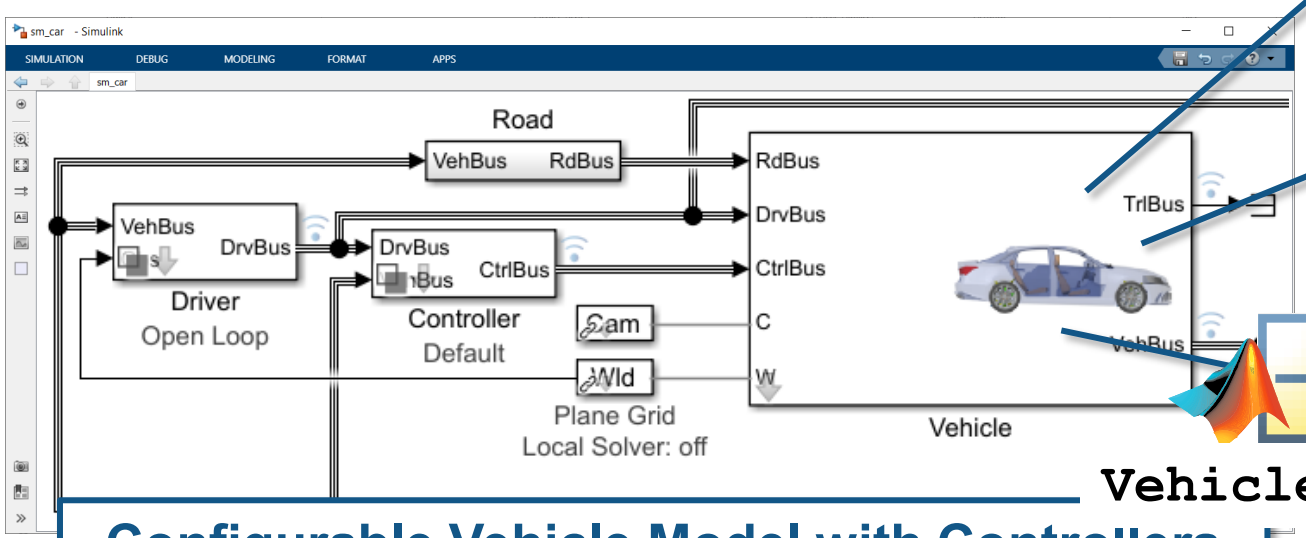




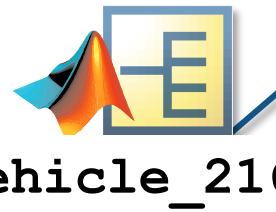


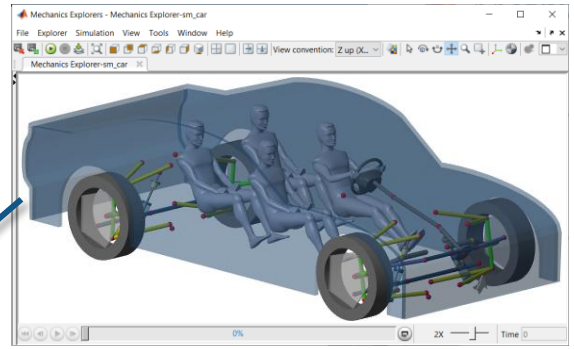
Parameterization

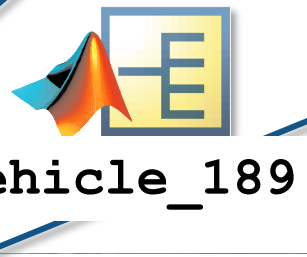
- Same model used for cars with different suspension, braking, etc.
- Vehicle data structure
  - Selects design and fidelity (double wishbone, etc.)
  - Sets parameter values (length, stiffness, etc.)

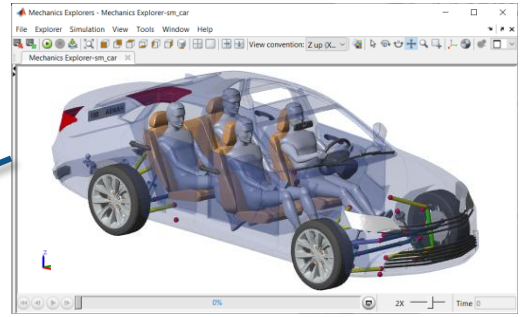


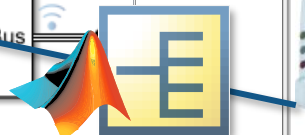
Configurable Vehicle Model with Controllers

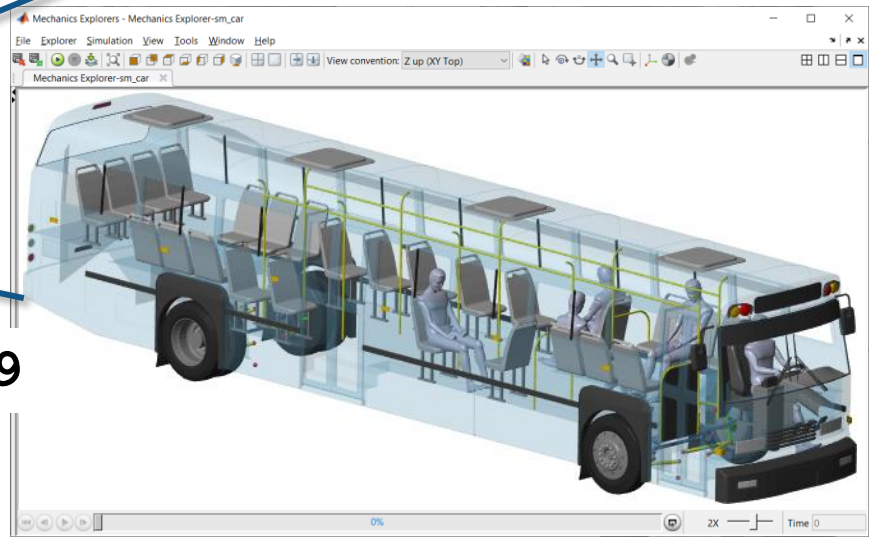






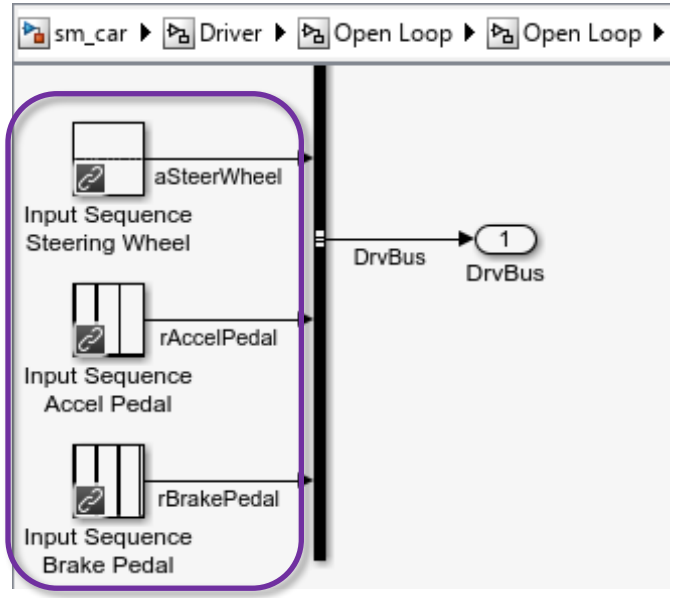


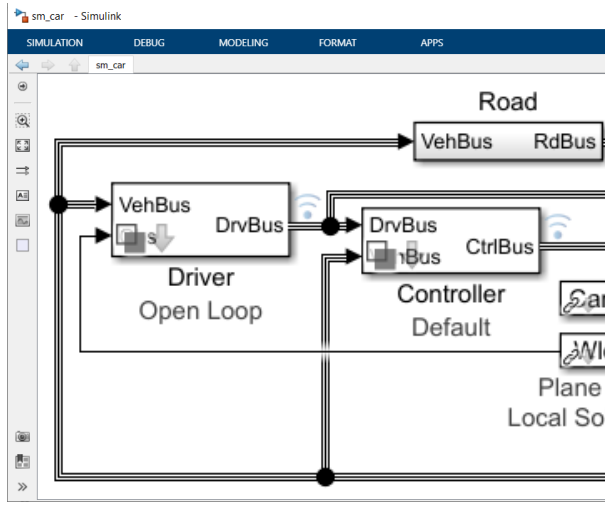




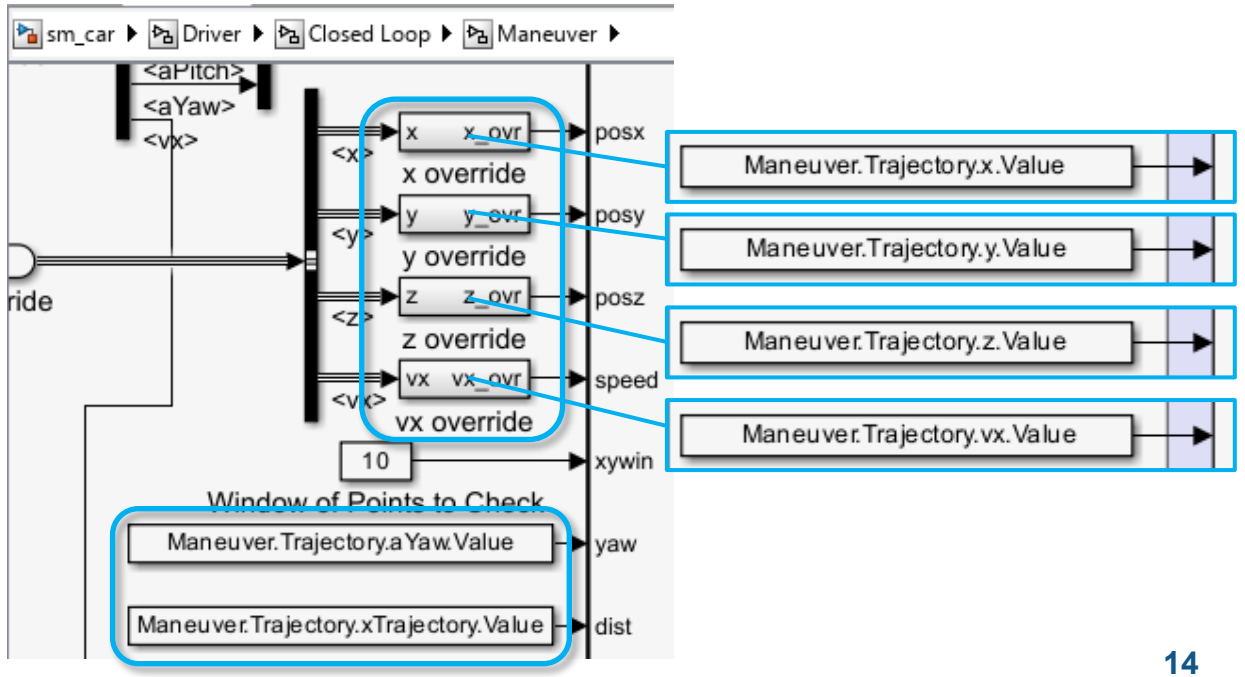
Model Inputs

- Open Loop Tests: **Driver Actions**
  - Steering wheel angle
  - Accelerator pedal position
  - Brake pedal position





- Closed Loop Test: **Driver Commands**
  - Target pose (x, y, z position, yaw angle) vs. distance along path
  - Driver model compares target pose to current vehicle pose and determines driver actions



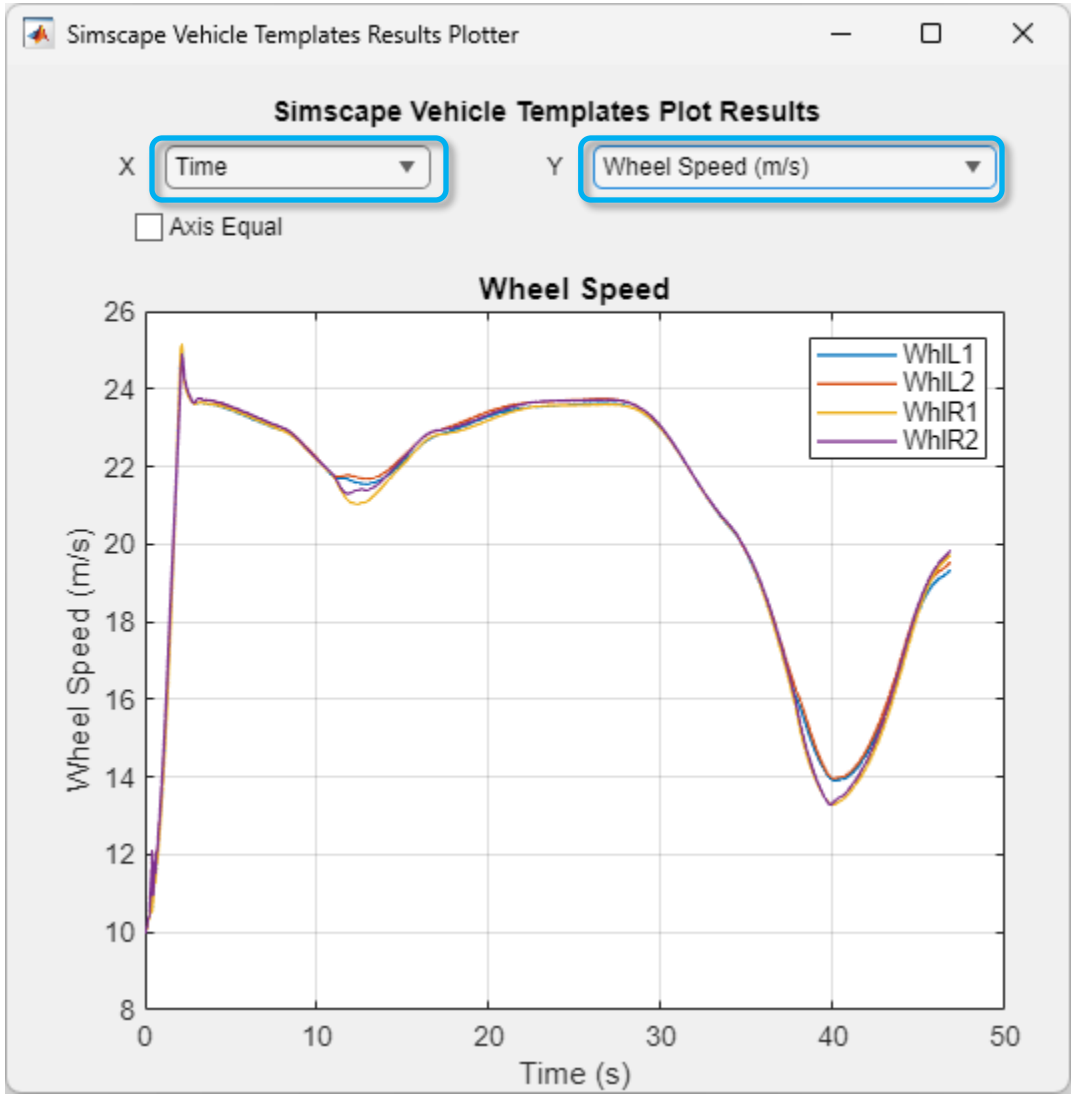
Model Outputs: Tools for 2D Plots

- A simple UI lets you select quantities to plot from dropdown menus

- Commands echoed to the Command Window help you generate plots outside of the App

```
sm_car_sim_res_plot('time','vWhl')
```

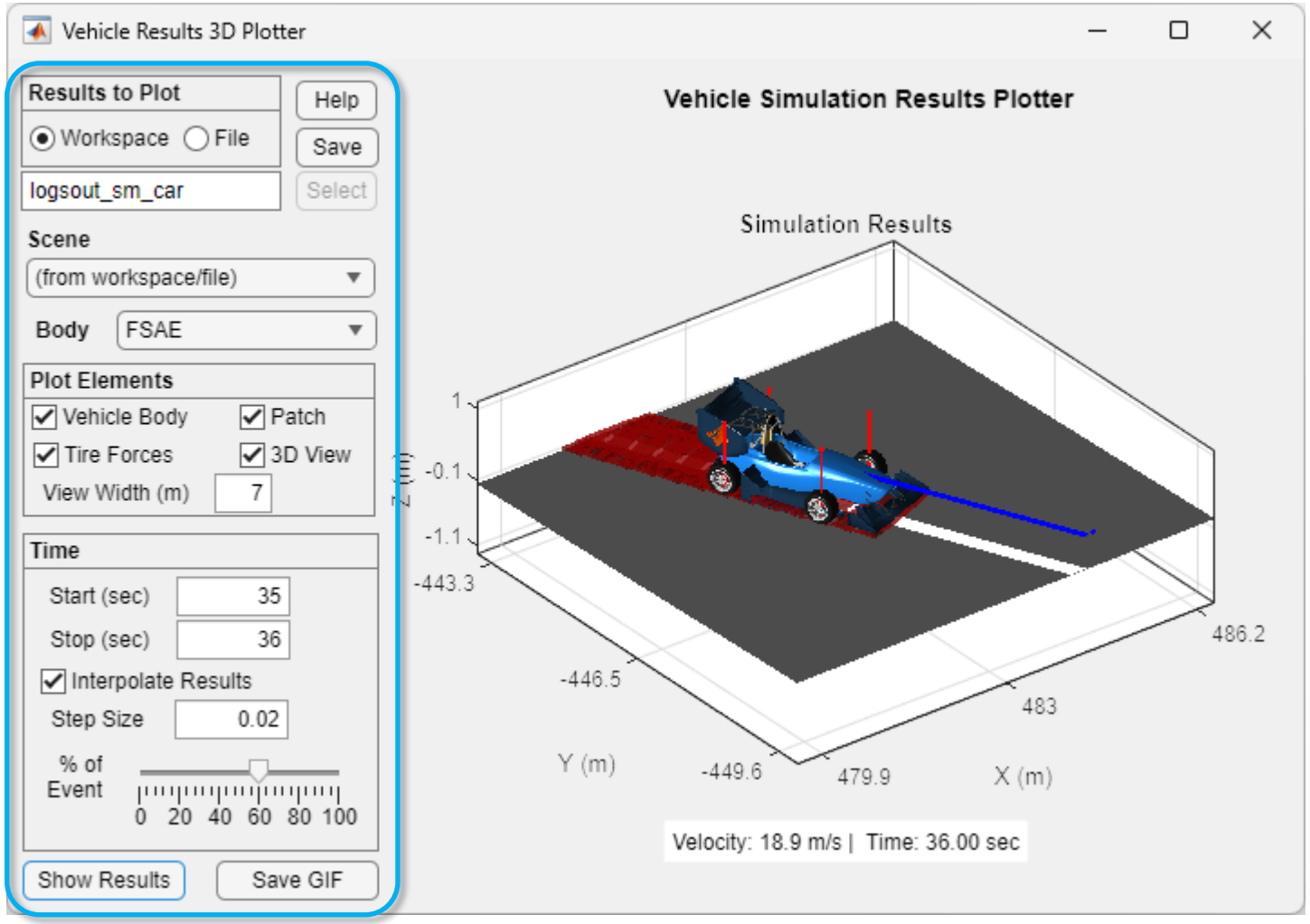
- A separate function helps you extract data from simulation results for custom plots




```
time = sm_car_sim_res_get(logsout_sm_car,simlog_sm_car,Vehicle,'time');
vWhl = sm_car_sim_res_get(logsout_sm_car,simlog_sm_car,Vehicle,'vWhl');
```

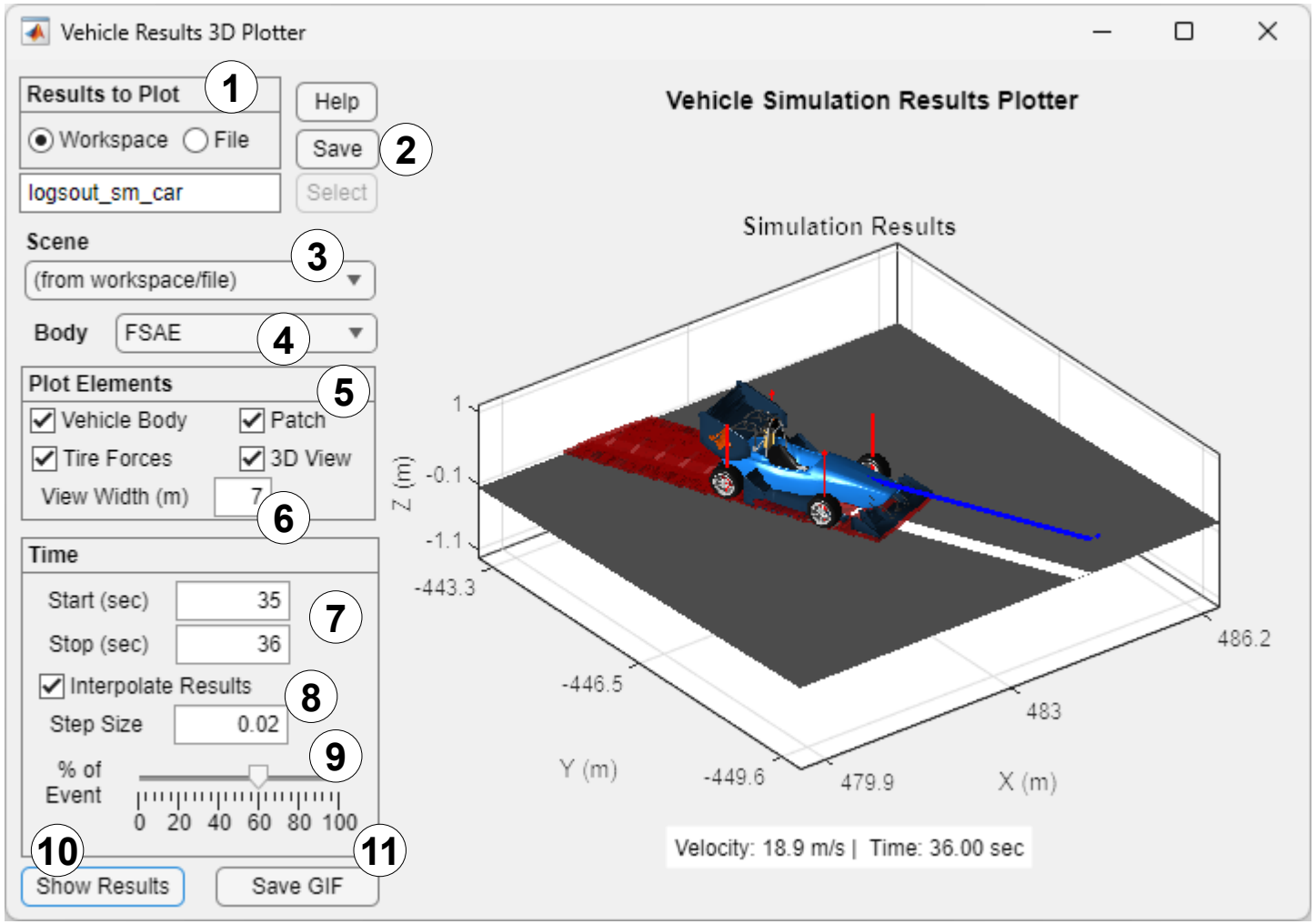
Tools for 3D Plots

- Plot or animate results
  - From a specific time window
  - Chassis and wheel positions
  - Tire forces
  - Vehicle velocity vector
- Save results for later use
  - Data or as an animated GIF
- Commands echoed to command window help you create your own plots



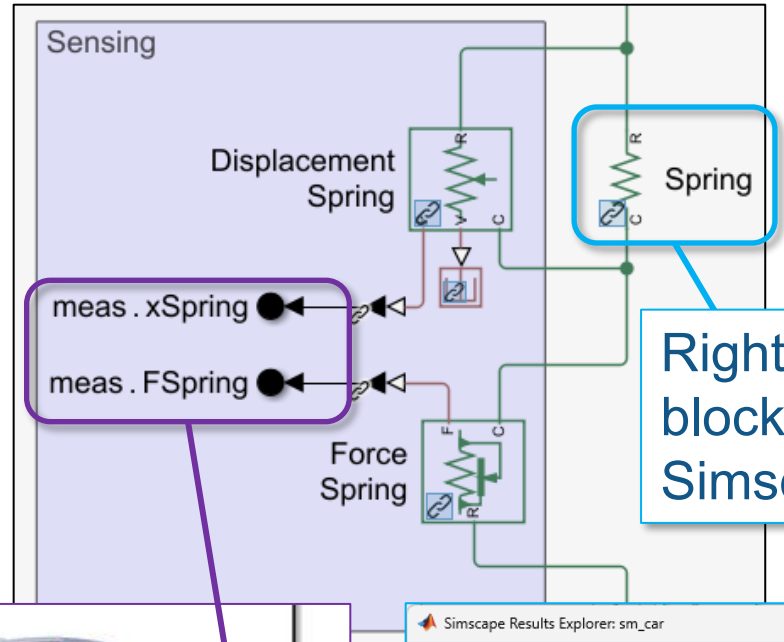
Tools for 3D Plots

1. Plot results from workspace or saved to a file
2. Save current results to file
3. Select scene or read from saved data
4. Select body visualization for plot
5. Select elements to view
6. Set length of plot area
7. Set time window to plot/animate
8. Interpolate results (good for animations)
9. Use slider to select time to plot
10. Plot/animate results
11. Save results to animated GIF

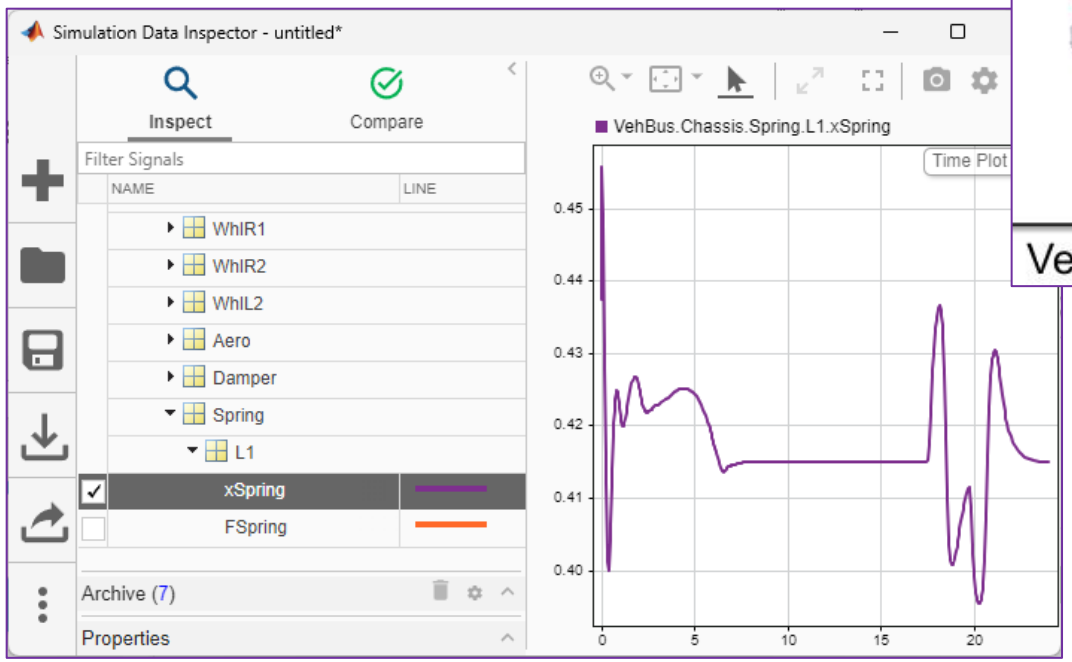


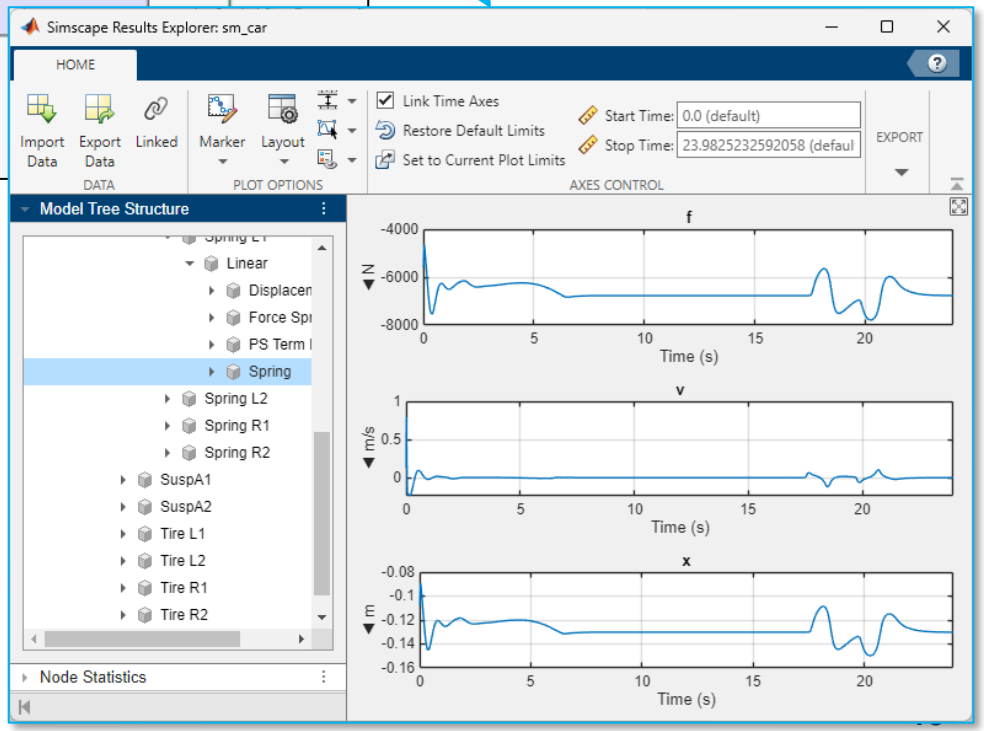
Interactive Investigation

- Access **results from Simscape** blocks via right-click in diagram
- Access **results from Simulink** Data Inspector for logged data



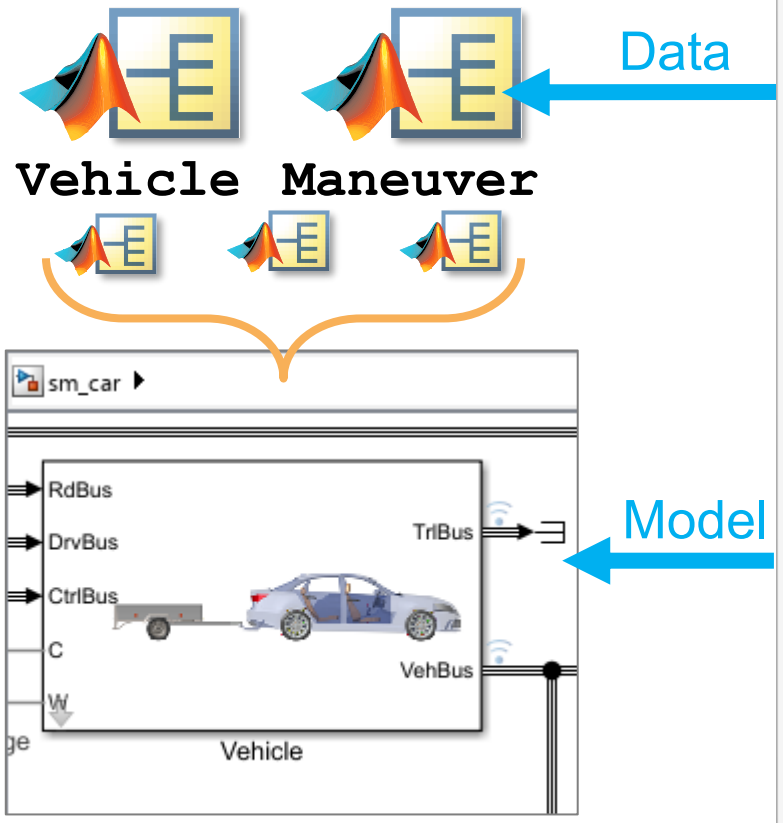
Right-click on any Simscape block to view results in the Simscape Results Explorer



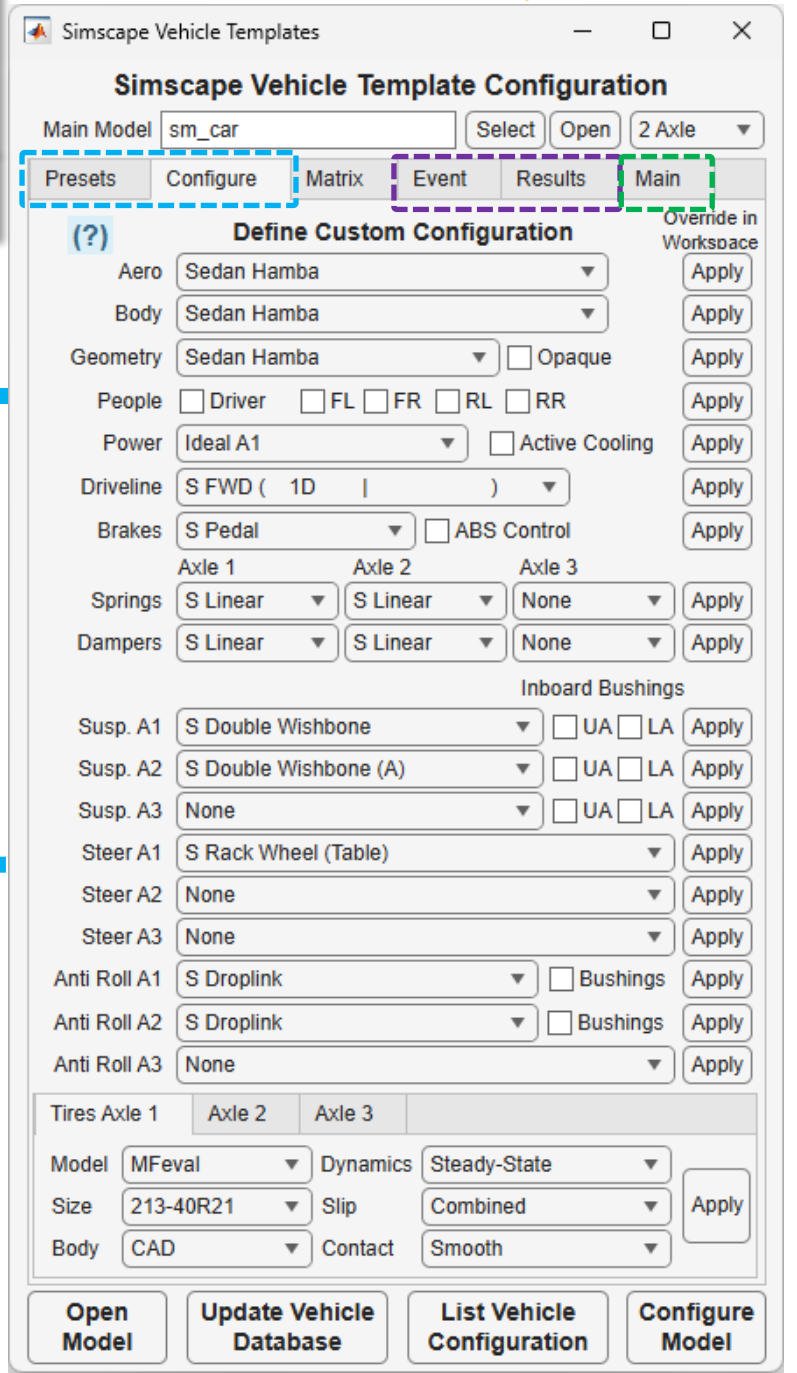


Configuration App

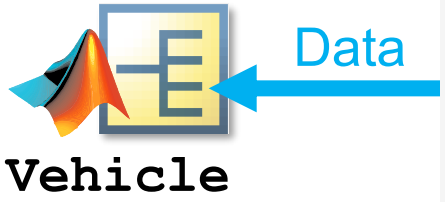
- App adjusts data in MATLAB workspace and changes model settings
  - Configure model
  - Select maneuver
- Call code to plot results
- Commands to do actions taken in App are shown in Command Window




```
Command Window
sm_car_config_maneuver('sm_car','Double Lane Change');
>>
```



Configuration App
Preset Tab

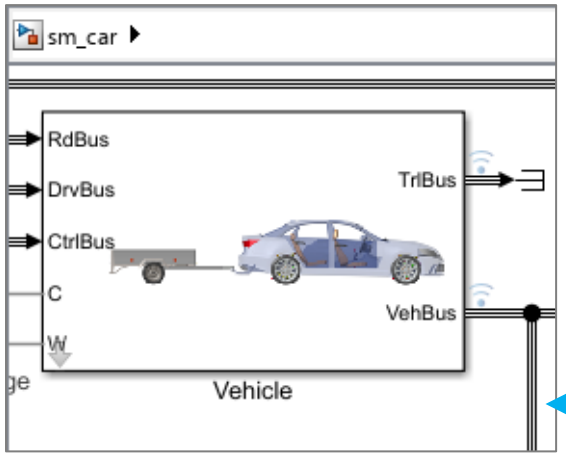


Data

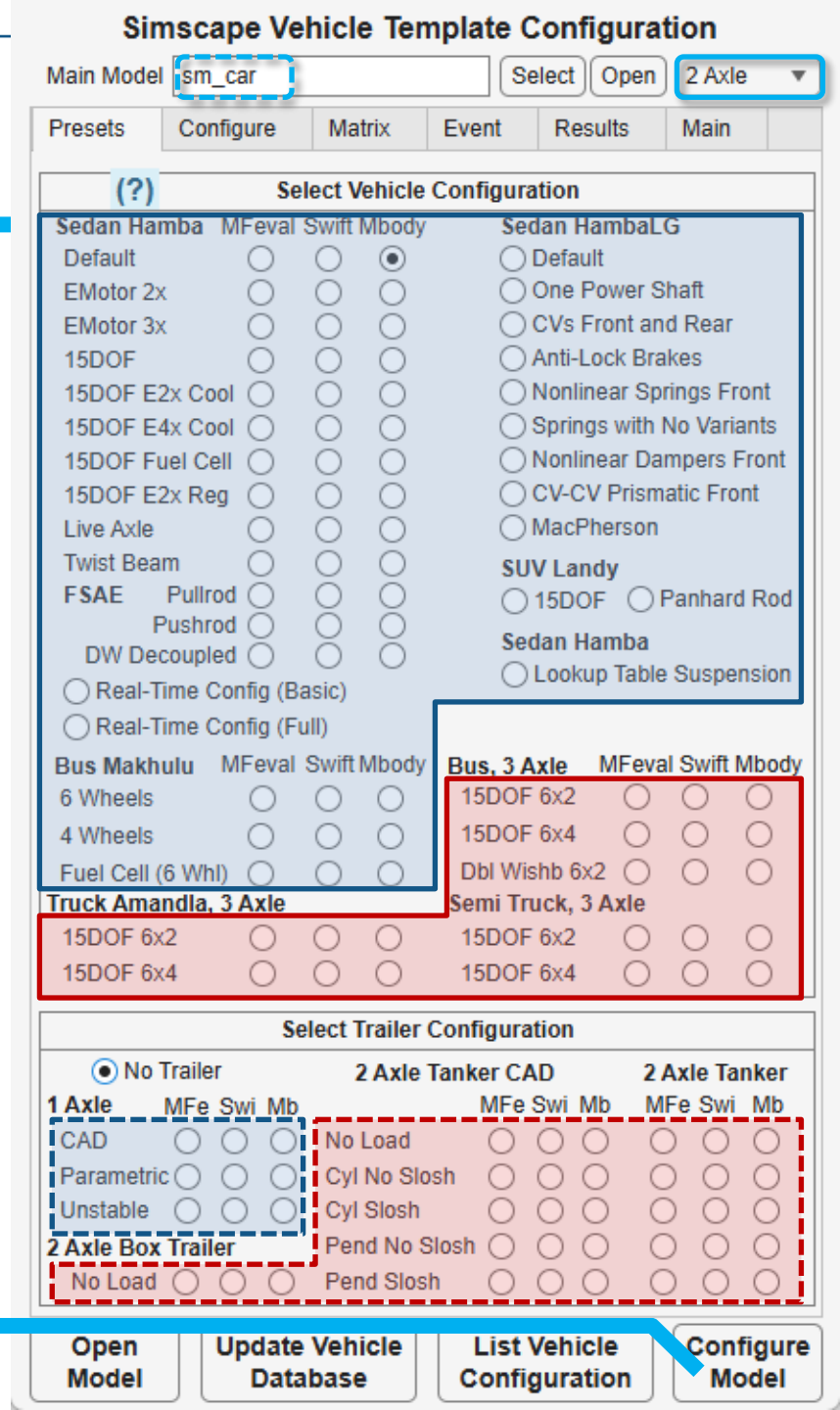
- Radio buttons load preset from .mat file
  - 2-axle vehicle and 1-axle trailer (sm_car.slx)
  - 3-axle vehicle and 2-axle trailer (sm_car_Axle3.slx)
  - Choose model from dropdown
- Radio button adjusts data
- Configure Model button updates model (optional)
  - Can simply run model

1 Axle Trailer — 2 Axle Vehicle

2 Axle Trailer — 3-Axle Vehicle

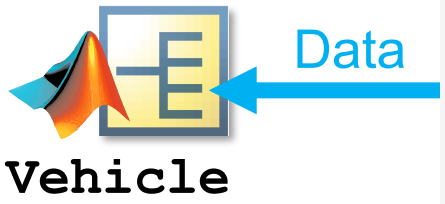


Model



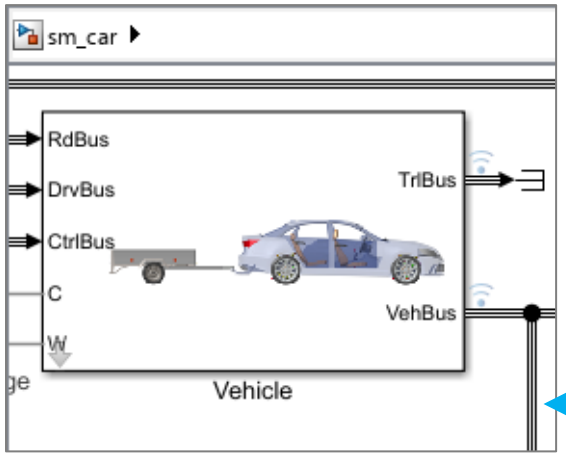
Configuration App
Matrix Tab

- Presets for 2-axle vehicle
  - Primarily for testing different configurations



Data

2 Axle Vehicle



Model

Simscape Vehicle Template Configuration

Main Model: sm_car | Select | Open | 2 Axle

Presets | Configure | Matrix | Event | Results | Main

Select a radio button below for a specific driveline, tire, and suspension preset. For testing specific configurations.

(?)

Configurations

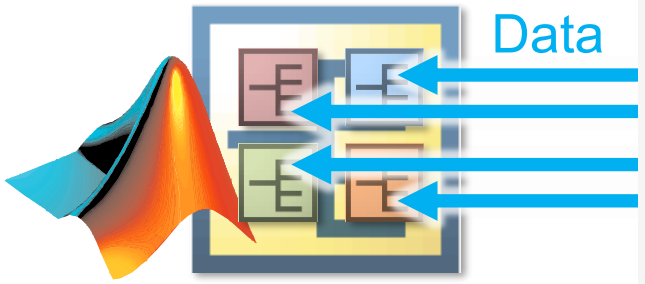
MS1	000	016	032	048	064	080	096	112
MSD	001	017	033	049	065	081	097	113
MS3	002	018	034	050	066	082	098	114
MSF	003	019	035	051	067	083	099	115
ML1	004	020	036	052	068	084	100	116
MLD	005	021	037	053	069	085	101	117
ML3	006	022	038	054	070	086	102	118
MLF	007	023	039	055	071	087	103	119
SS1	008	024	040	056	072	088	104	120
SSD	009	025	041	057	073	089	105	121
SS3	010	026	042	058	074	090	106	122
SSF	011	027	043	059	075	091	107	123
SL1	012	028	044	060	076	092	108	124
SLD	013	029	045	061	077	093	109	125
SL3	014	030	046	062	078	094	110	126
SLF	015	031	047	063	079	095	111	127
	DW	DWa	S2LAF	S2LAR	5S2LAF	5S2LAR	5CS2LAF	15DOF

Row abbreviation indicates tire and driveline configuration

MF / Swift Steady State / Linear Transient 1D/1D3D/3D/3DFlex

Open Model | Update Vehicle Database | List Vehicle Configuration | Configure Model

Configuration App
Configure Tab

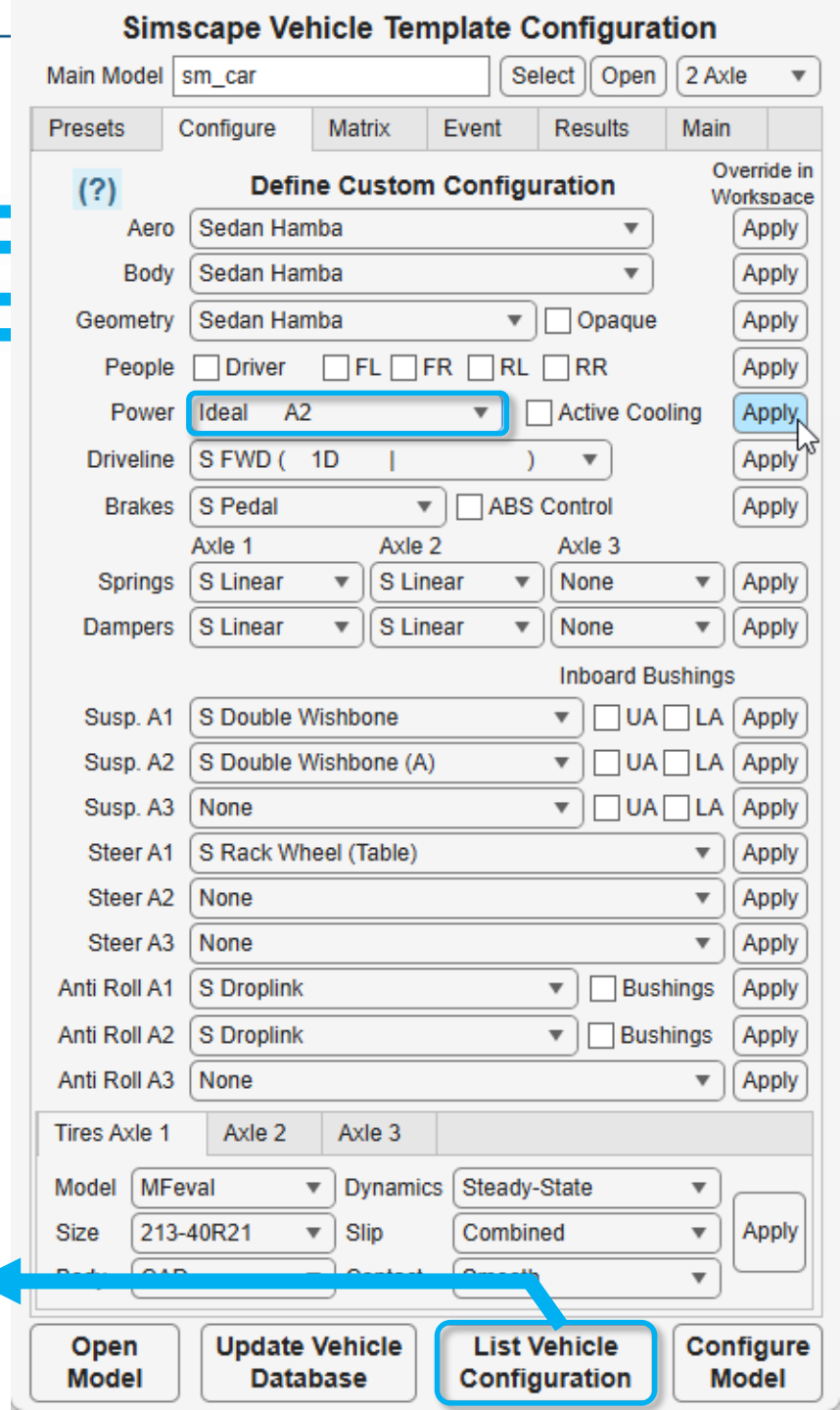


- Adjusts vehicle configuration
  - Overwrites portion of `Vehicle` pulling data from `VDatabase`
  - Select option, then press Apply. Updates `Vehicle` with new data
  - Commands echoed to command window

```
Command Window
Vehicle = sm_car_vehcfg_setPower(Vehicle,'Ideal_A2_default');
Vehicle = sm_car_vehcfg_setPowerCooling(Vehicle,'None');
```

  - Configure Model button updates model (optional)
- List Vehicle Configuration button prints current configuration to Command Window

```
Command Window
                Aero: 'sedan'
                Body: 'Body_Axle2'
        BodyGeometry: 'CAD_Sedan_Hamba'
               SuspA1: 'DoubleWishbone_Sedan
SuspA1_AntiRollBar: 'DroplinkRod_Sedan_H
           SuspA1_Steer: 'RackWheel'
               SuspA2: 'DoubleWishboneA_Seda
```



Configuration App
Event Tab

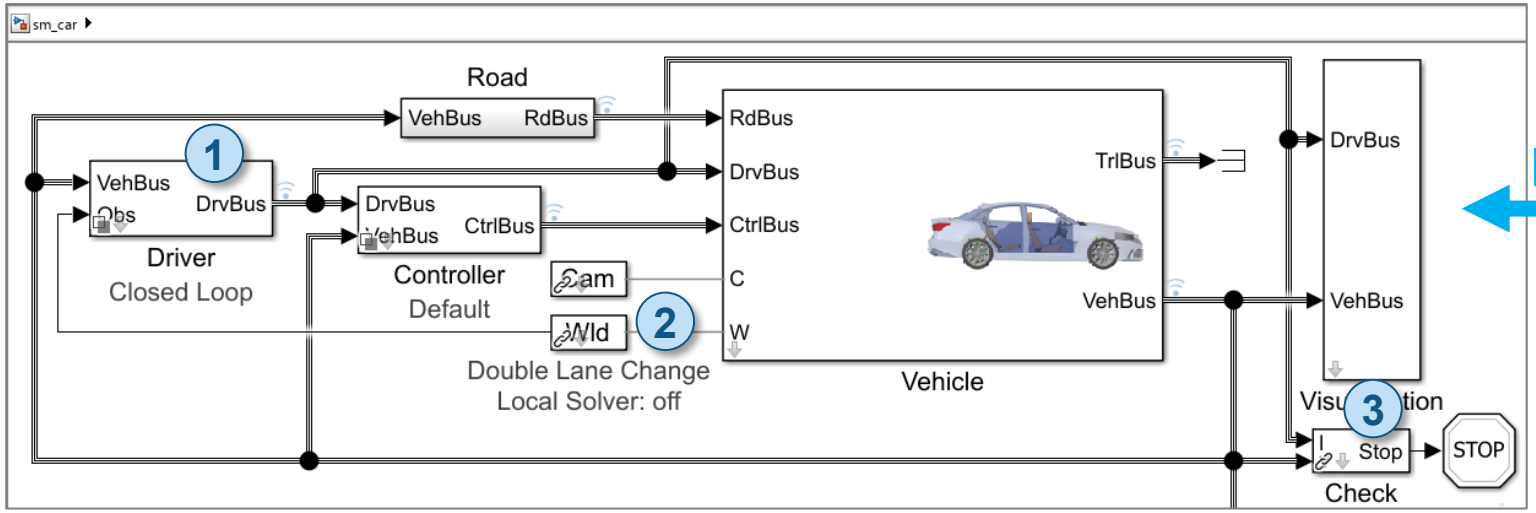


Data

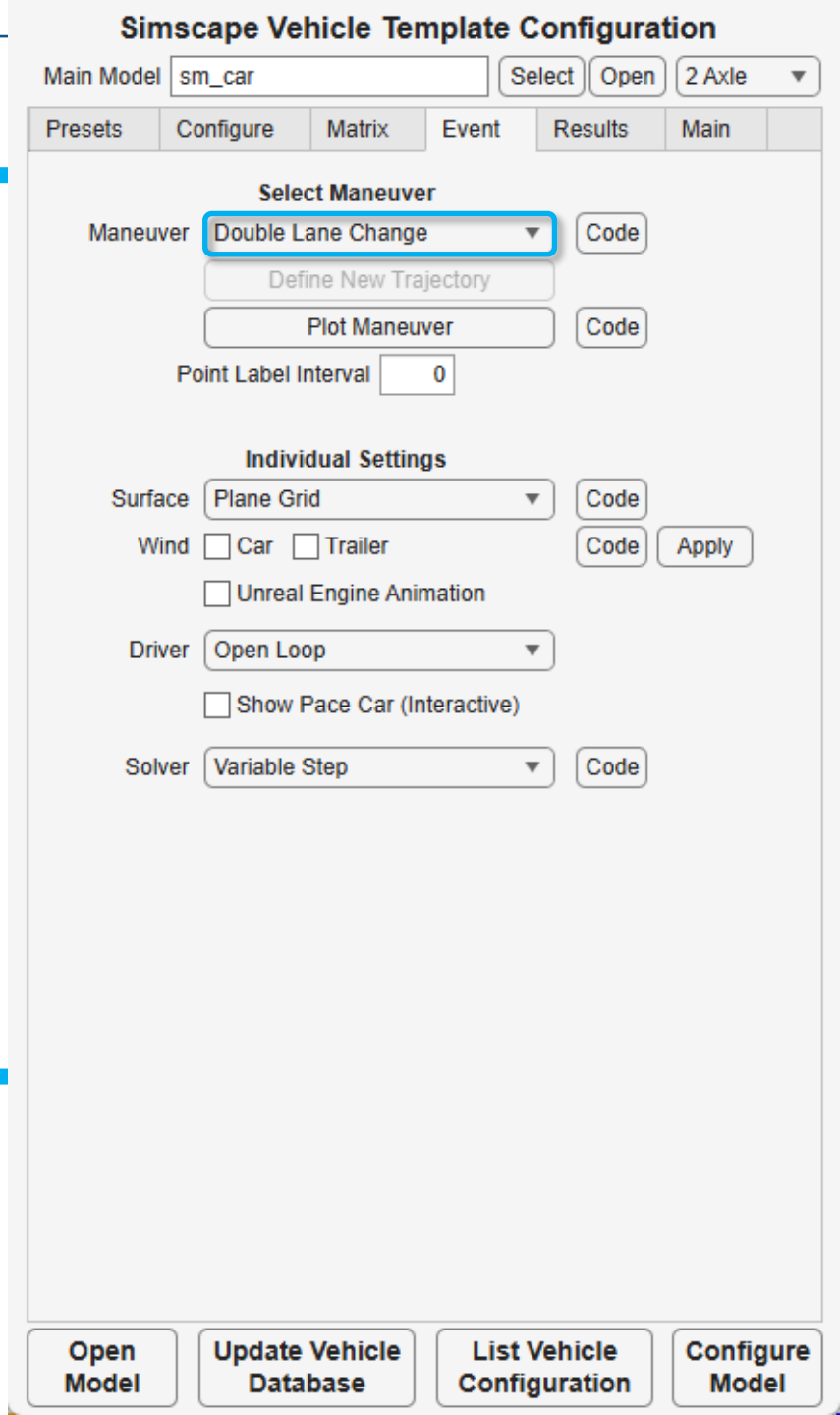
- Selects maneuver
  - Overwrites `Maneuver` and `Init`
  - Commands echoed to command window

```
Command Window
sm_car_config_maneuver('sm_car','Double Lane Change');
```

  - Adjust (1) driver model, (2) scene, and (3) stop conditions in model



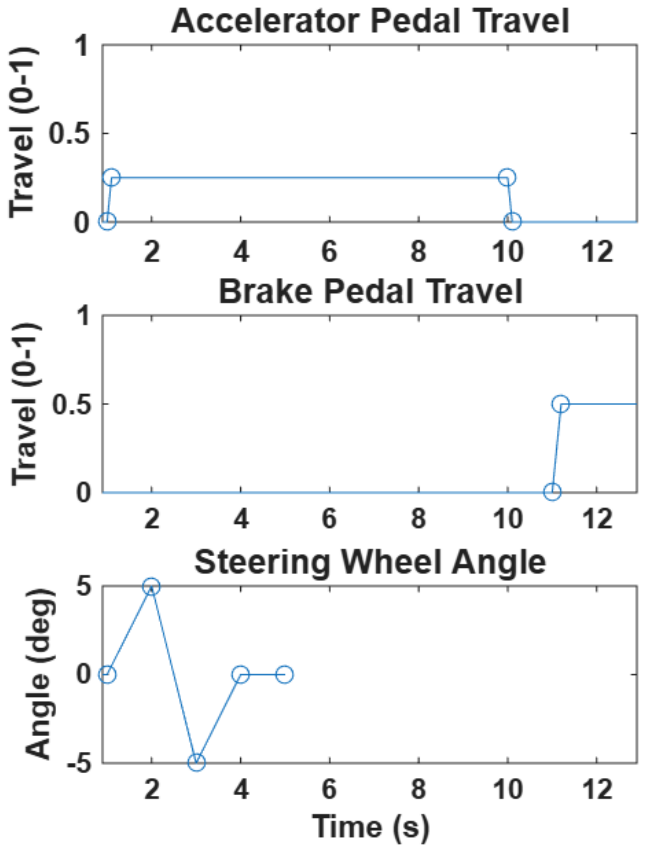
Model



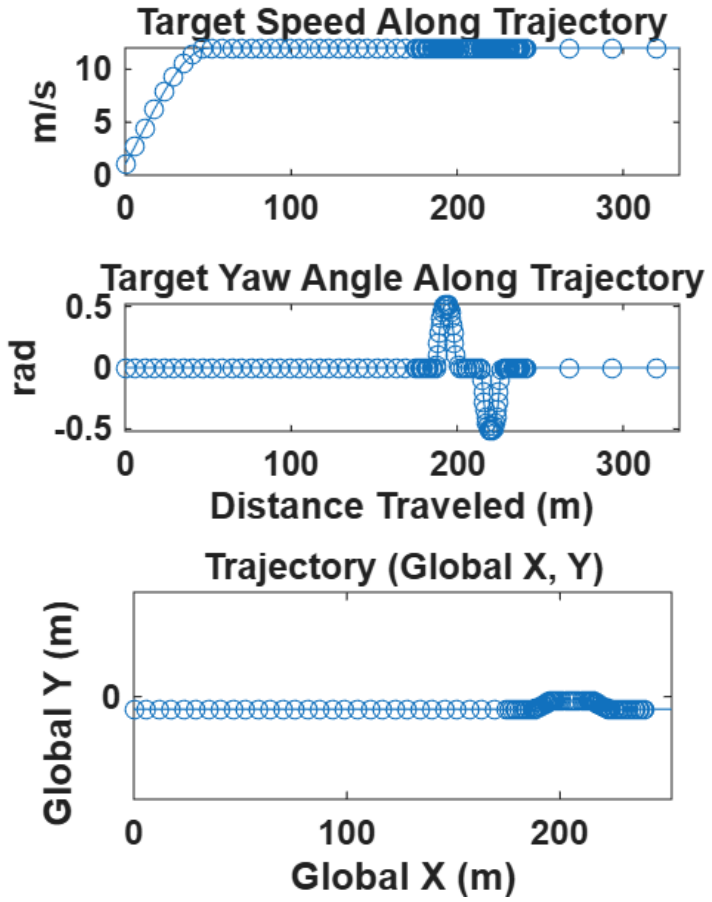
Configuration App

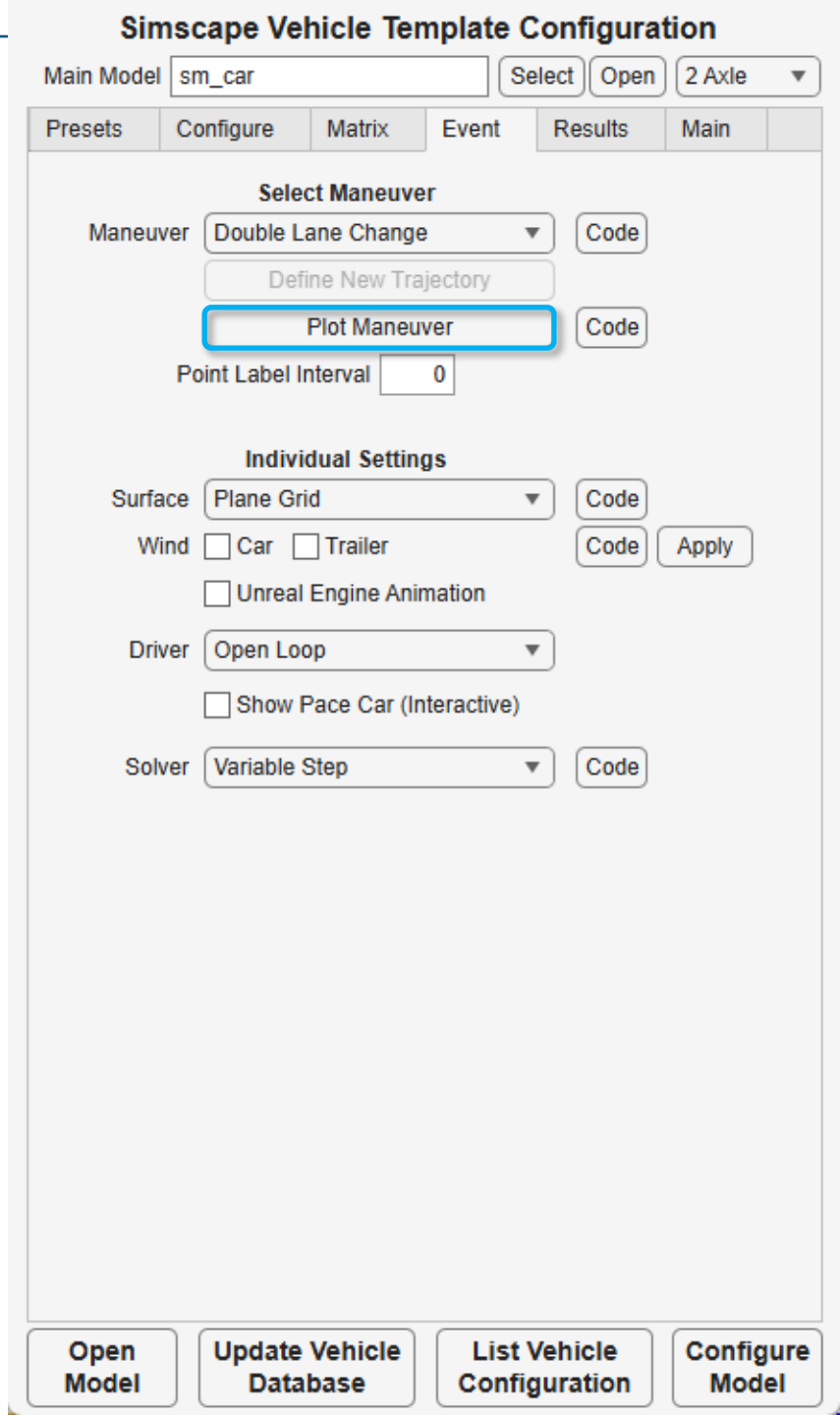
Event Tab, Plot Maneuver

- Driver actions for open-loop maneuvers



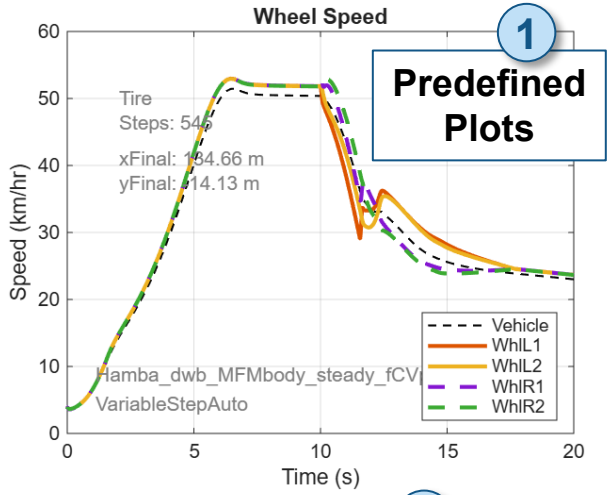
- Driver commands for closed-loop maneuvers



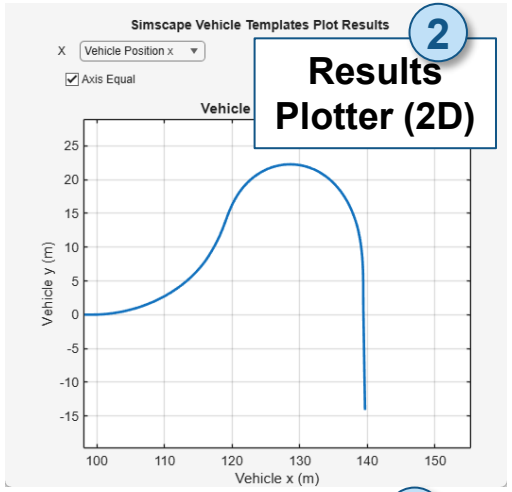


Configuration App
Results Tab

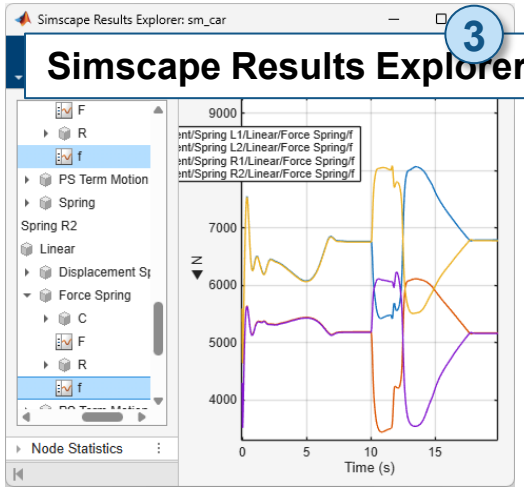
- Predefined plots, interactive plotting tools, and code



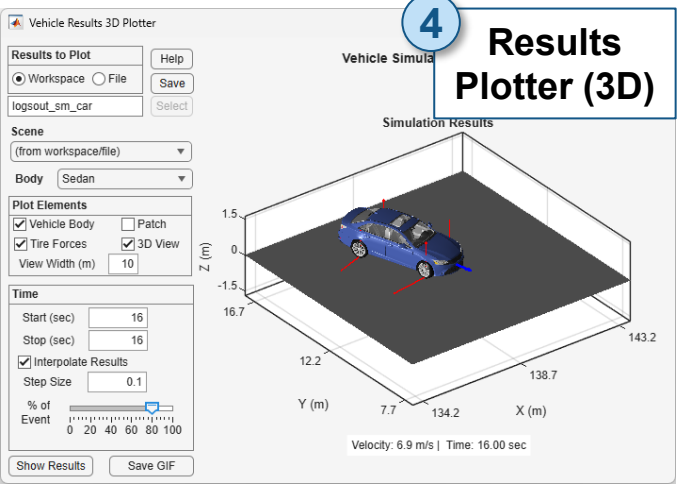
1 **Predefined Plots**



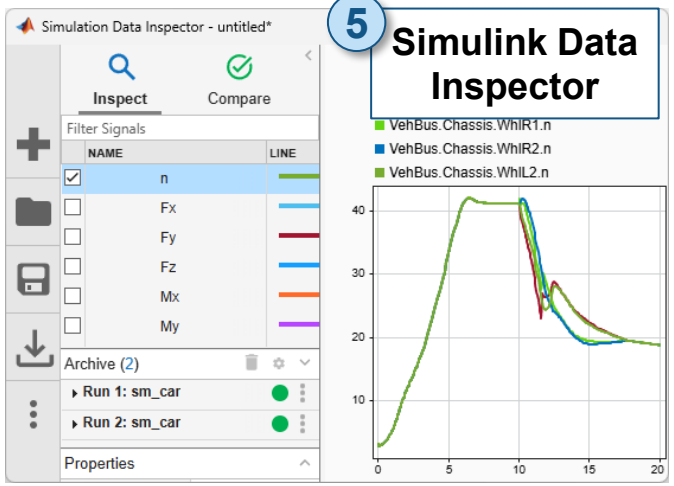
2 **Results Plotter (2D)**



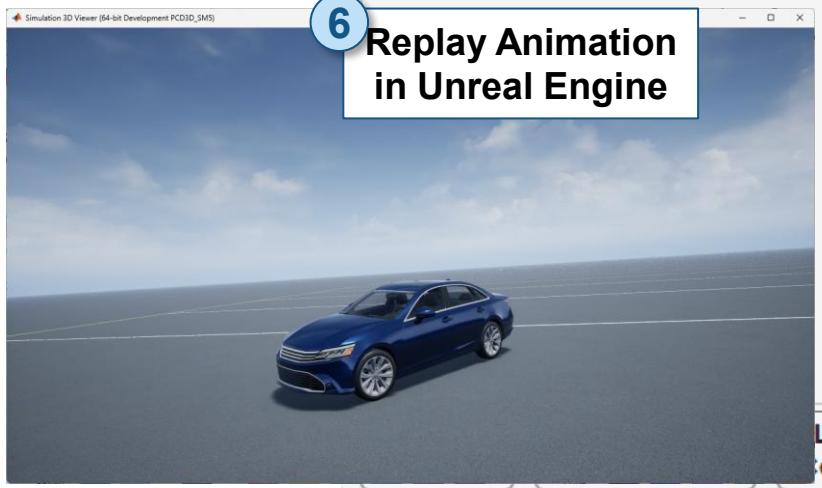
3 **Simscape Results Explorer**



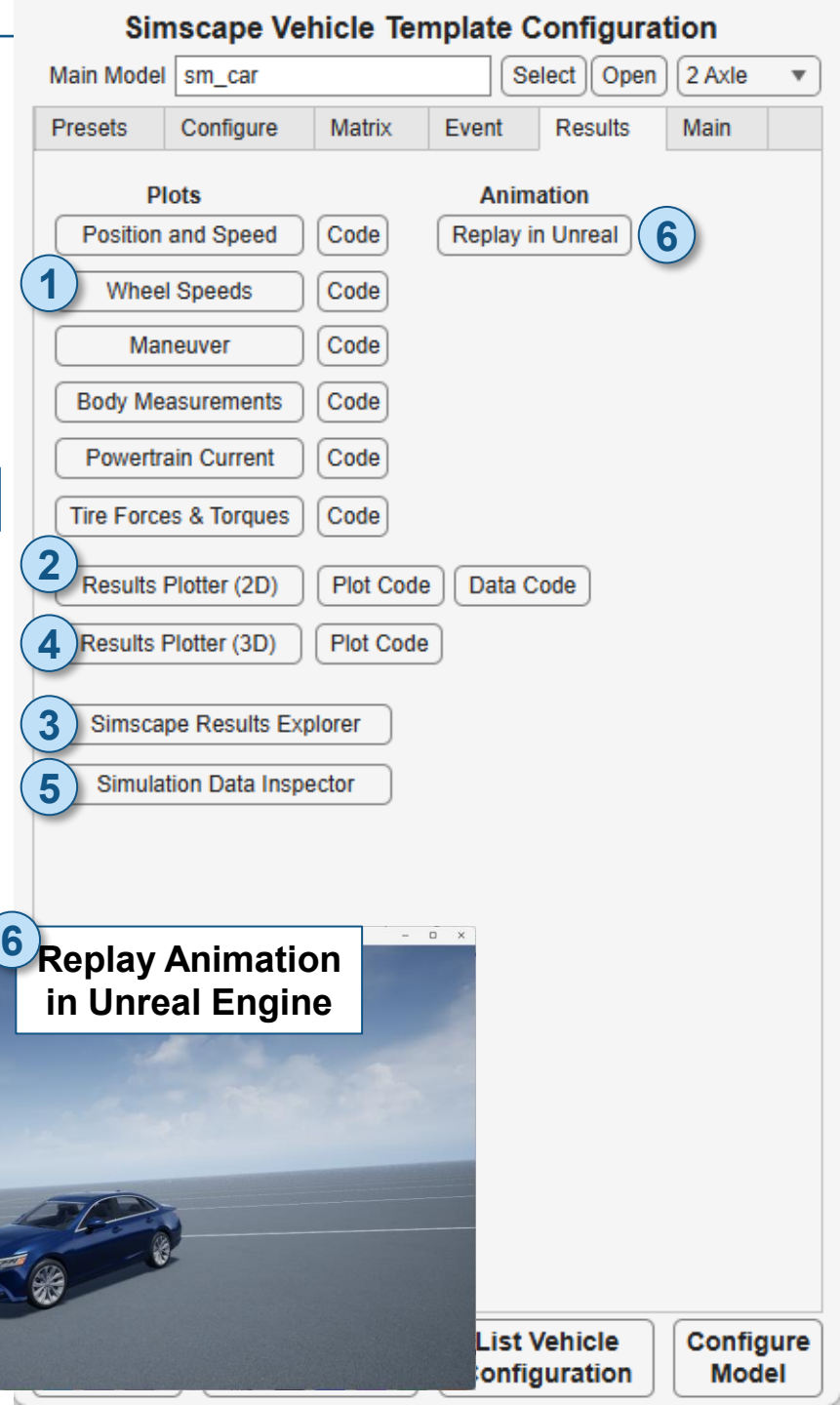
4 **Results Plotter (3D)**



5 **Simulink Data Inspector**

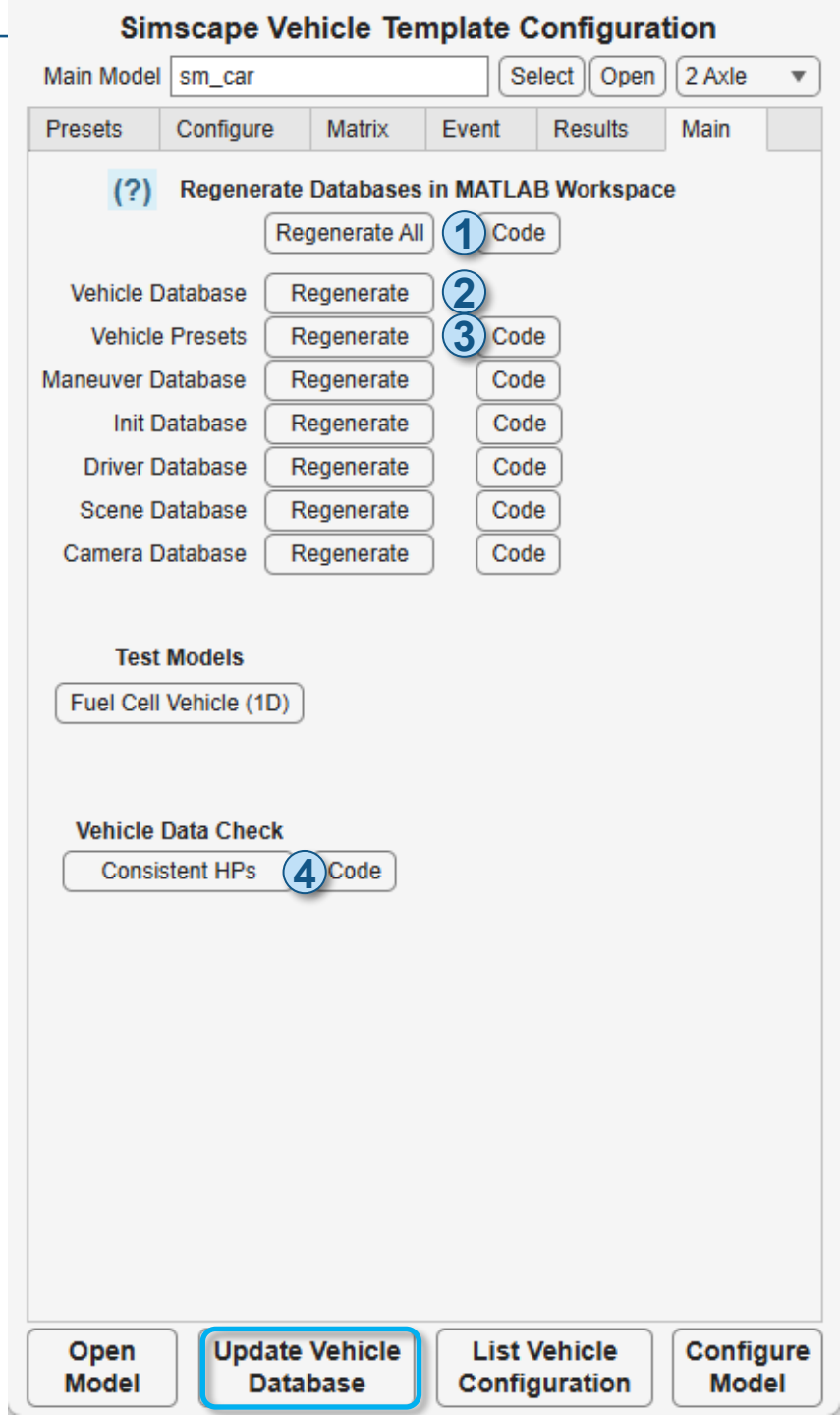


6 **Replay Animation in Unreal Engine**



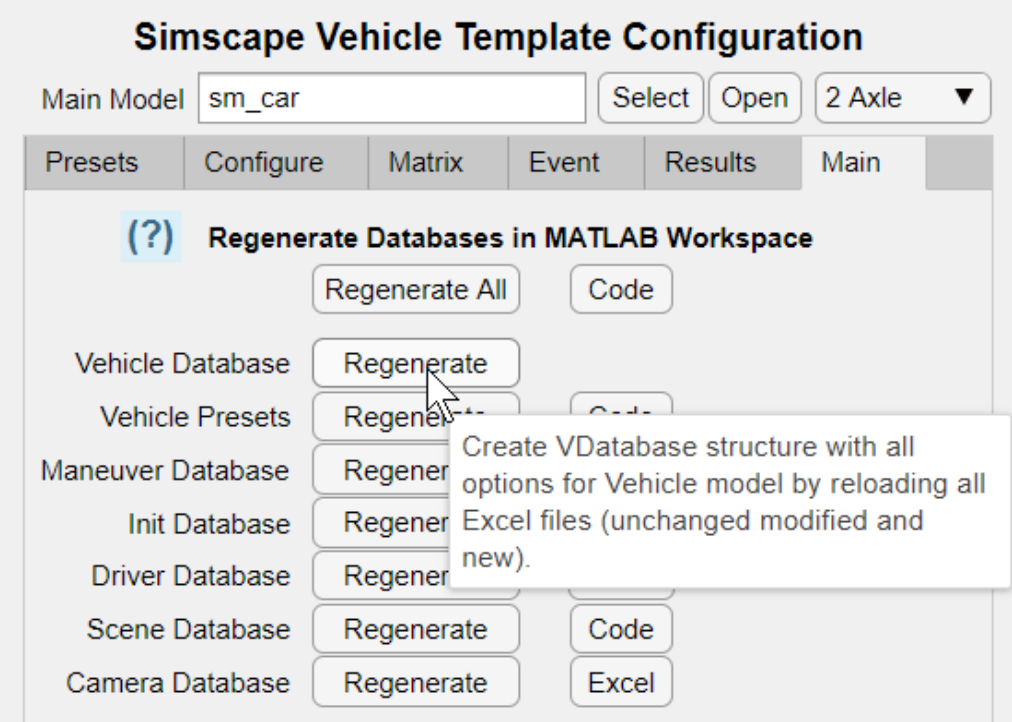
Configuration App
Main Tab

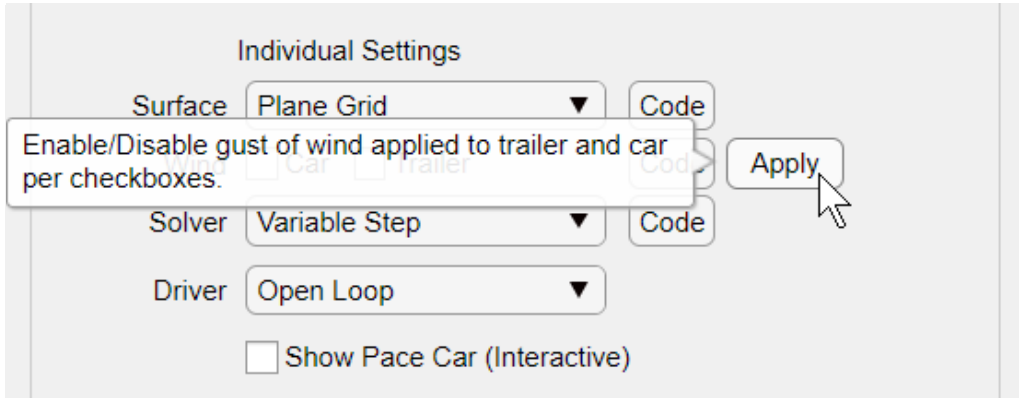
1. Regenerate all databases and presets
2. Regenerate `Vdatabase` (long)
   - Reload all vehicle parameters from Excel
   - Only necessary if new Excel files are added
     - If you only make changes to Excel, use “Update Vehicle Database” button at bottom. Only modified Excel files will be reloaded
3. Regenerate all presets from database
   - Necessary if you update `VDatabase`
4. Check for inconsistent hardpoint values in `Vehicle`
   - Certain hardpoints appear twice within `Vehicle` structure
   - This function checks to see that the values are identical

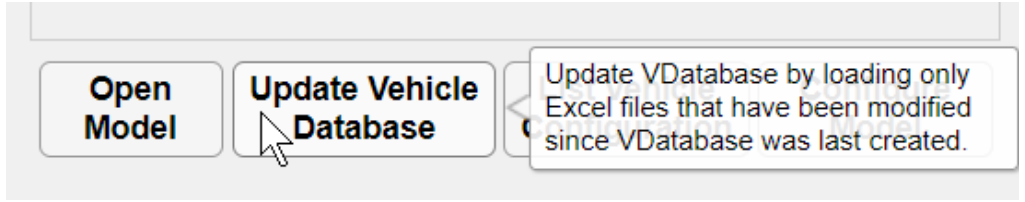


Configuration App

Other Functions

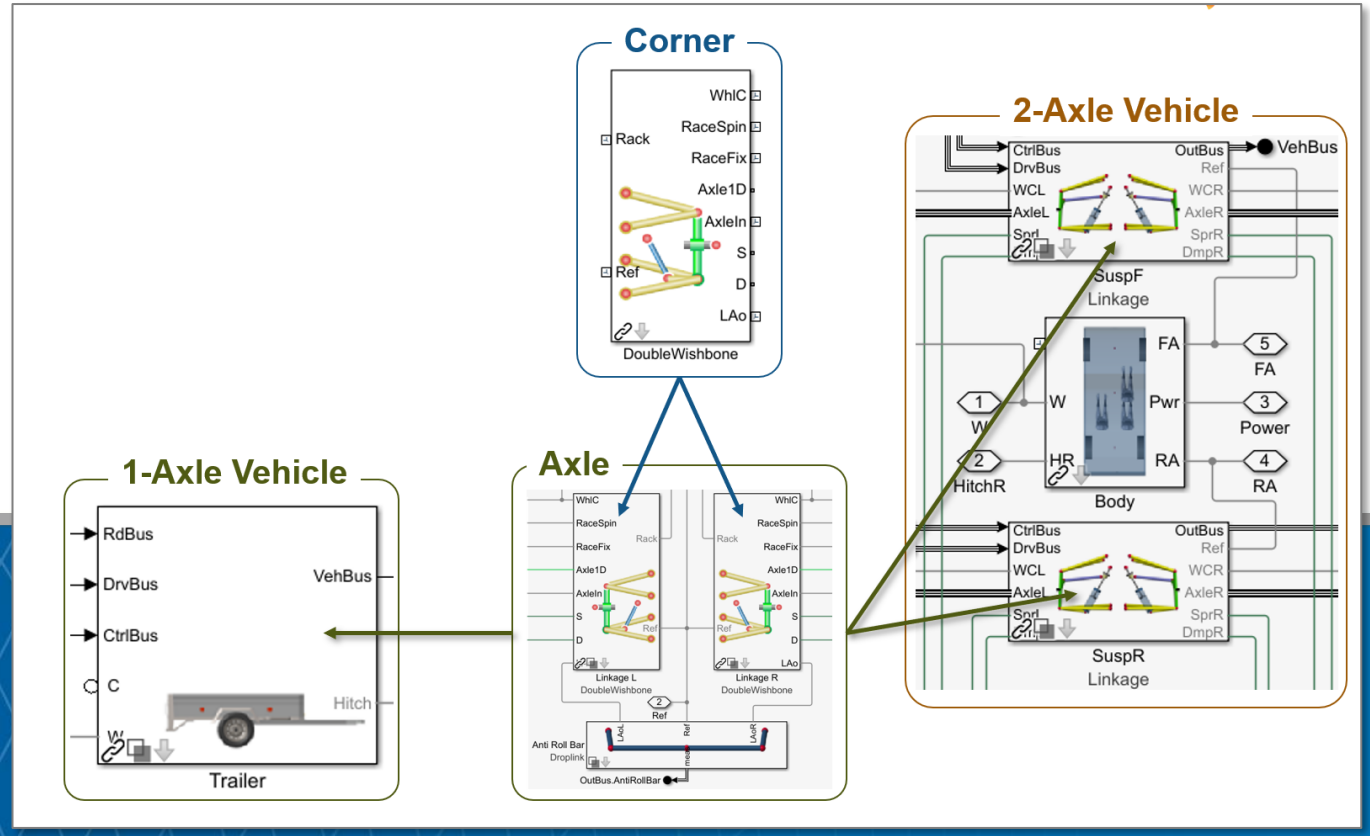






- Hover your mouse over the UI feature to see a tip about that feature‘s function

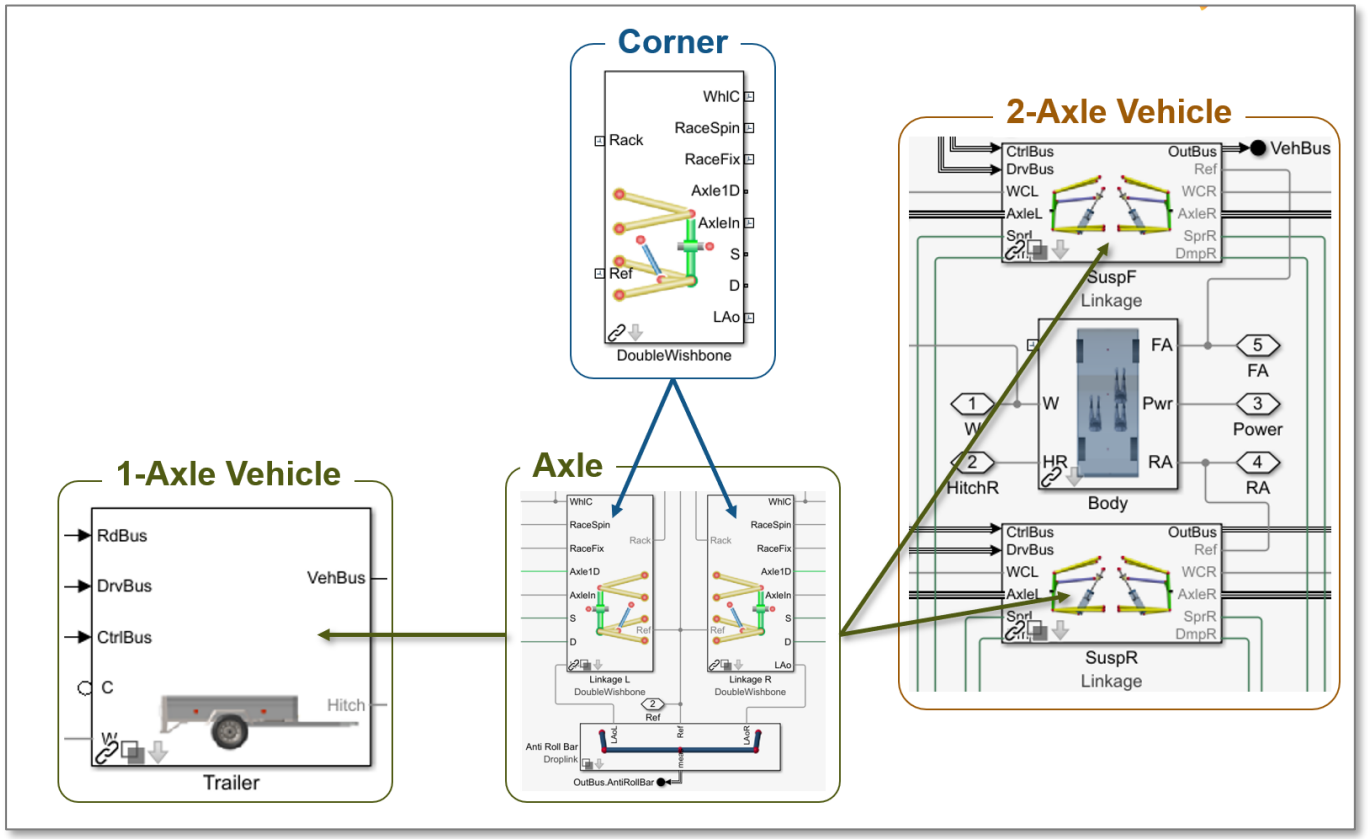
Simscape Vehicle Templates Subsystem Models



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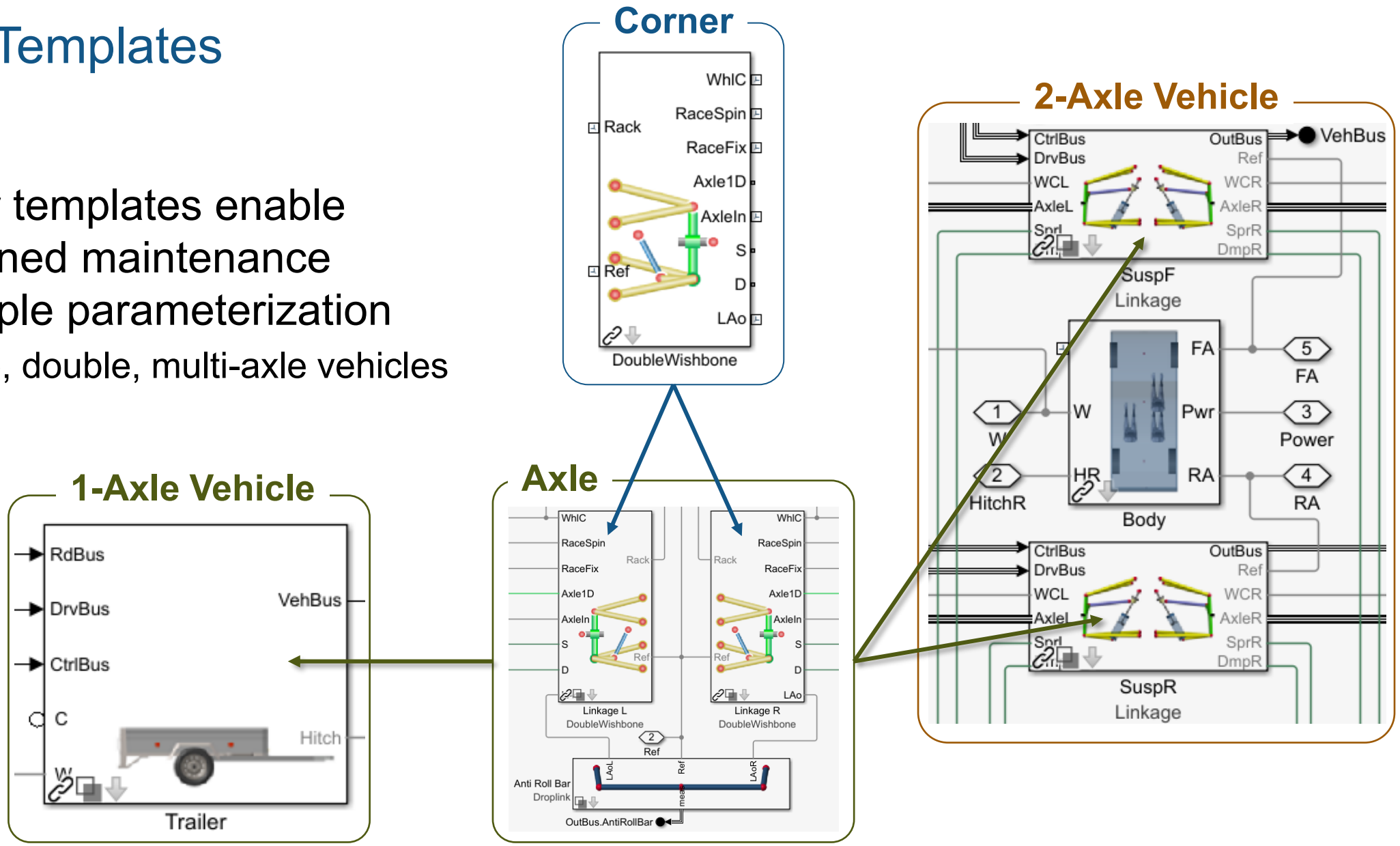
Subsystem Models

- Modular Approach
- Subsystem Interface



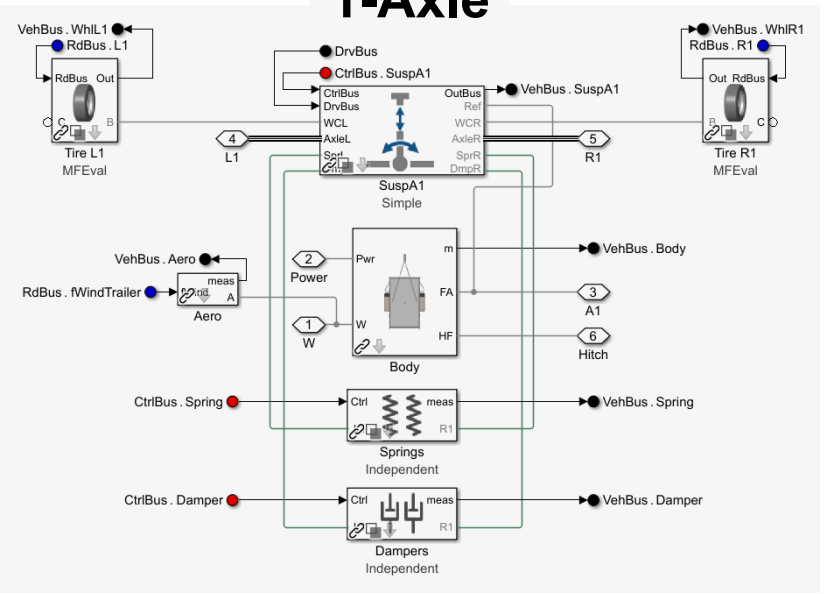
Modular Templates

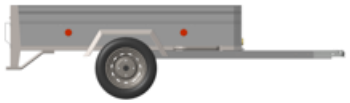
- Modular templates enable streamlined maintenance and simple parameterization
  - Single, double, multi-axle vehicles



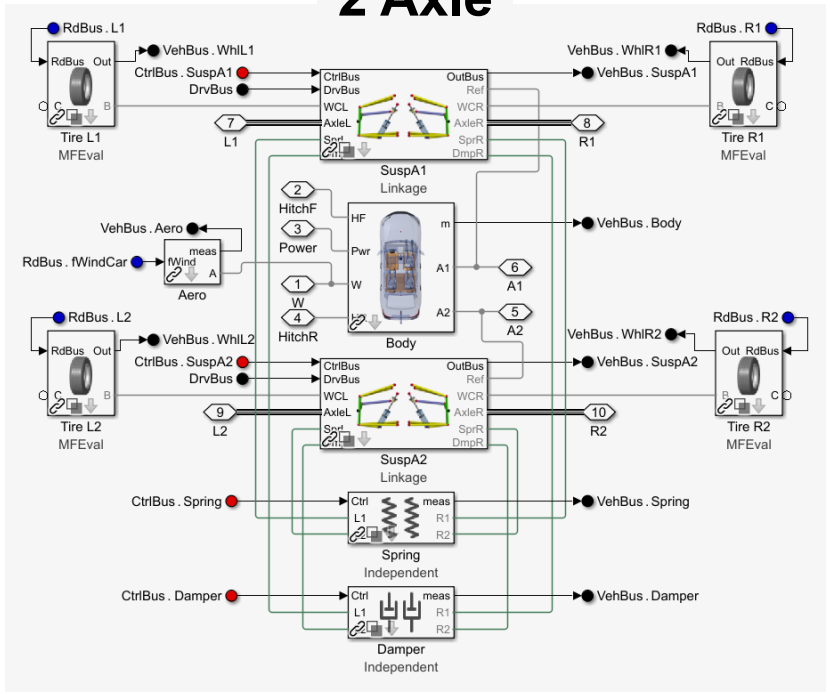
Multi-axle vehicles including trailers

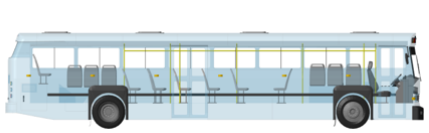
1-Axle

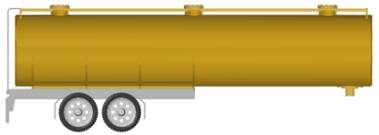




2 Axle



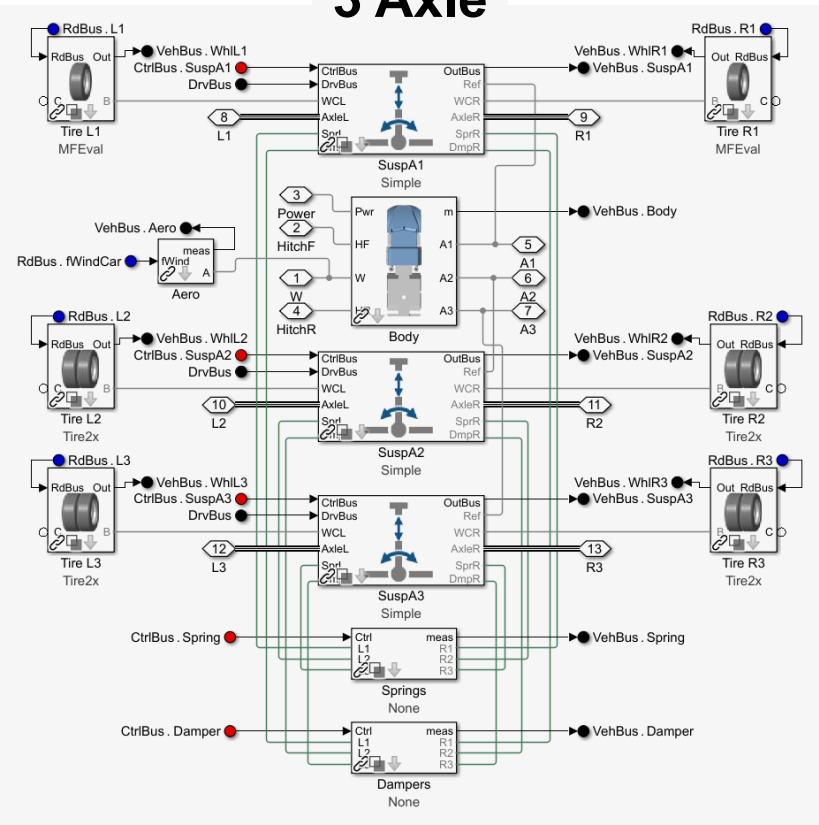




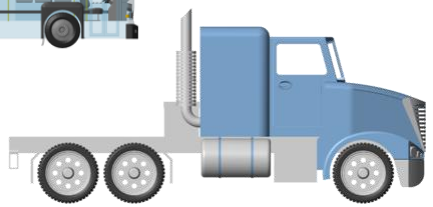




3 Axle



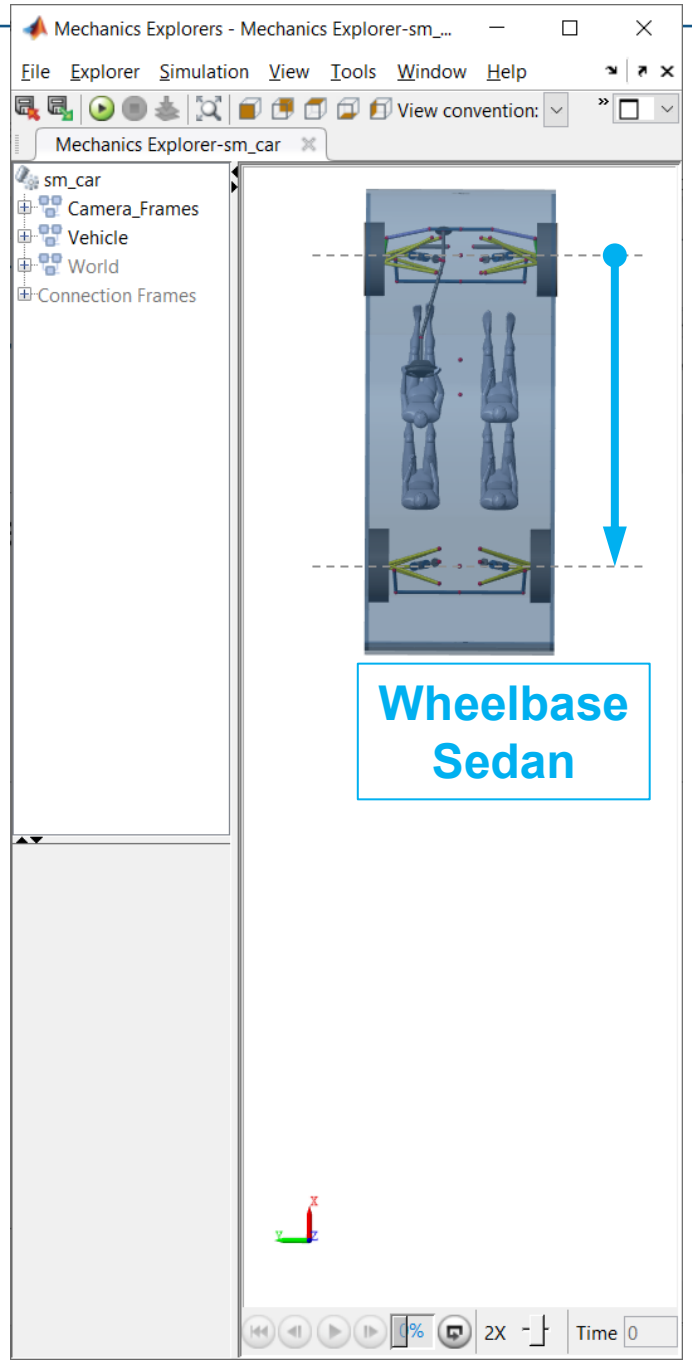


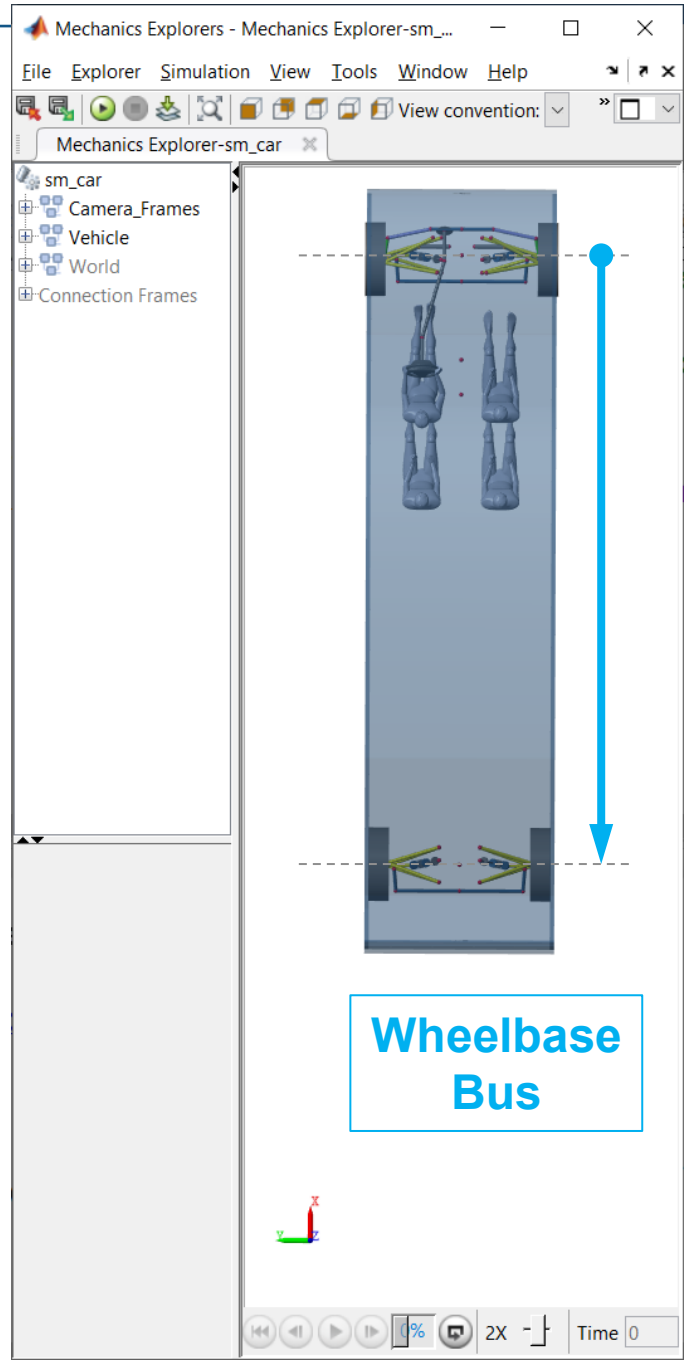


- Same components
- Copy/paste for vehicles with >3 axles

Modular Templates

- Modular templates enable streamlined maintenance and simple parameterization
  - Single, double, multi-axle vehicles
- Modular parameterization enables rapid adjustment of vehicle-level design parameters
  - Wheelbase can be changed by adjusting a single value

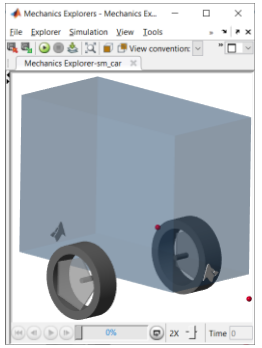


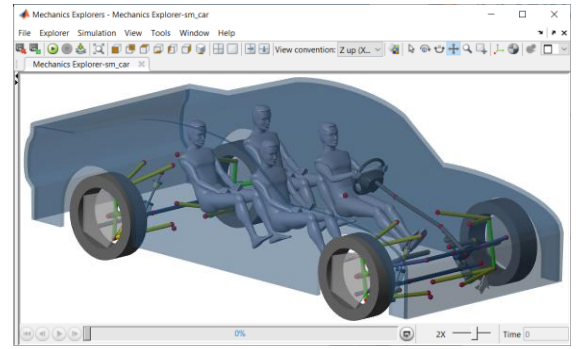


Modular Templates

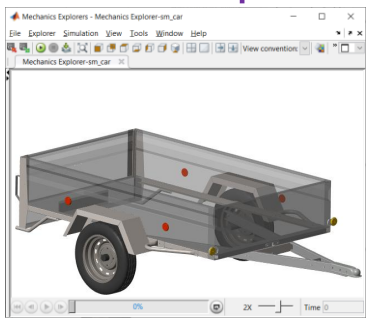
- Modular templates enable streamlined maintenance and simple parameterization
  - Single, double, multi-axle vehicles
- Modular parameterization enables rapid adjustment of vehicle-level design parameters
  - Wheelbase can be changed by adjusting a single value
- One set of templates can be used to define many vehicles

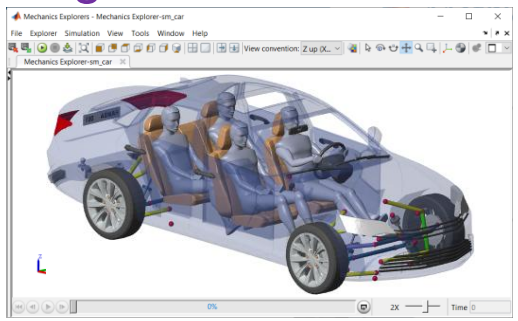
Parameterized Models

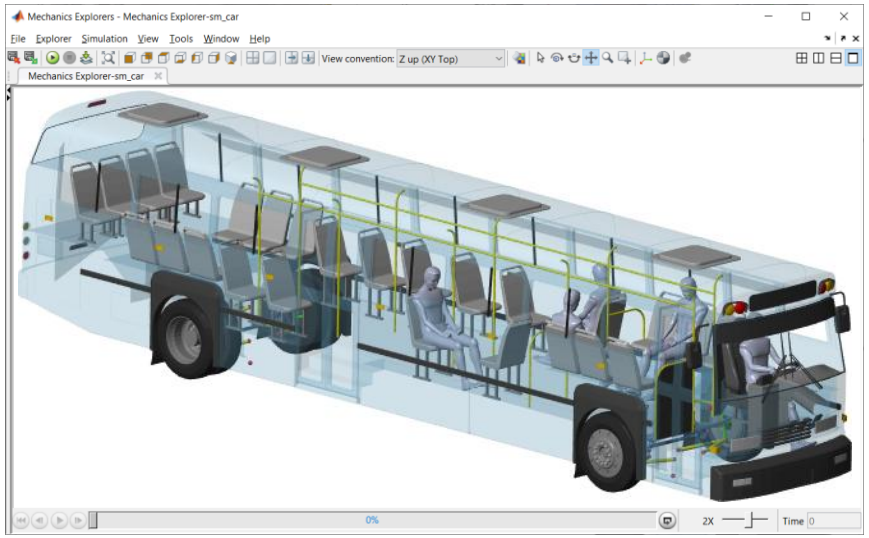




Specific Designs



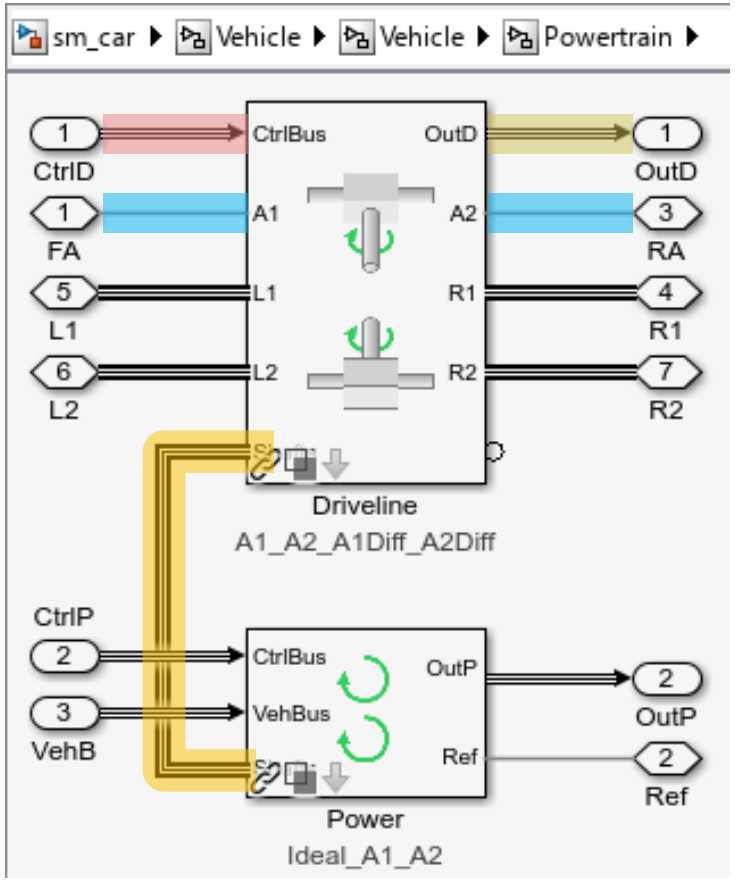




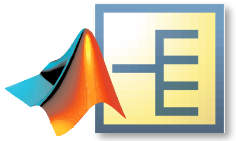
Subsystem Interface

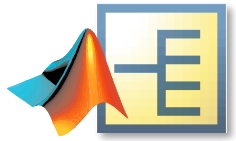
- Modular design requires carefully selected interfaces
  - Accommodates many designs

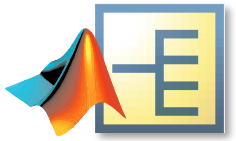
- Composed of
  - Physical connections
  - Simscape busses for bundles of connections
  - Input signals via Simulink bus
  - Output signals via Simulink bus

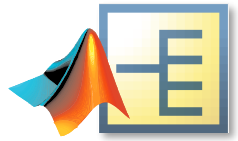


Simscape Vehicle Templates Parameters





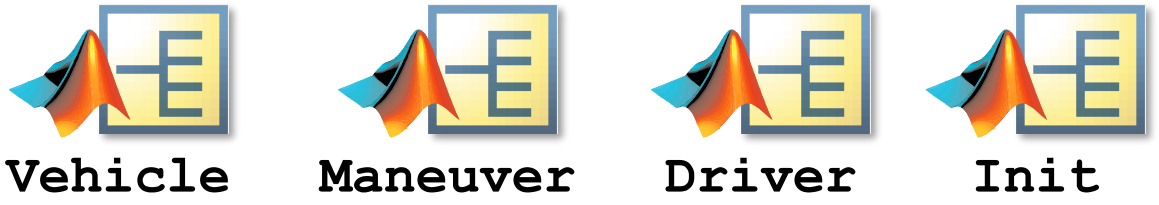


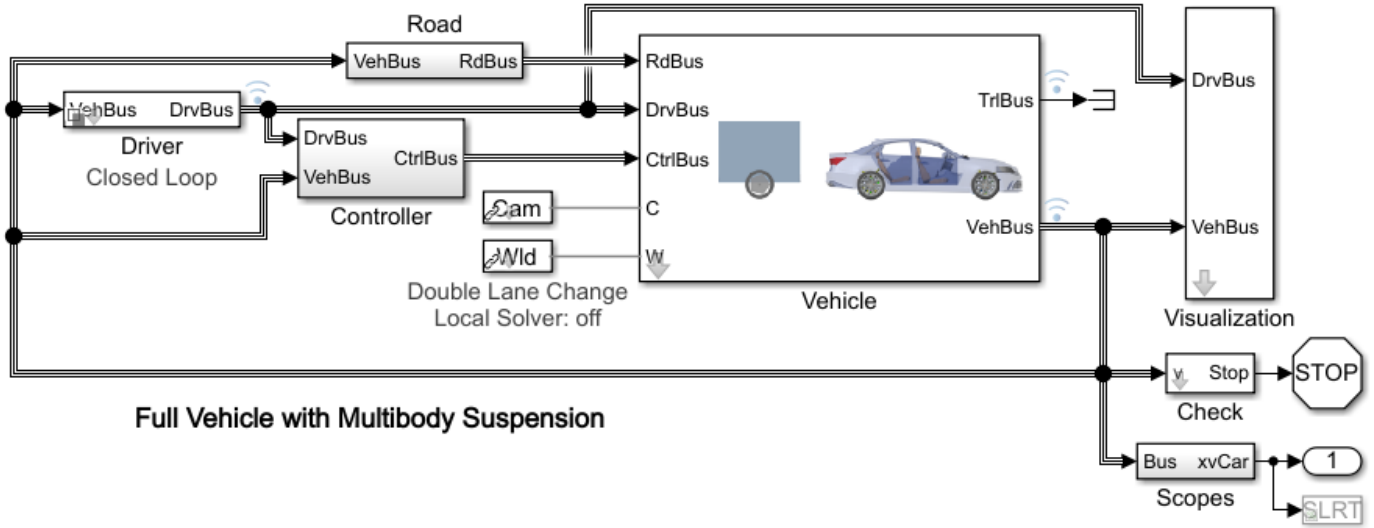


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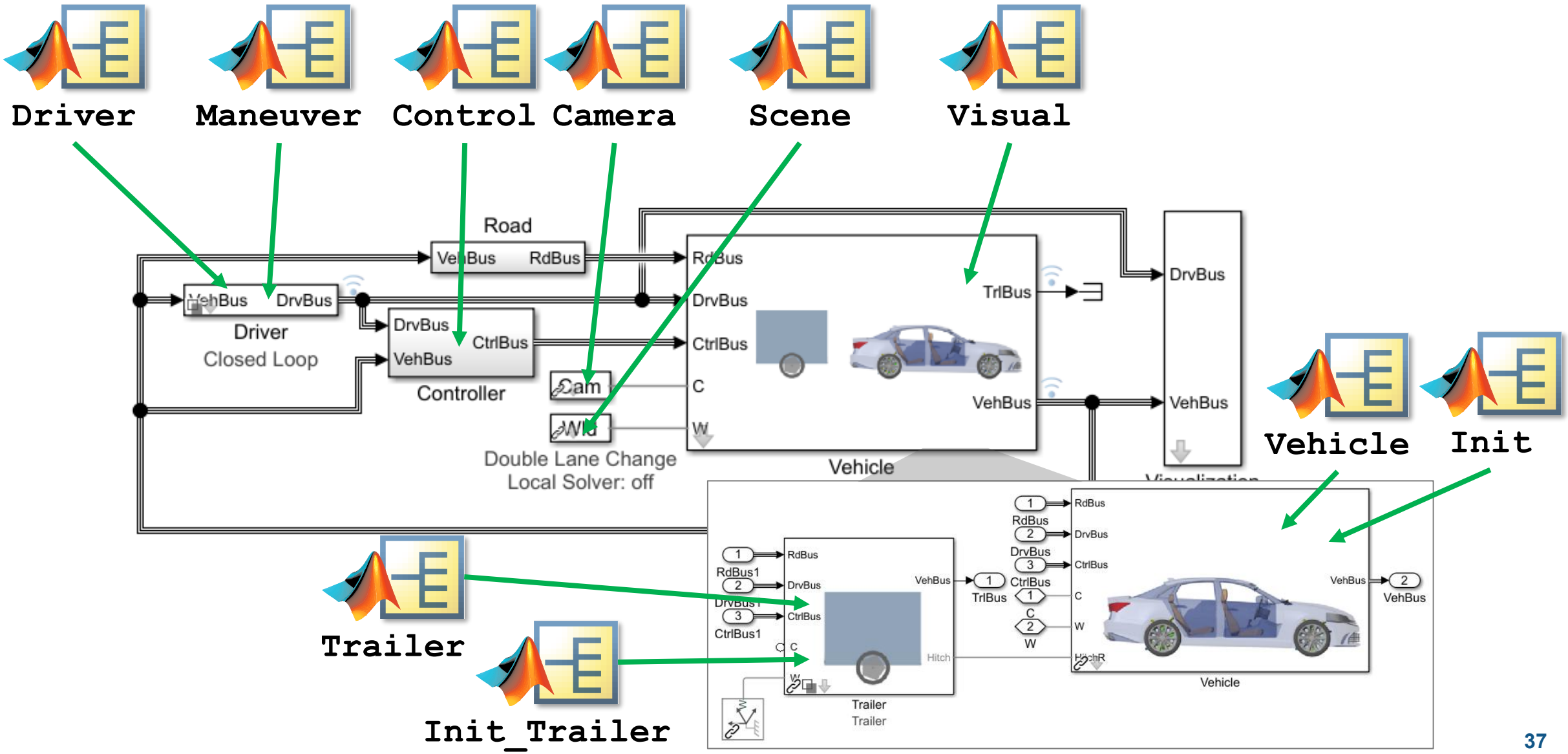
Parameters

- Parameter Database
- Model Parameters



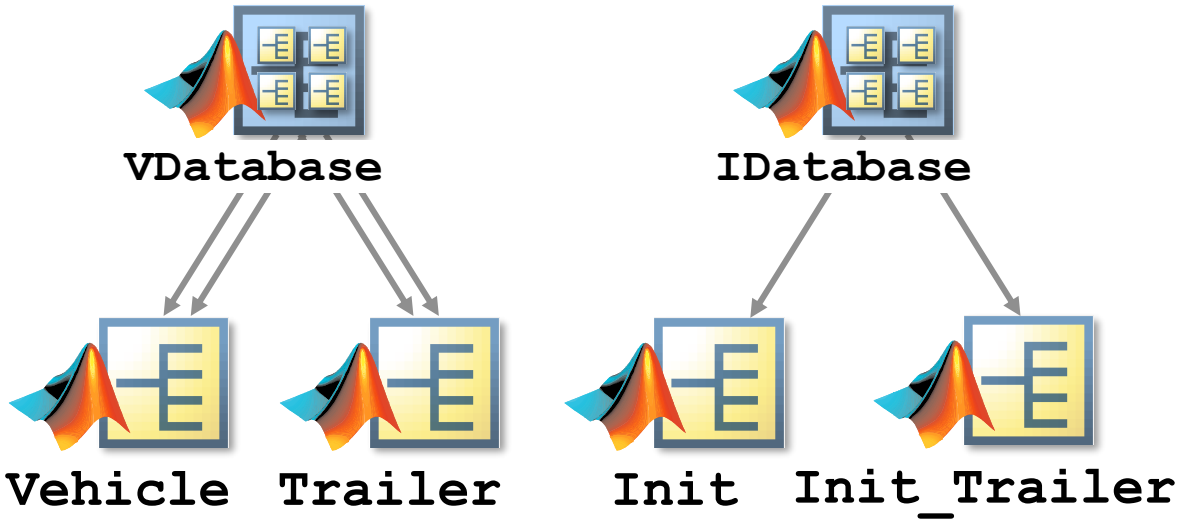


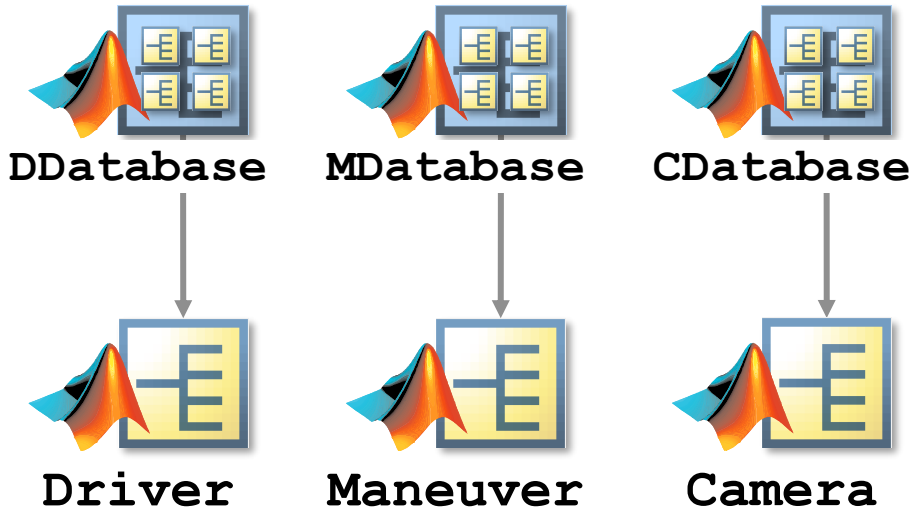
Model Parameterization: 10 MATLAB Data Structures

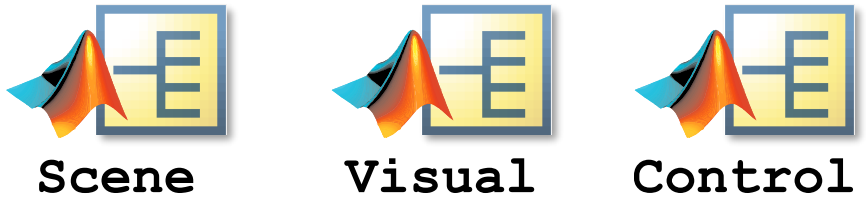


Database Structures Contain All Settings

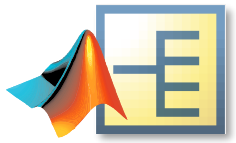
- Five *Database structures are used to populate the data structures for the model
  - Many elements are combined to define Vehicle, Trailer
  - Individual elements for the others







Model Parameterization













- Data structure mirrors model structure
- Contains parameter values and selected variants
  - Structure varies based on selected variants
  - Variant selection done by mask initialization
- Assembled by copying fields from MATLAB structure VDatabase

Data Structure

```
Vehicle.
        .Brakes
        .Chassis.
                 .Aero
                 .Body
                 .Damper
                 .Spring
                 .SuspA1
                 .SuspA2
                 .TireA1

                 .TireA2

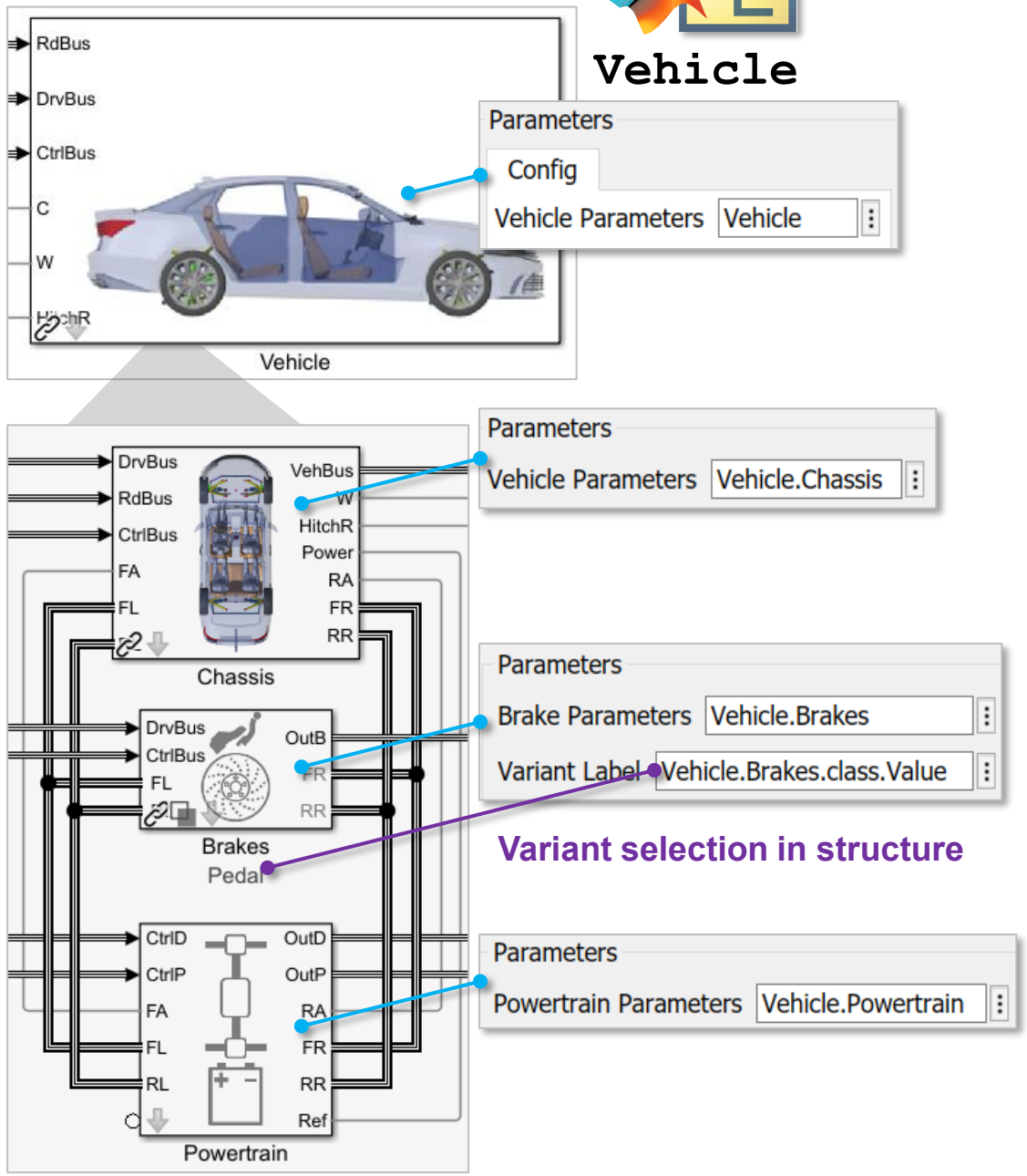
    .Powertrain.
                 .Driveline
                 .Power
```

Model

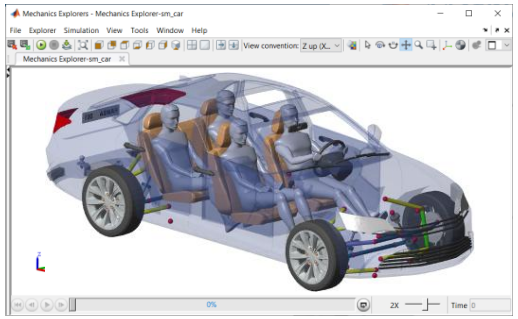


Defining a Full Vehicle

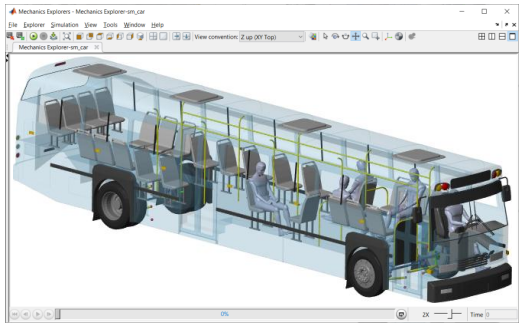
- Must specify:
  - Selected variants
    - Powertrain type, suspension type, etc.
  - Parameter values for selected variants
    - Variants and values must be synchronized
- Simscape Vehicle Templates use a single MATLAB structure to define variants and values




```
>>Vehicle = Vehicle_Sedan
```



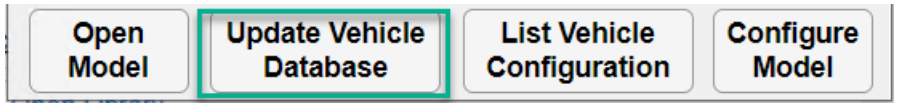
```
>>Vehicle = Vehicle_Bus
```



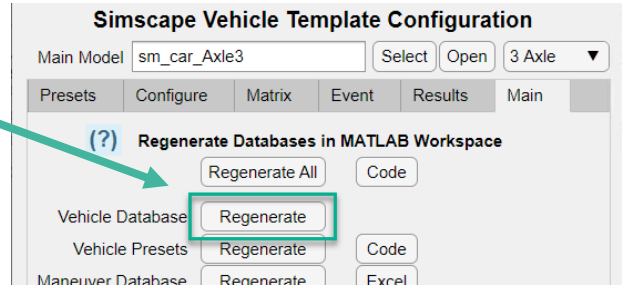
Model Parameterization

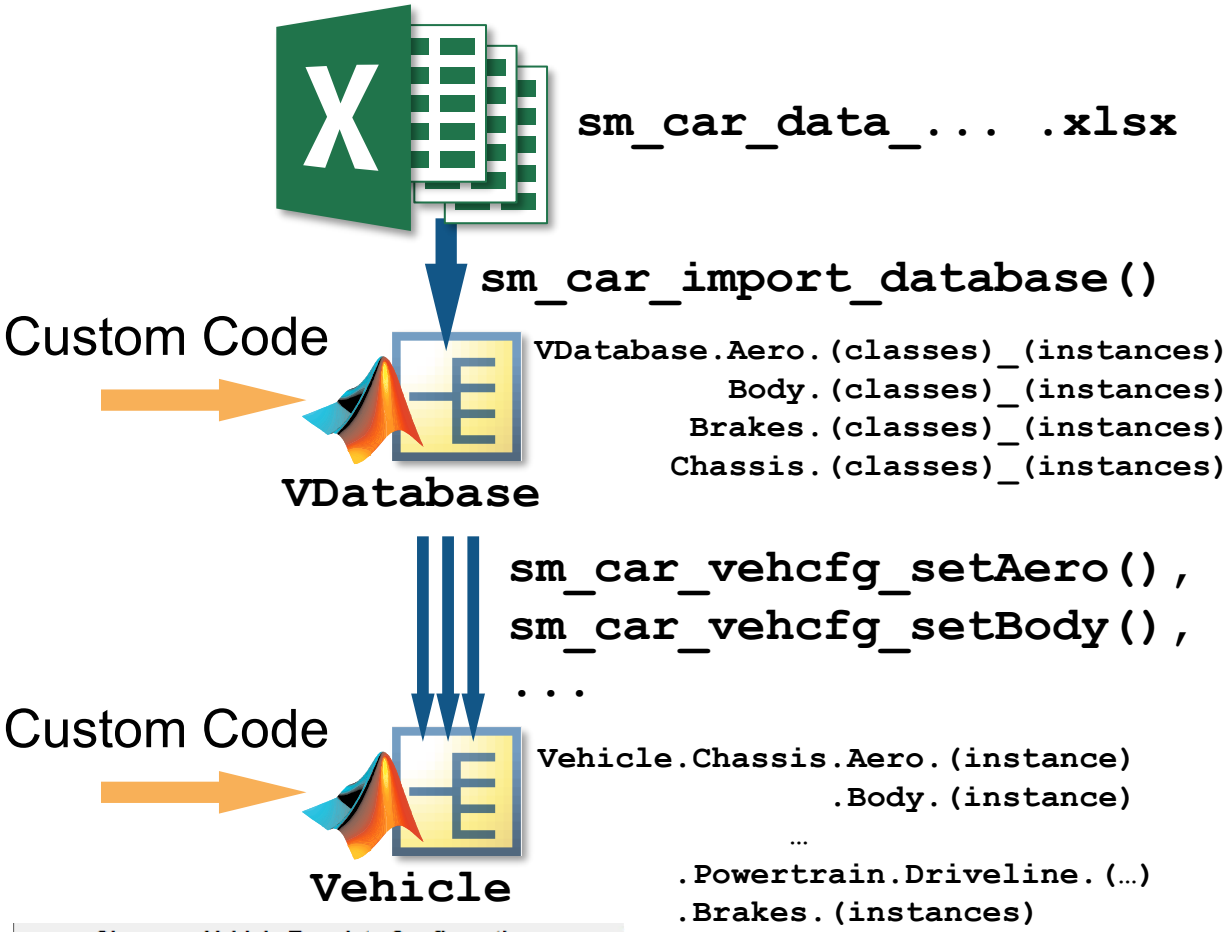


- Model uses **`Vehicle`**
- **`Vehicle`** parameters can come from
  - **`VDatabase`**
  - Overwrite **`Vehicle`** using MATLAB Code
- **`VDatabase`** saved in a .mat file
  - Updated if a .xlsx file has a newer "Modified Date"
- If.xlsx files are only modified, simply press "Update Vehicle Database" button at bottom of User Interface



- If a new `sm_car_data...xlsx` file is added, you must reload the entire Vehicle database
  - Update will scan for files that are named sm_car_data**.xlsx

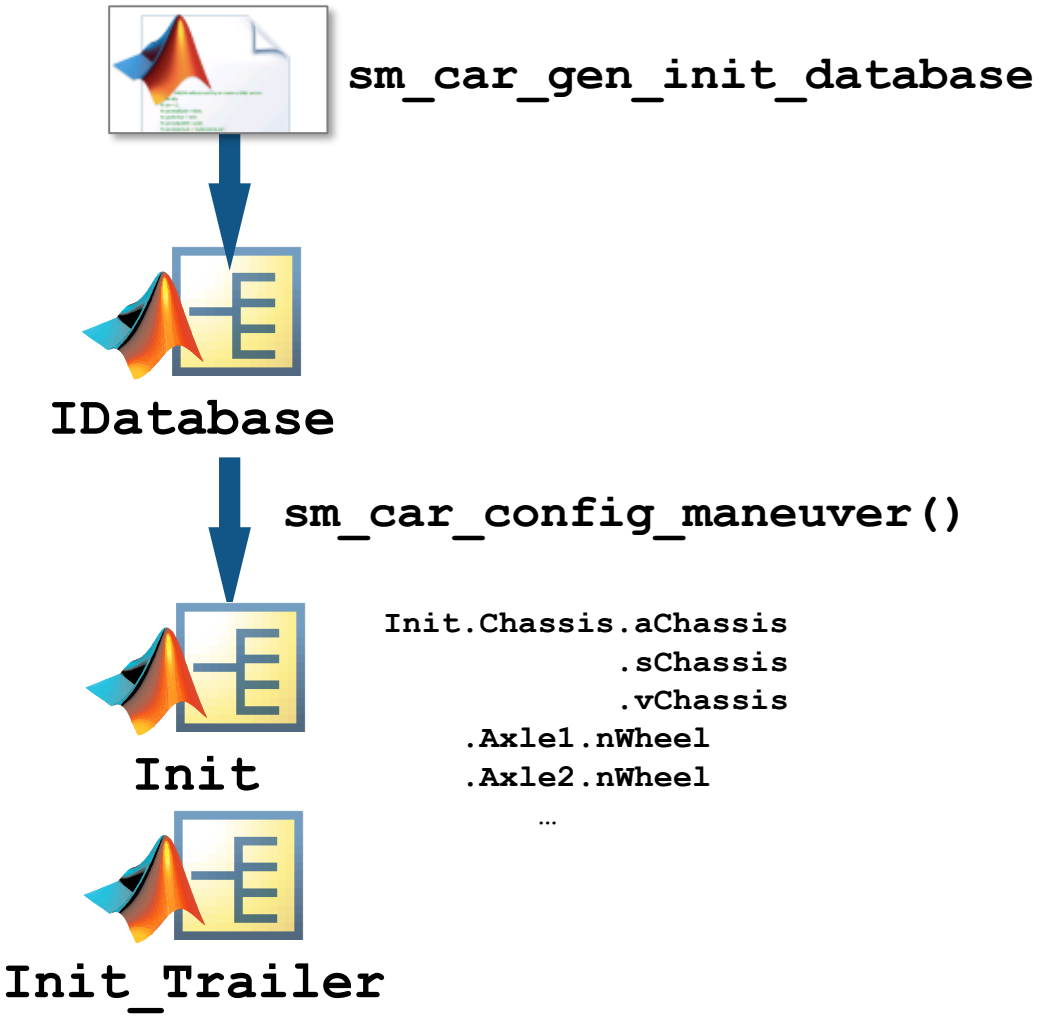




Model Parameterization



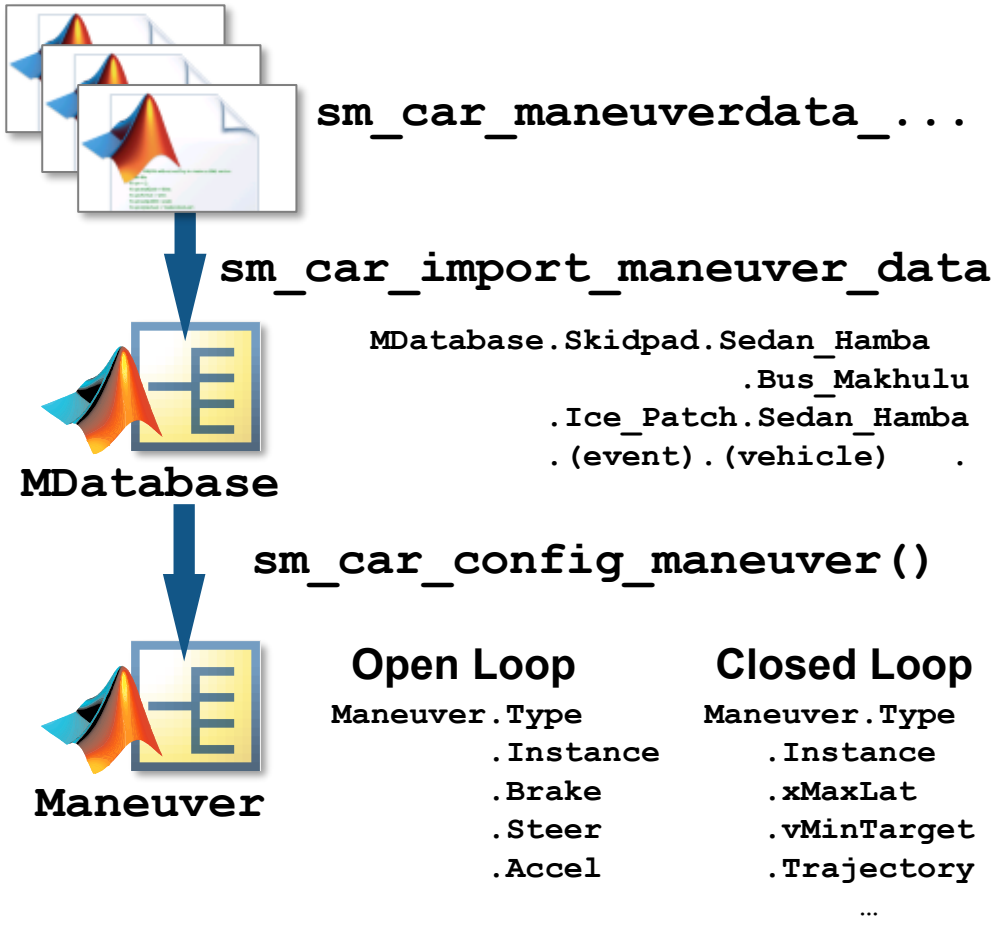
- MATLAB code defines database of initial states for the vehicle
  - Vehicle position, orientation
  - Chassis speed, wheel angular speed
- Initial state is extracted from database based on event and vehicle type



Model Parameterization



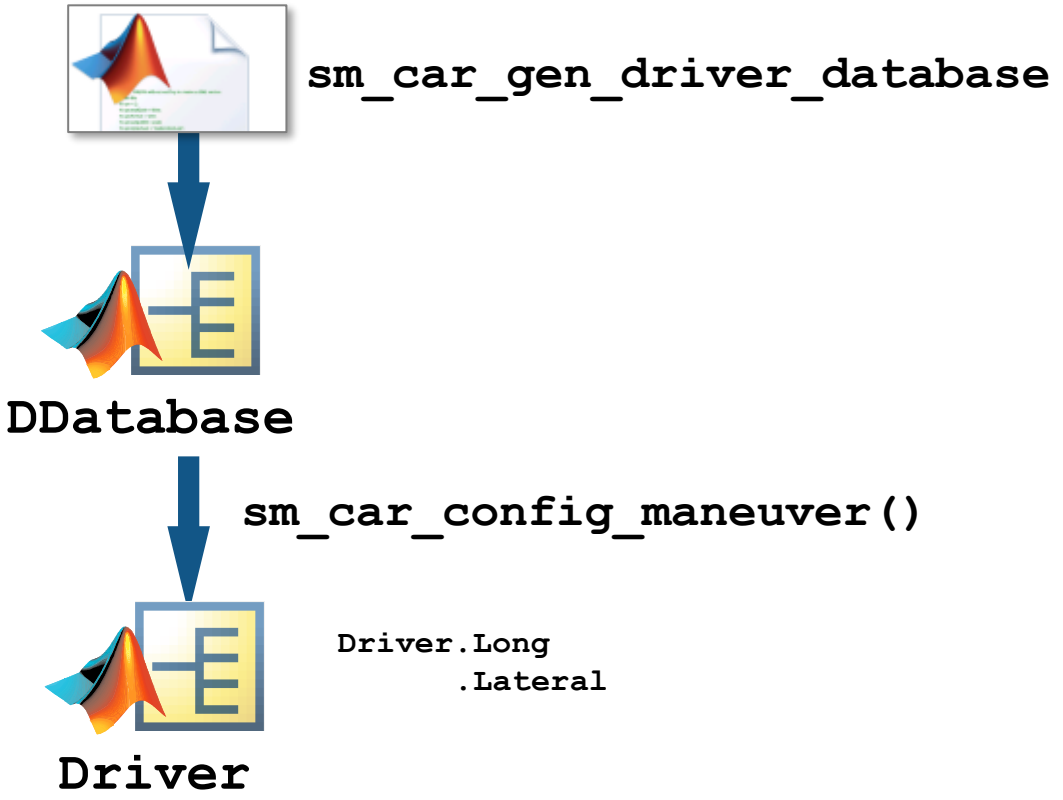
- MATLAB code defines database of maneuvers for vehicles
  - Specific to event, vehicle type, and open-loop or closed-loop driver
- Structure `Manuever` extracted from `MDatabase` and used by the model



Model Parameterization



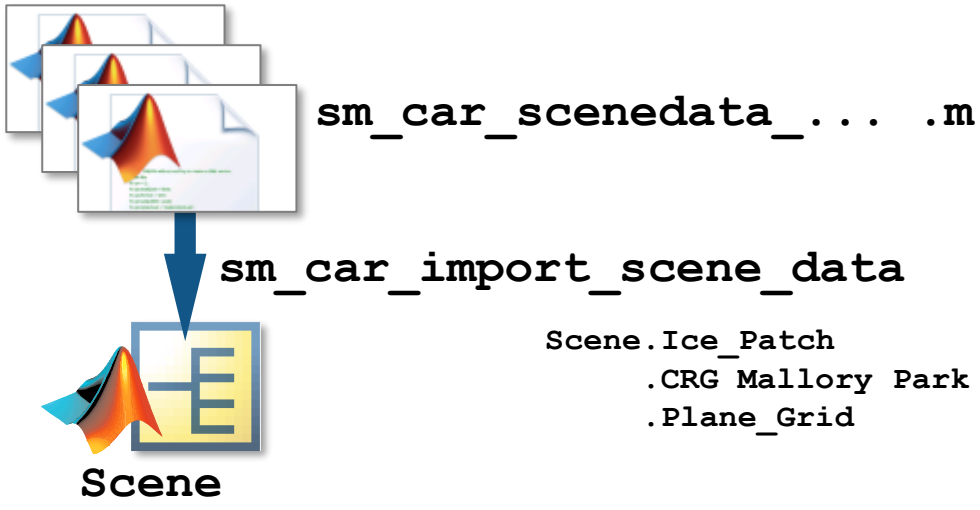
- MATLAB code defines database of drivers for vehicles
  - Closed-loop driver parameters specific to maneuver and vehicle type
- Structure `Driver` is extracted from `DDatabase` and used by the model

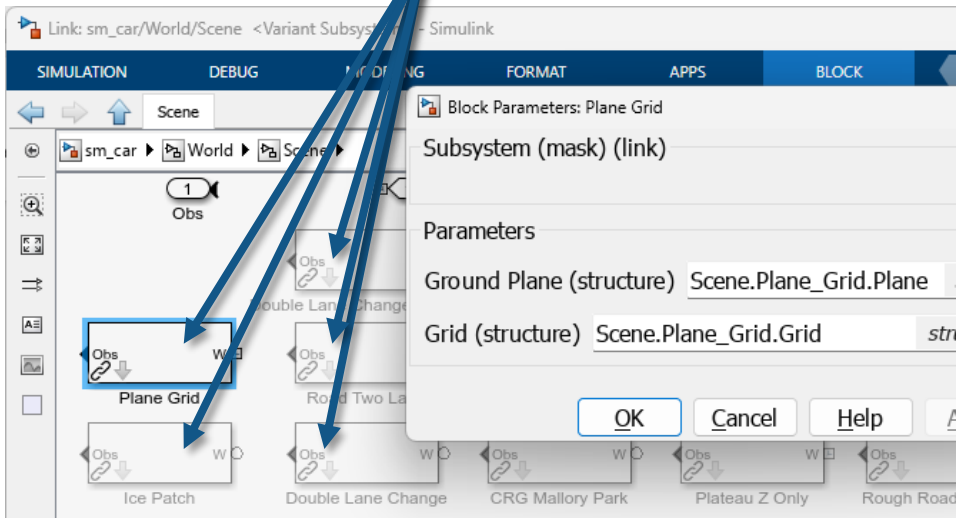


Model Parameterization



- MATLAB code defines database of parameters for each scene
  - Locations, sizes, colors, files, and other data to parameterize environment
- Structure `Scene` is directly referenced by variant subsystems that define each scene

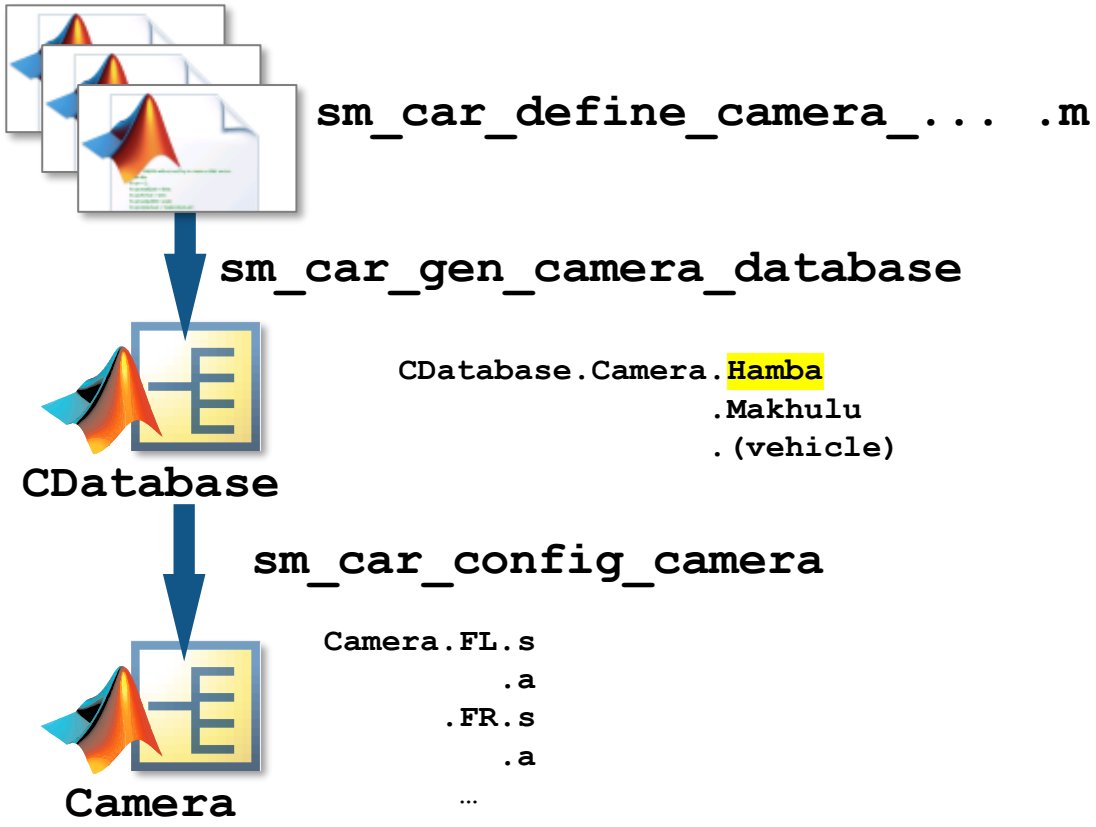




Model Parameterization



- MATLAB code defines database of frame locations and orientations to position dynamic cameras
  - Depends on vehicle chassis size
- Structure `Camera` extracted from `CDatabase` and used by the model
  - `sm_car_config_camera` is called during model initialization
  - Uses first portion of `Vehicle.config` to determine which set of frame locations to use

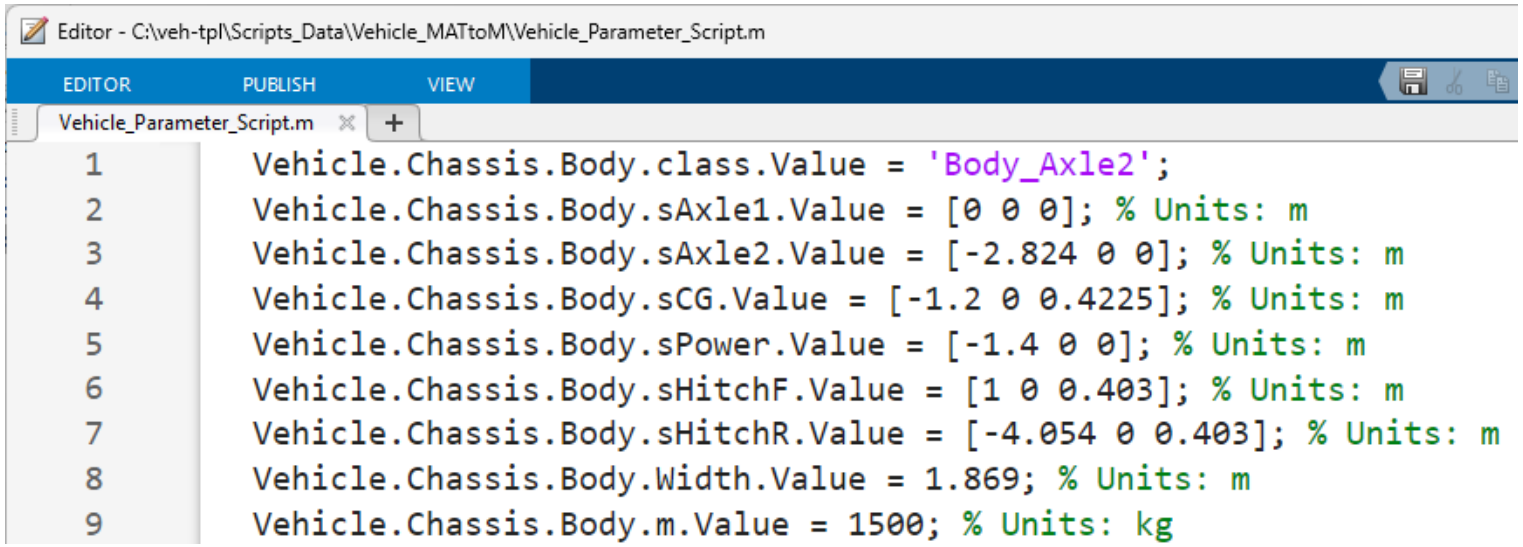



```
>> Vehicle.config
ans =
    'Hamba_dwb_MFMbody_steady_fCVpCVr1D'
```

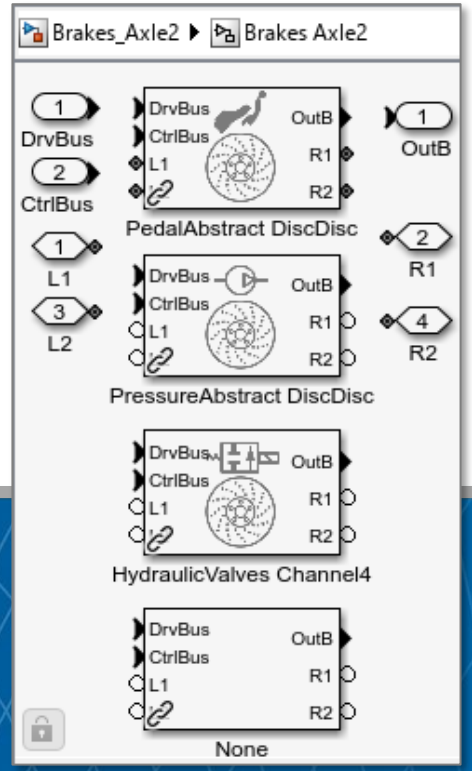
Converting Data Structure to Code

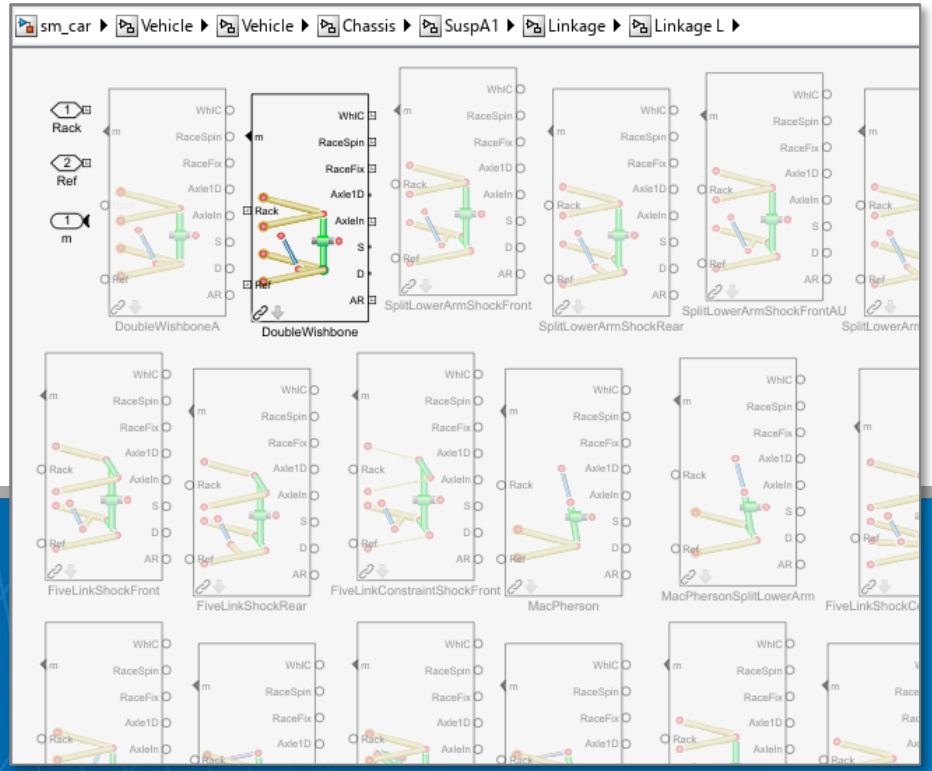
- Code is provided that will convert a `Vehicle` structure to MATLAB code
  - Single file to track parameters
  - Text is useful for diff, version control

```
sm_car_create_vehicle_values_only(Vehicle,'Vehicle_Parameter_Script.m')
```



Simscape Vehicle Templates Variants

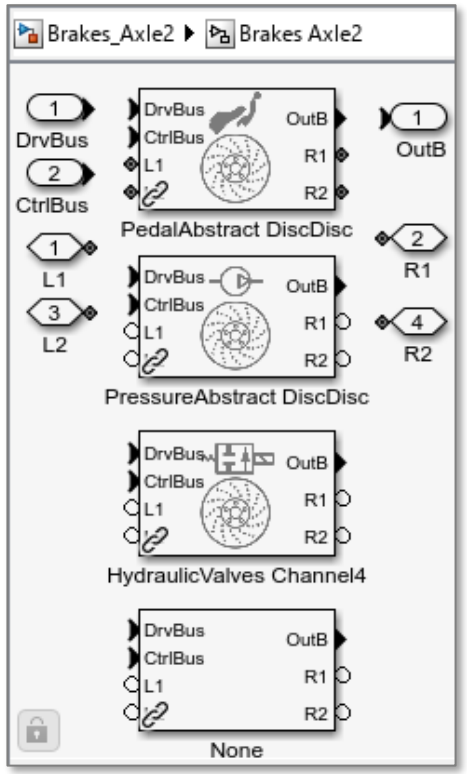


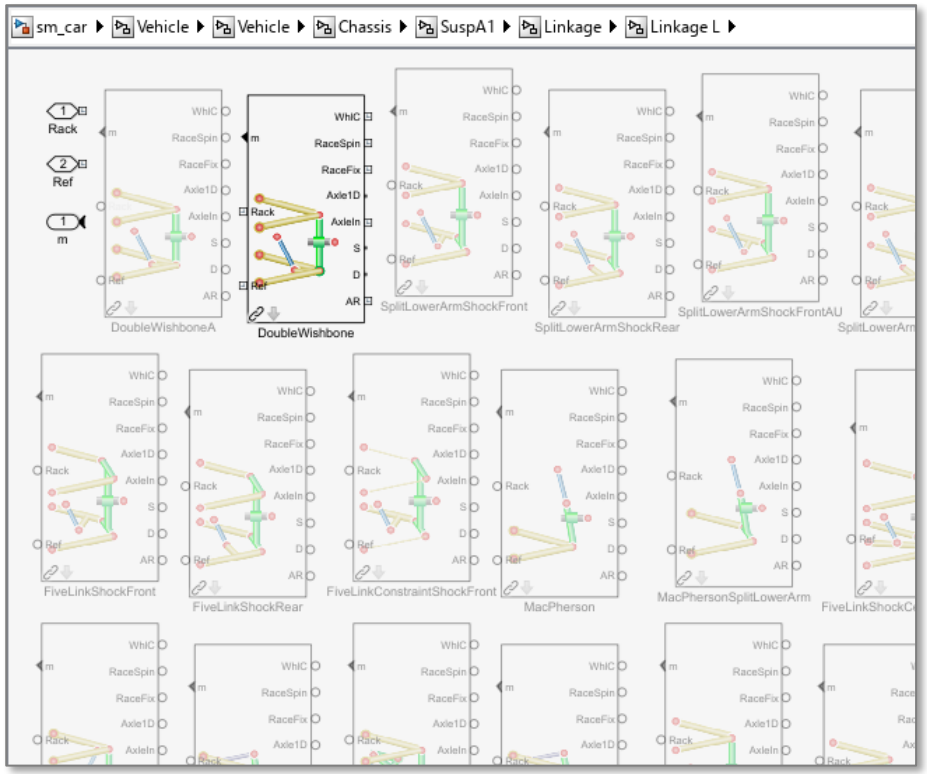


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Variants

- Selecting Variants
- Defining Variants

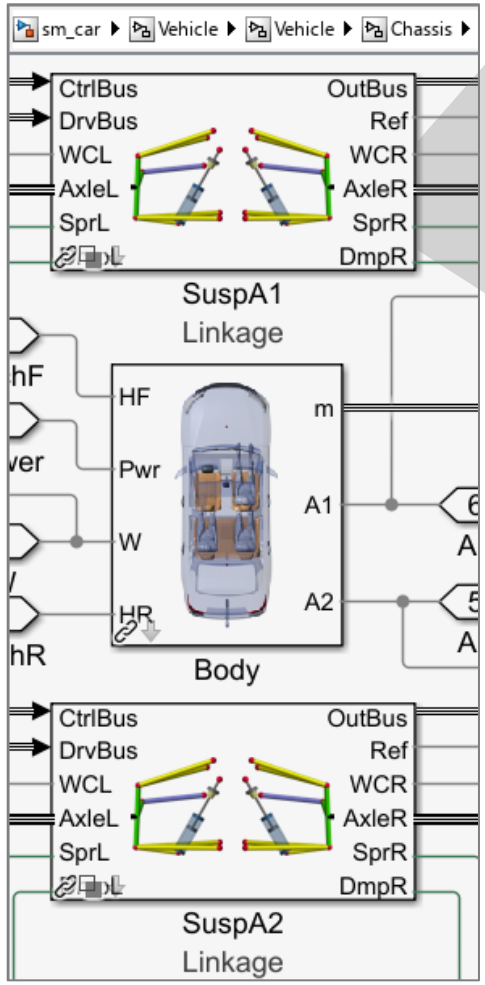


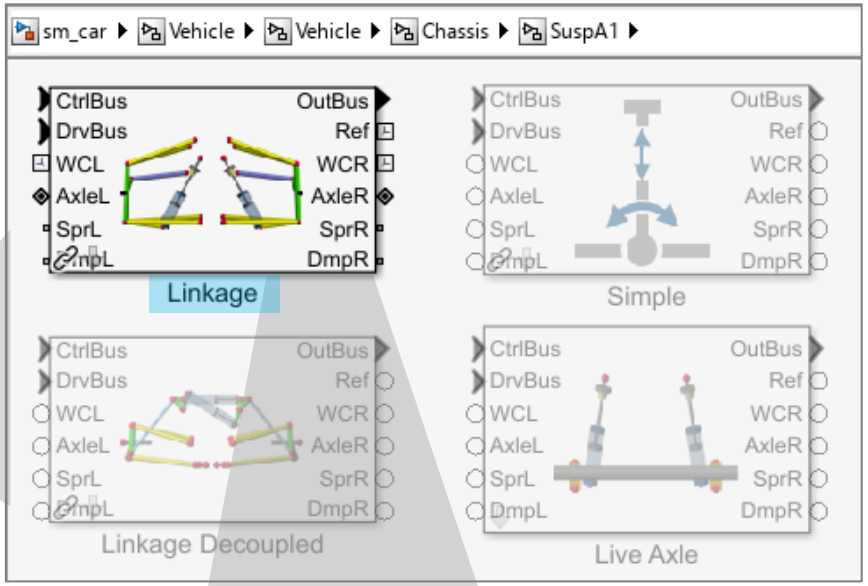


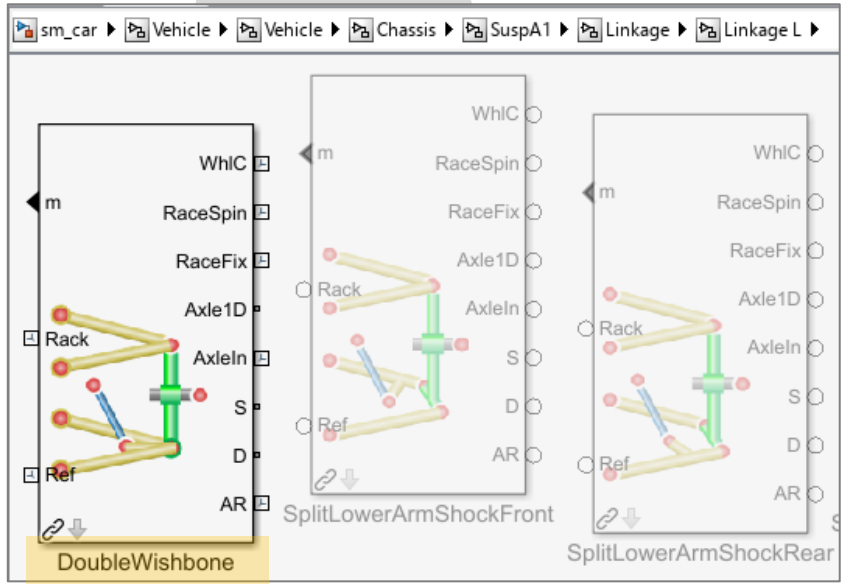
Configuration Using Variants

```
Vehicle.Chassis.SuspA1.class.Value = 'Linkage'
```

- Variants Subsystems let you configure your vehicle model to have the designs and effects that you need
- Variant selection is controlled by fields in the Vehicle structure
- Careful coordination of Simulink settings, mask parameters and Vehicle field settings is required





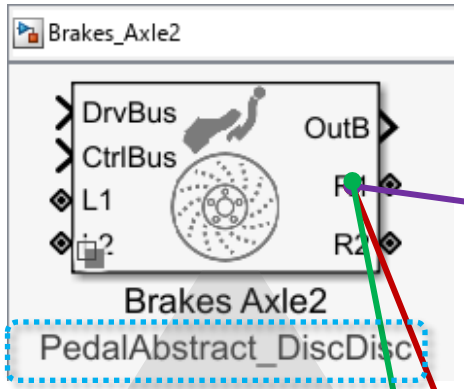


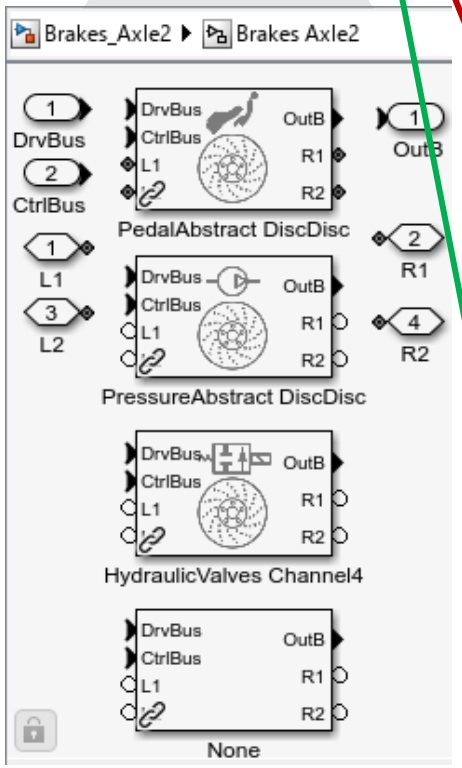

```
Vehicle.Chassis.SuspA1.Linkage.class.Value = 'DoubleWishbone'
```

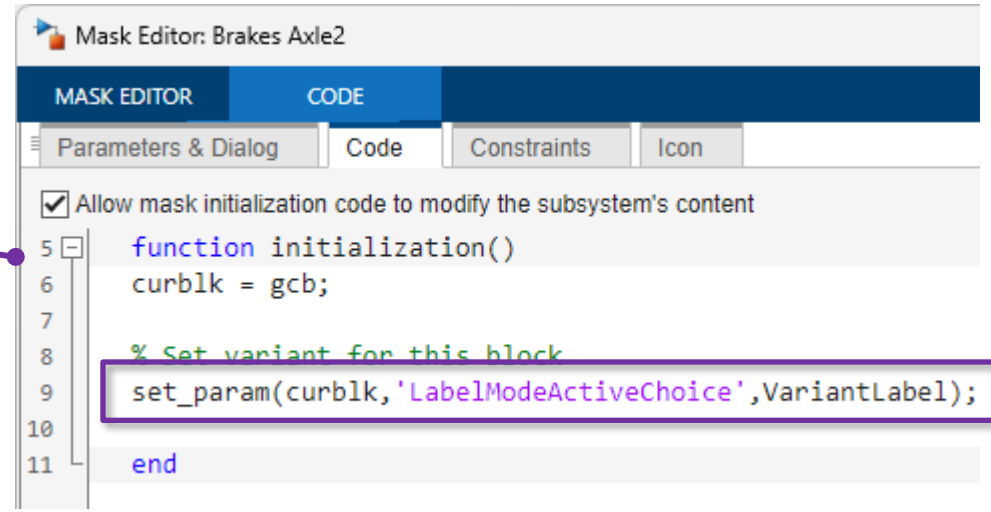
Selecting Variants

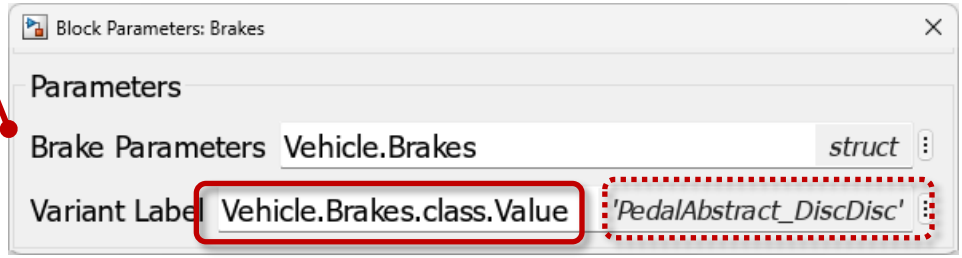
- **Variant selection** is done in **mask initialization code**
  - Sets block parameter `'LabelModeActiveChoice'` to string in **`...class.Value`** field
  - Parameter setting selects the **variant assigned that label**

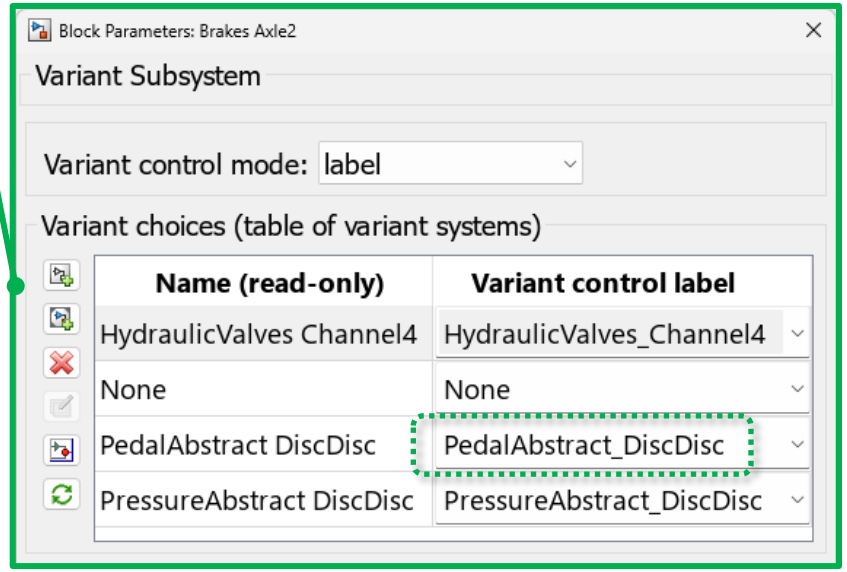
- Changing variants should only be done by setting the `class.Value` field





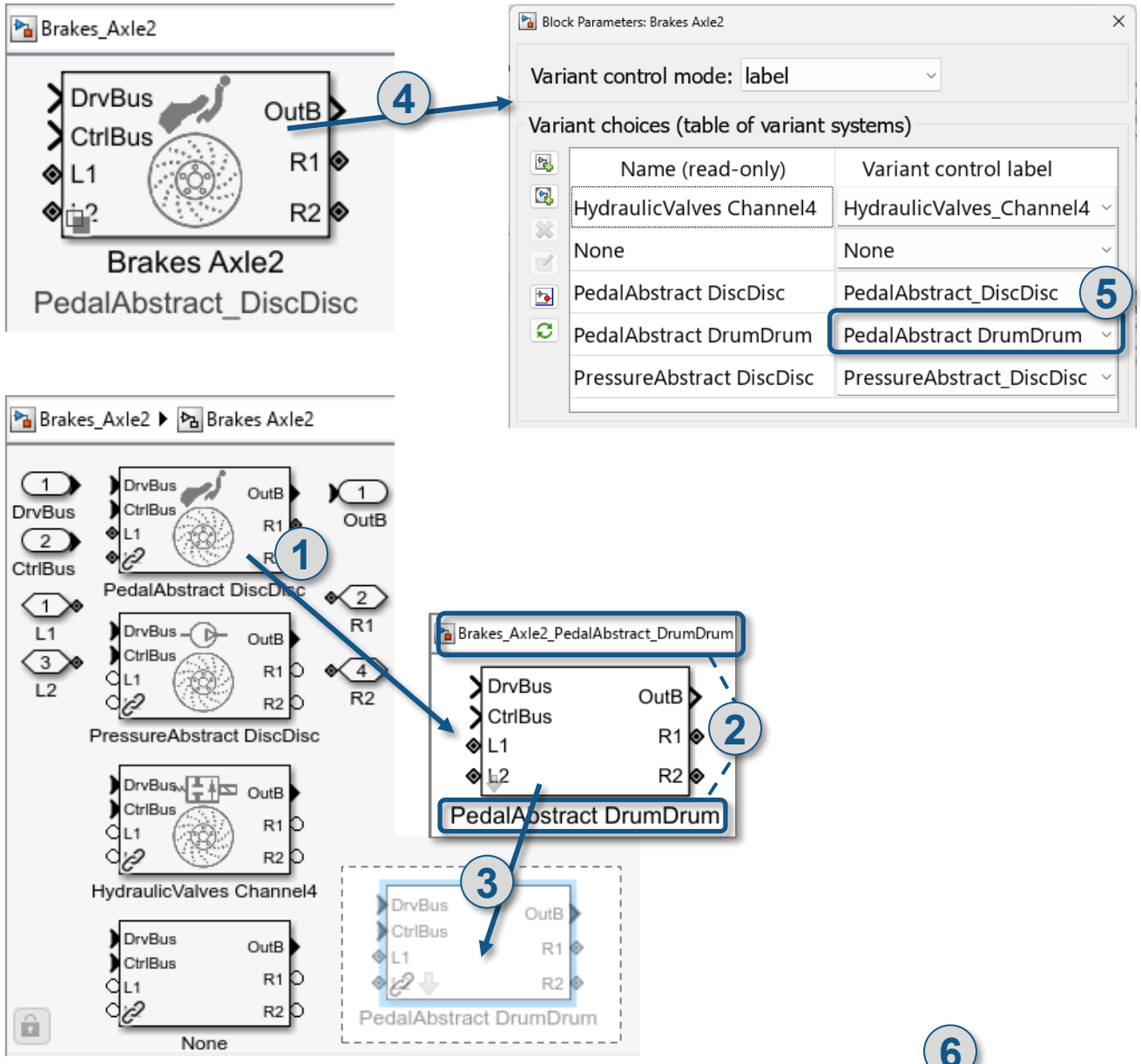






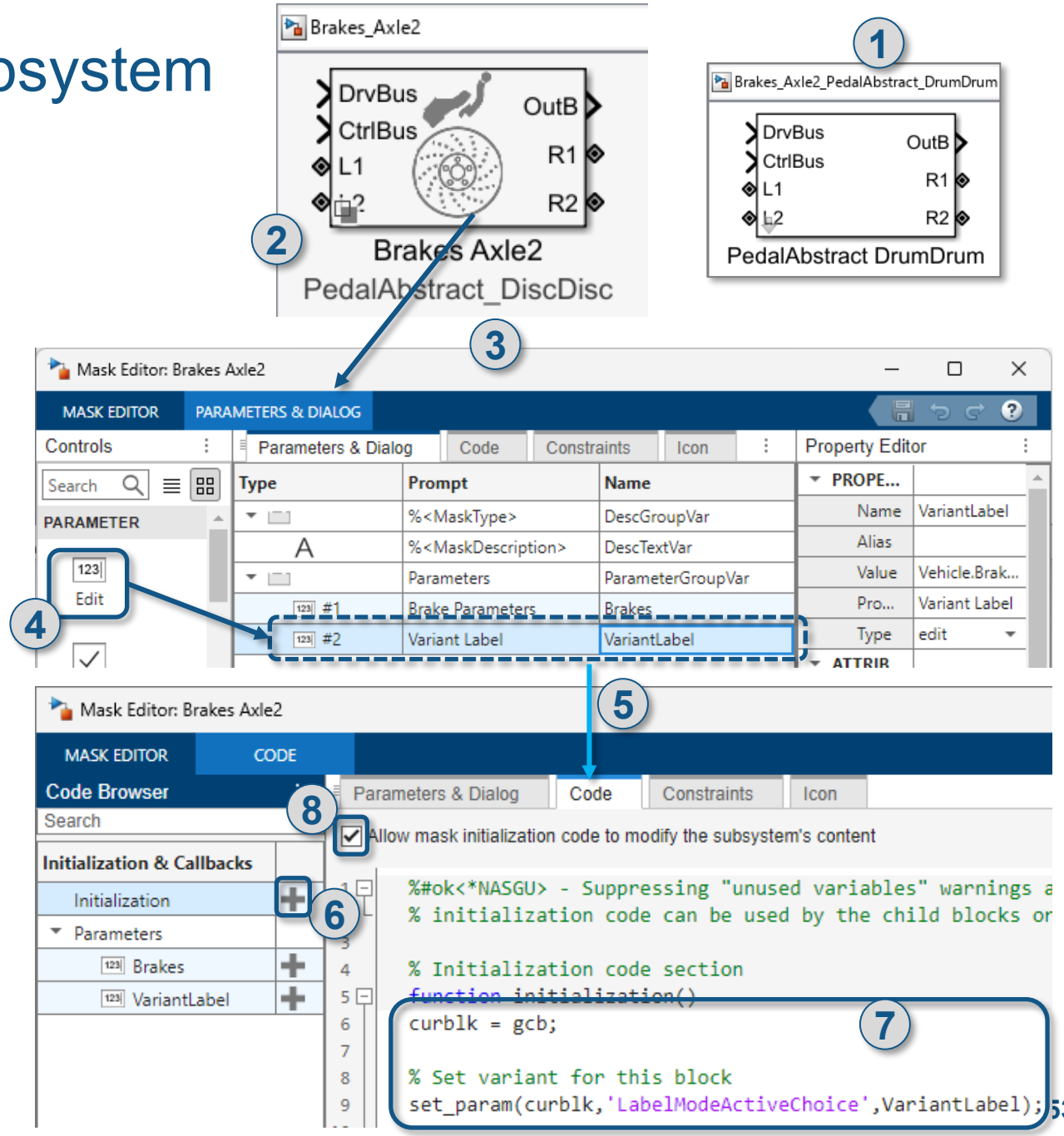
Steps to Add a New Variant to Existing Variant Subsystem

1. Open an existing variant
2. Save file under a new name
3. Add block directly (not wrapped) to Variant Subsystem
4. Right click on Variant Subsystem to access “Block Parameters”
5. Assign label to new variant
6. Field …`class.Value` must have the label as a string

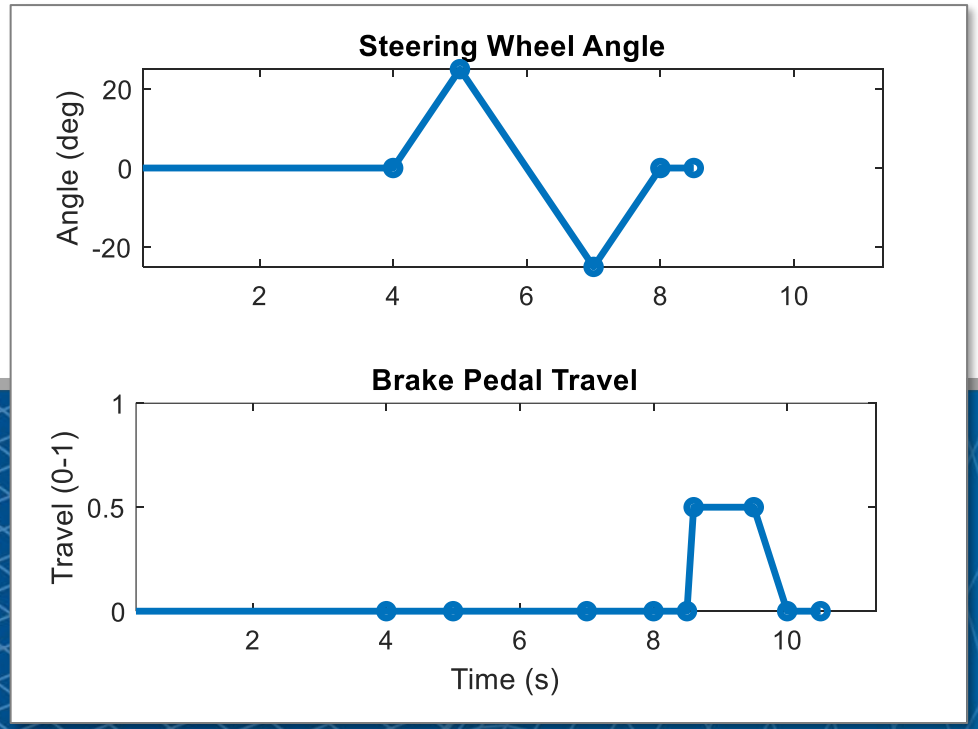


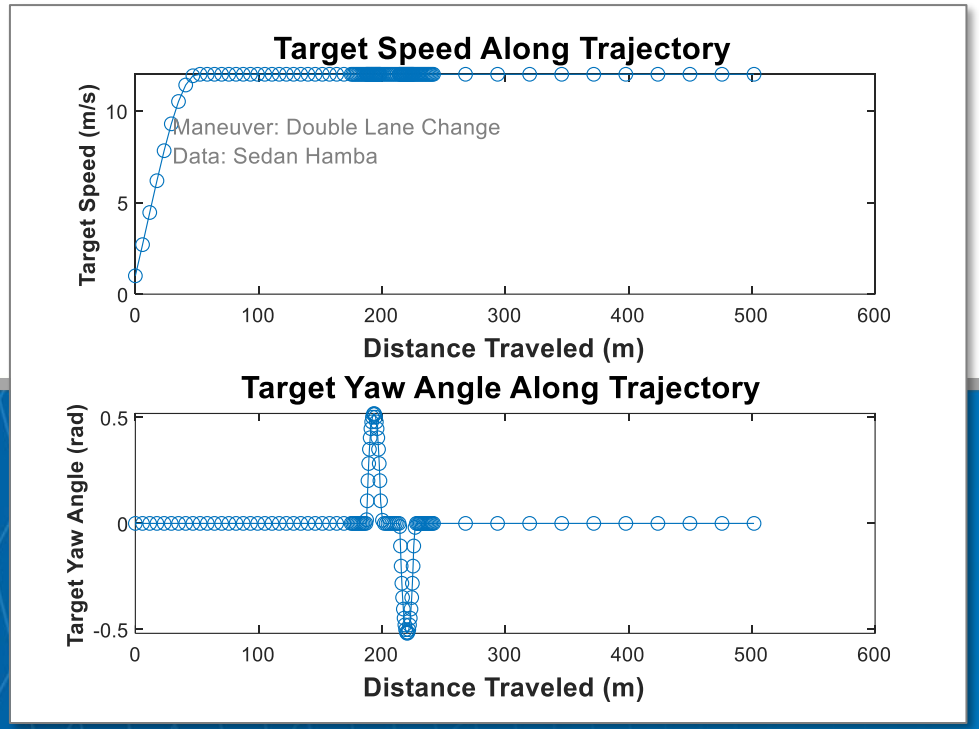
Steps to Create New Variant Subsystem

1. Create a variant, save in a library
2. Create a new library with a variant subsystem in it
3. Add a Mask the variant subsystem
4. Add parameter VariantLabel
5. Switch to the Code tab
6. Add an Initialization function
7. Add the set_param() command
8. Set the “Allow mask…” checkbox
9. Save mask
10. Add variants to this subsystem



Simscape Vehicle Templates Maneuvers

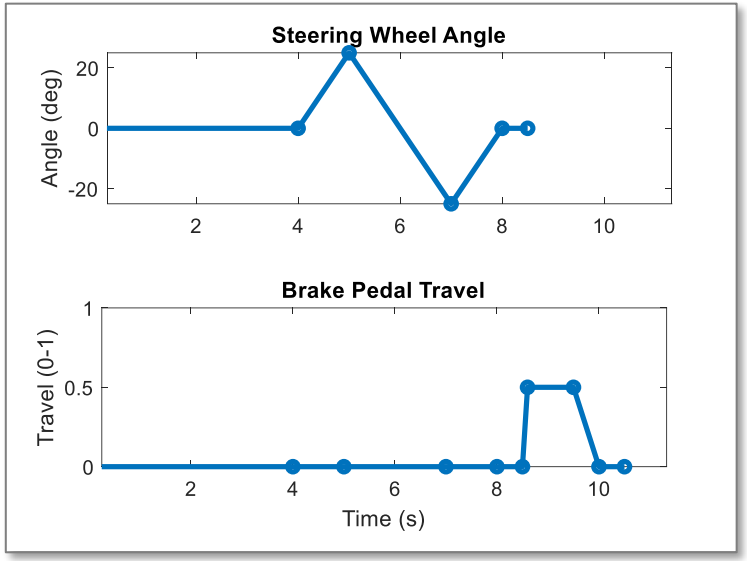


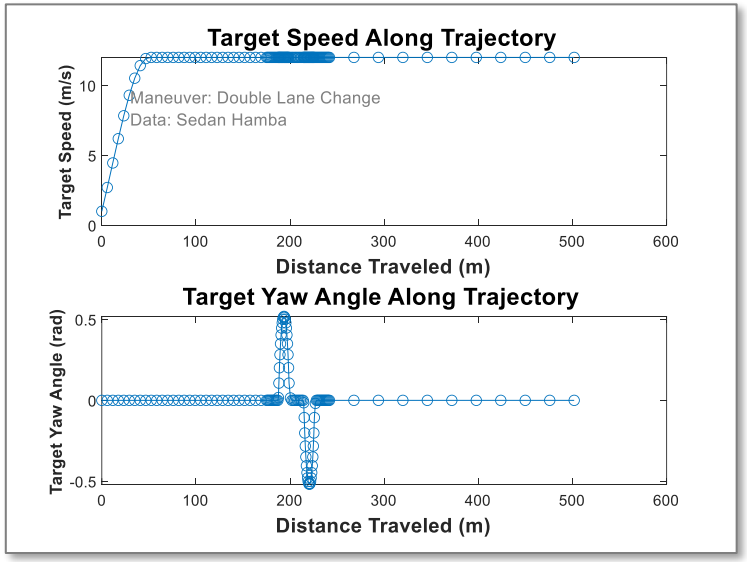


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Maneuvers

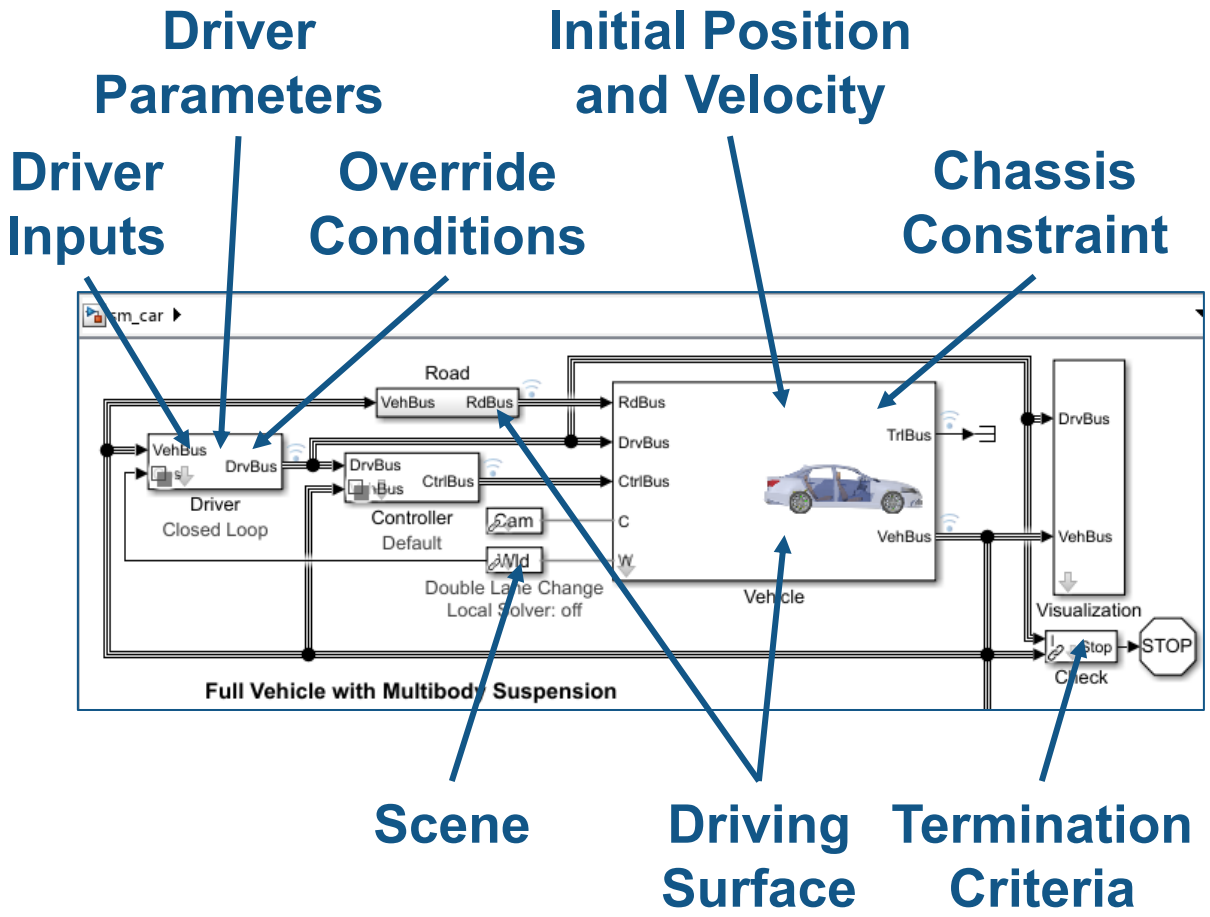
- Elements of a Maneuver
- Manveuvers: Moving Vehicle
  - Fishhook, Sine with Dwell, Slalom, and more
- Trajectory Definition
- Maneuvers: KnC



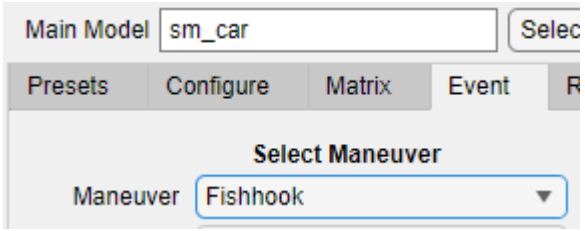


Elements of a Maneuver

- These items must be properly configured to define a maneuver
  - **Scene**
  - **Driving surface**
  - **Initial position and speed** of vehicle
  - **Inputs** for driver (Actions OR Commands)
  - **Termination criteria** for simulation
- Closed-loop maneuvers need
  - **Driver Parameters**
  - **Override Conditions** (optional)
- KnC test requires **chassis constraint**
- Everything can be configured using MATLAB



Fishhook



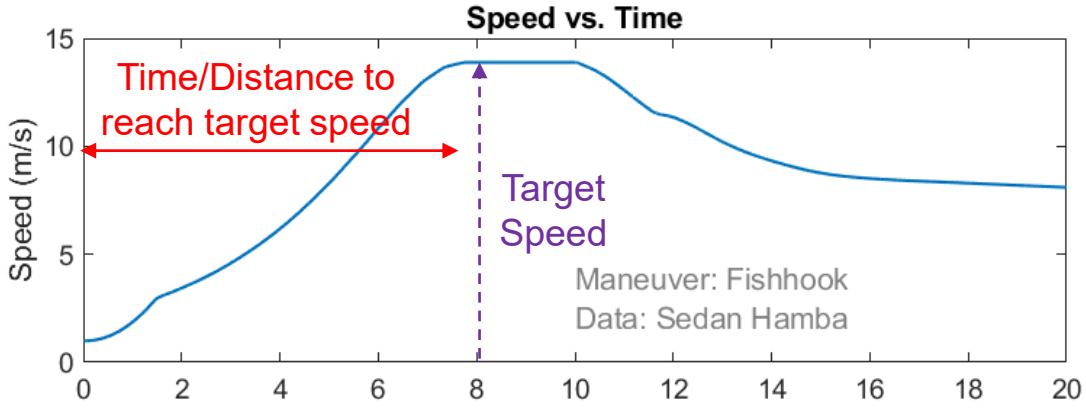
- Test Sequence
  - Vehicle accelerated to target speed
  - Release accelerator
  - Turn steering wheel to target angle and hold
  - Turn steering wheel to opposite angle and hold
  - Return to Center

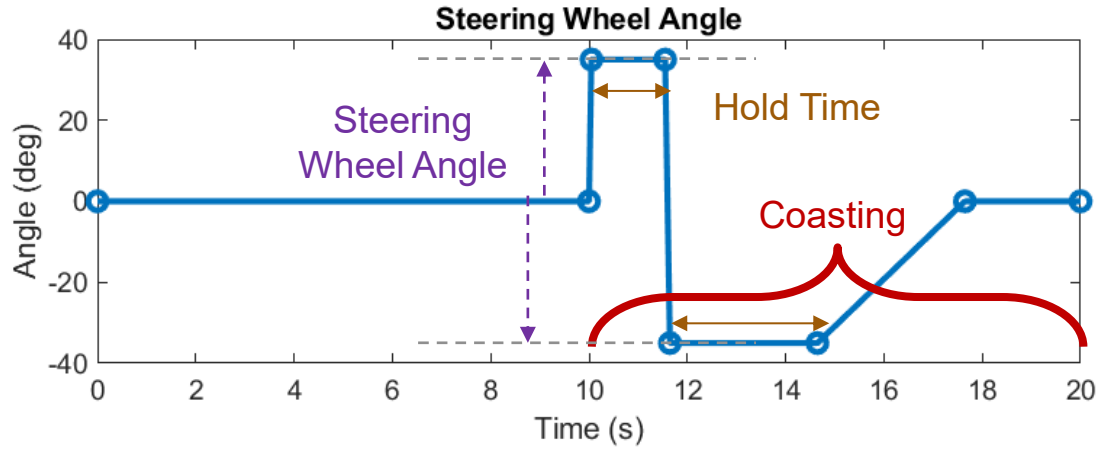
```
sm_car_config_maneuver('sm_car','Fishhook');
```

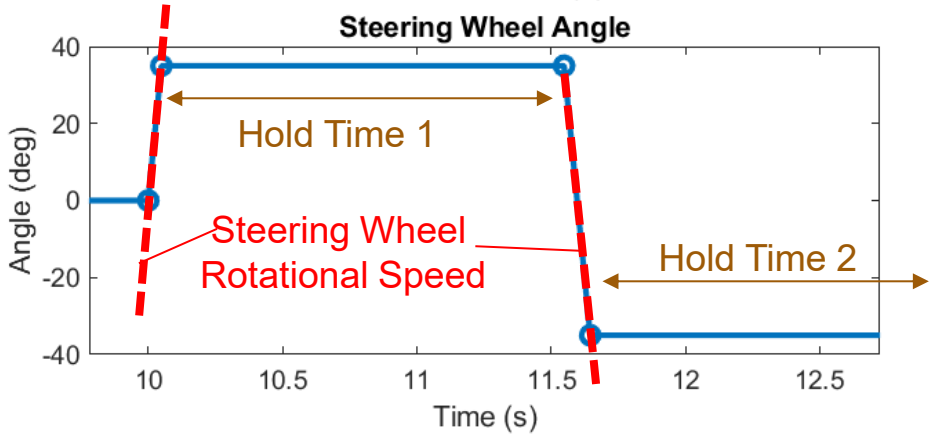
- For custom settings

```
Maneuver = sm_car_maneuverdata_fishhook(
50, 8, NaN, 35, 720, 1.5, 3, 10, 3)
```

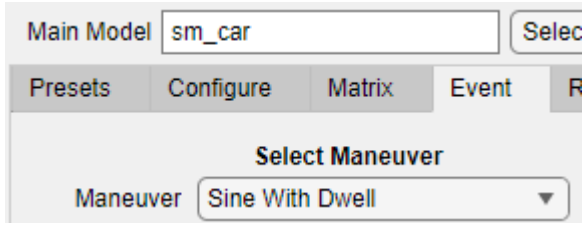
Target Speed (km/h)	Target Time (sec)	Target Distance (m)	Steering Angle (deg)	Steering Rate (deg/s)	Hold Time 1 (sec)	Hold Time 2 (sec)	Event Start (sec)	Return Time (sec)







Sine With Dwell



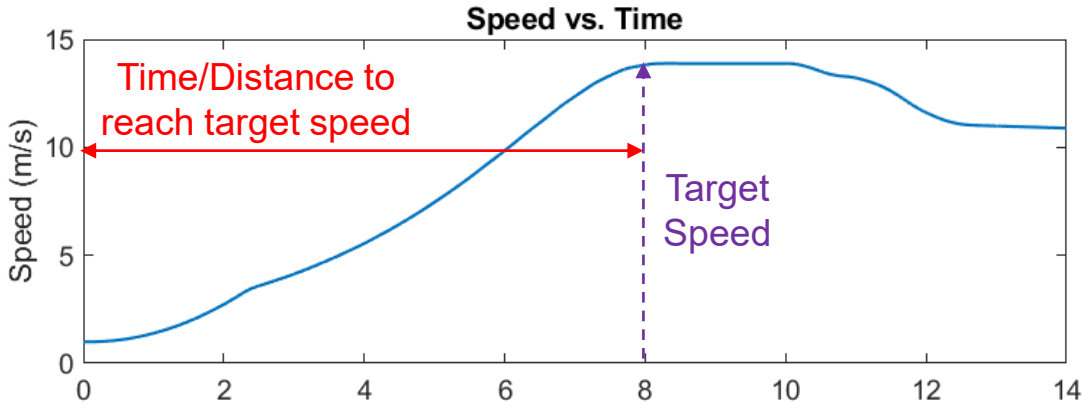
- Test Sequence
  - Vehicle accelerated to target speed
  - Release accelerator
  - Turn steering wheel through 3/4 of sine wave
  - Hold for dwell period
  - Return to Center

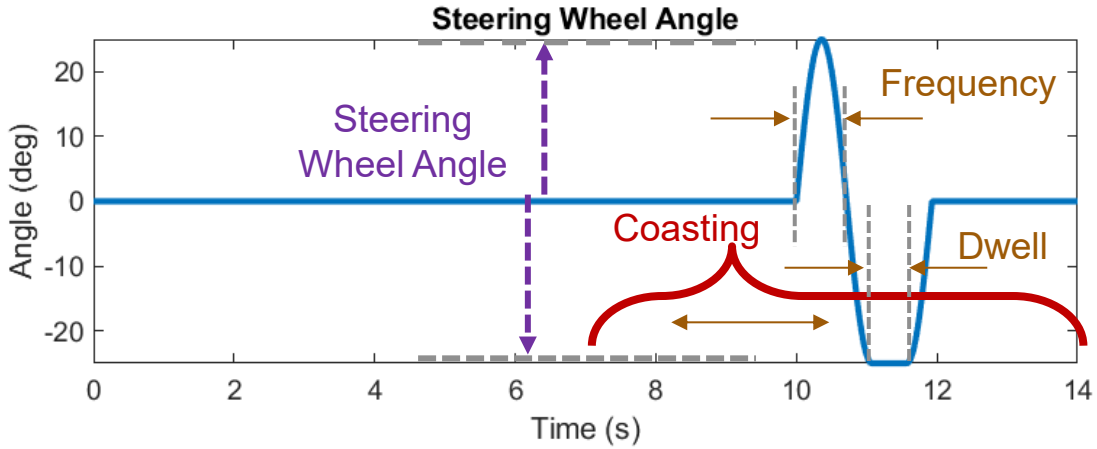
```
sm_car_config_maneuver('sm_car','Sine with Dwell');
```

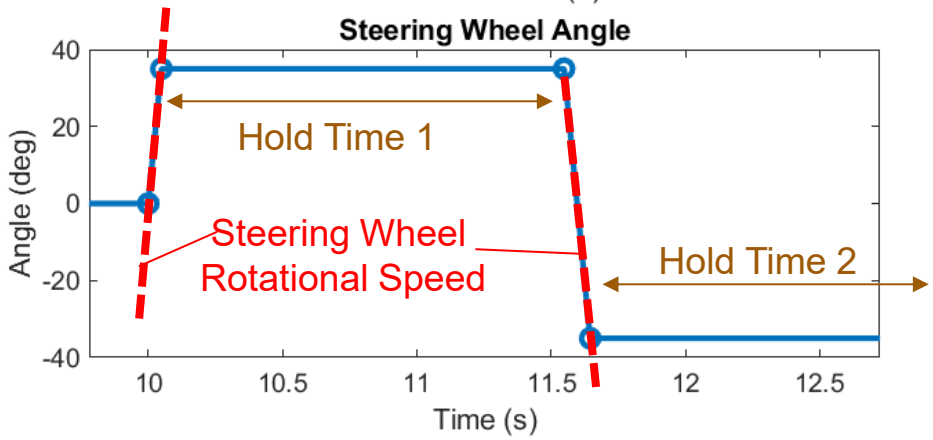
- For custom settings

```
Maneuver = sm_car_maneuverdata_sine_dwell (
50, 8, NaN, 25, 0.7, 0.5, 10, 3)
```

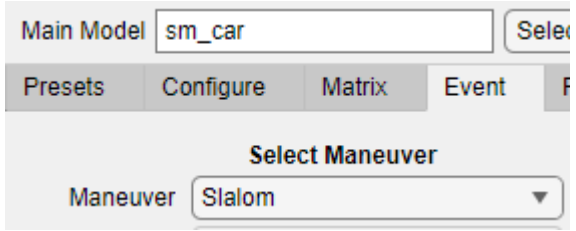
Target Speed (km/h)	Target Time (sec)	Target Distance (m)	Steering Angle (deg)	Sine Frequency (Hz)	Dwell Time (sec)	Event Start (sec)	End Time (sec)





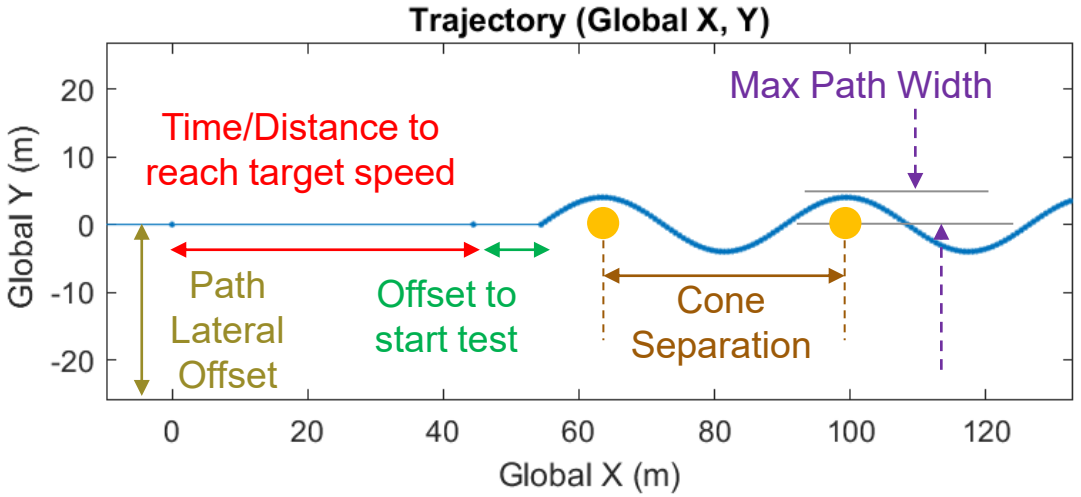


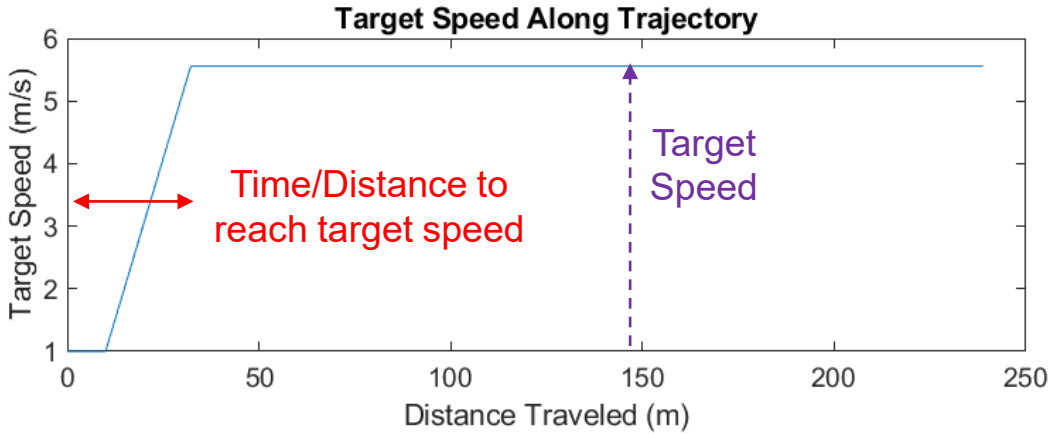
Slalom



- Test Sequence
  - Vehicle accelerated to target speed
  - Hold target speed
  - Follow slalom path around cones

```
sm_car_config_maneuver('sm_car','Slalom');
```



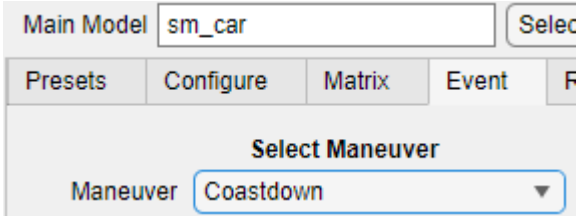


- For custom settings

```
Maneuver = sm_car_maneuverdata_slalom (
40, 8, NaN, 10, 0, 8, 18, 4, 1, 'sine', true)
```

Target Speed (km/h)	Target Time (sec)	Target Distance (m)	Start Offset (deg)	Lateral Offset (m)	Cone Separation (m)	Path Width (sec)	Left or Right (1, -1)	Path Shape	Show Plots

Coastdown



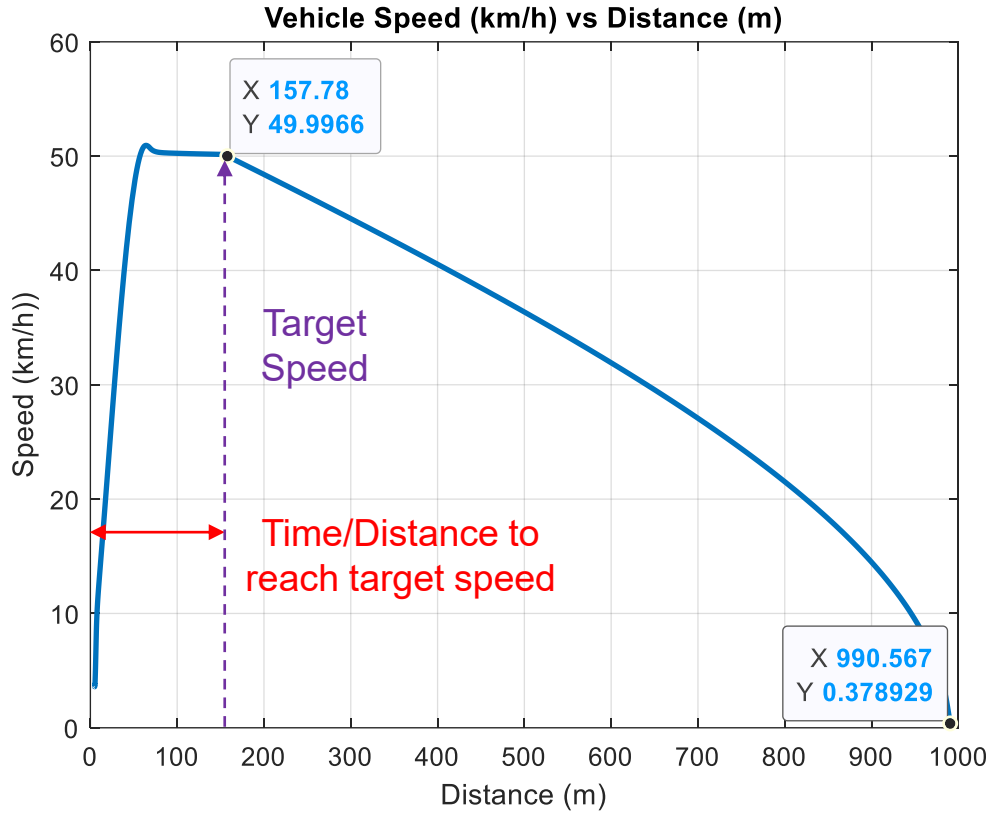
- Test Sequence
  - Vehicle accelerated to target speed
  - Hold target speed
  - Release accelerator, steer to hold straight line
  - Coast to stop

```
sm_car_config_maneuver('sm_car',Coastdown');
```

- For custom settings

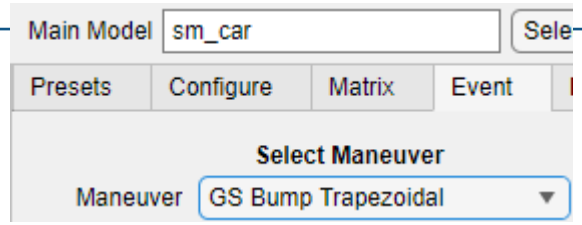
```
Maneuver = sm_car_maneuverdata_coastdown(
50, 10, NaN, 15)
```

Target Speed (km/h) | Target Time (sec) | Target Distance (m) | Event Start (sec)



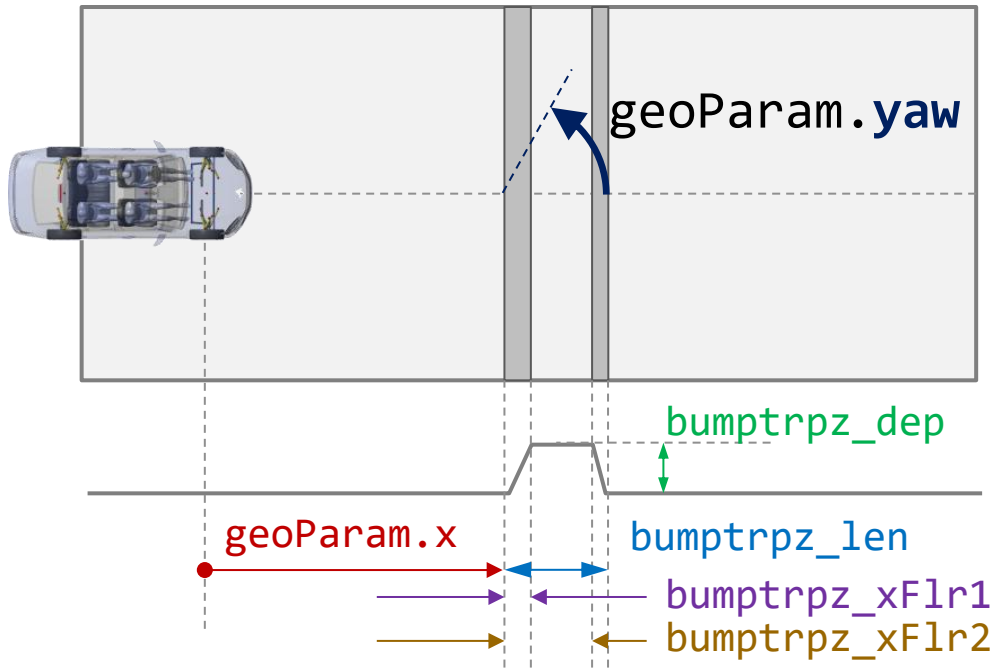
Coastdown distance = 990m-157m

Trapezoidal Bump




```
sm_car_config_maneuver('sm_car','GS Grid Surface BumpTrpz');
```

- Test Sequence
  - Vehicle accelerated to and holds target speed
  - Straight line along y=0
  - Uses Grid Surface
- For custom surface




```
Scene.GS_Grid_Surface = sm_car_scenedata_gsd_bump_trpz ...
                        (geoPar,gridPar);

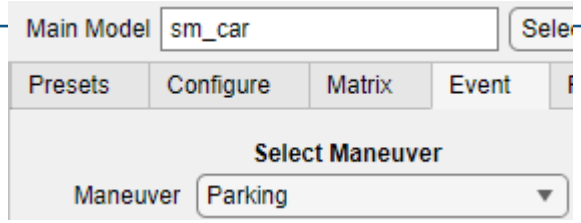
% Offsets - applied yaw-pitch-roll, then x-y-z.
% Rotation done first to simplify asymmetrical bumps
(left/right only)
geoParam.x = 50; % m (pothole distance)
geoParam.y = 0; % m
geoParam.z = 0; % m
geoParam.yaw = 15*pi/180; % rad (rotation of surface)
geoParam.pitch = 0; % rad
geoParam.roll = 0; % rad
```

```
gridPar.bumptrpz_len = 4;
gridPar.bumptrpz_xFlr1 = 0.25;
gridPar.bumptrpz_xFlr2 = 3.75;
gridPar.bumptrpz_dep = -0.15;
gridPar.bumptrpz_side = 'both';
```

'both' : Bump left and right side
'left' : Bump left side only, right is flat
'right': Bump right side only, left is flat

See example `sm_car_test_bumptrpz.m`

Parking

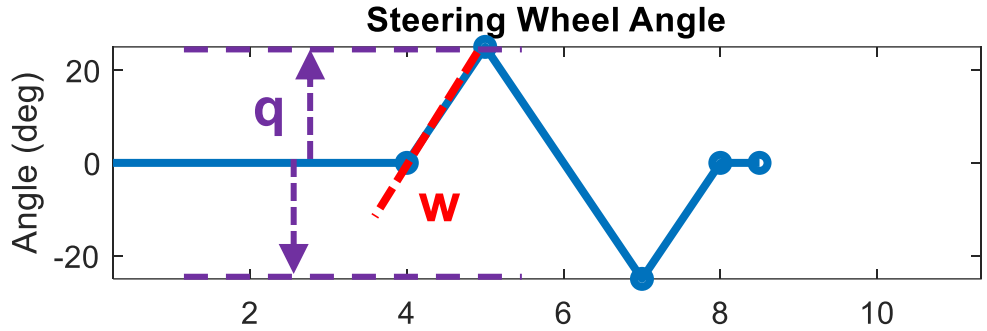


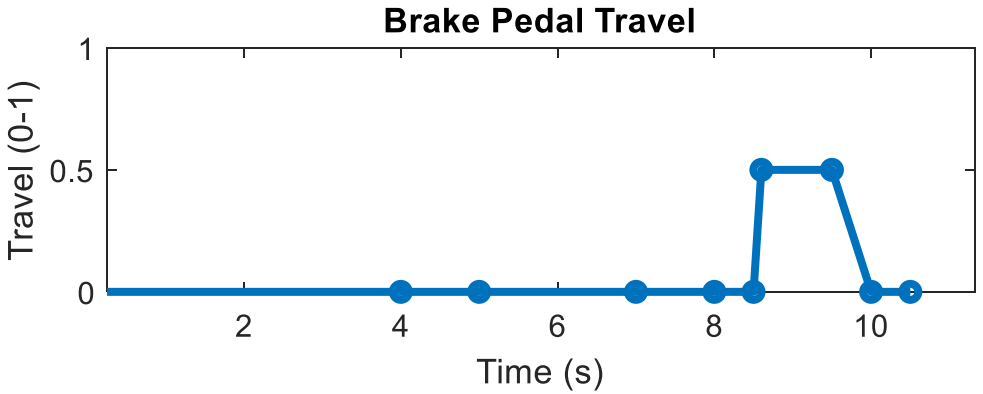

```
sm_car_config_maneuver('sm_car', 'Parking');
```

- Test Sequence
  - Vehicle drives at target speed (closed-loop)
  - Steering sequence: [0 q -q 0]
    Steering wheel turned at target speed (w)
  - Apply brakes to stop vehicle
- For custom settings

```
Maneuver = sm_car_maneuverdata_parking (
                -7, 25, 25)
```

Target Speed (km/h) | Max Steering Angle (deg) | Steering Angle Rate (deg/s)

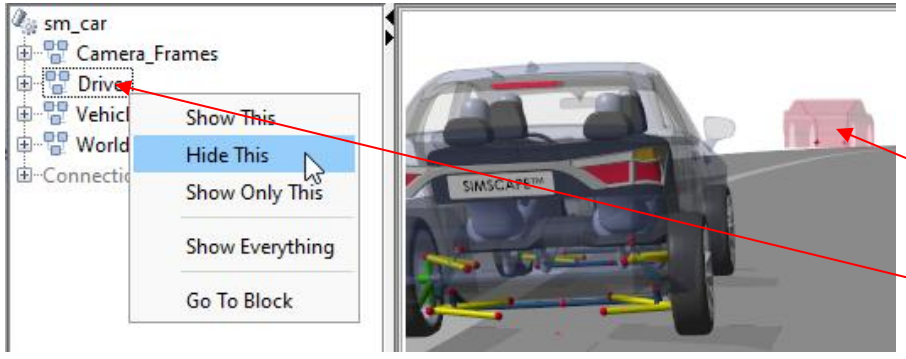




Trajectory Definition

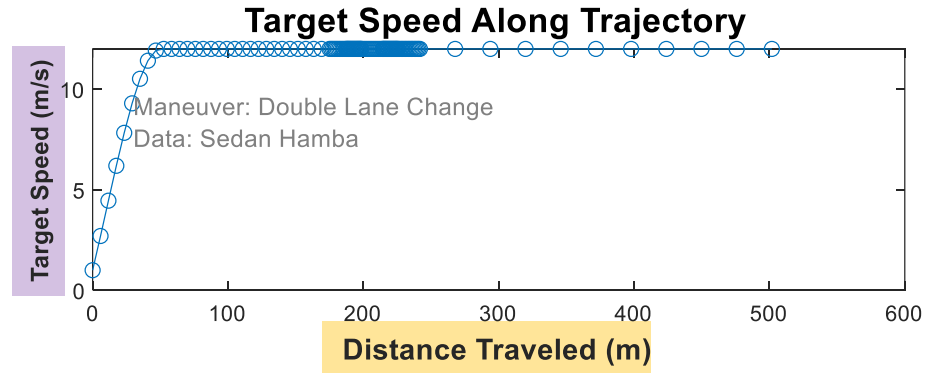
- Closed Loop Test: **Driver <u>Commands</u>**
  - **Target pose** (x, y, z position, yaw angle) vs. <u>distance</u> along path (`xTrajectory`)
  - Driver model compares **target pose** to current vehicle pose and determines driver actions

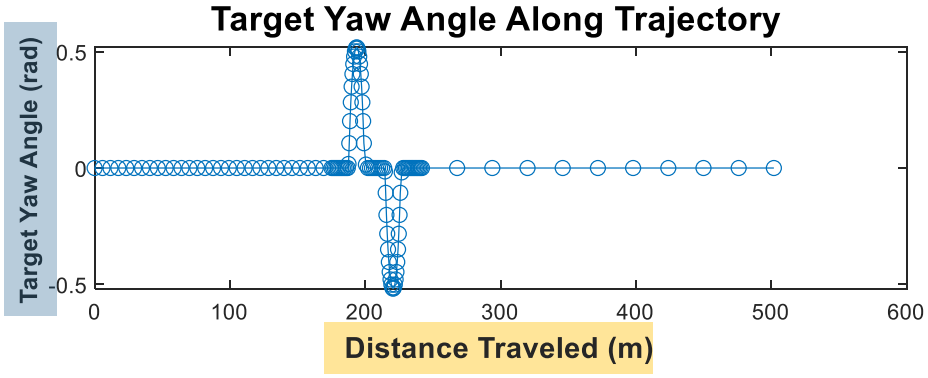
```
Maneuver.Trajectory
                x: [1×1 struct]
                y: [1×1 struct]
                z: [1×1 struct]
      xTrajectory: [1×1 struct]
               vx: [1×1 struct]
             aYaw: [1×1 struct]
```

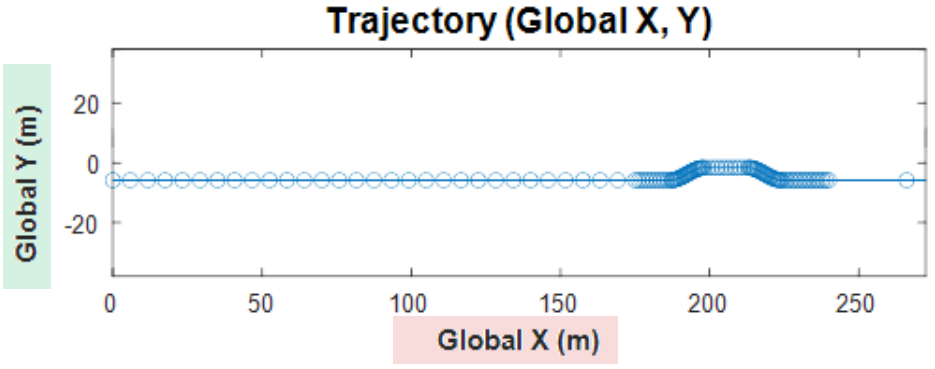


Target pose

Can be hidden in Explorer Tree

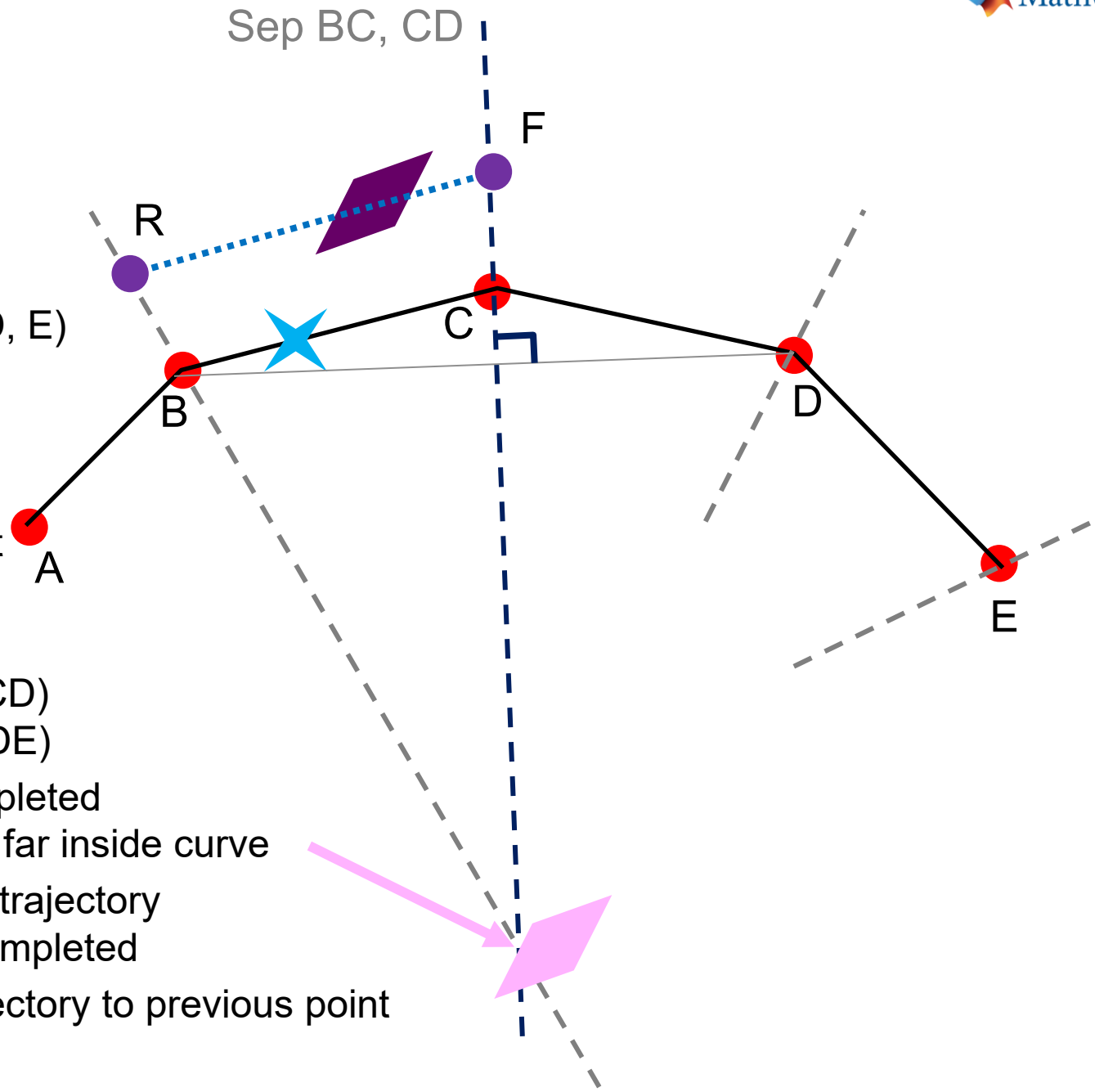






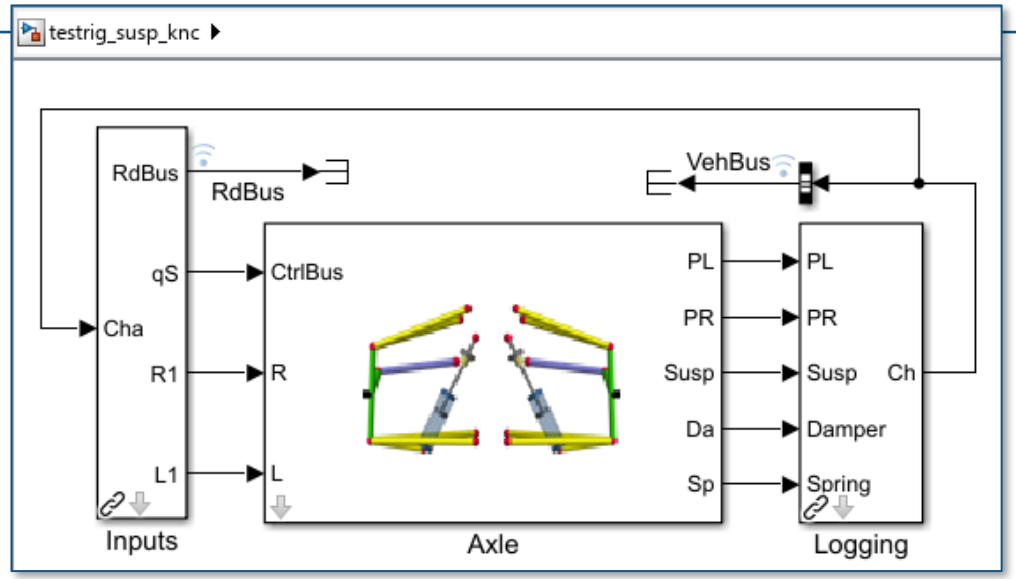
Path Following

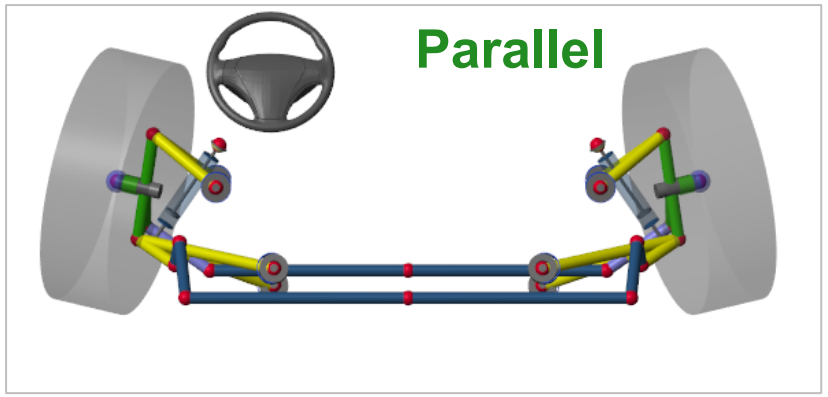
1. Find point closest to **car position** (pt C)
2. Find 2 points ahead and behind (A, B, D, E)
3. Find orientation of line separating segments BC, CD
4. Is car on BC side or CD side?
5. Find line parallel to BC (or CD) segment upon which car lies (FR)
6. Find intersection of line FR with line separating AB & BC (or BC & CD) and line separating BC & CD (or CD & DE)
7. Find percentage of line FR car has completed
   Limit percentage to 100% in case car is far inside curve
8. Use percentage to determine portion of trajectory along current segment that has been completed
9. Add that distance to distance along trajectory to previous point

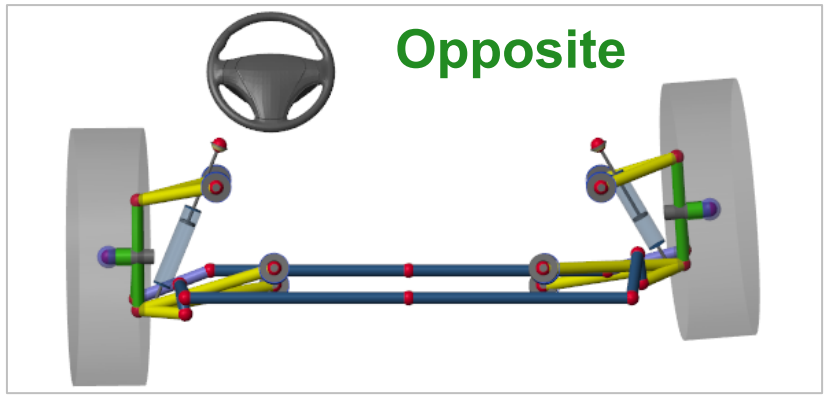


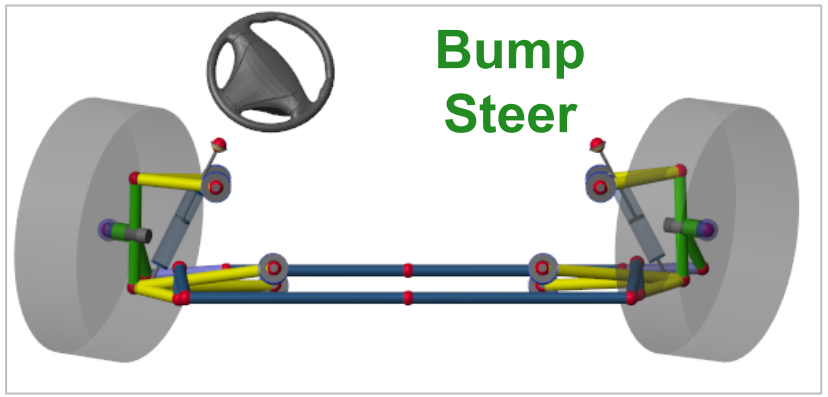
KnC Tests

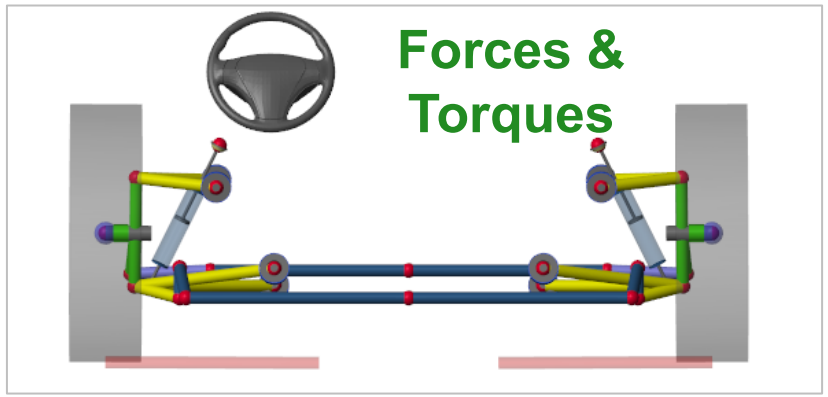
- Can be performed on quarter car, half car and full vehicle
- Most refined test setup is on half-car testrig
  - Test inputs
  - Calculated metrics







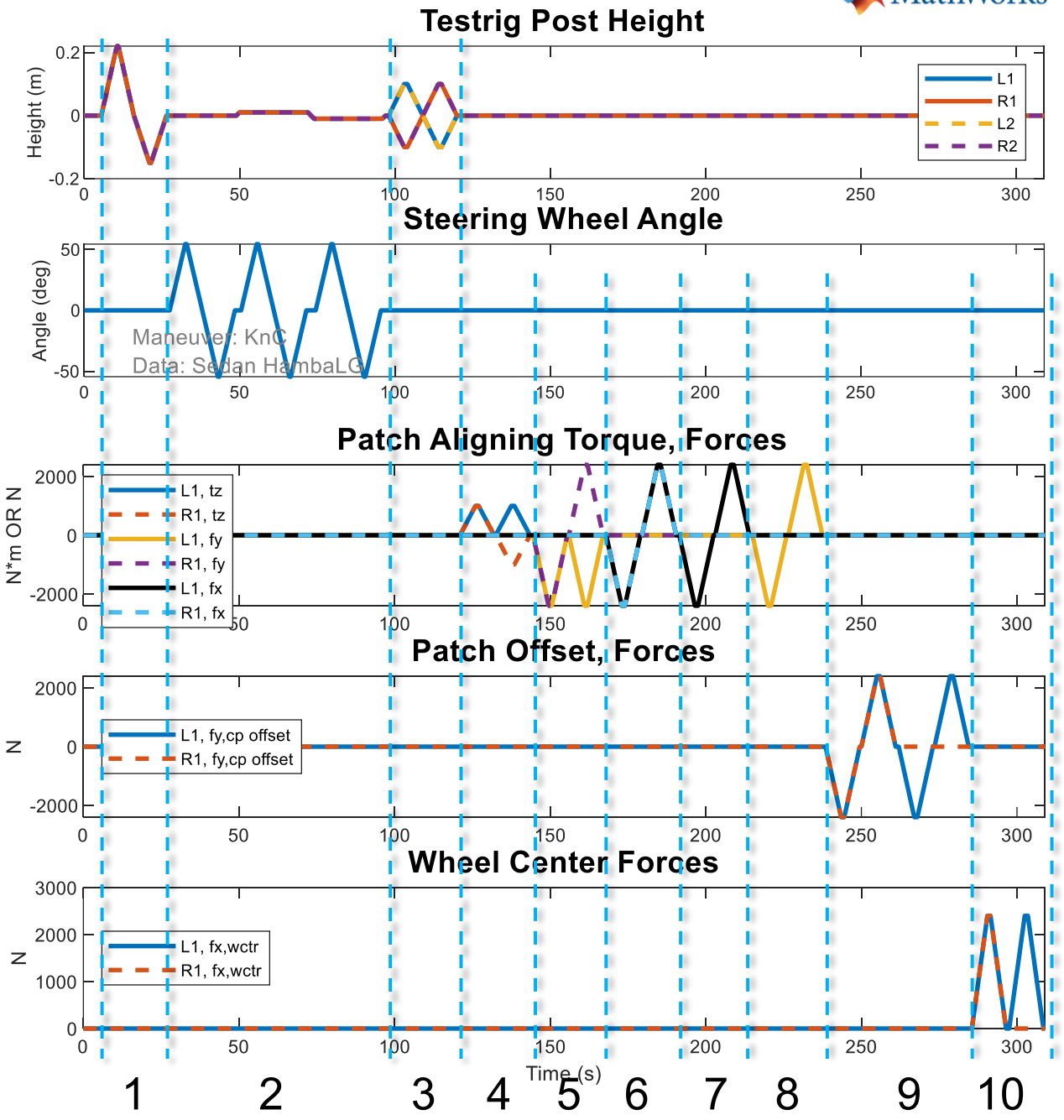




KnC Half Car

Test Sequence

1. Wheel center up and down, parallel
2. Bump Steer, Camber, Caster, KP)
3. Wheel center up and down, opposite
4. Contact patch, aligning torque, parallel, opposite
5. Contact patch, lateral force, parallel, opposite
6. Contact patch, longitudinal force, parallel
7. Contact patch, longitudinal force, left only
8. Contact patch, lateral force, left only
9. Contact patch offset, lateral force, parallel, left only
10. Wheel center, longitudinal force, parallel, left only

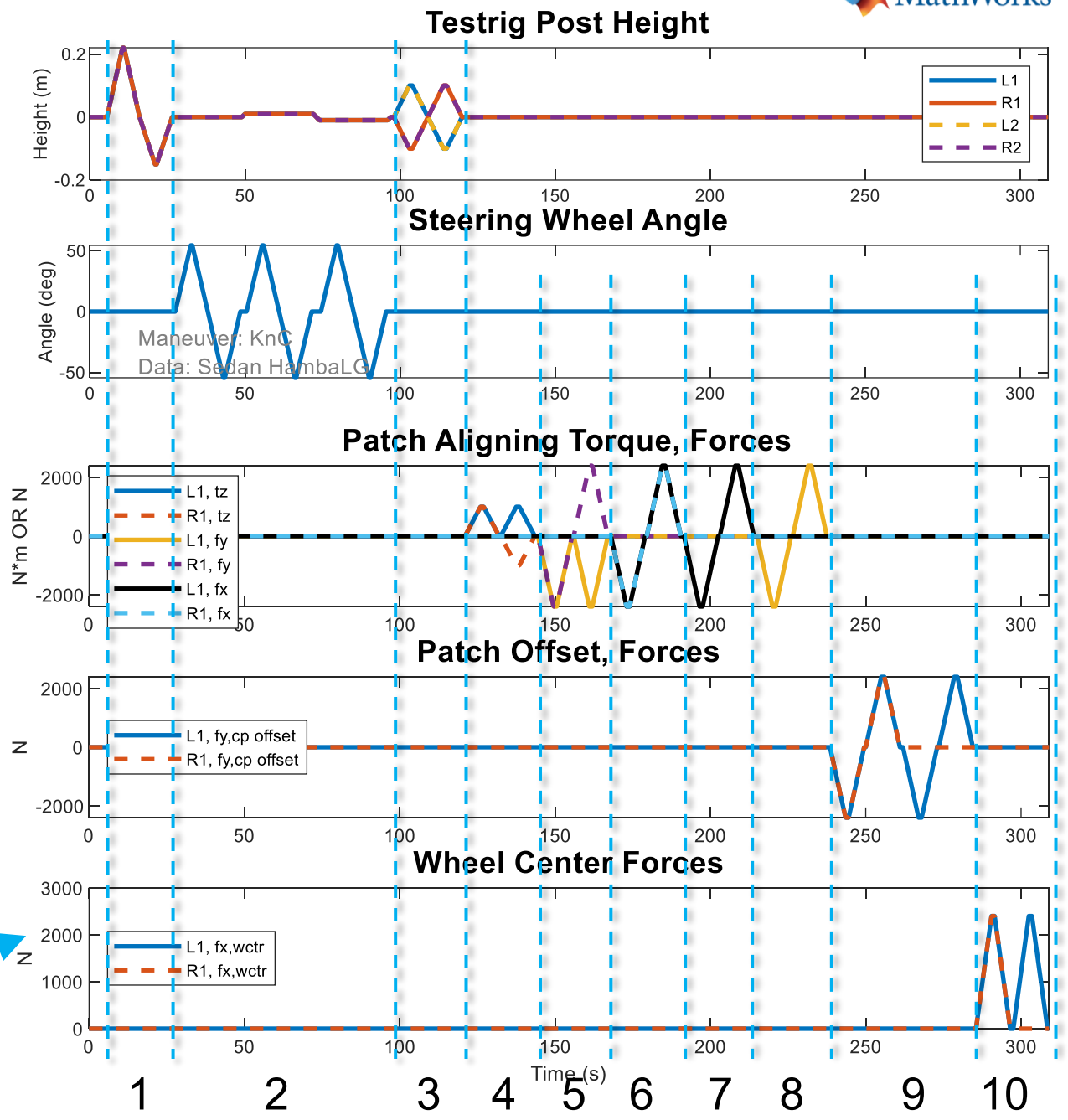


KnC Half Car
Test Sequence

- Set levels for test sequence via MATLAB command

```
Maneuver = sm_car_maneuverdata_knc(...
    0.22,... % Post Height for Jounce (m)
   -0.15,... % Post Height for Rebound (m)
    0.01,... % Height for KPI samples (m)
    1.0, ... % Steering Wheel Angle (rad)
    0.10, ...  % Post Height for Roll (m)
  500, ... % Aligning Torque (N*m)
1200, ... % Longitudinal force at patch (N)
1200, ... % Lateral force at patch (N)
1200, ... % Lateral force offset from patch (N)
1200, ... % Longitudinal force at wheel center (N)
-0.3);    % Longitudinal offset from patch (m)
```

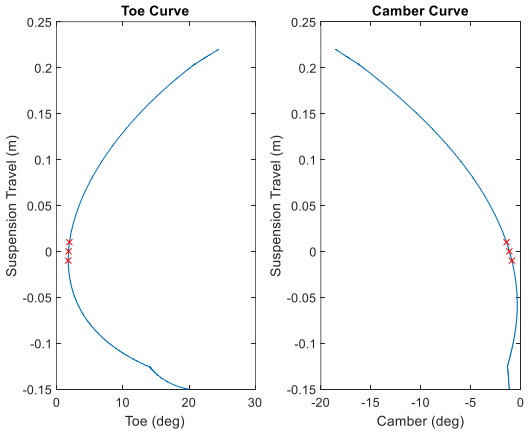
```
sm_car_plot_maneuver(Maneuver)
```

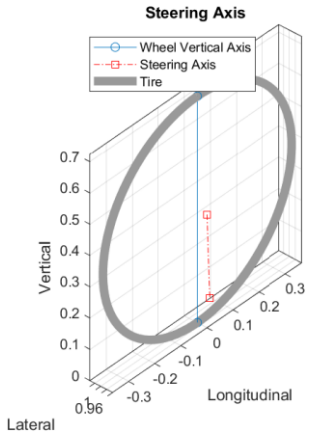


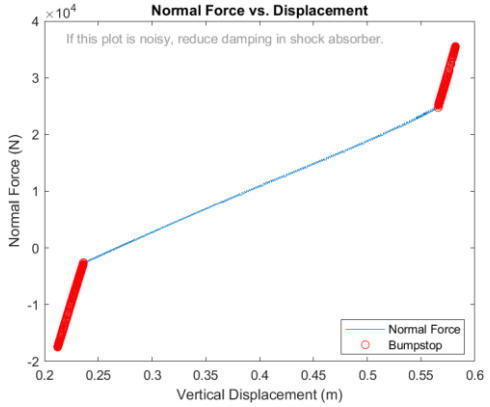
KnC Half Car

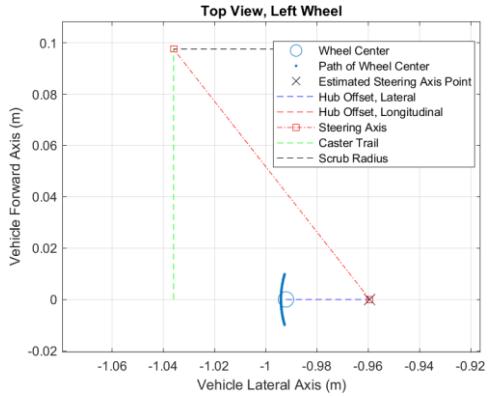
Outputs

- Table with plots to illustrate the calculations







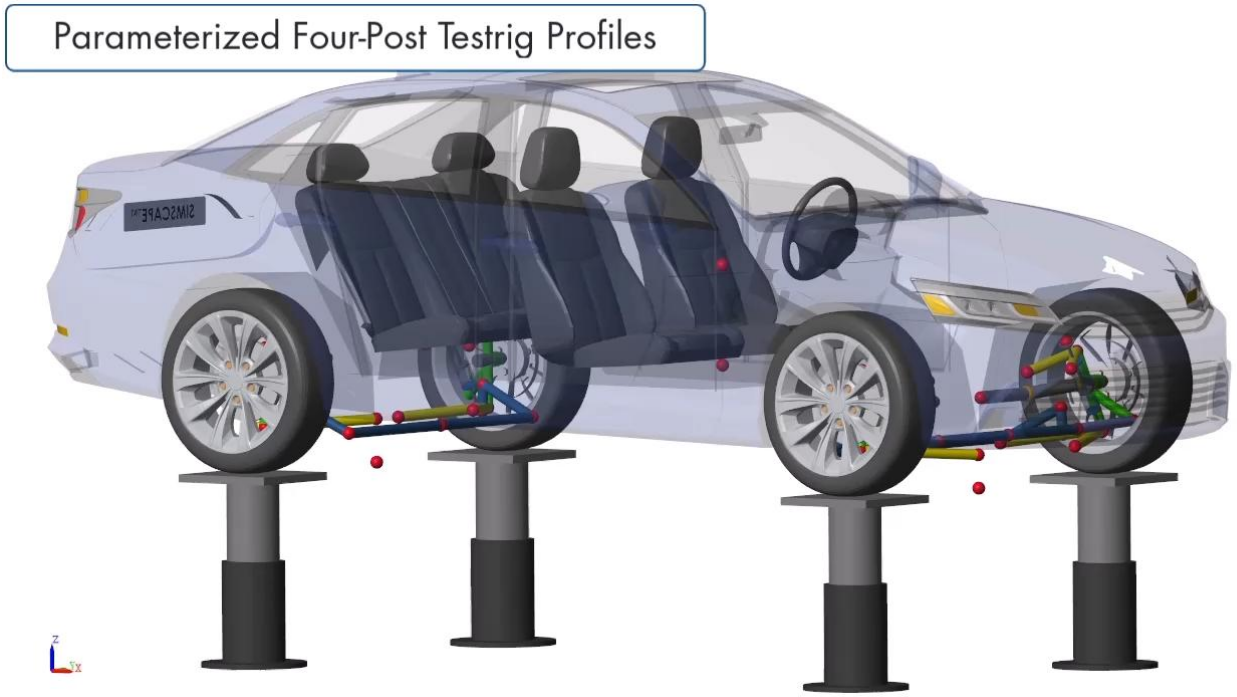


Names	Values	Units	Description
"Toe"	1.8003	"deg"	"+Toe In"
"Camber"	-1.1304	"deg"	"+Top Out"
"Caster"	15.073	"deg"	"+Top Rear of WC"
"KPI"	11.952	"deg"	"+Top Inside Bottom"
"Bump Steer"	8.3447	"deg/m"	"+/- 10mm, +Toe In"
"Bump Camber"	-26.442	"deg/m"	"+/- 10mm, +Top Out"
"Bump Caster"	27.988	"deg/m"	"+/- 10mm, +Caster Decrease"
"Caster Trail"	97.562	"mm"	"+Forward of Contact Patch"
"Scrub Radius"	-43.895	"mm"	"+Inside Wheel Center"
"Hub Long Offset"	0.028968	"mm"	"+Forward of WC"
"Hub Lat Offset"	32.764	"mm"	"+Inside Wheel Center"
"Suspension Rate"	78.789	"N/mm"	"+/-20mm, Wheel Center"
"Wheel Travel Bump"	0.20404	"mm"	"Up"
"Wheel Travel Rebound"	-0.1261	"mm"	"Down"
"Wheel Travel Total"	0.33014	"mm"	"Up+Down"
"WC Recession Long"	0.053421	"mm/m"	"+Backwards"
"WC Recession Lat"	-0.0040977	"mm/m"	"+In"
"Spring Ratio"	0.74939	"1"	"Spring Disp/WC Disp"
"Damper Ratio"	0.74939	"1"	"Damper Disp/WC Disp"
"Roll Steer"	0.61443	"deg"	"+/-2deg Roll, +Toe In"
"Roll Camber"	-1.9239	"deg"	"+/-2deg Roll, +Top Out"
"Roll Stiffness"	1736.7	"N*m/deg"	"+/-2deg Roll, Wheel Center"
"Max Rack Travel"	57.643	"mm"	"Rack Travel at Max Steer"
"Steering Ratio"	1.5837	"1"	"Handwheel Angle/Toe Angle, 20 deg Toe"
"Max Toe Out"	-31.151	"deg"	"+Toe In"
"Max Toe In"	38.336	"deg"	"+Toe In"
"Toe/Align Torque (In)"	-1.6736	"deg/kN"	"In Phase"
"Toe/Align Torque (Out)"	-1.6783	"deg/kN"	"Out Phase"
"Toe/Lat Force (In)"	-0.11366	"deg/kN"	"In Phase"
"Toe/Lat Force (Out)"	-0.11385	"deg/kN"	"Out Phase"
"Camber/Lat Force (In)"	0.039126	"deg/kN"	"In Phase"
"Camber/Lat Force (Out)"	0.039276	"deg/kN"	"Out Phase"
"Lat Compliance WCtr/Lat Force (In)"	1.4071	"mm/kN"	"In Phase"
"Lat Compliance WCtr/Lat Force (Out)"	1.407	"mm/kN"	"Out Phase"
"Lat Compliance CP/Lat Force (In)"	1.4071	"mm/kN"	"In Phase"
"Lat Compliance CP/Lat Force (Out)"	1.407	"mm/kN"	"Out Phase"

```
>> TSuspMetrics = sm_car_knc_plot1toecamber(logsout_sm_car_testrig_quarter_car,true,true,true)
```

Four-Post Testrig (and Six-Post)

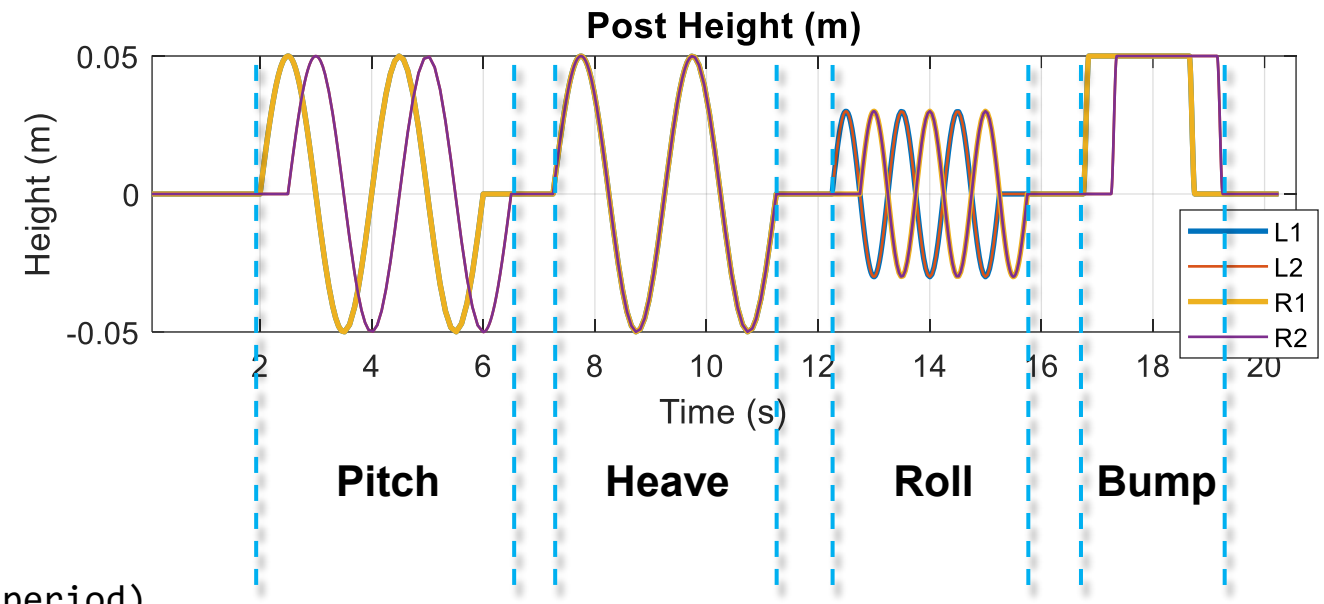
- Test suspensions using post-driven testrigs
  - Specify height with time
  - Parameterized tests sequences (pitch, heave, roll, bump)
  - Use custom test data
  - Include/neglect tire compliance
- Can be used on two and three axle vehicles



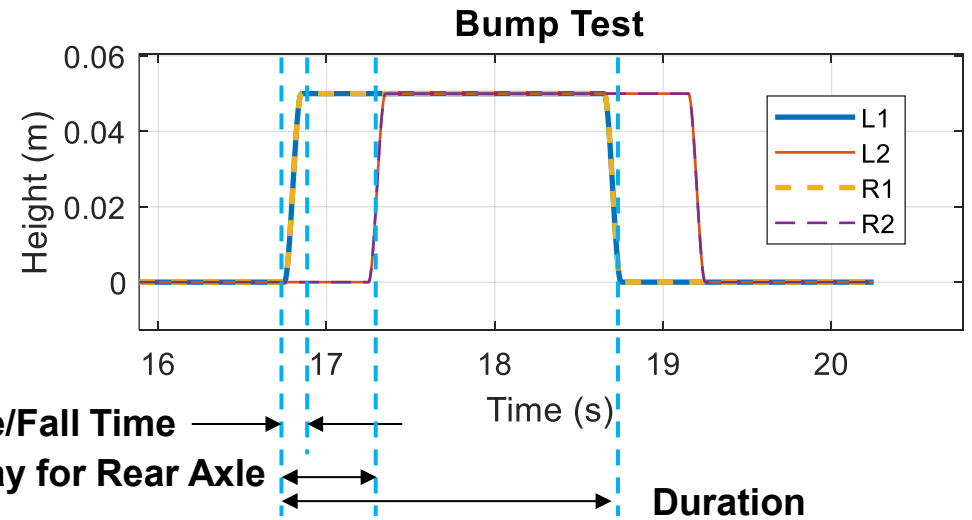
Four-Post Testrig (and Six-Post)

Cycle 1: Pitch, Heave, Roll, Bump

- Set levels for test sequence using MATLAB command



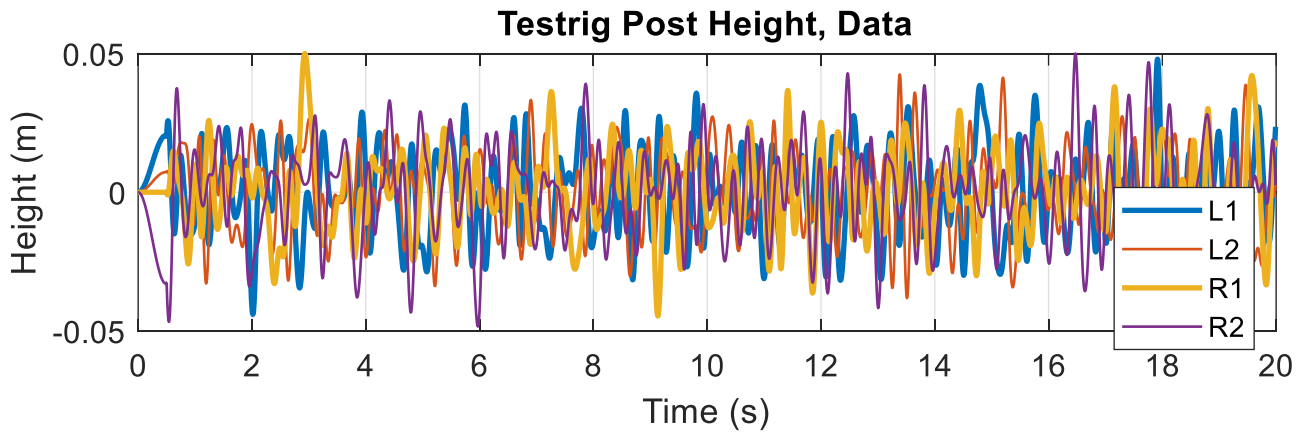

```
Maneuver = sm_car_maneuverdata_testrig4Post_cycle1(...
    2,    ... % Pitch test period (sec)
    2,    ... % Pitch test number of cycles
    0.05, ... % Pitch test amplitude (m)
    0.25, ... % Pitch delay for rear axle (fraction of period)
    2,    ... % Heave test period (sec)
    2,    ... % Heave test number of cycles
    0.05, ... % Heave test amplitude (m)
    1,    ... % Roll test period (sec)
    3,    ... % Roll test number of cycles
    0.03, ... % Roll test amplitude (m)
    0.1,  ... % Bump test rise and fall time (sec)
    2,    ... % Bump test duration (sec)
    0.05, ... % Bump test amplitude (m)
    0.5); ... % Bump test delay for rear axle (sec)
```




```
>> sm_car_config_maneuver('sm_car','Testrig 4 Post Cycle 1');
```

Four-Post Testrig (and Six-Post) Data

- Set levels for test sequence using MATLAB structure
  - Function creates random input
  - Replace data with your own input




```
Maneuver = sm_car_maneuverdata_testrig4Post_randFreqLim(...
   100,    ... % Sampling frequency (Hz)
    20,    ... % Desired duration (sec)
     2,    ... % Extra seconds at each end (sec)
     0.5,  ... % Lower cutoff frequency (Hz)
     5,    ... % Upper cutoff frequency (Hz)
     0.05, ... % Desired amplitude (m)
     4);       % Filter order
```

```
Maneuver.PostL1.t.Value  % Time vector (s)
Maneuver.PostL1.z.Value  % Height vector (m)
Maneuver.PostL1.InterpMethod.Value = 'Cubic spline';

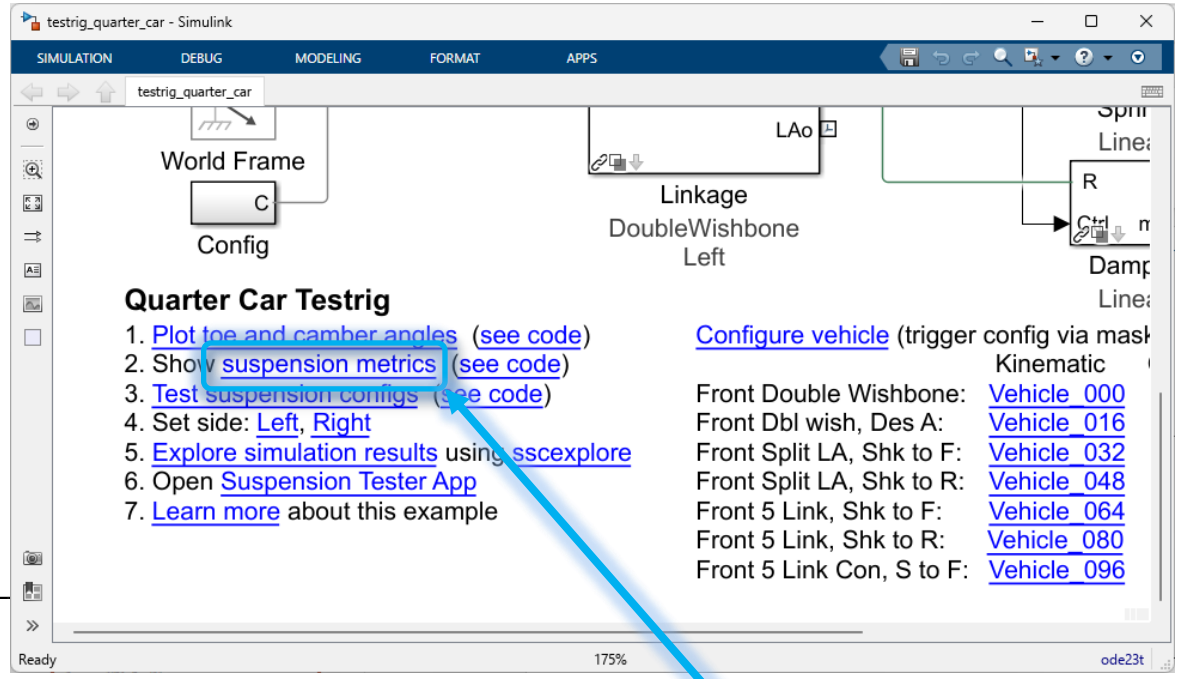
% Same for PostR1, PostL2, PostR2, PostL3, PostR3
```

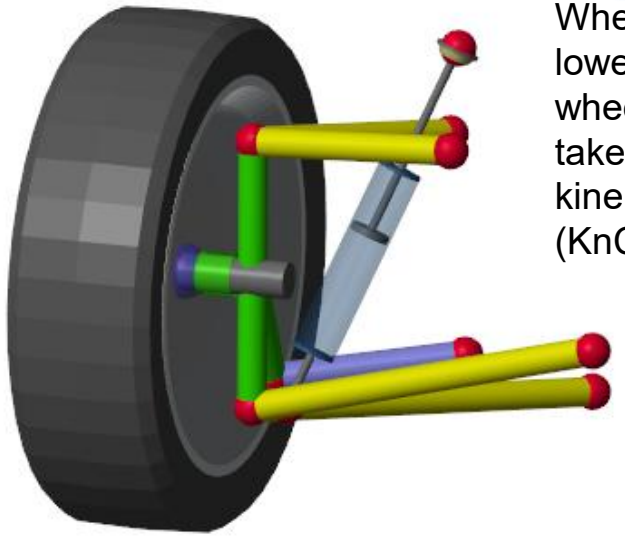
```
>> sm_car_config_maneuver('sm_car','Testrig 4 Post Data');
```

KnC Quarter Car

- Many standard KnC metrics are calculated using quarter car testrig

Names	Values	Units	Description
"Toe"	-6.5148e-14	"deg"	"+Toe In"
"Camber"	6.3611e-14	"deg"	"+Top Out"
"Caster"	15.329	"deg"	"+Top Rear of WC"
"KPI"	1.4744	"deg"	"+Top Inside Bottom"
"Bump Steer"	4.3307	"deg/m"	"+/- 10mm, +Toe In"
"Bump Camber"	-24.198	"deg/m"	"+/- 10mm, +Top Out"
"Bump Caster"	22.156	"deg/m"	"+/- 10mm, +Caster Decrease"
"Caster Trail"	96.113	"mm"	"+Forward of Contact Patch"
"Scrub Radius"	77.89	"mm"	"+Inside Wheel Center"
"Hub Long Offset"	-3.1539	"mm"	"+Forward of WC"
"Hub Lat Offset"	87.211	"mm"	"+Inside Wheel Center"
"Suspension Rate"	74.981	"N/mm"	"+/-20mm, Wheel Centre"
"Wheel Travel Bump"	0.20332	"mm"	"Up"
"Wheel Travel Rebound"	-0.13121	"mm"	"Down"
"Wheel Travel Total"	0.33453	"mm"	"Up+Down"
"WC Recession Long"	0.033205	"mm/m"	"+Backwards"
"WC Recession Lat"	-0.0043338	"mm/m"	"+In"
"Spring Ratio"	0.74744	"1"	"Spring Disp/WC Disp"
"Damper Ratio"	0.74744	"1"	"Damper Disp/WC Disp"



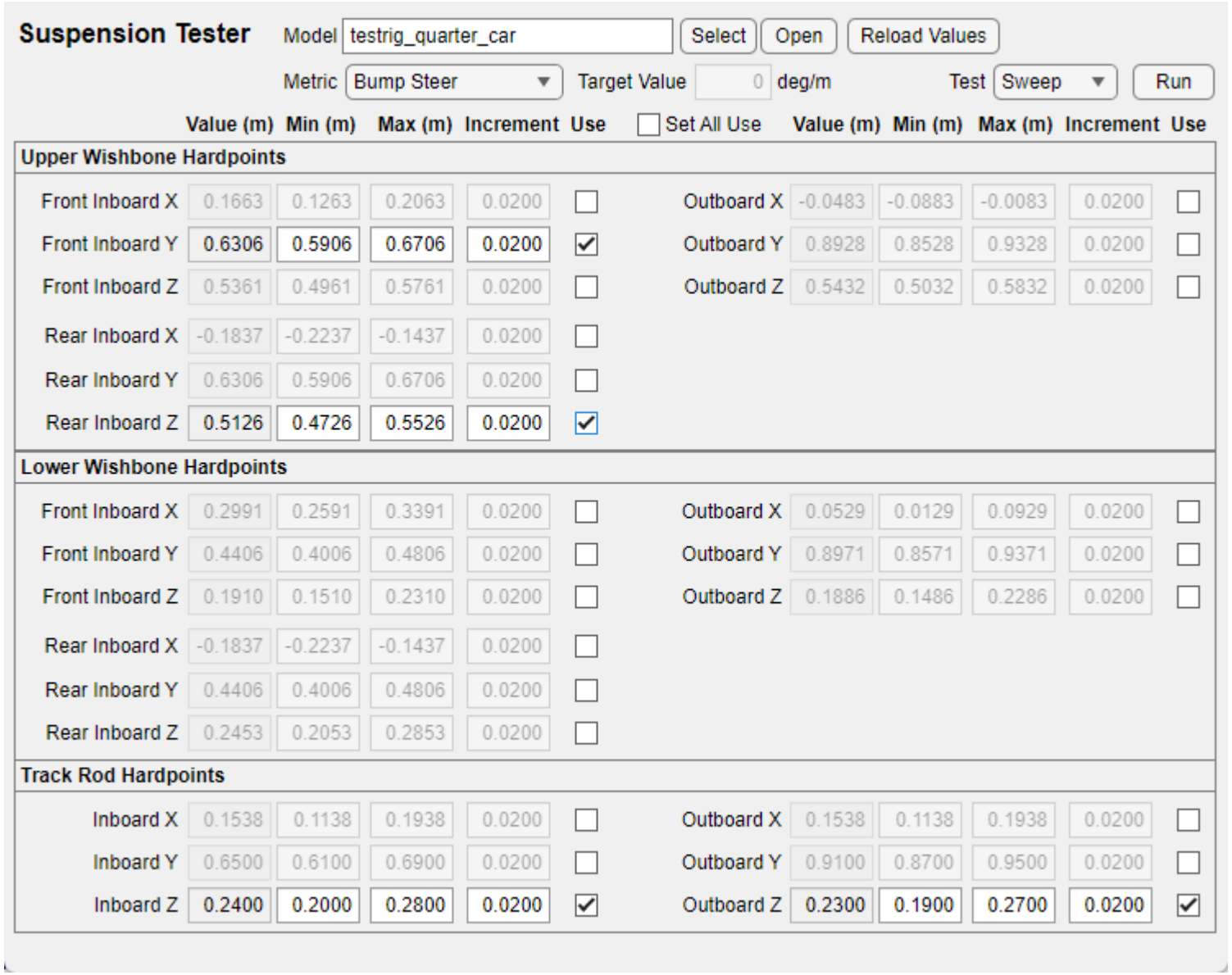


Wheel center is raised and lowered, rack is moved to steer wheel. Measurements are taken to calculate various kinematics and compliance (KnC) metrics.

KnC Quarter Car

Suspension Tester App

- App to configure sweeps and optimizations of hardpoint coordinates
  - Works on many suspension types
  - Apply to any combination of hardpoint coordinates
  - Use sweep to explore design space
  - Use optimization to find combination that achieves performance metric

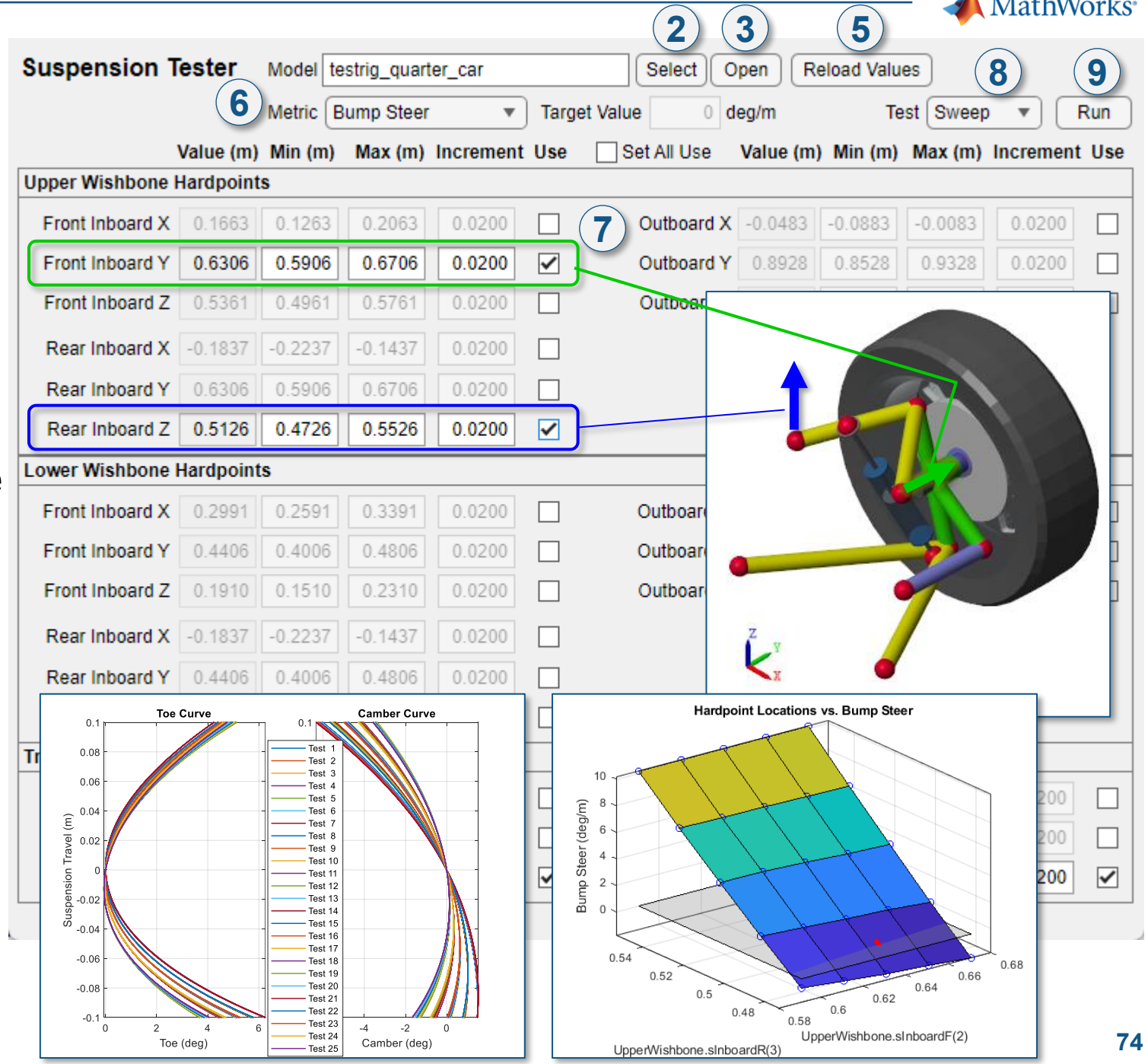



```
>> testrig_quarter_car_mlapp_sweep_optim
```

KnC Quarter Car

Suspension Tester App

1. Open App
2. Select model for test
3. Open model
4. Adjust values in workspace
5. Load values into App
6. Choose metric
7. Select coordinates for test and set valid ranges
8. Choose sweep/optim
9. Click Run



MATLAB R2022a

HOME PLOTS APPS

Search Documentation

Steve

New Script | New Live Script | New | Open | Find Files | Compare — FILE

Import Data | Clean Data | Variable | Save Workspace | Clear Workspace — VARIABLE

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Simulink — SIMULINK

Layout | Preferences | Set Path | Parallel | Add-Ons — ENVIRONMENT

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Web Browser - Simscape™ Vehicle Templates

Simscape™ Vehicle Templates

Location: hers/smiller/ssvt_srcR22a/Scripts_Data/Simscape_Vehicle_Library_Demo_Script.html

Simscape™ Vehicle Templates

Simscape™ Vehicle Templates are designed to help engineers model and simulate vehicles using Simscape products. This includes one-axle, two-axle, multi-axle vehicles, and trailers that are integrated with powertrain and chassis controls.

The templates include a vehicle model that you can configure for different powertrain architectures, driveline concepts, and suspension designs. You can also use a MATLAB® app to help you configure the vehicle and the test scenario. A modular library of components gives you a starting point to create custom vehicle models.

Full Vehicle

1. Vehicle: Open Model, Documentation
2. Library: Open Library

Workflow Examples

1. Vehicle Simulation Exercises
2. Lap Time Optimization: Overview, Edit Code
3. Lap Time + Battery SOC Optimization: Overview, Edit Code

Project - sm_car

Views

Files

Dependency Analyzer

Labels

Git

Current branch: master
Branch status: Normal
Coincident with

All | Project (2110) | Modified (86) | Layout: Tree

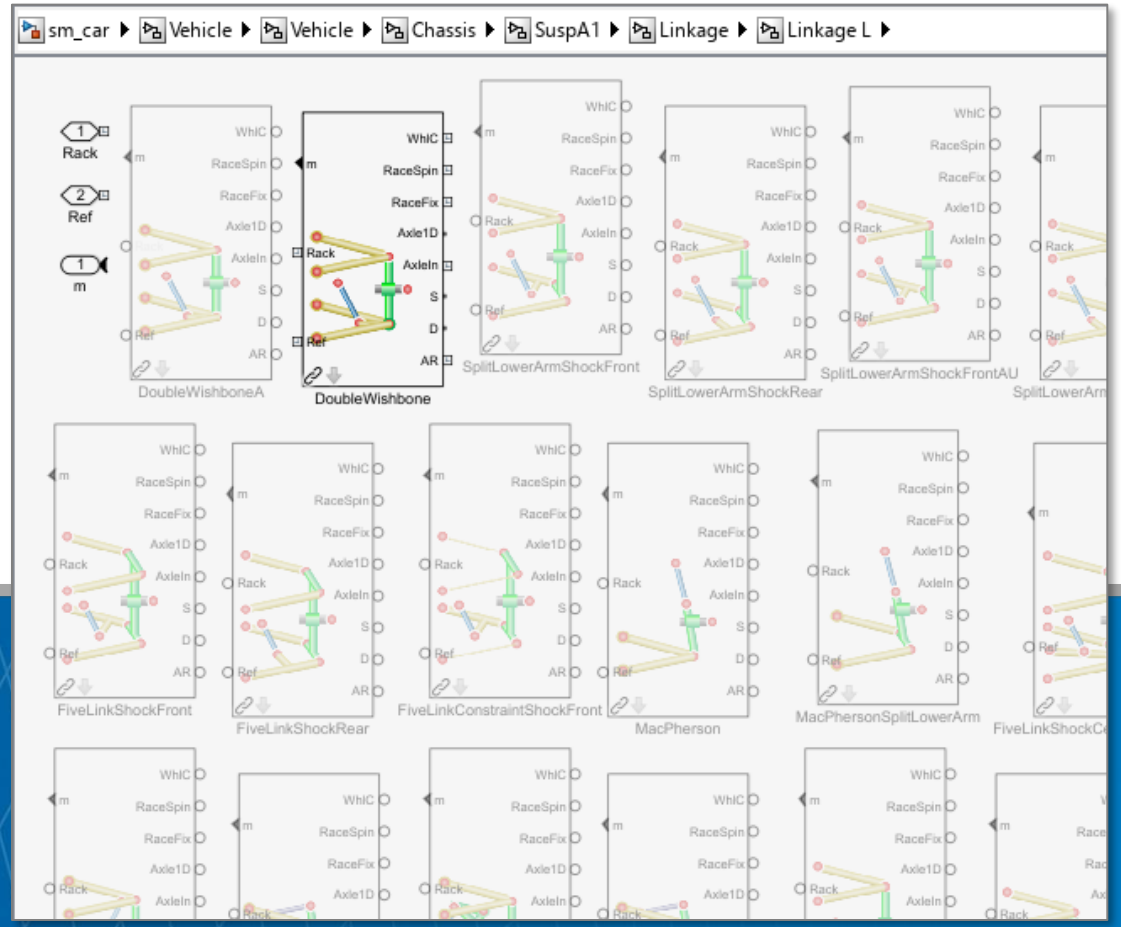
Name	Classification	Status	Path
.SimulinkProject			S\
Libraries	ForUser	✓	S\
Overview	ForUser	✓	S\
Scripts_Data	ForUser	✓	S\
SimResults	ForUser	✓	S\
Slides_Videos	ForUser	✓	S\
Testrigs	ForUser	✓	S\
work		·	S\
Workflows	ForUser	✓	S\
.gitattributes		✓	S\
.gitignore		✓	S\
fixMaskTunablePar.m		·	S\
LICENSE.md	ForUser	✓	S\
README.md	ForUser	✓	S\
SECURITY.md	ForUser	✓	S\
sm_car.slx	ForUser	✓	S\
sm_car_Axle3.slx	ForUser	✓	S\

Scripts_Data (Folder) — 1 labels

Command Window | Workspace

fx >>

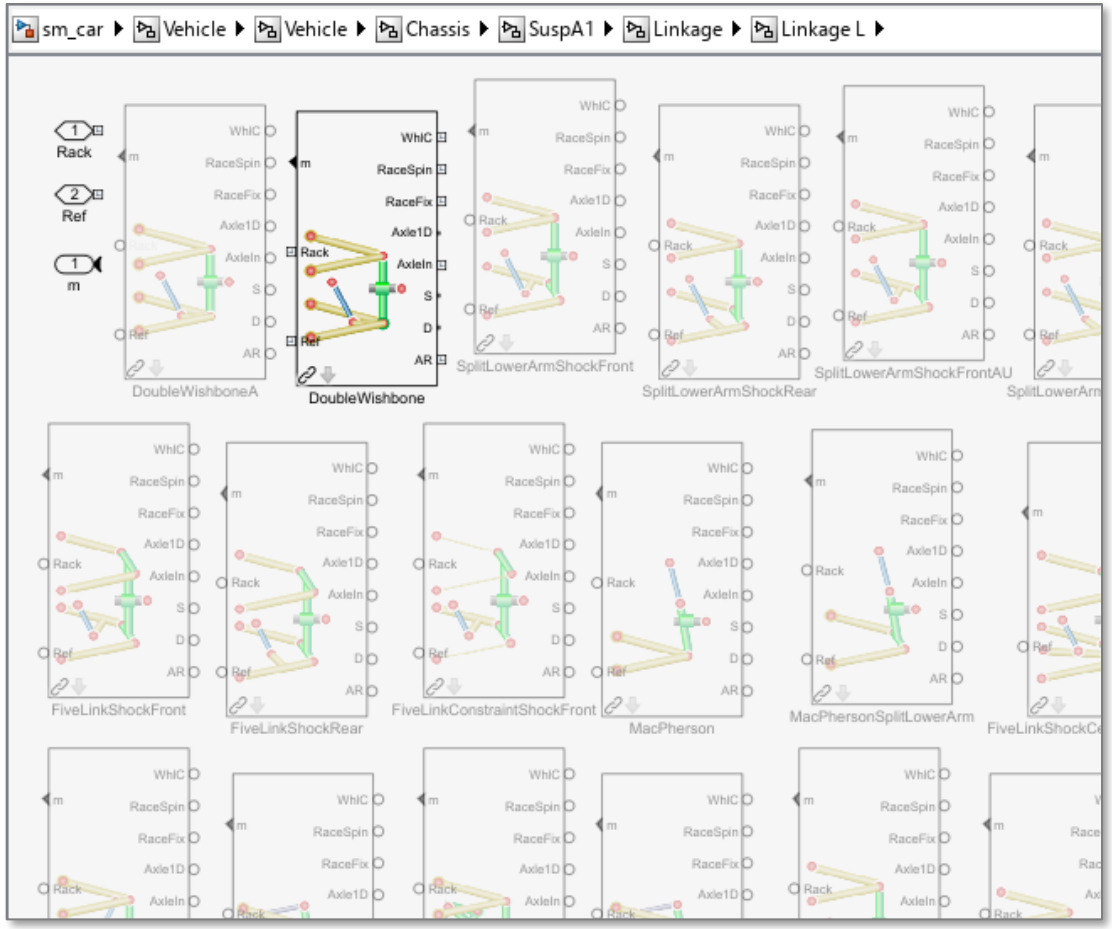
Simscape Vehicle Templates
Suspension (Linkages)



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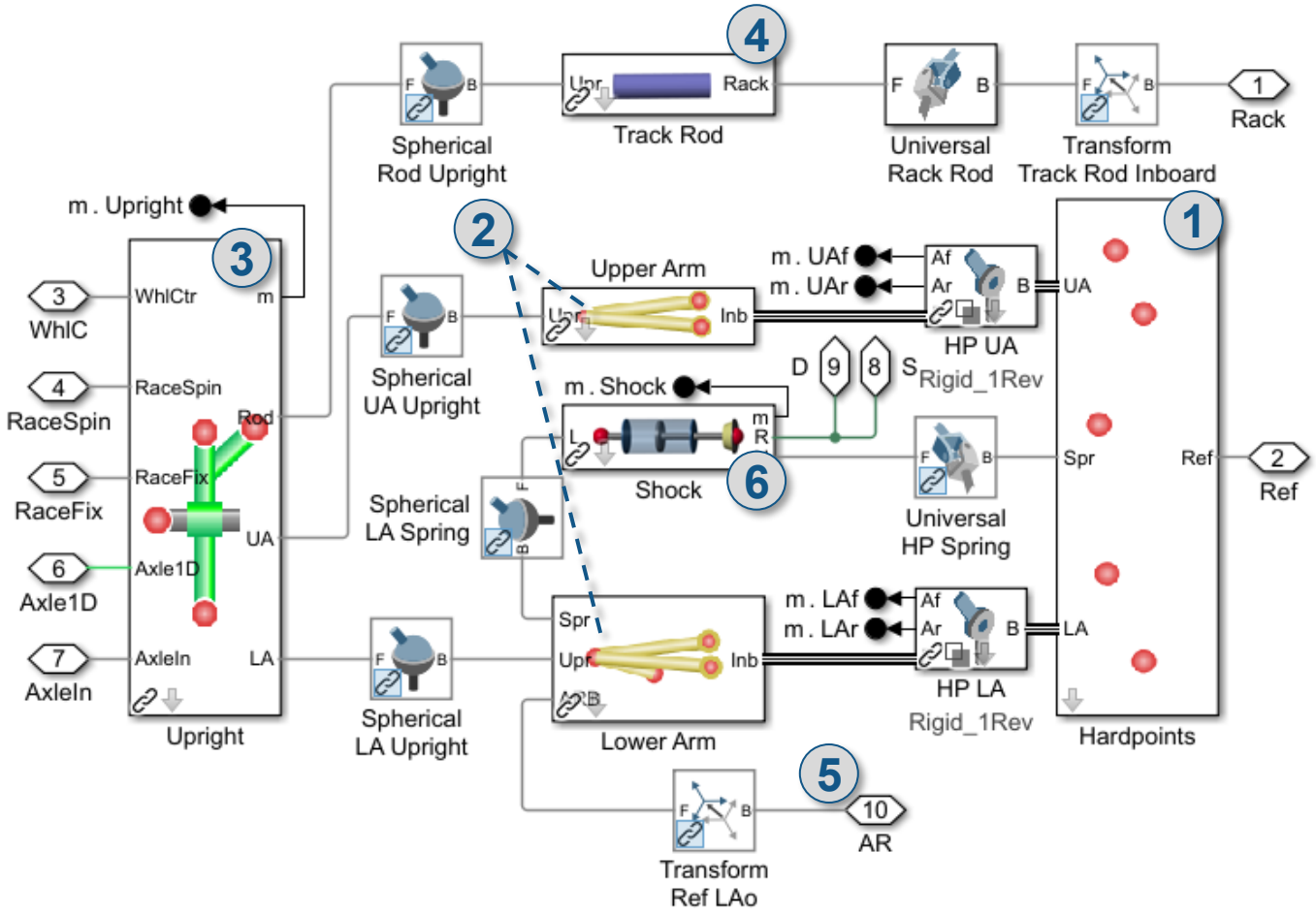
Suspension (Linkages)

- Suspension Structure
- Reference Frames
- Hardpoints
- Joints
- Springs and Dampers



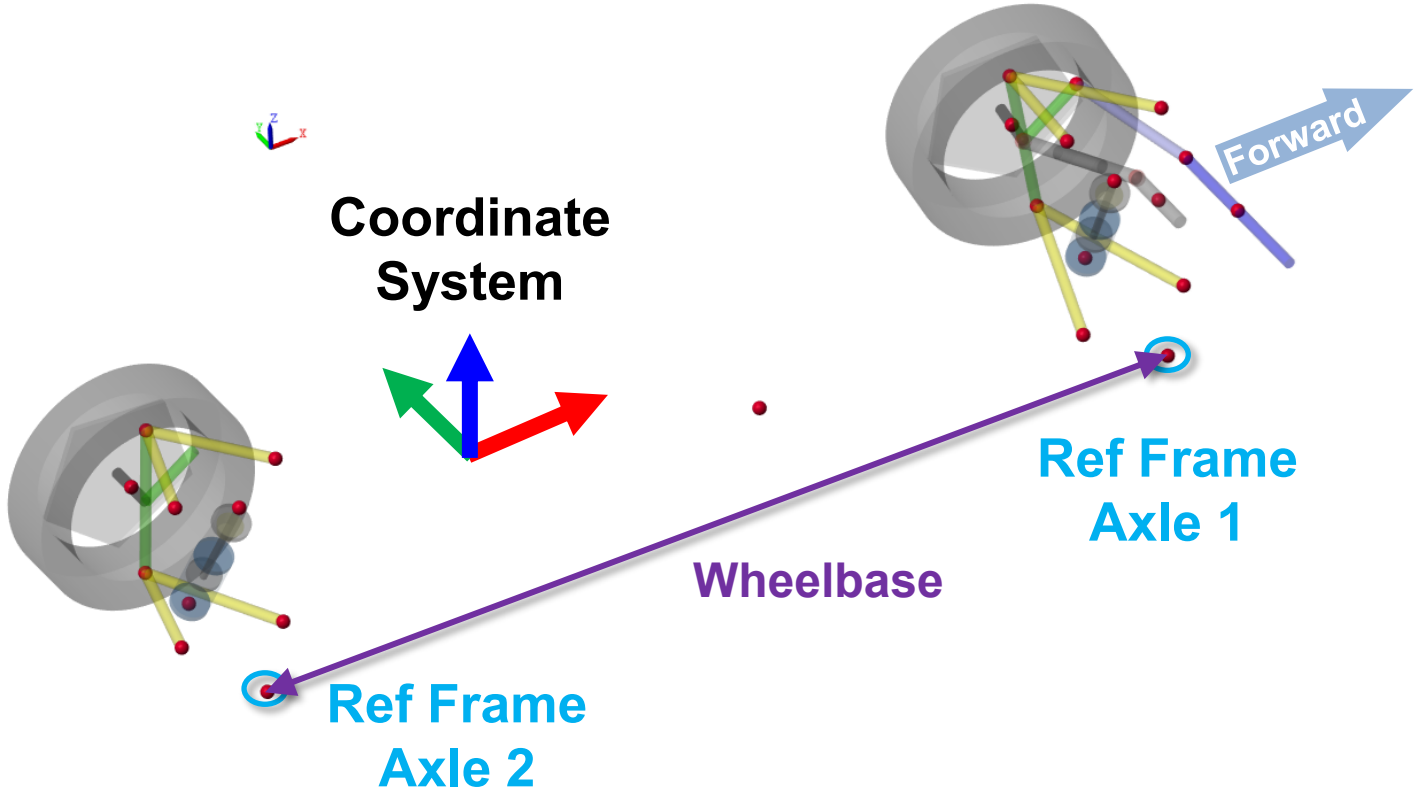
Suspension Structure

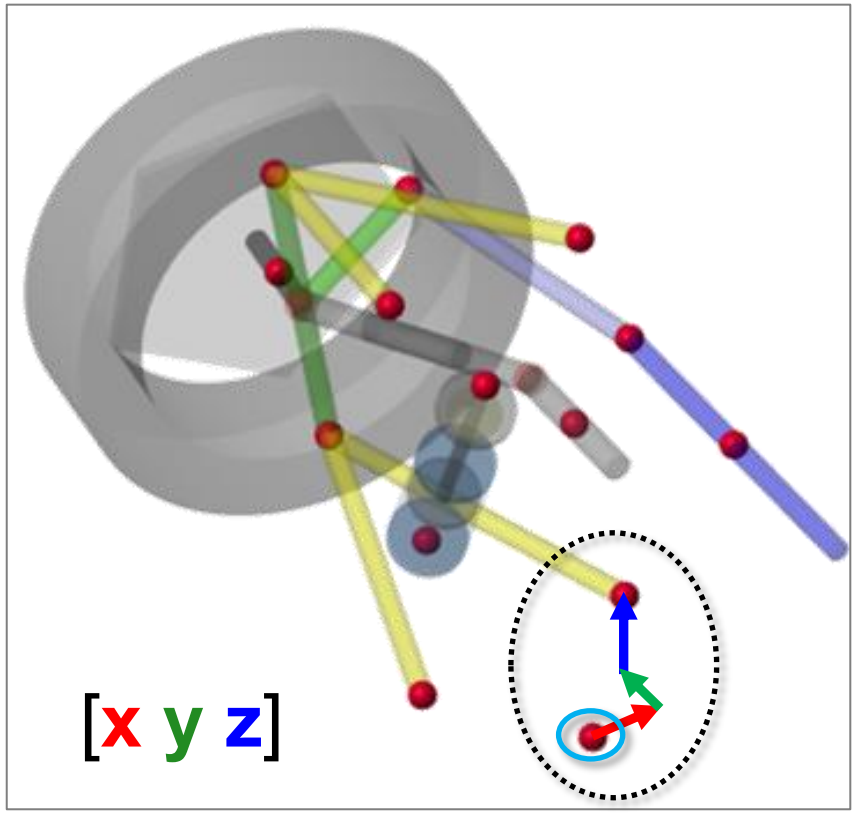
- Suspension linkages are composed of
  1. **Hardpoints** on chassis
  2. **Arms** connects chassis to upright
  3. **Upright** holds wheel bearing
  4. **Track Rod** connects to steering
  5. **ARB frame** connects to anti-roll bar
  6. **Shock** interface to force laws for spring and damping



Reference Frames

- Hardpoints are specified relative to a **reference frame at each axle**
  - Provided parameter sets use point on ground between wheels
  - Axle reference frames offset by a scalar distance from front axle (**wheelbase**)





Double-Wishbone

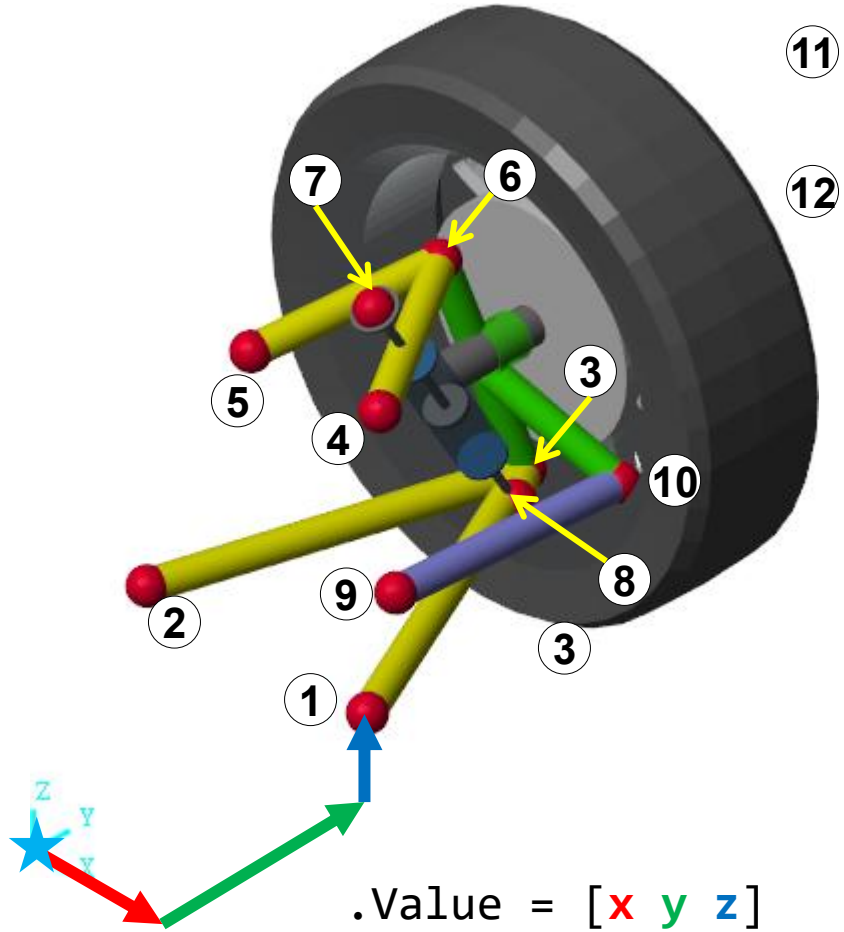
```
Vehicle.Chassis.SuspA1.Linkage.
      LowerWishbone.sInboardF      (1)
                   .sInboardR      (2)
                   .sOutboard      (3)

      UpperWishbone.sInboardF      (4)
                   .sInboardR      (5)
                   .sOutboard      (6)

            Shock.sTop             (7)
                 .sBottom          (8)

         TrackRod.sInboard         (9)
                 .sOutboard       (10)

          Upright.sCG             (11)
                 .sWheelCentre    (12)
```



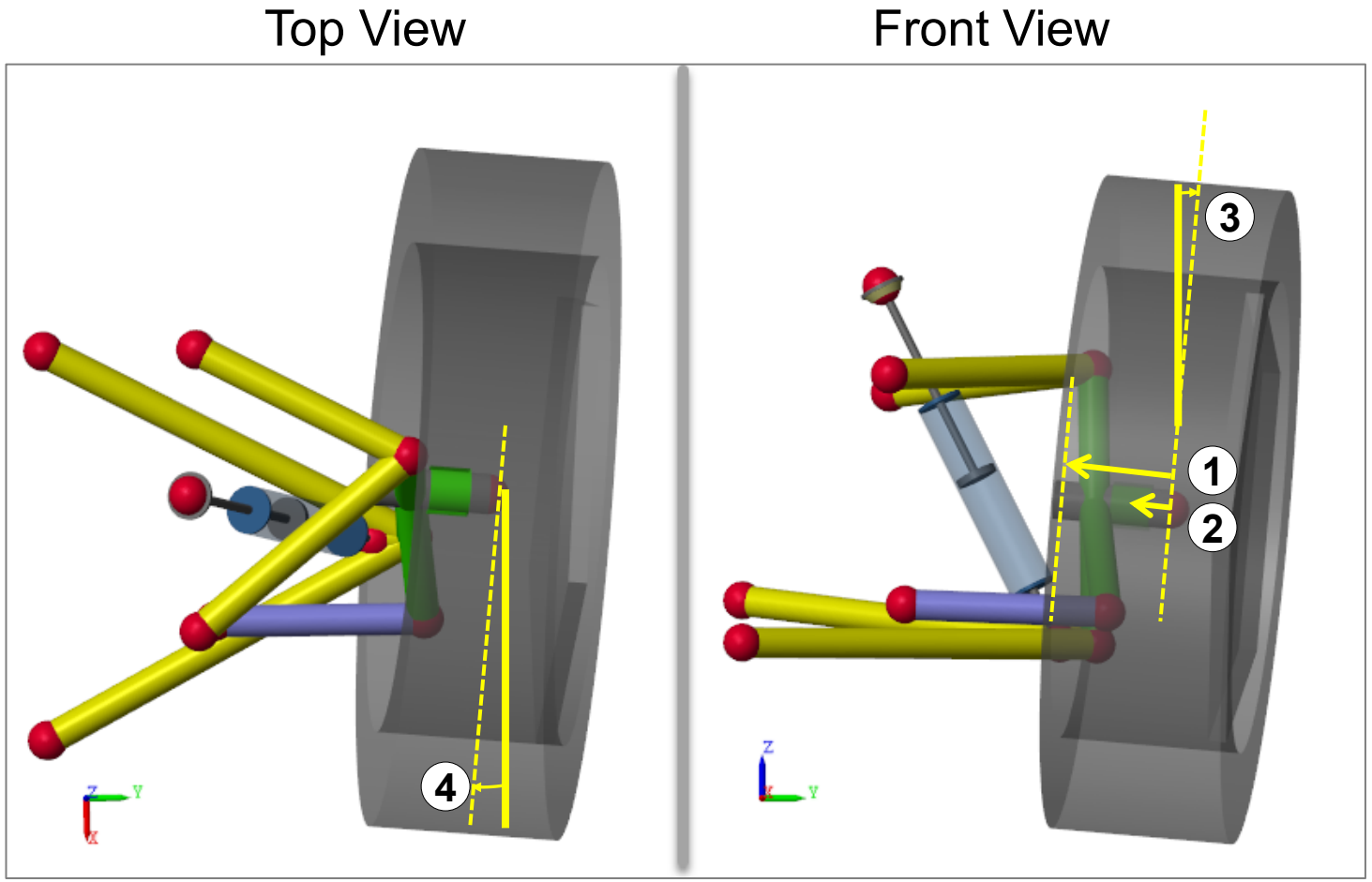
(11) Upright center of gravity (not shown)

(12) Center of Wheel (not shown)

`.Value = [x y z]`
In meters, relative to reference frame of axle ★

Double-Wishbone

```
Vehicle.Chassis.SuspA1.Linkage.
            Upright.xAxle          (1)
                   .xWCtoBearing   (2)
                   .aCamber        (3)
                   .aToe           (4)
```



(1) Length of axle shaft connecting wheel center to driveshaft

(2) Distance from wheel center to location of wheel revolute joint

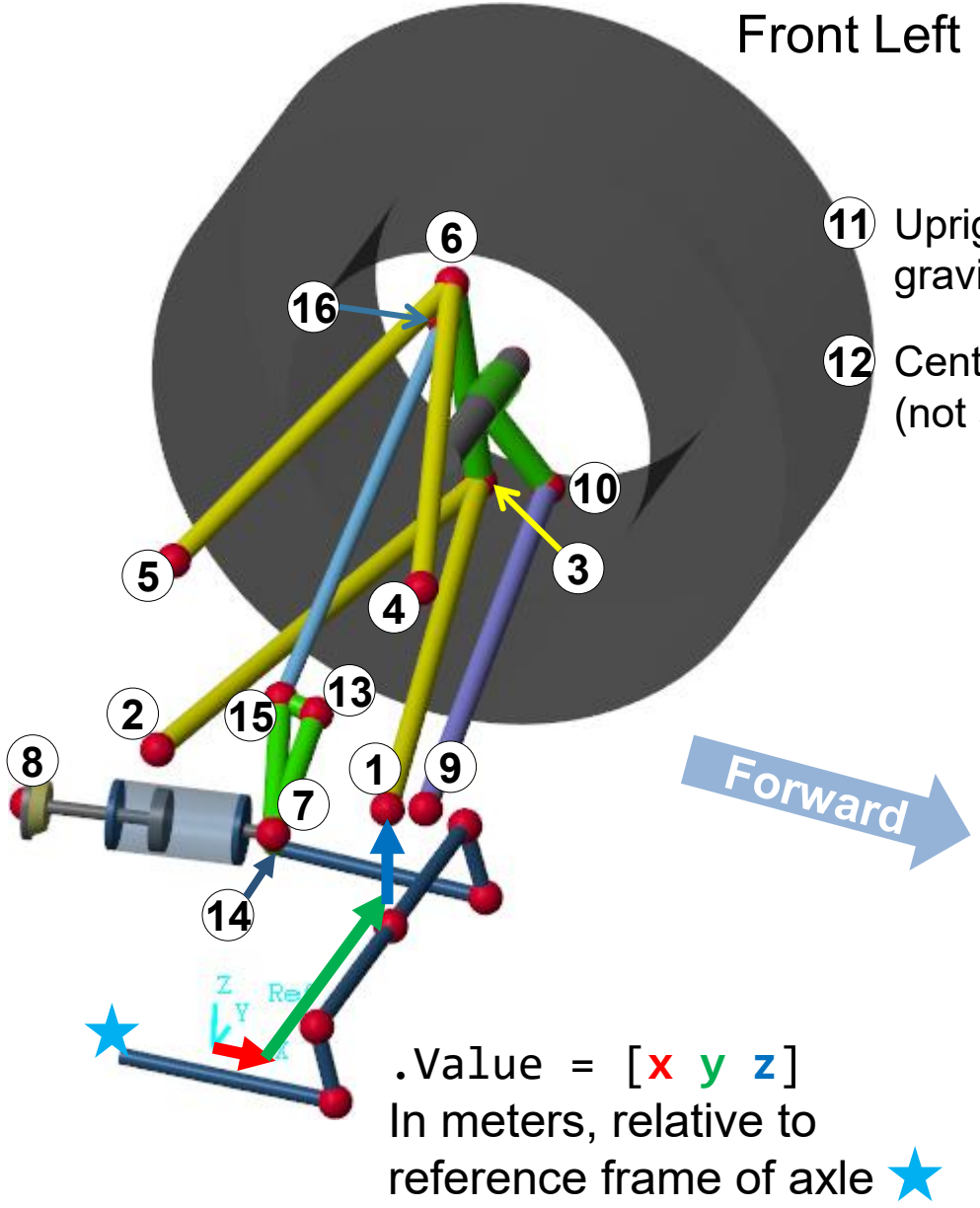
Double Wishbone Pullrod

```
Vehicle.Chassis.SuspA1.Linkage.
      LowerWishbone.sInboardF      (1)
                   .sInboardR      (2)
                   .sOutboard      (3)
      UpperWishbone.sInboardF      (4)
                   .sInboardR      (5)
                   .sOutboard      (6)

            Shock.sTop             (7)
                 .sBottom          (8)

          TrackRod.sInboard        (9)
                  .sOutboard      (10)
           Upright.sCG            (11)
                  .sWheelCentre   (12)

        Bellcrank.sPivot          (13)
                 .sAntiRollBar    (14)
         Pullrod.sBellcrank       (15)
                 .sUpperArm       (16)
```

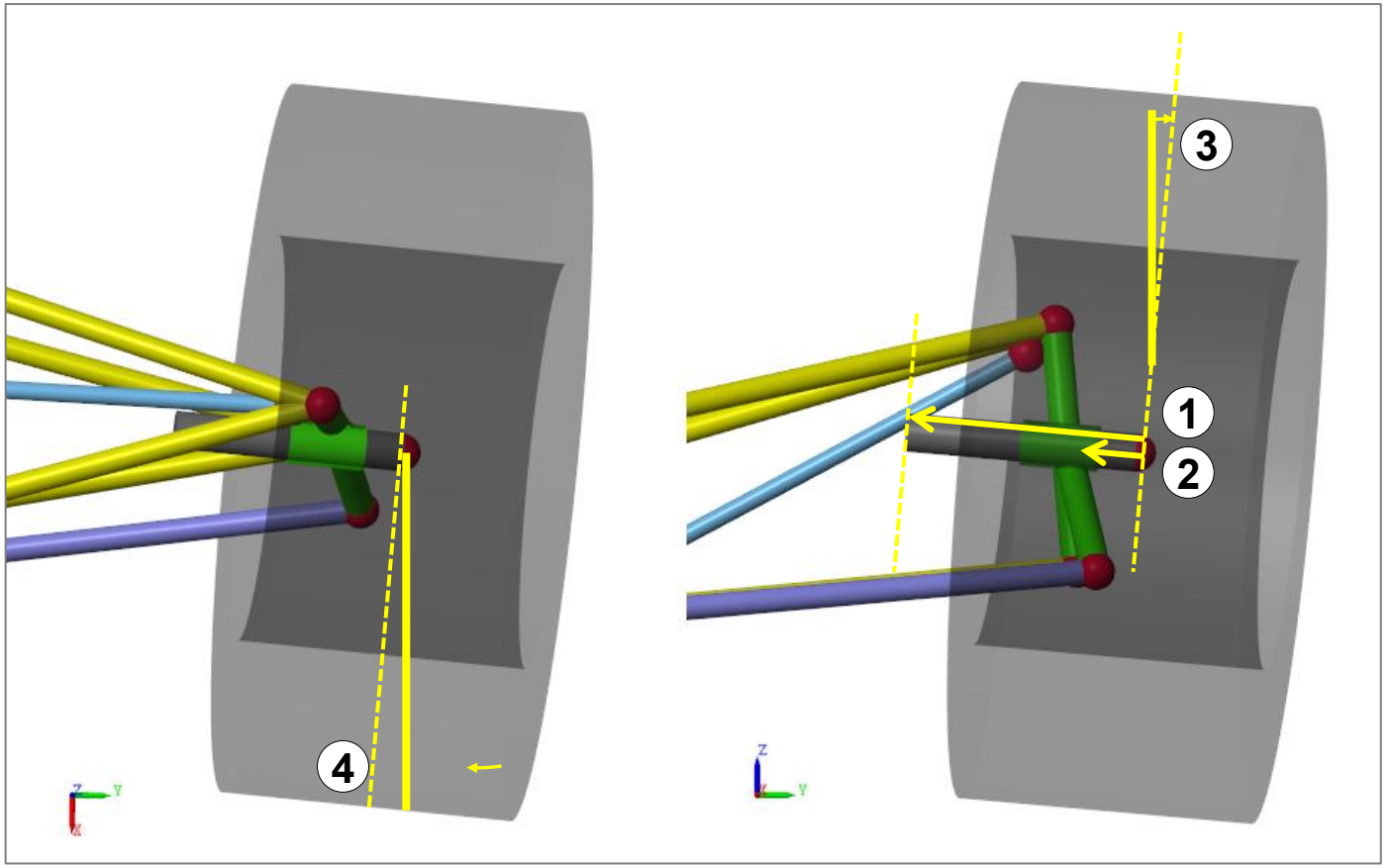


Double Wishbone Pullrod

```
Vehicle.Chassis.SuspA1.Linkage.
            Upright.xAxle          (1)
                   .xWCtoBearing   (2)
                   .aCamber        (3)
                   .aToe           (4)
```

Top View

Front View



(1) Length of axle shaft connecting wheel center to driveshaft

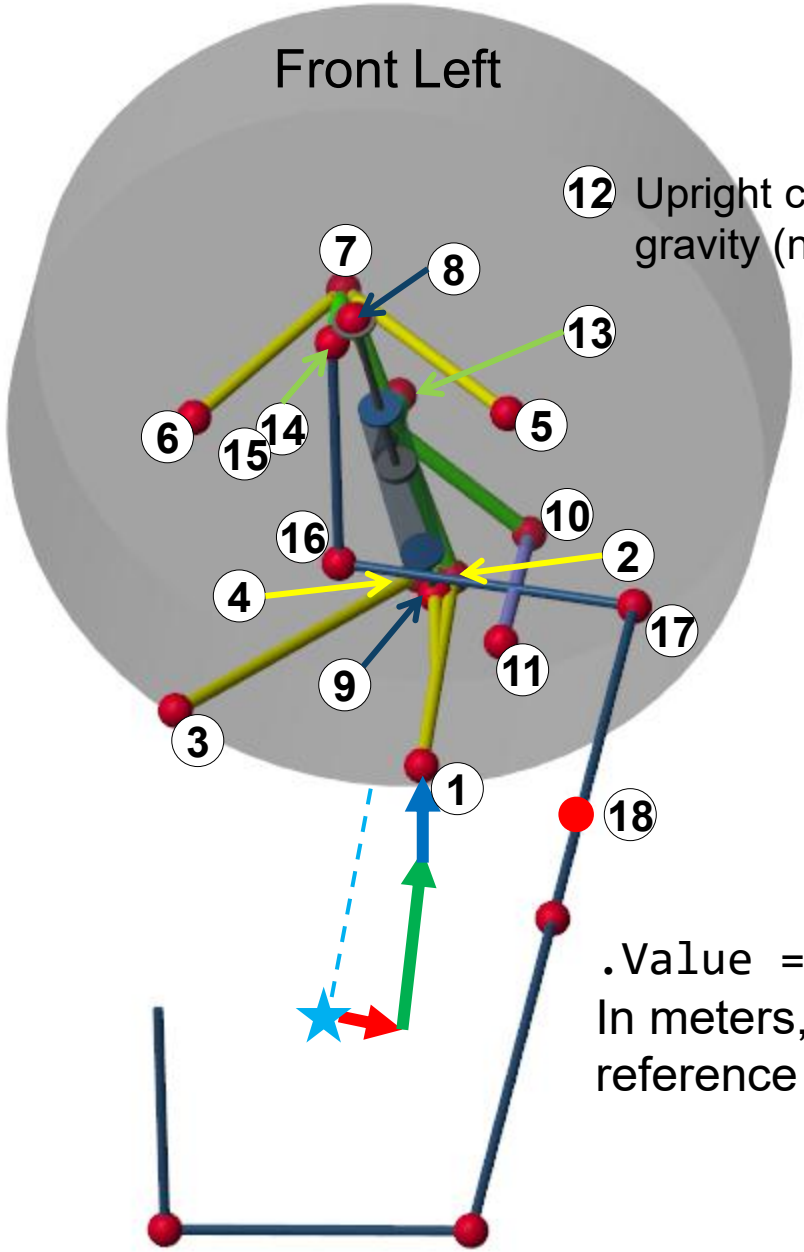
(2) Distance from wheel center to location of wheel revolute joint

Split Lower Arm, Shock to Front ARB to Upright

```
Vehicle.Chassis.SuspA1.Linkage.
        LowerArmF.sInboard       (1)
                 .sOutboard      (2)
        LowerArmR.sInboard       (3)
                 .sOutboard      (4)
    UpperWishbone.sInboardF      (5)
                 .sInboardR      (6)
                 .sOutboard      (7)
            Shock.sTop           (8)
                 .sBottom        (9)
         TrackRod.sInboard      (10)
                 .sOutboard     (11)
          Upright.sCG           (12)
                 .sWheelCentre  (13)
                 .sARB          (14)

Vehicle.Chassis.SuspA1.AntiRollBar.
                 .sSuspension   (15)
                 .sOutboard     (16)
                 .sInboard      (17)
                 .sMount        (18)
```

Must be identical

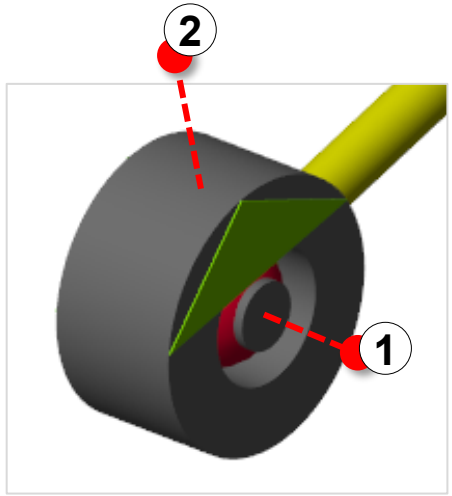


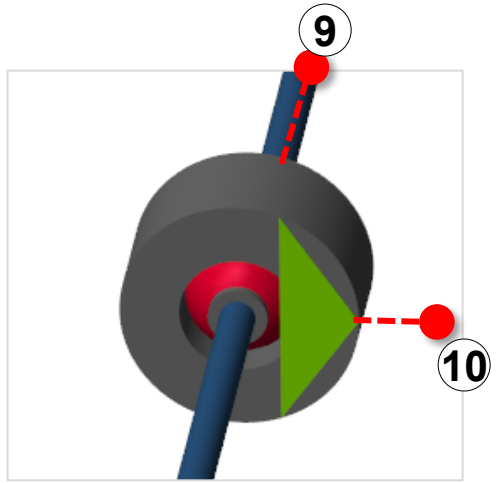
Split Lower Arm, Shock to Front ARB to Upright

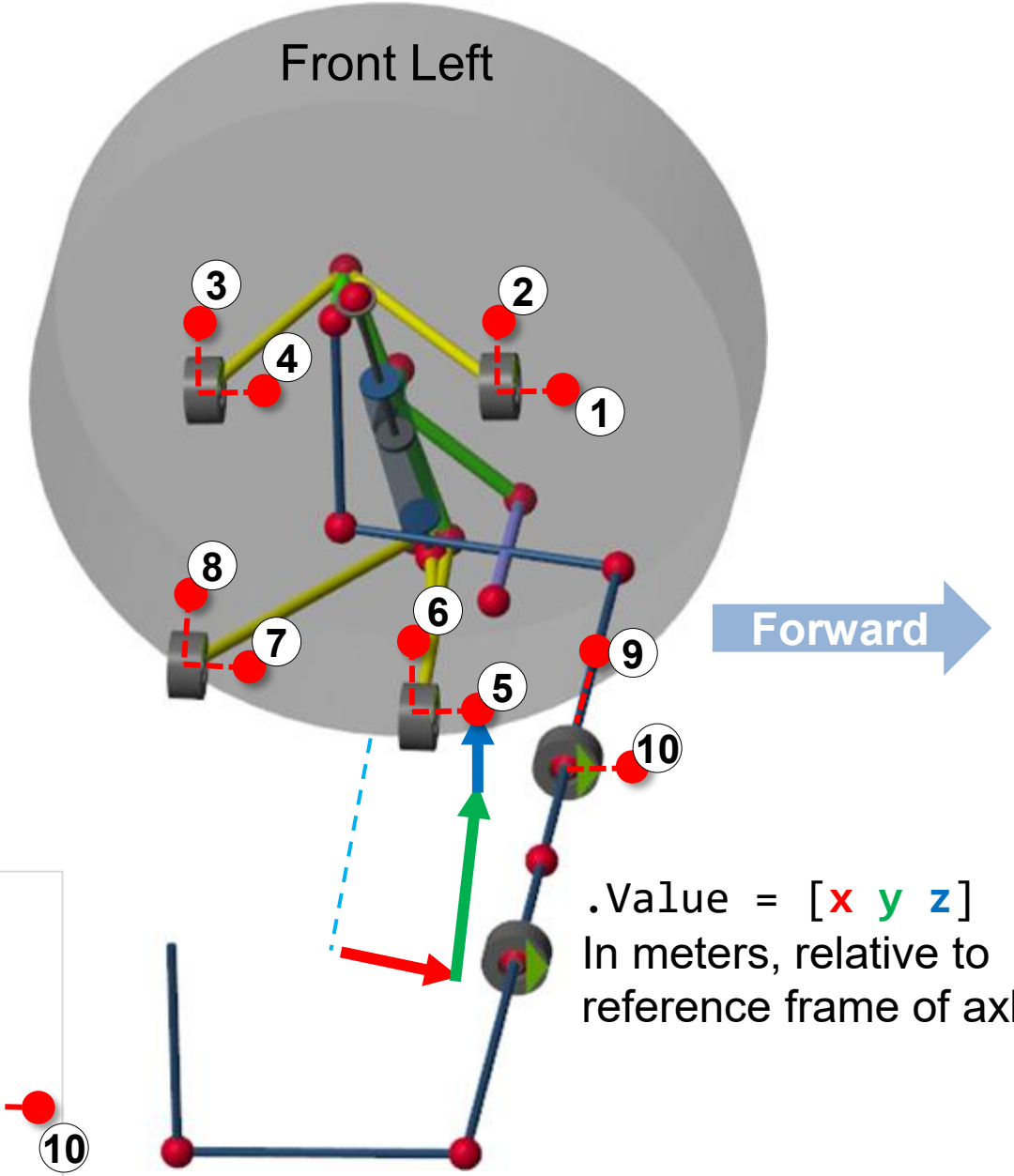
```
Vehicle.Chassis.SuspA1.Upper_Arm_to_Subframe.
      sFAxialOrientationZ (1)
      sFAxialOrientationX (2)
      sRAxialOrientationZ (3)
      sRAxialOrientationX (4)

Vehicle.Chassis.SuspA1.LowerArmF_to_Subframe.
      sAxialOrientationZ  (5)   (7) and (8) on
      sAxialOrientationX  (6)   LowerArmR_to_Subframe

Vehicle.Chassis.SuspA1.AntiRollBar.SubframeConnection
      sAxialOrientationZ  (9)
      sAxialOrientationX  (10)
```







`.Value = [x y z]`
In meters, relative to reference frame of axle ★

Split Lower Arm, Shock to Front ARB to Upright

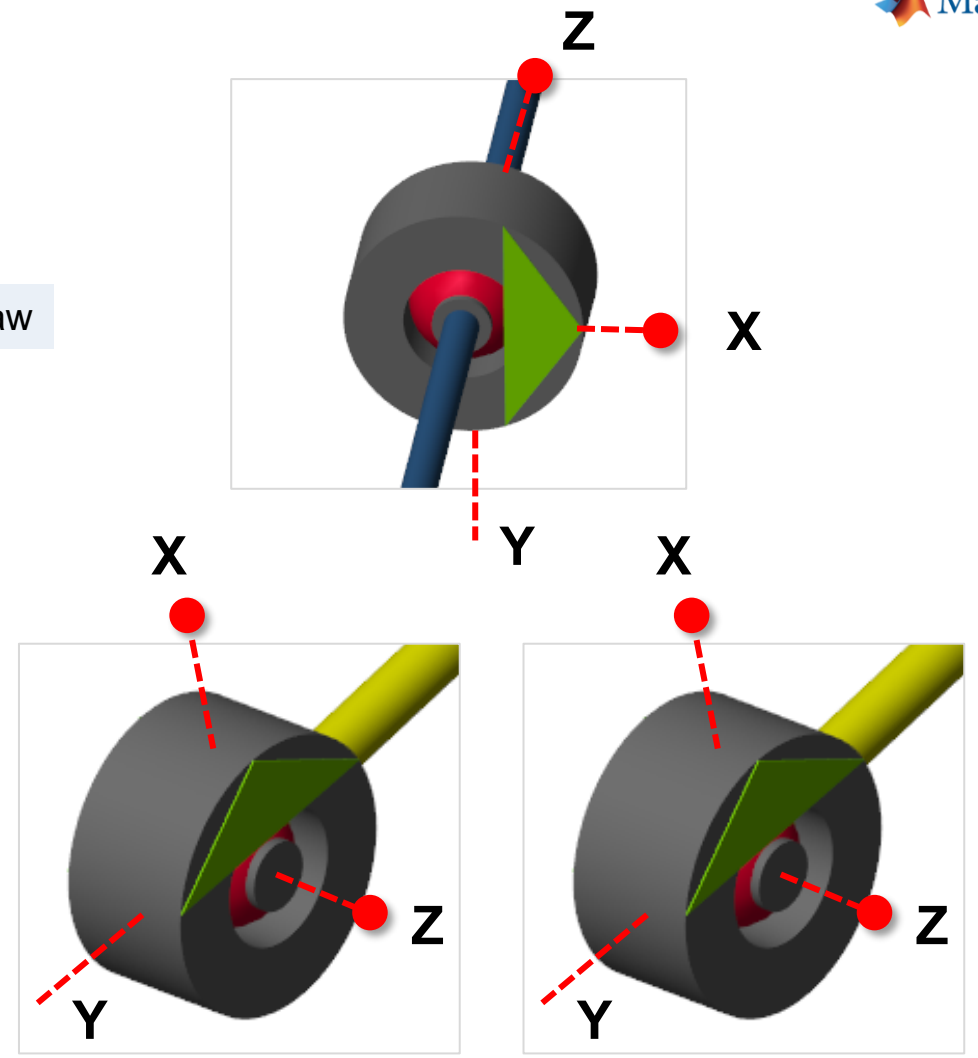
```
Vehicle.Chassis.SuspA1.AntiRollBar.SubframeConnection.
      SpringX.Law                'Linear' or 'Table'
             .Linear.K.Value
             .Table.x.Value
                   .f.Value
      (same for SpringY, SpringZ)

      DamperX.Law                'Linear' or 'Table'
             .Linear.d.Value
             .Table.v.Value
                   .f.Value
      (same for SpringY, SpringZ)

Vehicle.Chassis.SuspA1.Linkage.Upper_Arm_to_Subframe.
      FSpringX, FSpringY, FSpringZ, …
      RSpringX, RSpringY, RSpringZ, …
      FDamperX, FDamperY, FDamperZ, …
      FDamperX, FDamperY, FDamperZ
```

Select Law

Define Behavior



Kinematic or Compliant Joint

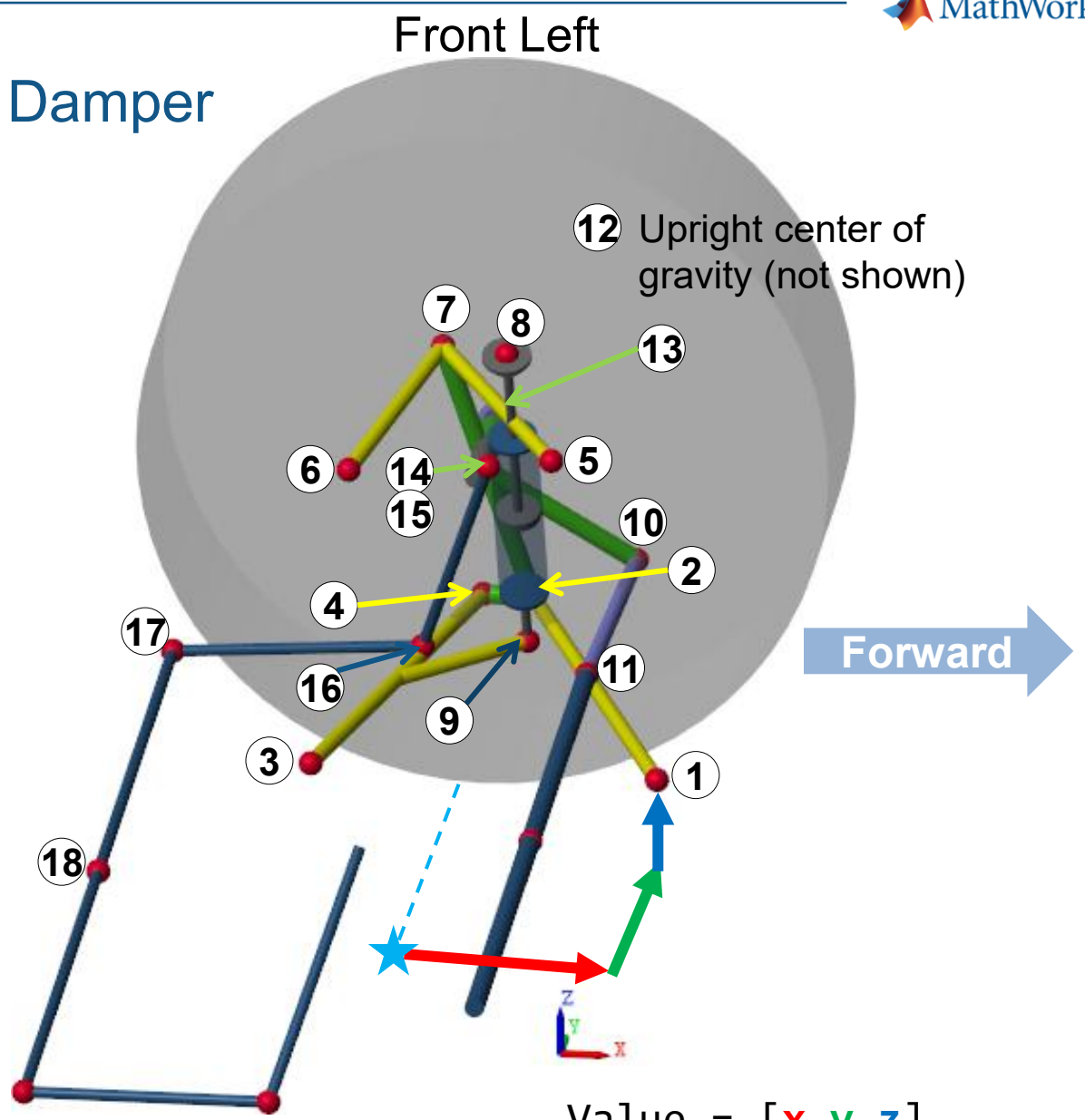
```
Vehicle.Chassis.SuspA1.AntiRollBar.SubframeConnection.class.Value      'Rigid_1Rev' or 'BushArm_Ax3'
Vehicle.Chassis.SuspA1.Linkage.Upper_Arm_to_Subframe.class.Value       'Rigid_1Rev' or 'BushArm_Ax3'
Vehicle.Chassis.SuspA1.Linkage.LowerArmF_to_Subframe.class.Value       'Rigid_UJ'   or 'BushArm_Ax3'
```

Split Lower Arm, Shock to Rear, ARB to Damper

```
Vehicle.Chassis.SuspA1.Linkage.
          LowerArmF.sInboard       (1)
                   .sOutboard      (2)
          LowerArmR.sInboard       (3)
                   .sOutboard      (4)
      UpperWishbone.sInboardF      (5)
                   .sInboardR      (6)
                   .sOutboard      (7)
             Shock.sTop            (8)
                   .sBottom        (9)
           TrackRod.sInboard      (10)
                   .sOutboard     (11)
            Upright.sCG           (12)
                   .sWheelCentre  (13)
                   .sARB          (14)

Vehicle.Chassis.SuspA1.AntiRollBar.
                   .sSuspension   (15)
                   .sOutboard     (16)
                   .sInboard      (17)
                   .sMount        (18)
```

Must be identical



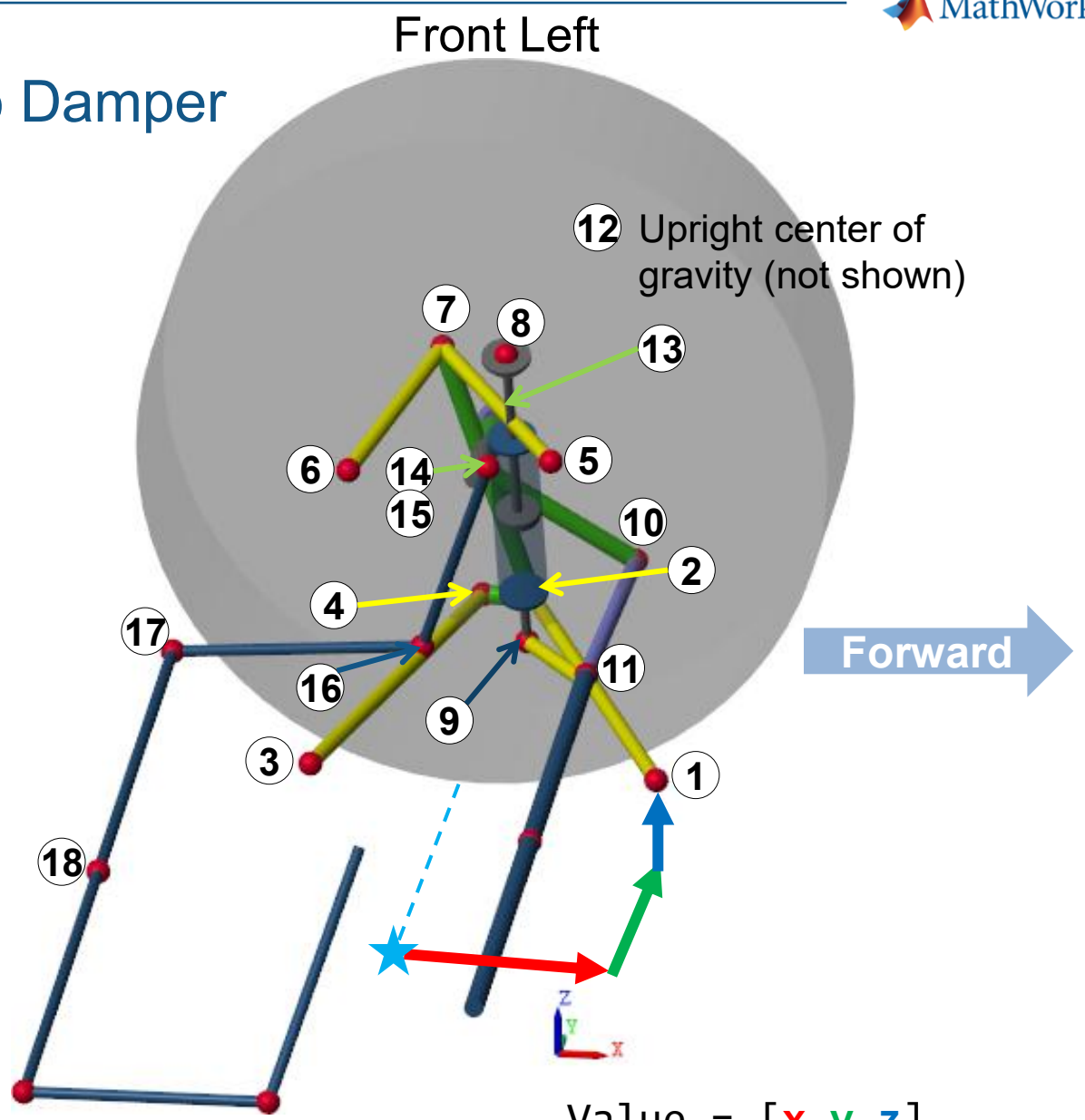
`.Value = [x y z]`
In meters, relative to reference frame of axle ★

Split Lower Arm, Shock to Front, ARB to Damper

```
Vehicle.Chassis.SuspA1.Linkage.
          LowerArmF.sInboard       (1)
                   .sOutboard      (2)
          LowerArmR.sInboard       (3)
                   .sOutboard      (4)
      UpperWishbone.sInboardF      (5)
                   .sInboardR      (6)
                   .sOutboard      (7)
             Shock.sTop            (8)
                   .sBottom        (9)
           TrackRod.sInboard      (10)
                   .sOutboard     (11)
            Upright.sCG           (12)
                   .sWheelCentre  (13)
                   .sARB          (14)

Vehicle.Chassis.SuspA1.AntiRollBar.
                   .sSuspension   (15)
                   .sOutboard     (16)
                   .sInboard      (17)
                   .sMount        (18)
```

Must be identical



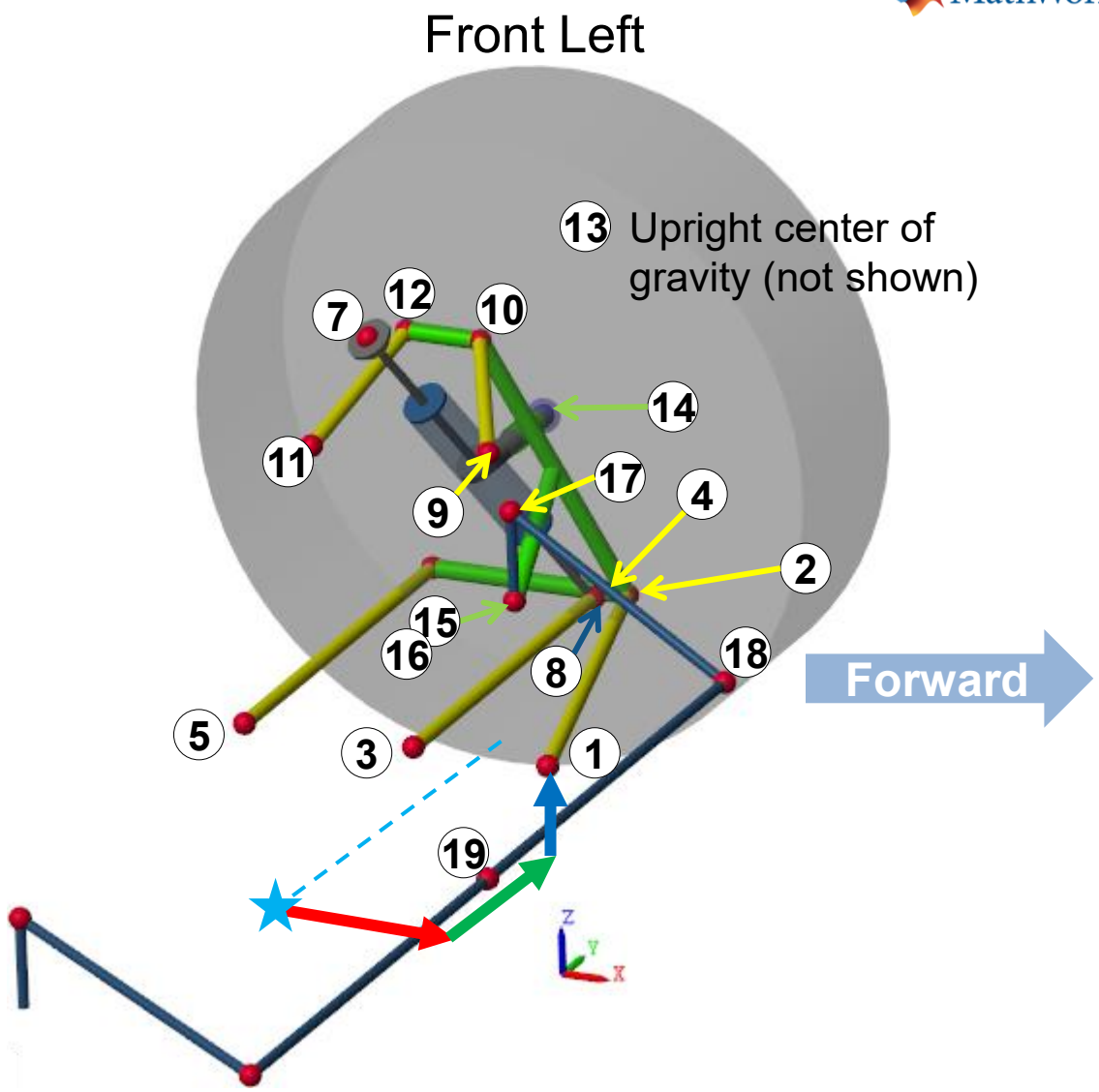
`.Value = [x y z]`
In meters, relative to reference frame of axle ★

5 Link, Shock to Center, ARB to Upright

```
Vehicle.Chassis.SuspA1.Linkage.
          LowerArmF.sInboard       (1)
                   .sOutboard      (2)
          LowerArmC.sInboard       (3)
                   .sOutboard      (4)
          LowerArmR.sInboard       (5)
                   .sOutboard      (6)
              Shock.sTop           (7)
                   .sBottom        (8)
          UpperArmF.sInboard       (9)
                   .sOutboard     (10)
          UpperArmR.sInboard     (11)
                   .sOutboard     (12)
            Upright.sCG          (13)
                   .sWheelCentre (14)
                   .sARB         (15)

Vehicle.Chassis.SuspA1.AntiRollBar.
                   .sSuspension  (16)
                   .sOutboard    (17)
                   .sInboard     (18)
                   .sMount       (19)
```

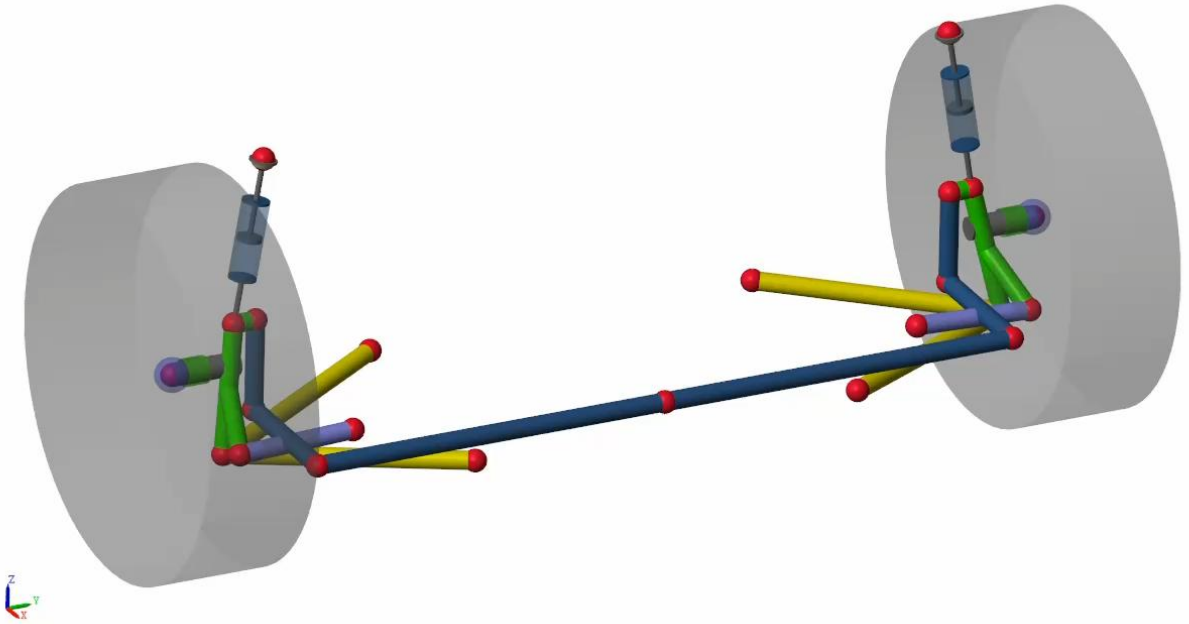
Must be identical

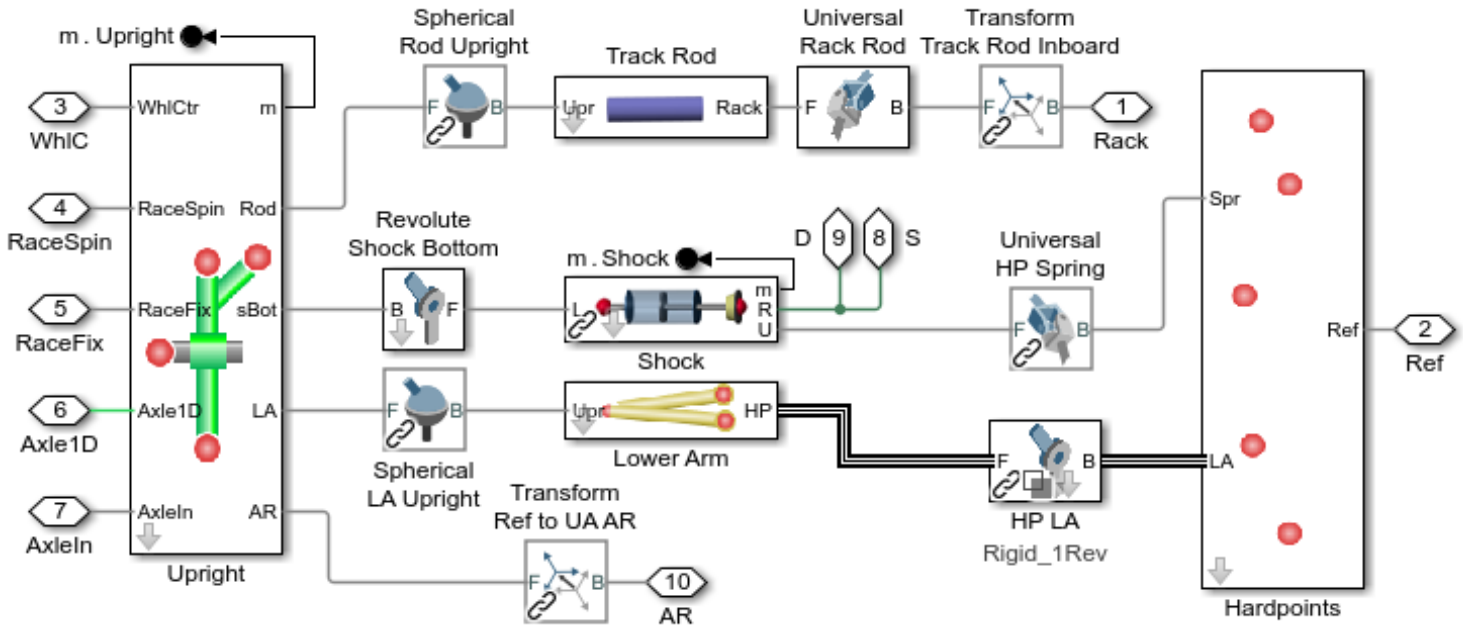


`.Value = [x y z]`
In meters, relative to reference frame of axle ★

MacPherson

- Assembled from standard and custom components
- Model clearly documents parts and constraints
- Extend to custom designs
  - Custom damper models
  - Steer-by-wire systems





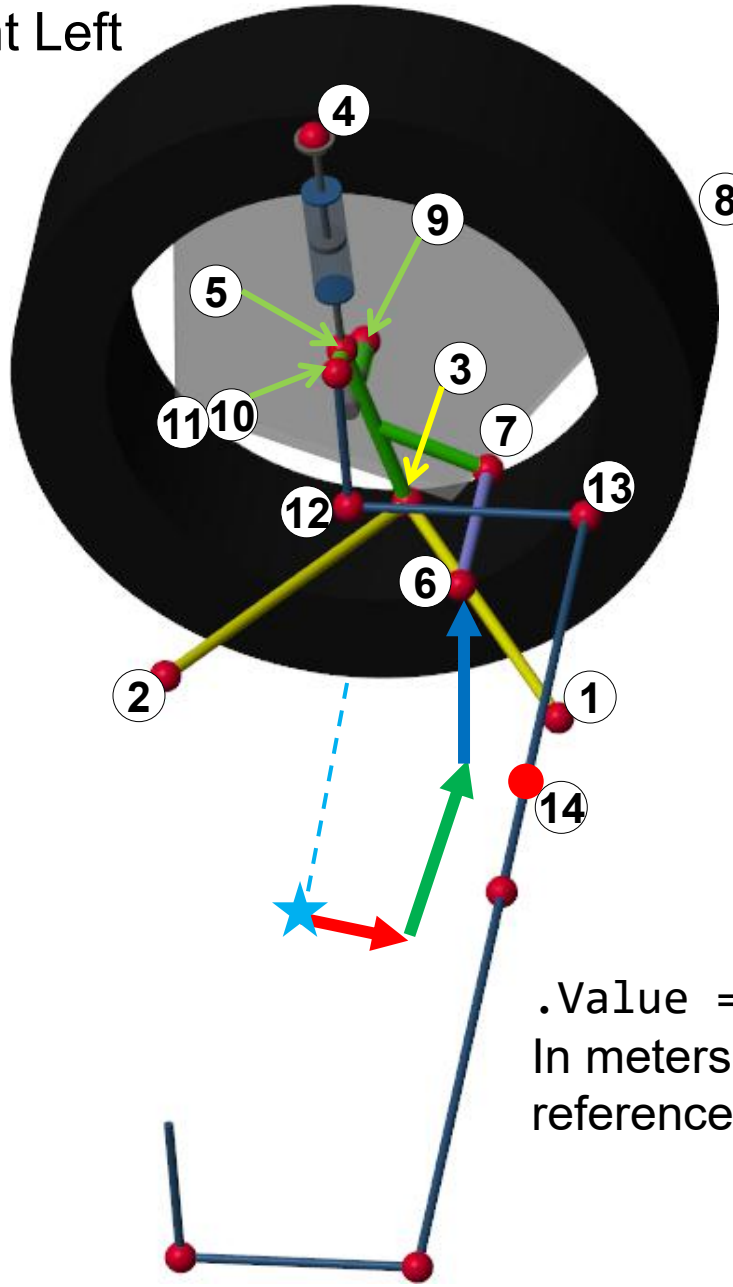
MacPherson

```
Vehicle.Chassis.SuspA1.Linkage.
      LowerWishbone.sInboardF      (1)
                   .sInboardR      (2)
                   .sOutboard      (3)
             Shock.sTop            (4)
                   .sBottom        (5)
          TrackRod.sInboard        (6)
                   .sOutboard      (7)
           Upright.sCG             (8)
                   .sWheelCentre   (9)
                   .sARB          (10)

Vehicle.Chassis.SuspA1.AntiRollBar.
                   .sSuspension   (11)
                   .sOutboard     (12)
                   .sInboard      (13)
                   .sMount        (14)
```

Must be identical

Front Left



8 Upright center of gravity (not shown)

Forward

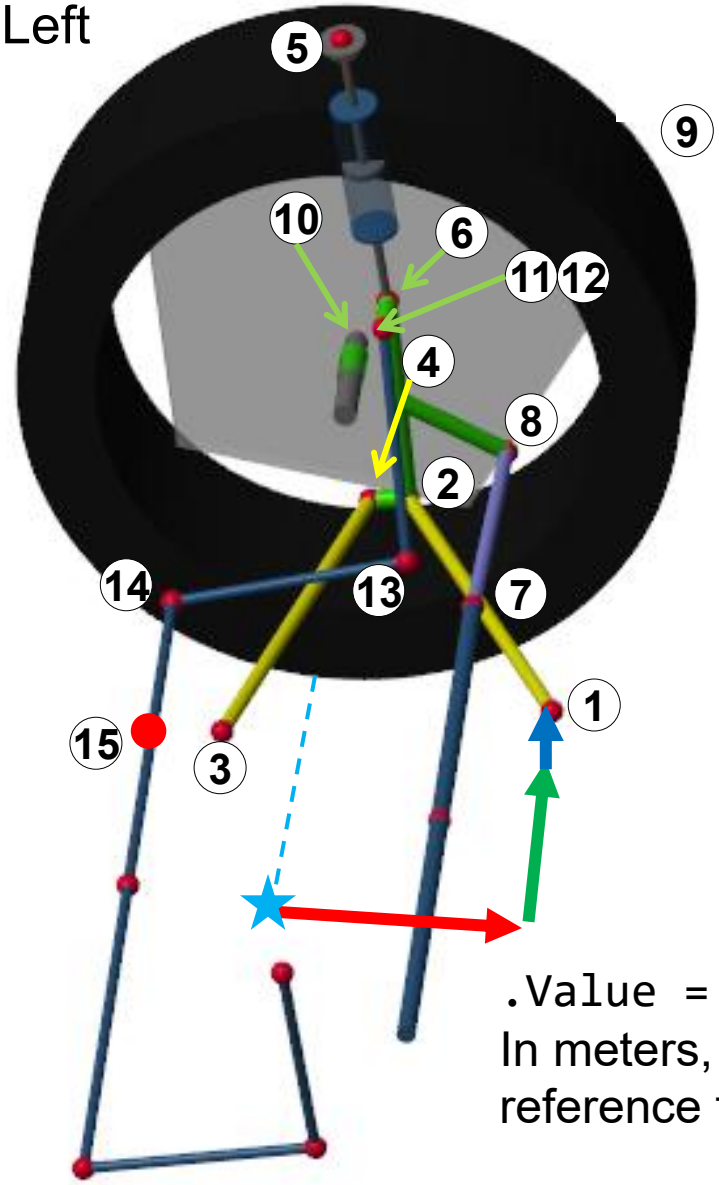
`.Value = [x y z]`
In meters, relative to reference frame of axle ★

MacPherson, Split Lower Arm

```
Vehicle.Chassis.SuspA1.Linkage.
      LowerArmF.sInboard       (1)
               .sOutboard      (2)
      LowerArmR.sInboard       (3)
               .sOutboard      (4)
          Shock.sTop           (5)
               .sBottom        (6)
       TrackRod.sInboard       (7)
               .sOutboard      (8)
        Upright.sCG            (9)
               .sWheelCentre  (10)
               .sARB          (11)
Vehicle.Chassis.SuspA1.AntiRollBar.
               .sSuspension   (12)
               .sOutboard     (13)
               .sInboard      (14)
               .sMount        (15)
```

Must be identical

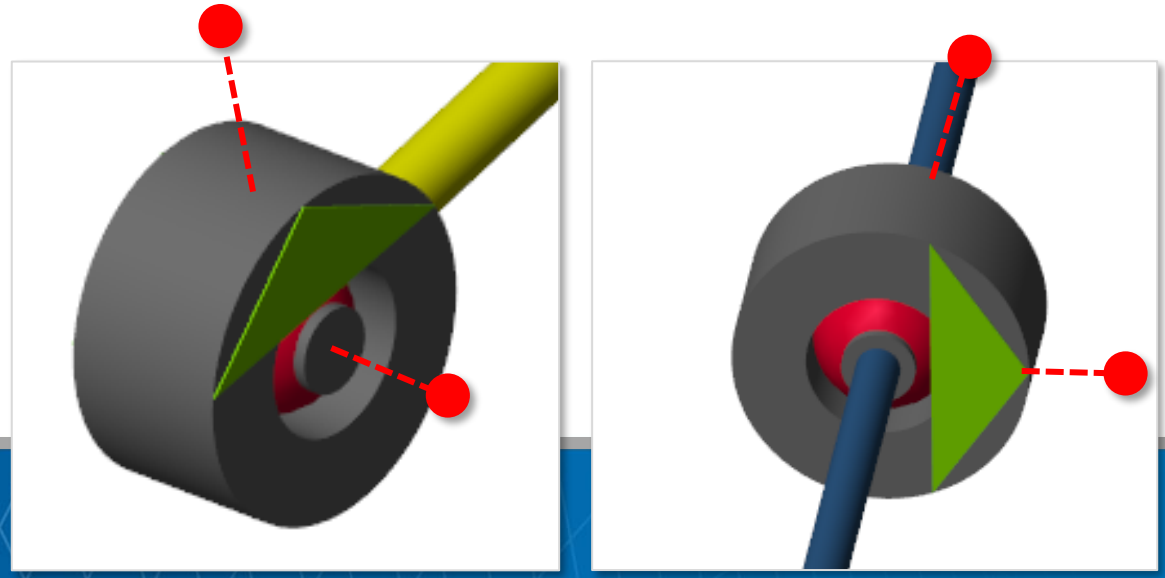
Front Left



9 Upright center of gravity (not shown)

`.Value = [x y z]`
In meters, relative to reference frame of axle ★

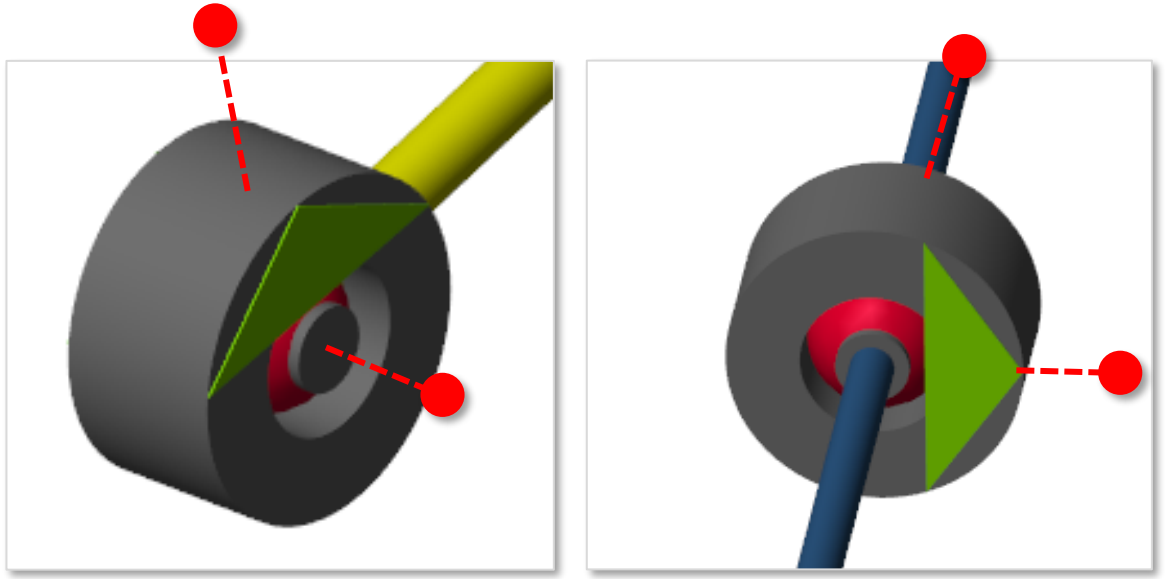
Simscape Vehicle Templates
Bushings



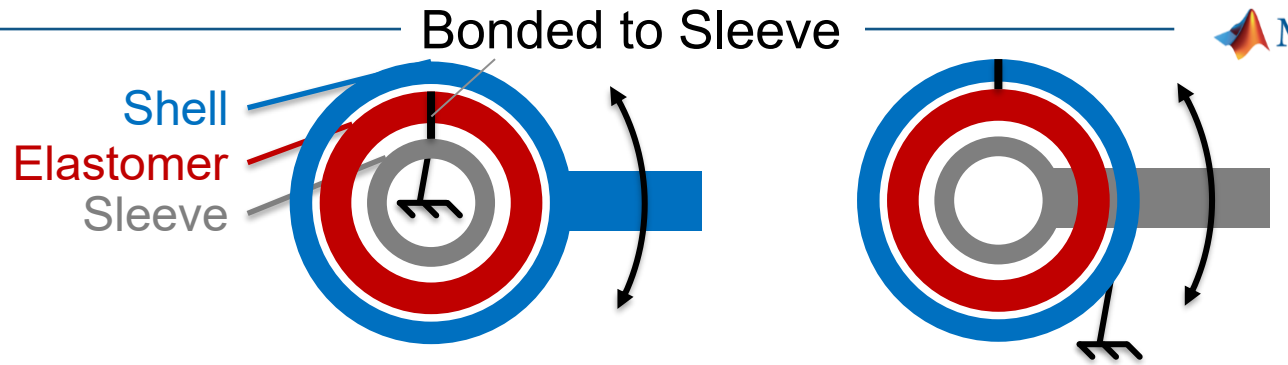
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Bushings

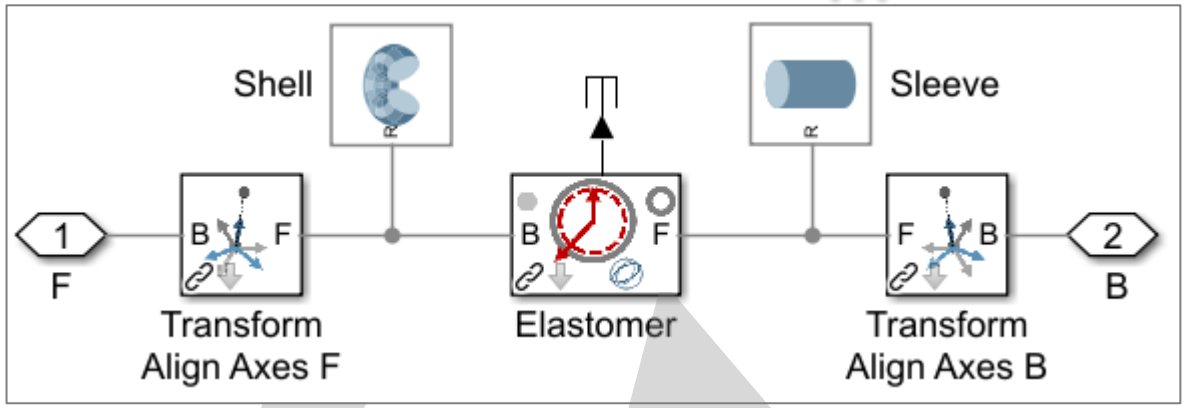
- Bushing Locations
- Bushing Structure
- Bushing Force Laws

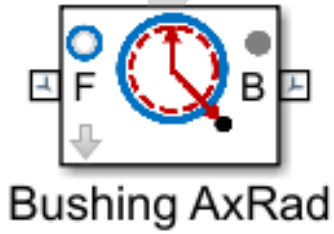


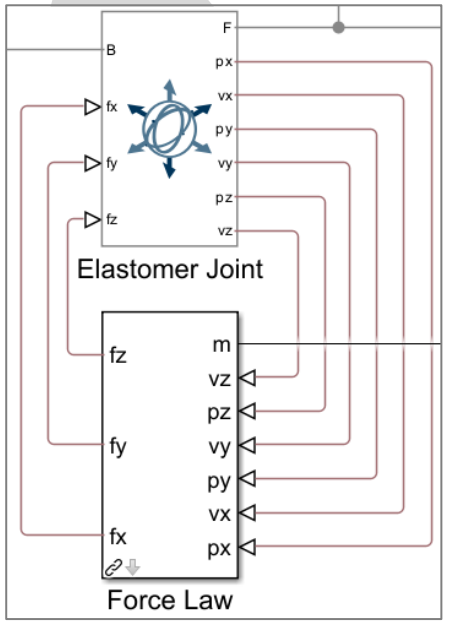
Bushings Structure



- Elements
  - Shell (visual)
  - Elastomer (force law along constraints)
  - Sleeve (visual)
- Options
  - Force law axes
  - Method for specifying compliance axes (see suspension section)
    - Two points for axial and in plane
    - Angle about z-axis
  - Compliance axes attached to Outer or Inner
    - Elastomer bonded to shell or sleeve
  - Optionally specify joint targets

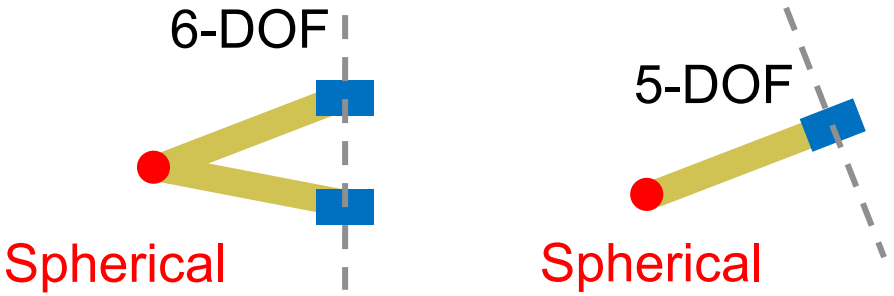


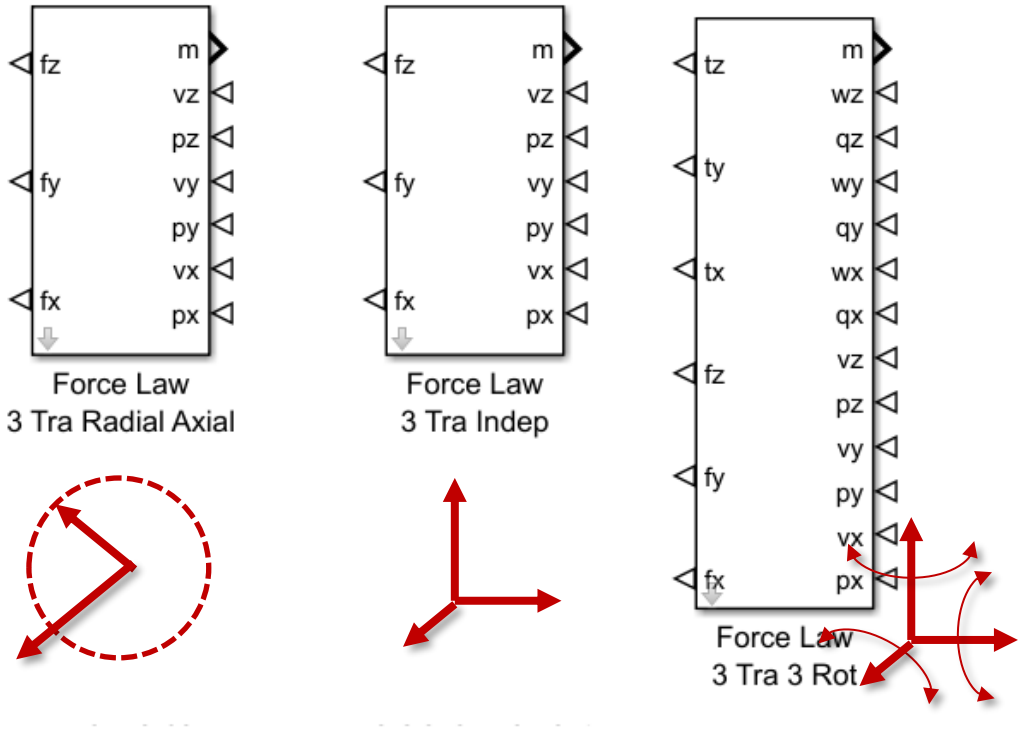




Bushings Library

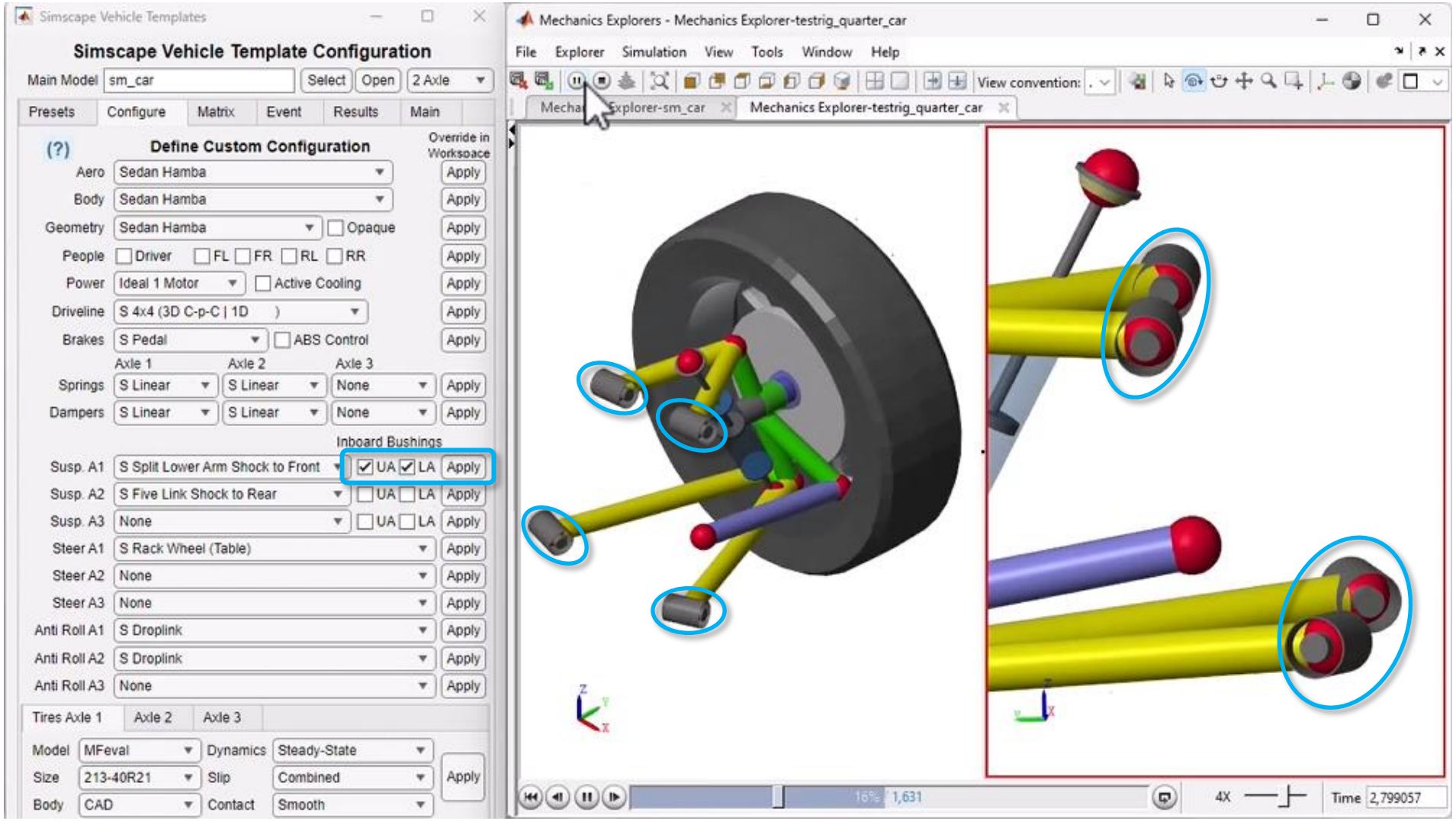
- Use 6-DOF bushings for connecting arms to chassis
- Use 5-DOF bushings at one end of links (no rotation about long axis)
- Force axes
  - 1 axial, 1 radial
  - 3 translation axes
  - 3 translation, 3 rotational
- Customizable force laws
  - Linear and table
  - Maxwell SLS on radial axis





Enable Compliance on Inboard Link Connections

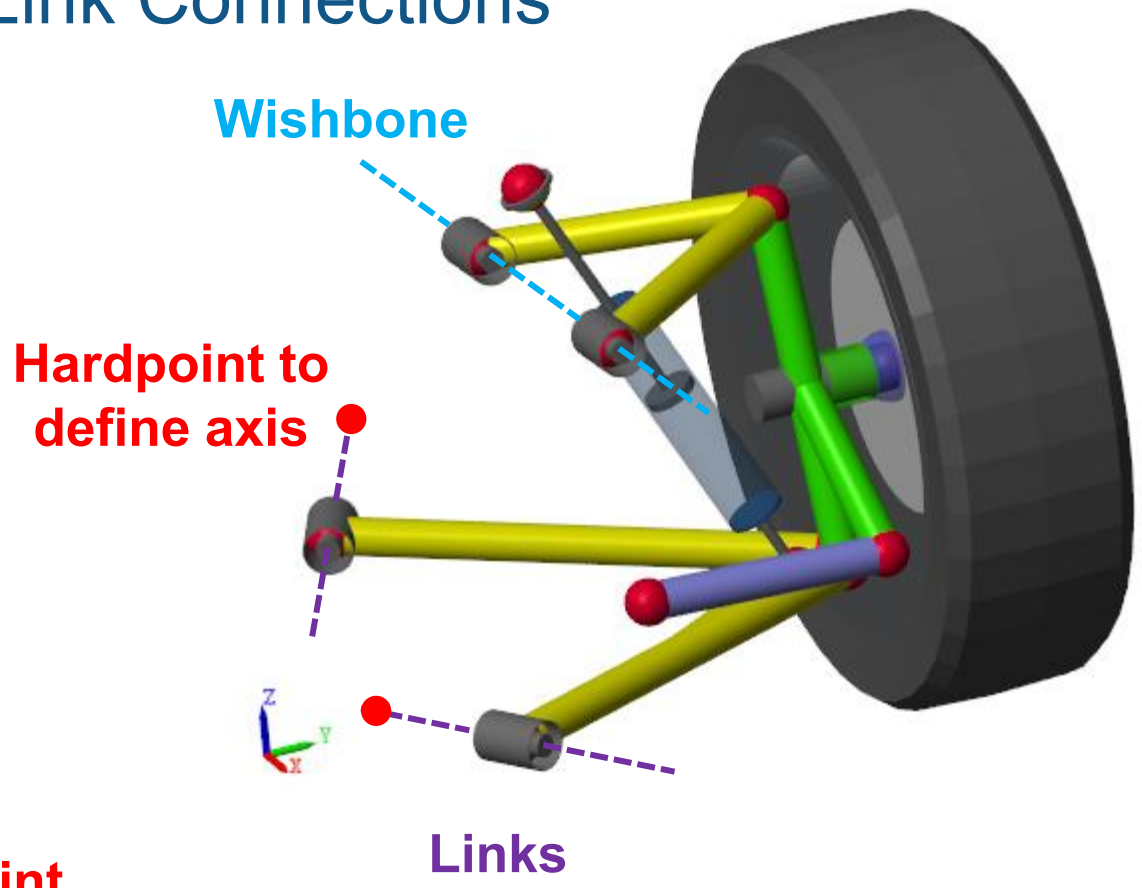
- All inboard connections can have kinematic or compliant connections
- Enable using UI or via MATLAB



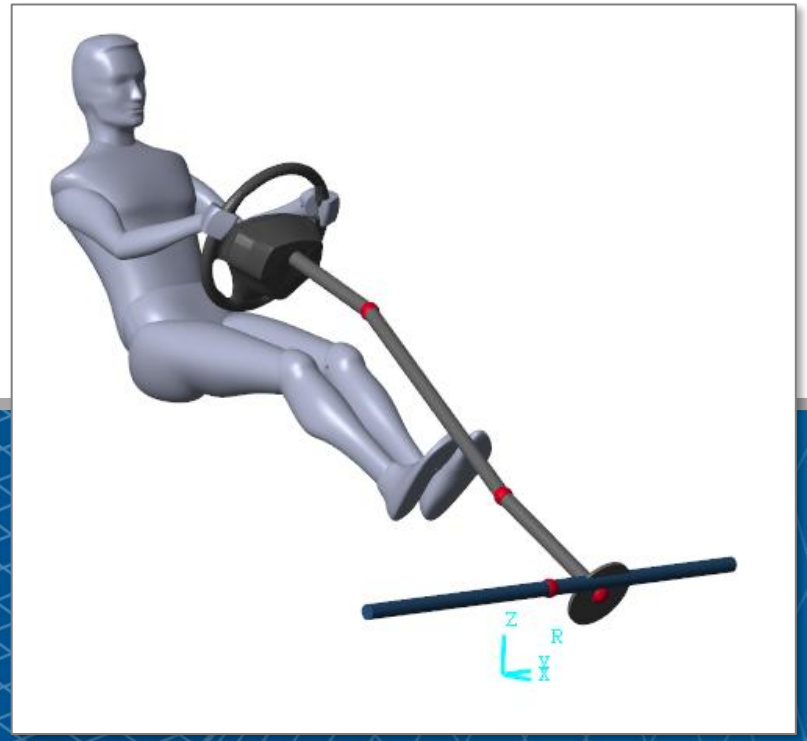
Enable Compliance on Inboard Link Connections

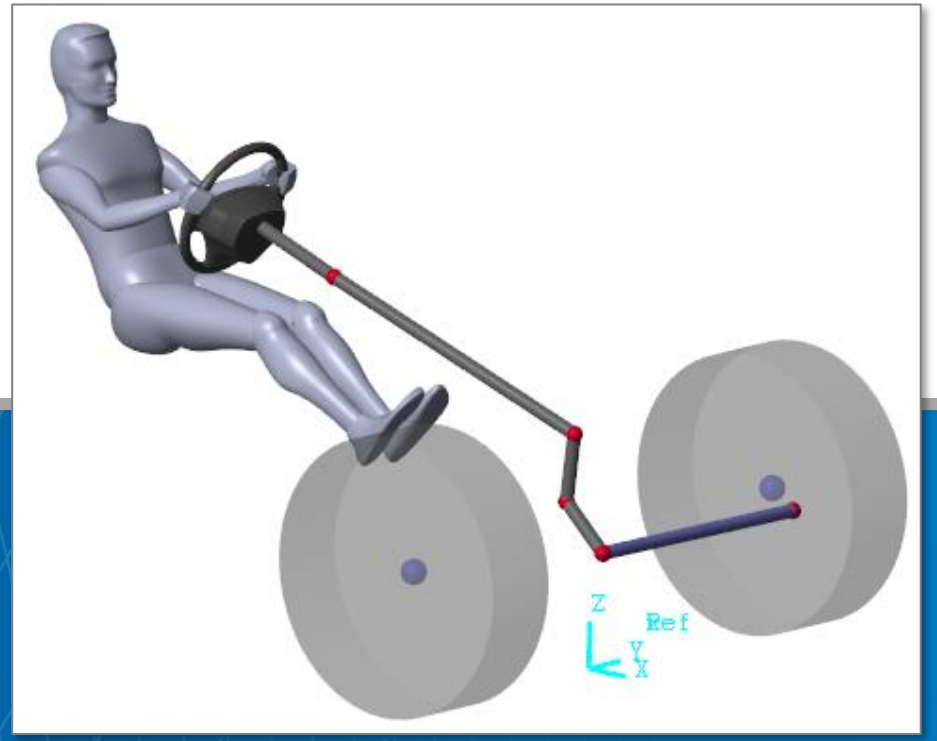
- Bushing placed at hardpoints
- Separate stiffness and damping models for radial and axial
  - Linear, lookup table
  - Nonlinear stiffness with hysteresis

- Axial direction
  - **Wishbone**: line through hardpoints
  - **Links**: Specified via additional **hardpoint**



Simscape Vehicle Templates
Steering

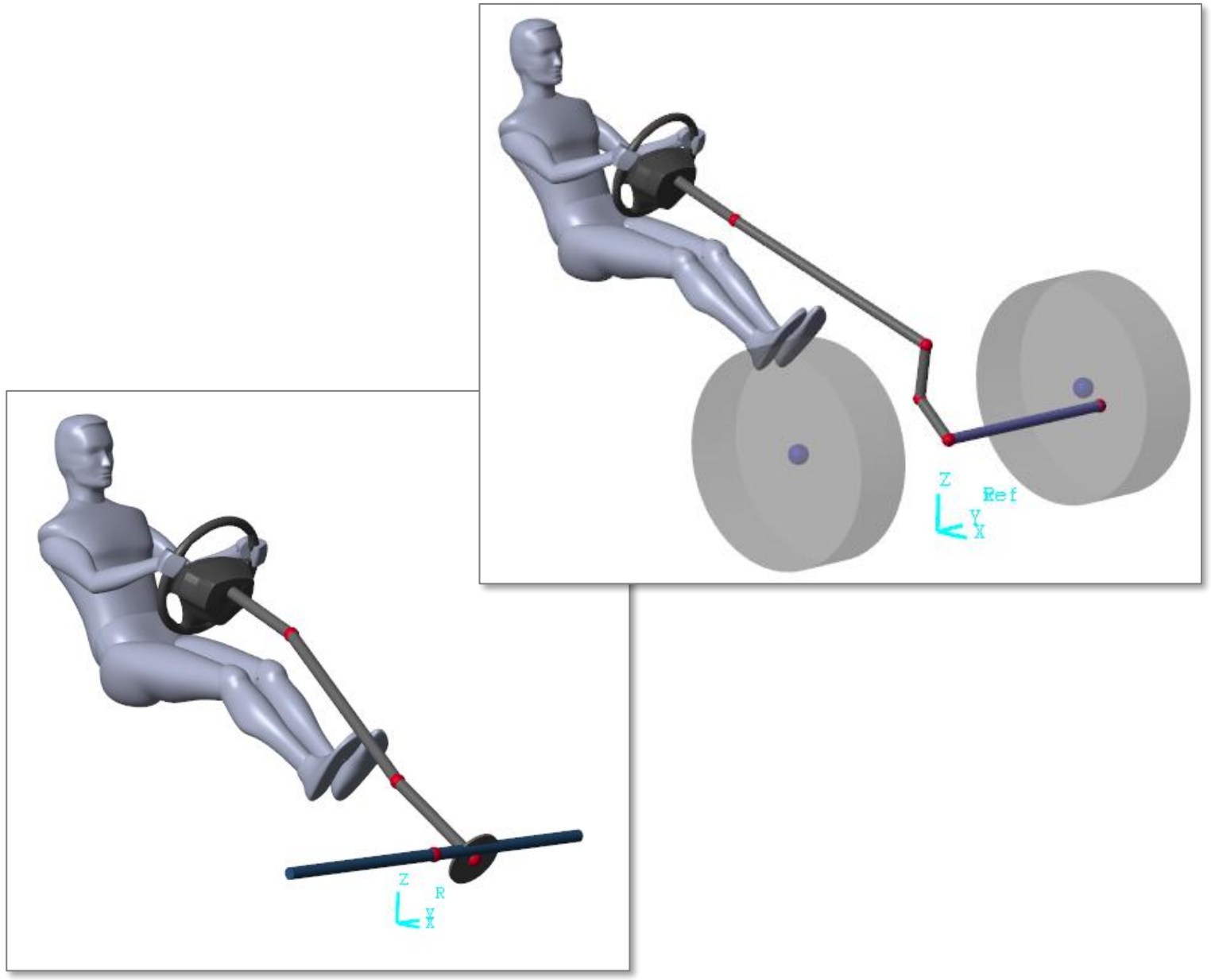




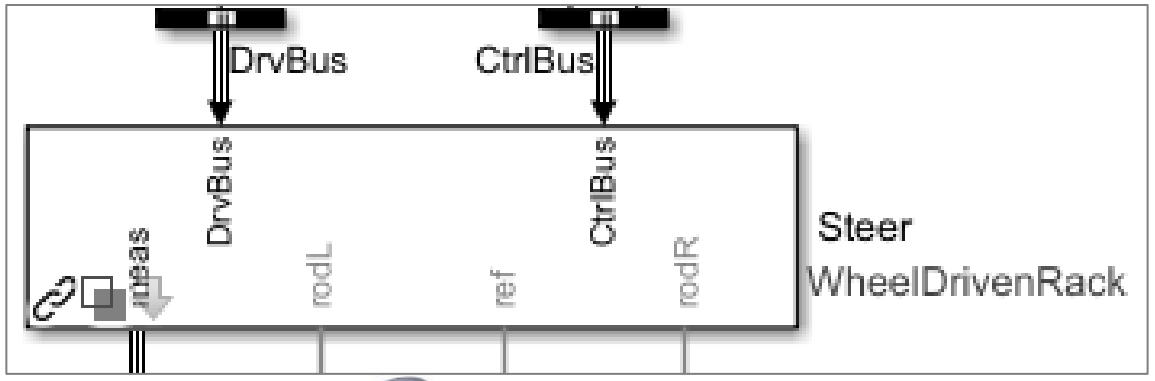
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Steering

- Rack and Pinion
- Drag Link
- Steer by Wire



Rack and Pinion Interface



- Three mechanical connections
  - Chassis (port ref)
    Steering wheel bearing, rack bearing
  - Left Track Rod (port rodL)
  - Right Track Rod (port rodR)
- Chassis connection is at axle reference
- Rack connections are at axle reference
  - No information about rack needs to be passed to the suspension subsystem
  - Ensures mechanism will always assemble
  - Rack ends in Steer structure and Track Rod ends in Linkage structure **must be same**



```
>> Vehicle.Chassis.SuspA1.Steer.Rack.sOutboard.Value
ans = 0.1371    0.4390    0.2050
>> Vehicle.Chassis.SuspA1.Linkage.TrackRod.sInboard.Value
ans = 0.1371    0.4390    0.2050
```

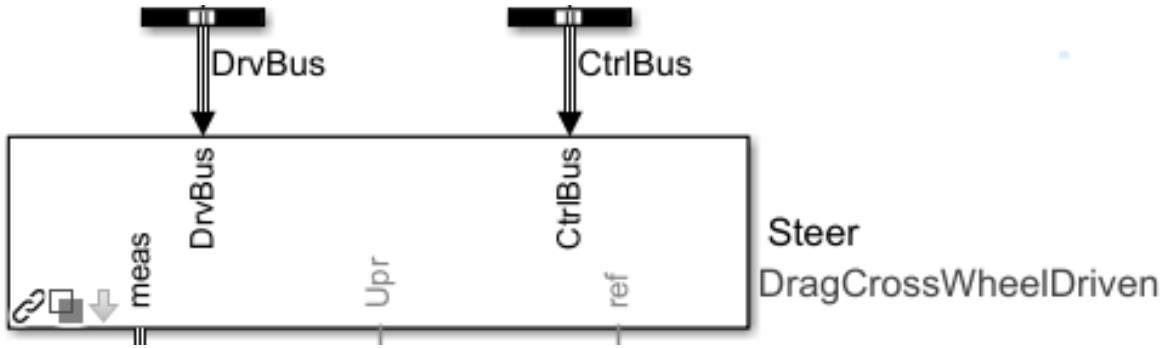
Rack and Pinion Options

- Options to tailor model to system and task
  - Motion ratio from wheel to rack
  - Ideal or with actuator
  - Feedback to hand wheel
- Modular design permits rear wheel and four-wheel steering

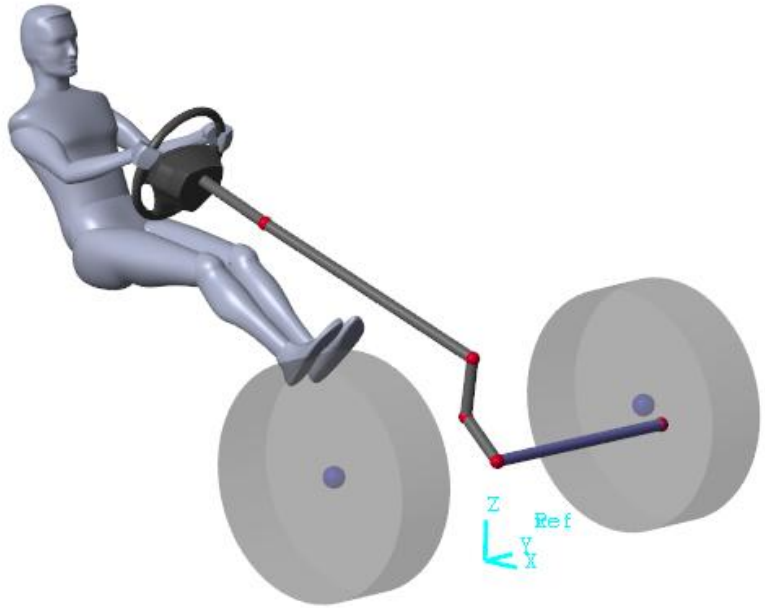
#	Name	Wheel Angle	Rack Position	Force Feedback
1	Rack Wheel Var Ratio	Input	Input[1]	Yes
2	Rack Wheel ByWire	Input	Input[2]	Yes
3	Rack Wheel Indep	Input	Input	No
4	Wheel Driven Rack	Input	Joints[3]	Yes
5	Wheel Driven Rack 1UJoint	Input	Joints[4]	Yes
6	Rack Static Shafts	Input	LUT	No
7	Rack Wheel	Input	LUT	No
8	Rack Driven Shafts	Joints	Input	Yes
9	Rack	n/a	Input	n/a
10	None	n/a	n/a	n/a

[1]Ideal actuator with filter
[2]Motor with controller
[3]Two Universal Joints
[4]One Universal Joint

Drag Link Interface



- Three mechanical connections
  - Chassis (port ref)
    Steering wheel bearing, Pitman arm bearing
  - Drag link connection (port Upr)
    Connects to left or right upright
- Chassis connection is at axle reference
- Drag link connection is end of link
  - Different than rack and pinion because link is contained in Steer system



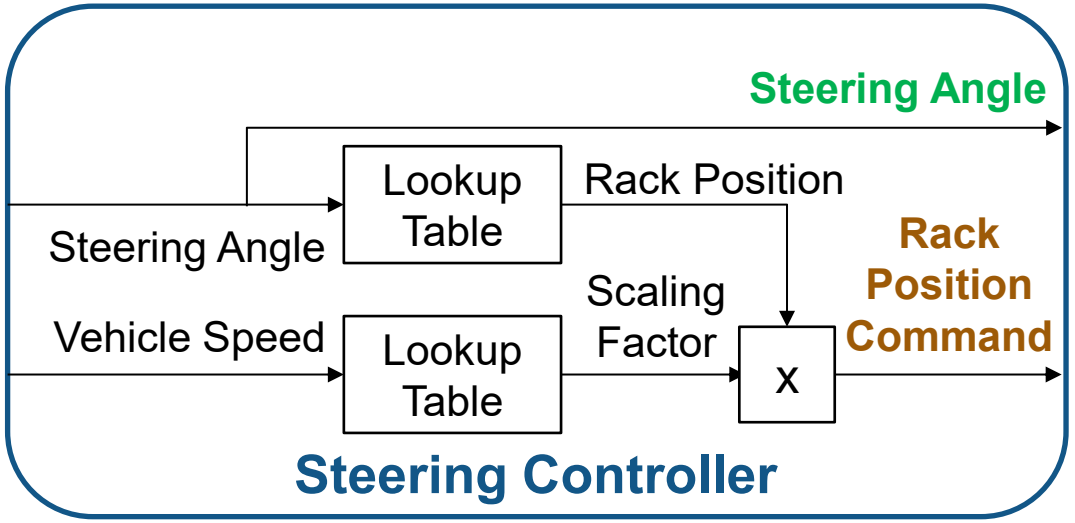
Drag Link Options

- Options to tailor model to system and task
  - With steering wheel (driver)
  - Actuated (rear-wheel steering)
- Modular design permits rear wheel and four-wheel steering

#	Name	Wheel Angle	Pitman Angle	Force Feedback
1	Cross Wheel Driven	Input	Joints	Yes
2	Cross Actuator	n/a	Joints	Yes

Steer-By-Wire
Ideal and Electrical Actuation

- Steering ratio depends upon steering wheel angle and vehicle speed
- Feedback torque to driver via steering wheel is based on effort to position rack
- See scripted example



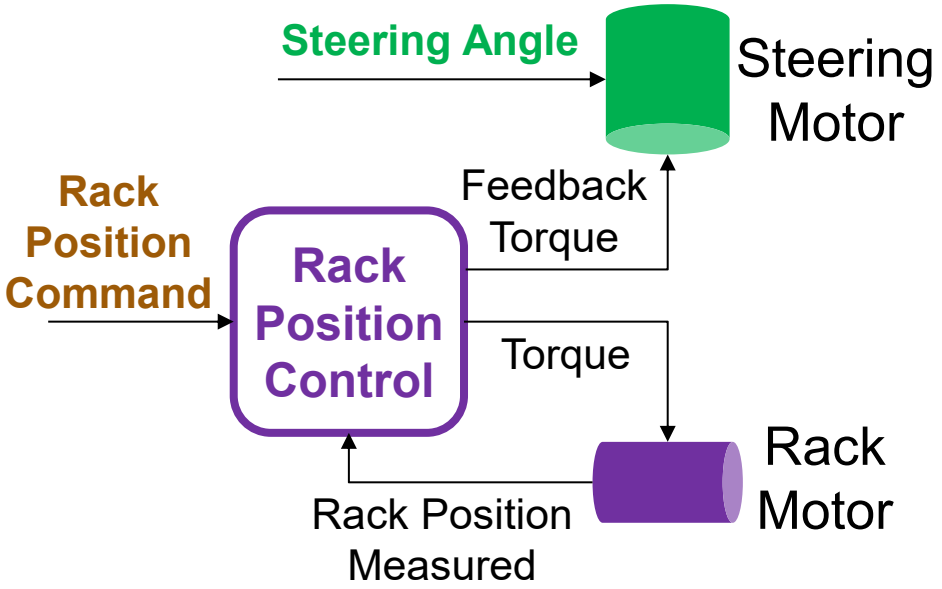
Simscape™ Vehicle Templates

Full Vehicle

Workflow Examples

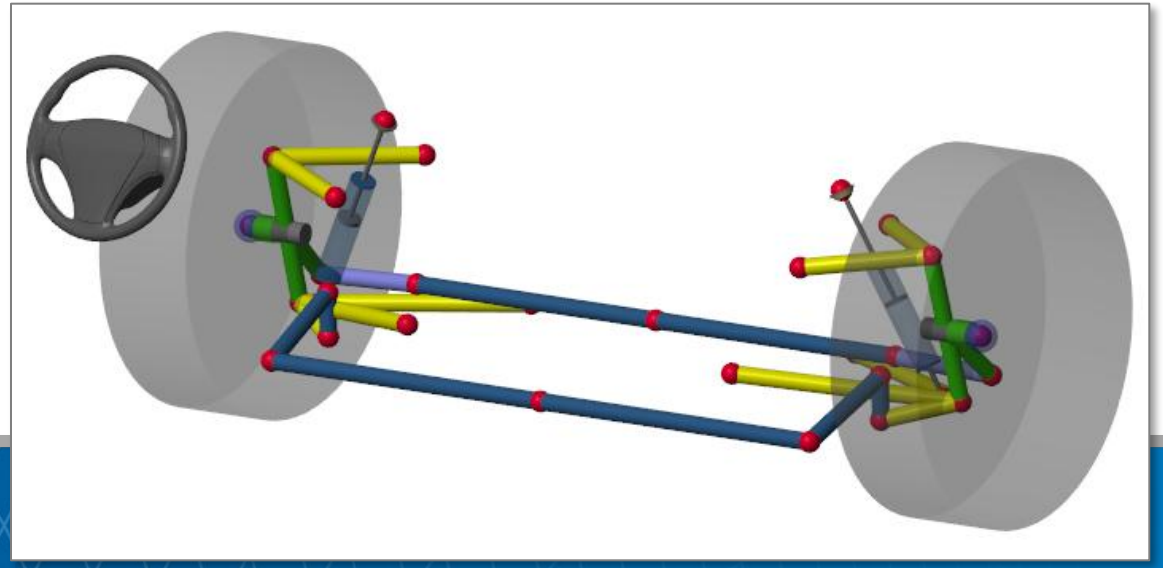
Testing Examples

1. Serial Regenerative Braking, 2 Motor: Run, Edit Code, Results
2. Torque Vectoring: Run, Edit Code, Results
3. Four-Wheel Steering: Run, Edit Code, Results
4. Steer-By-Wire: Run, Edit Code, Results




```
>> open('Simscape_Vehicle_Library_Demo_Script.html')
```

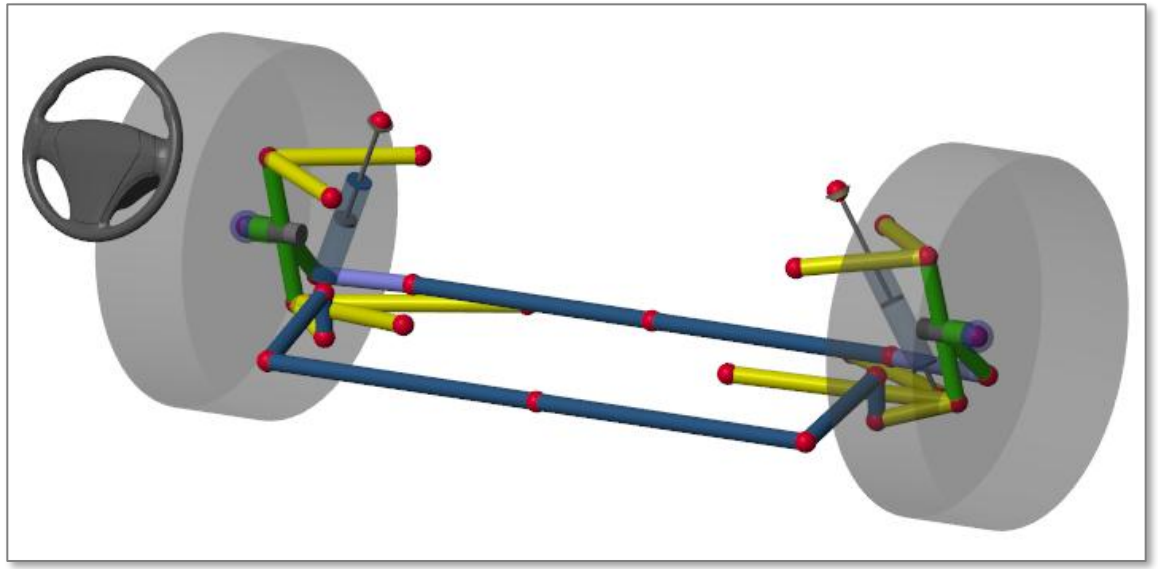
Simscape Vehicle Templates
Anti-Roll Bar



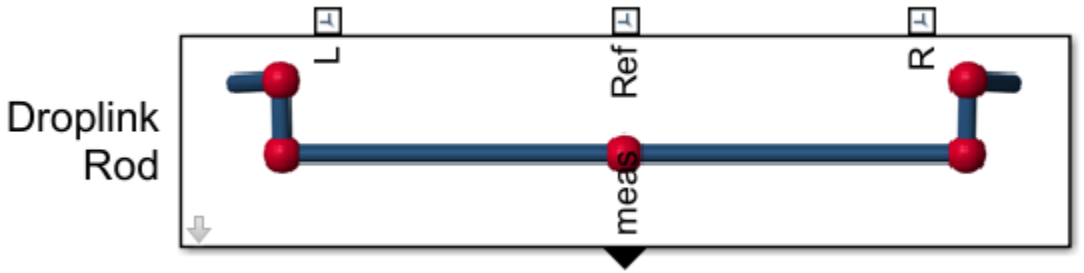
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Anti-Roll Bar

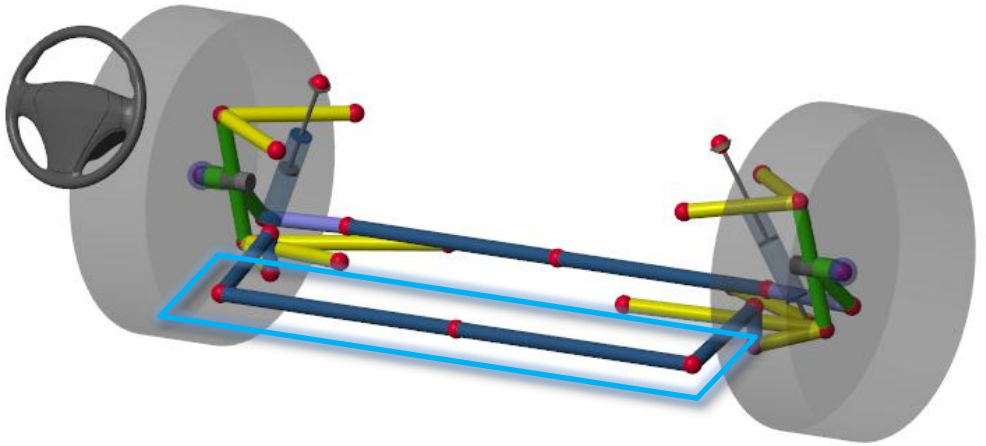
- Interface
- Flexible Element
- Bushings



Anti-Roll Bar Interface



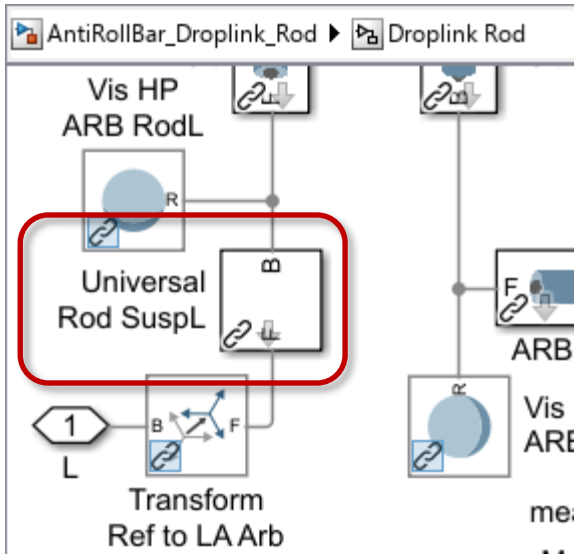
- Three mechanical connections
  - Chassis (port ref)
    For mount connection, joint or bushing
  - Left and right ends (ports L and R)
    Connects to unspecified part on linkage
- Chassis connection is at axle reference
- Left and right connections are axle reference
  - No information about ARB needs to be passed to the linkage subsystem
  - Ensures mechanism will always assemble
  - ARB connection in Linkage and Drop Link ends in AntiRollBar **must be same**

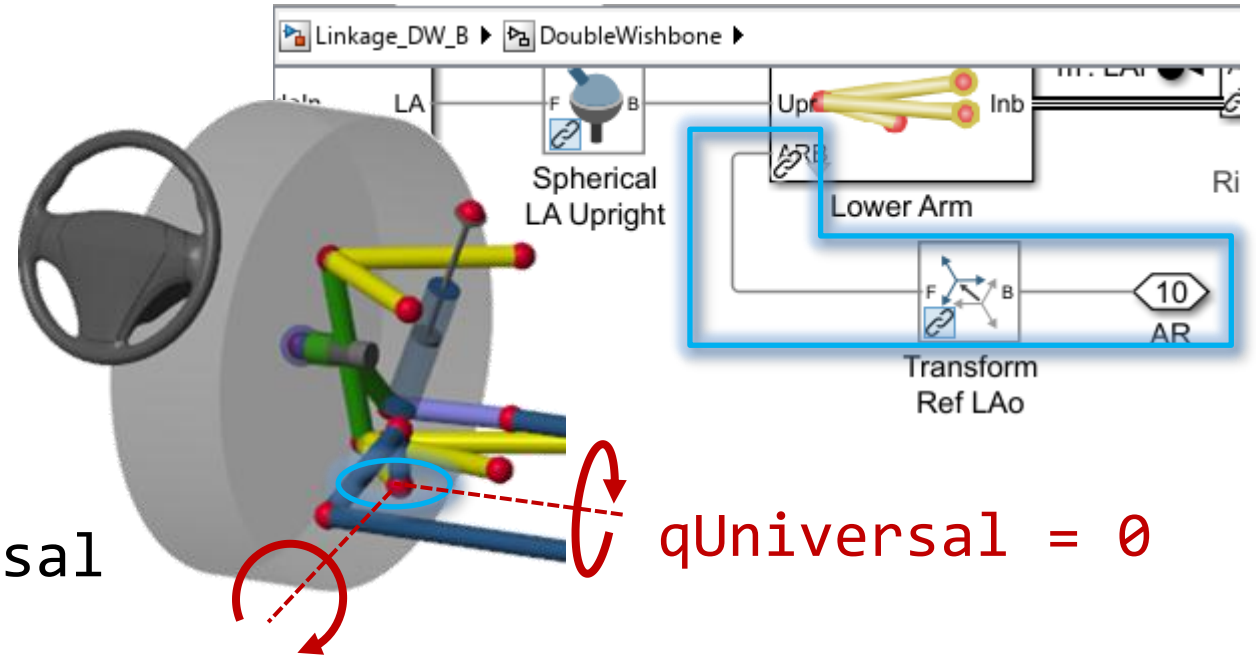


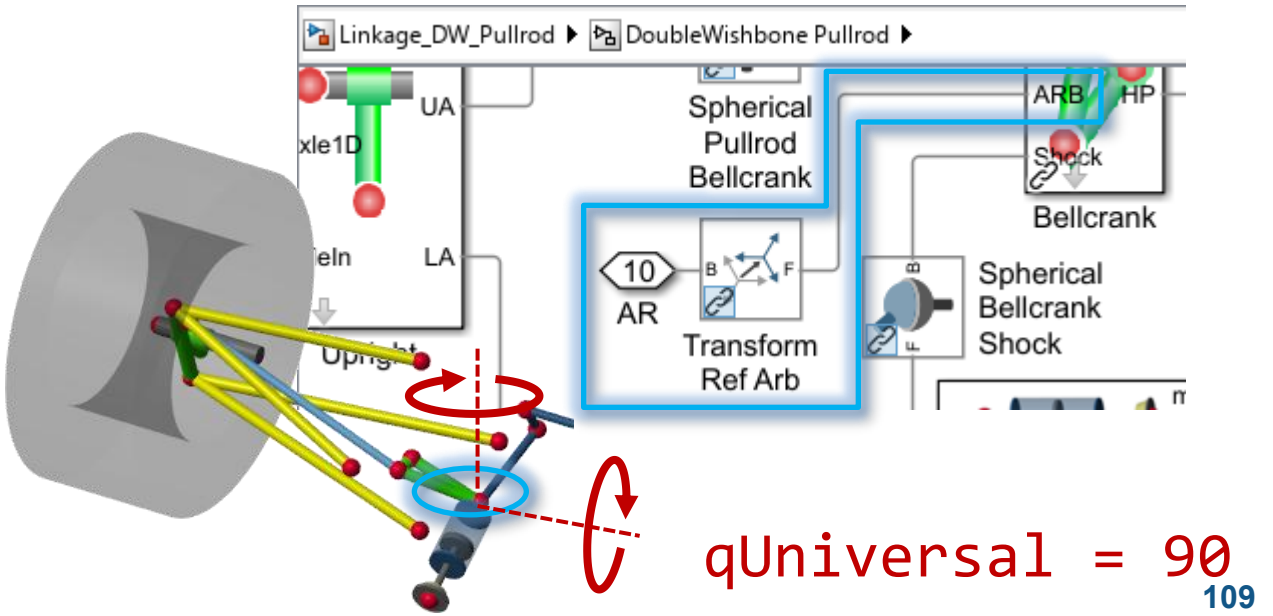
```
>> Vehicle.Chassis.SuspA1.Linkage.LowerWishbone.sARB.Value
ans = -0.0500    0.6000    0.1500
>> Vehicle.Chassis.SuspA1.AntiRollBar.sSuspension.Value
ans = -0.0500    0.6000    0.1500
```

Anti-Roll Bar Interface

- Connect to any part in linkage
- Orientation of Universal Joint between ARB and linkage adjusted by parameter `AntiRollBar.qUniversal`
  - Orientation of joint important rapid re-orientation of joint DOFs

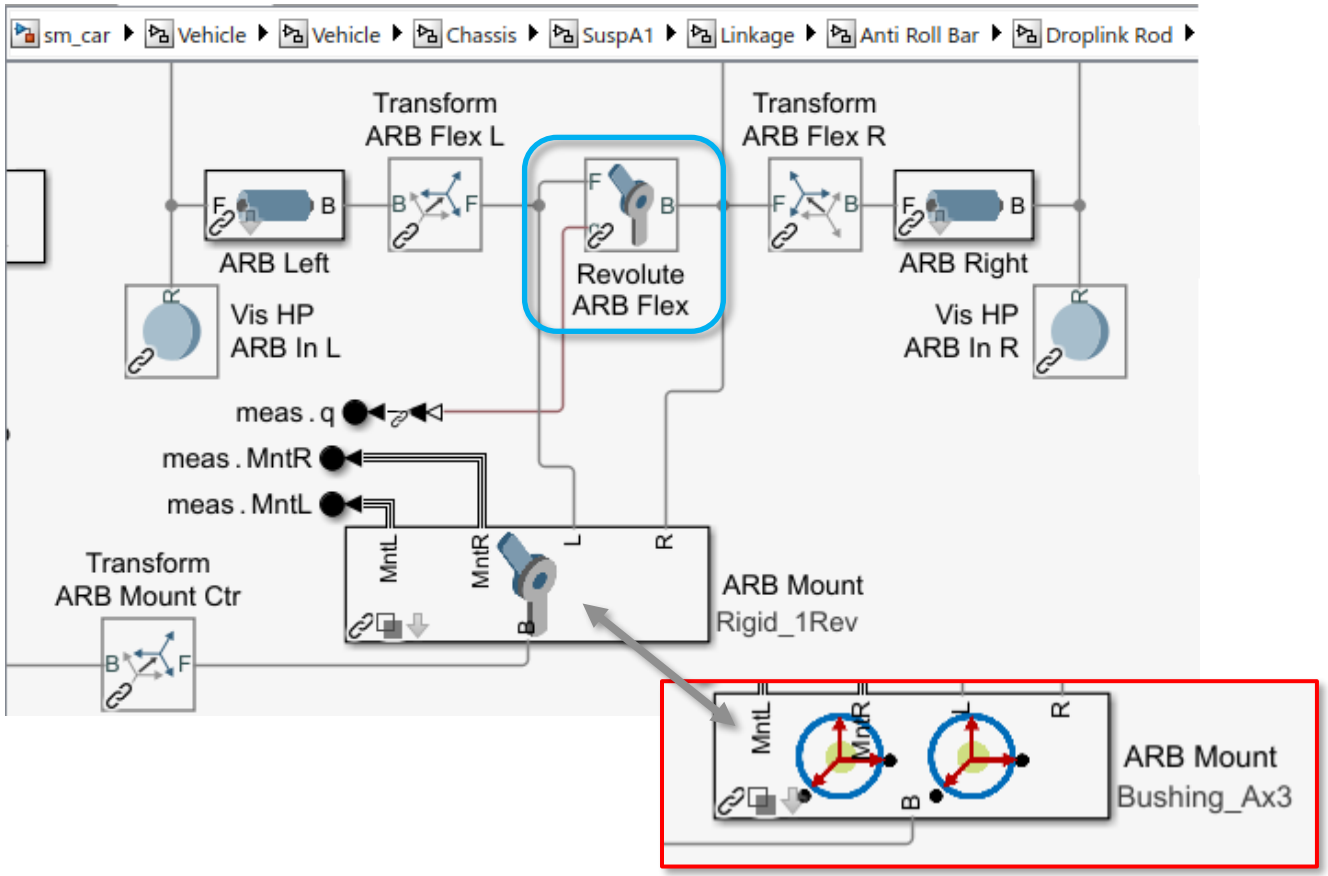


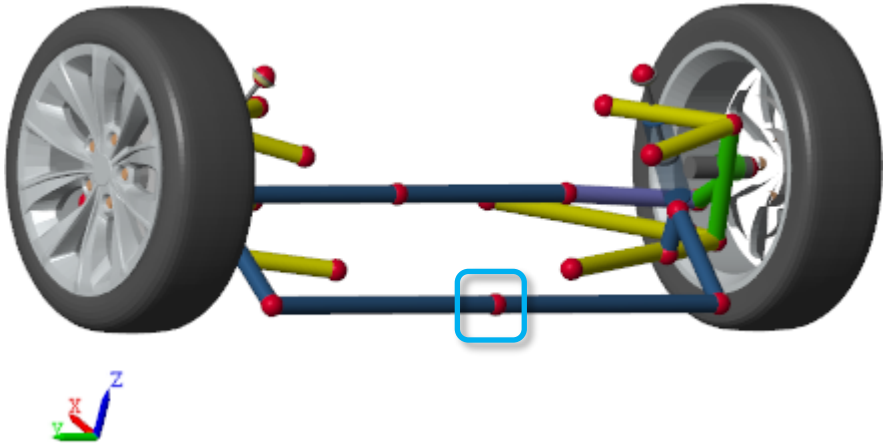


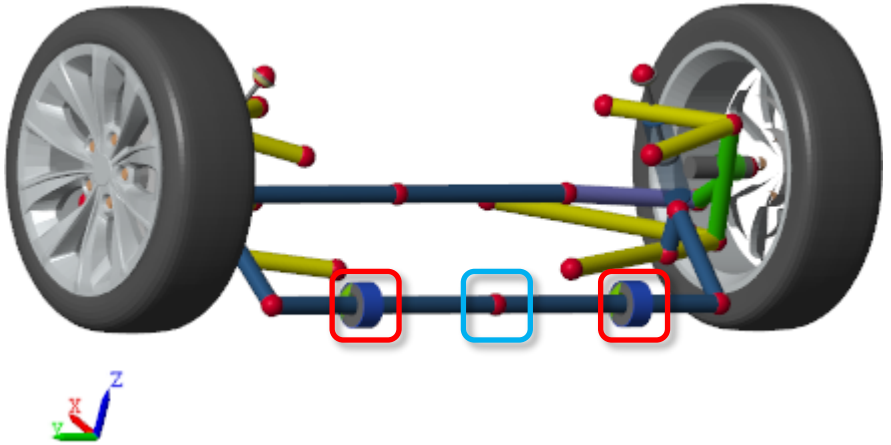


Flexible Element

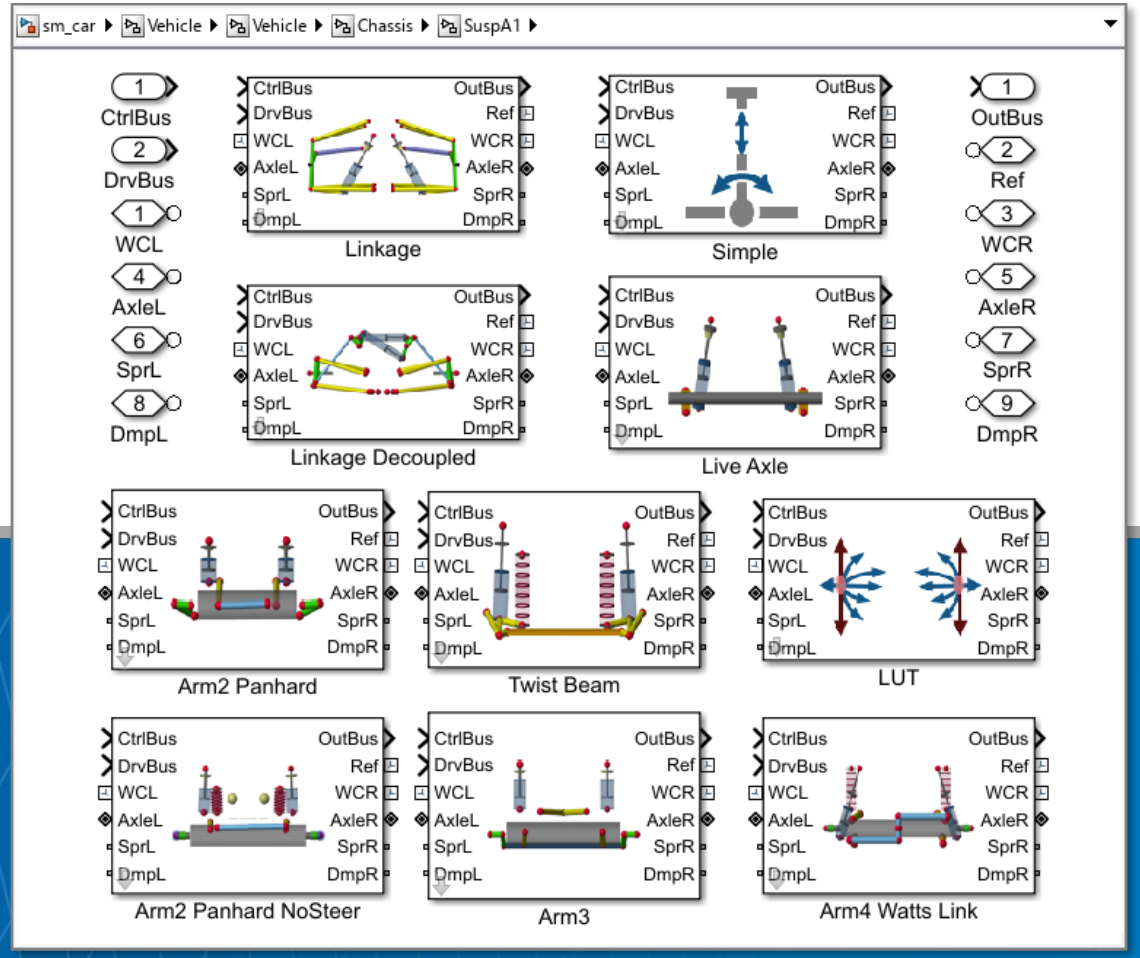
- A Revolute Joint at the midpoint of the ARB with an integrated spring damper permits twisting
- Connection to chassis can be rigid or compliant (bushings)







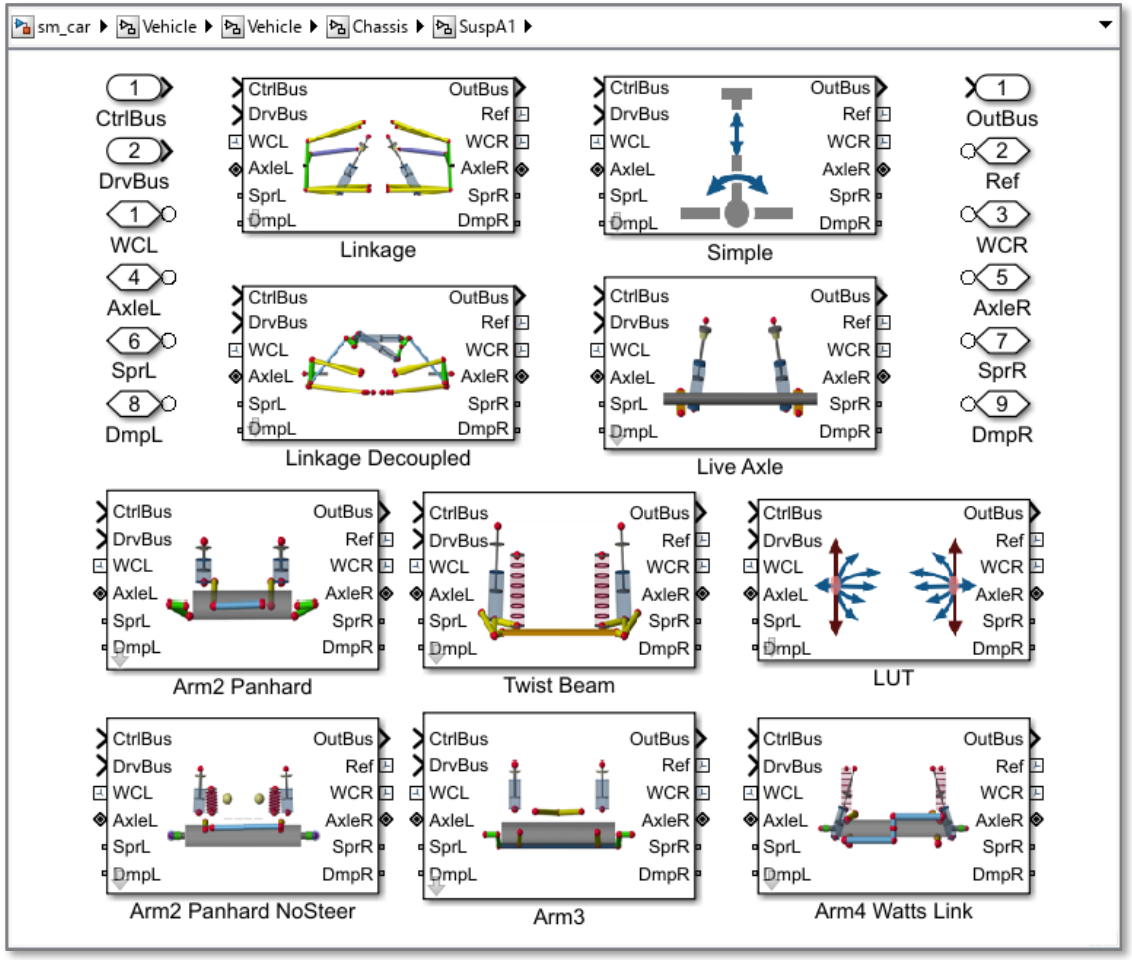
Simscape Vehicle Templates
Suspension (Axle)



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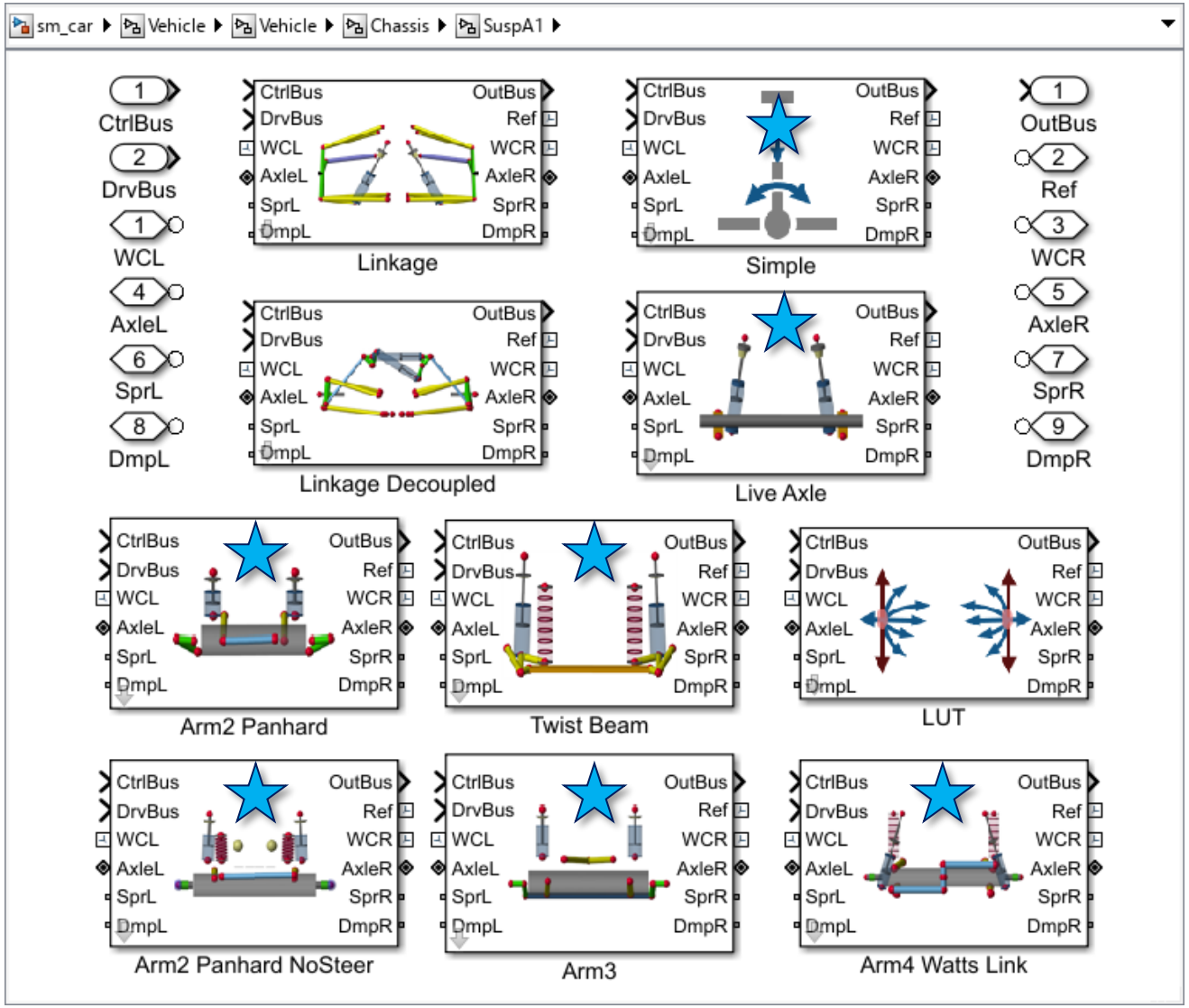
Suspension Axle

- Interface
- Models
  - Twist Beam
  - Panhard Rod, with steering
  - Panhard Rod, no steering
  - Watts Link
- Lookup-Table Based Suspension



Suspension Axle Interface

- Suspension designs with parts that connect the left and right sides with no joints on the path
- Interface is same at axle level as linkage suspensions
  - Plugs into modular infrastructure
- Also useful for abstract models



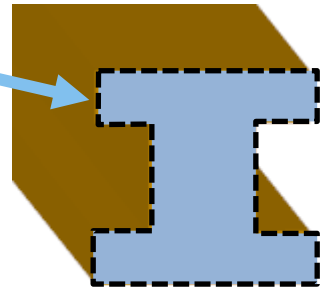
Twist Beam

```
Vehicle.Chassis.SuspA2.TwistBeam

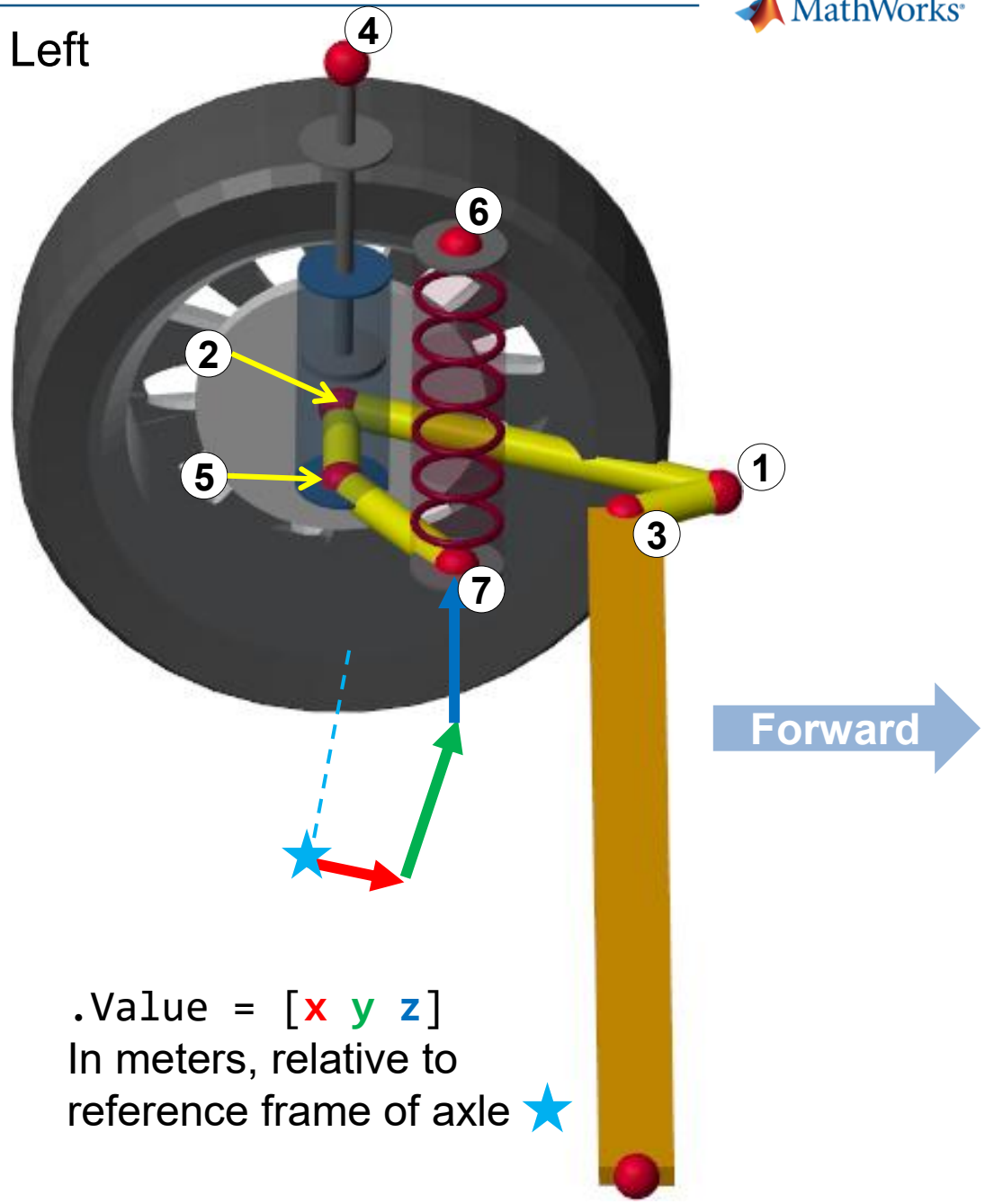
      TrailingArm.sBody          (1)
                 .sWheelCentre   (2)
           Beam.sOutboard        (3)
                 .xWidth         (3)
          Damper.sTop            (4)
                 .sBottom        (5)
          Spring.sTop            (6)
                 .sBottom        (7)

           Beam.xcX
                .xcZ
```

Defines extrusion for beam



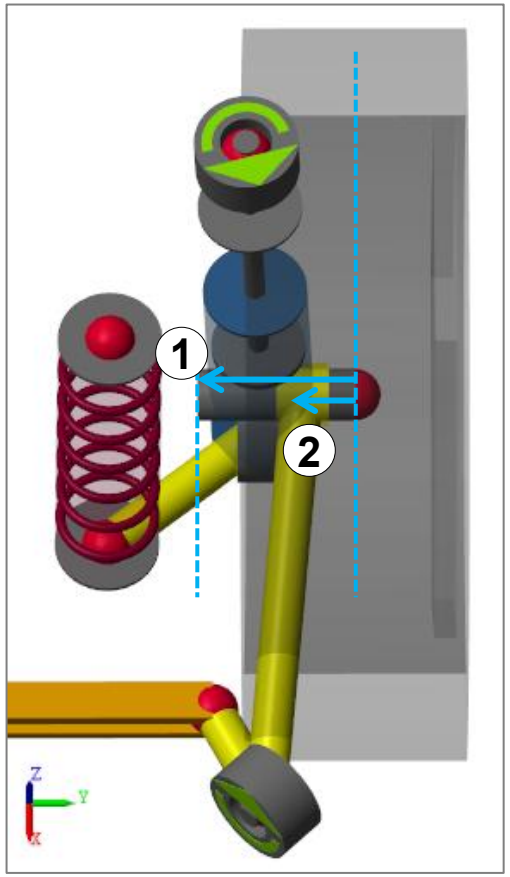
Front Left



`.Value = [x y z]`
In meters, relative to reference frame of axle ★

Twist Beam

```
Vehicle.Chassis.SuspA2.TwistBeam
      TrailingArm.xAxle          (1)
                  .xWCtoBearing  (2)
```

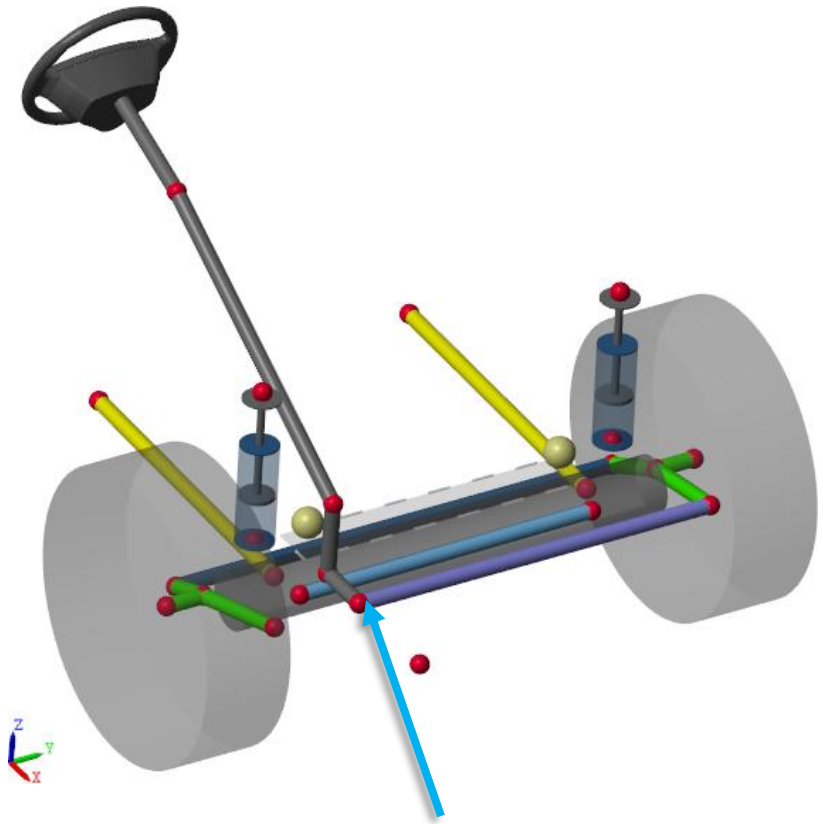


(1) Length of axle shaft connecting wheel center to driveshaft

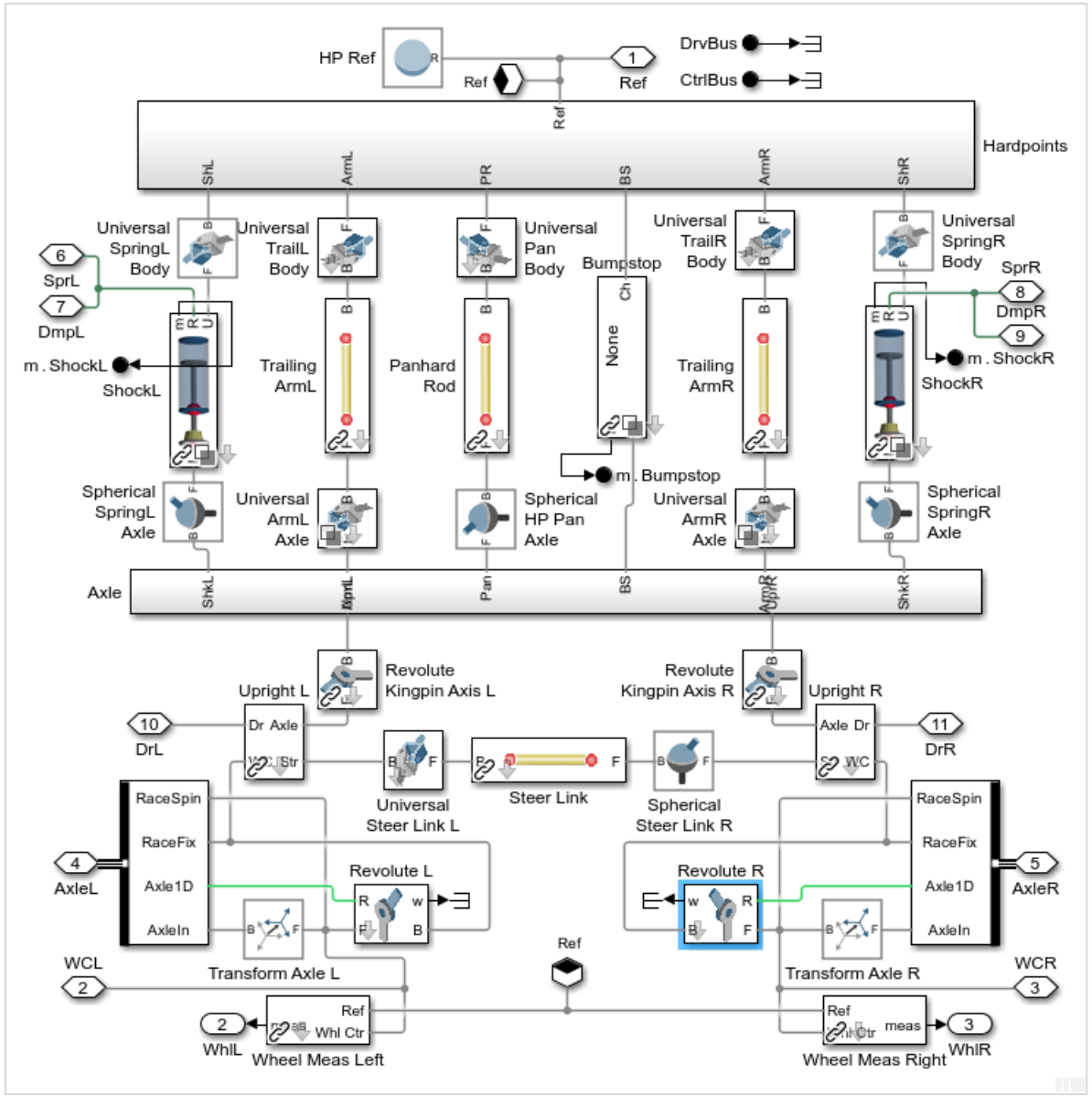
(2) Distance from wheel center to location of wheel revolute joint

Two Trailing Arm with Panhard Rod

- Connects to steering templates that use a drag link



Optional Drag Link Steering via Pitman Arm



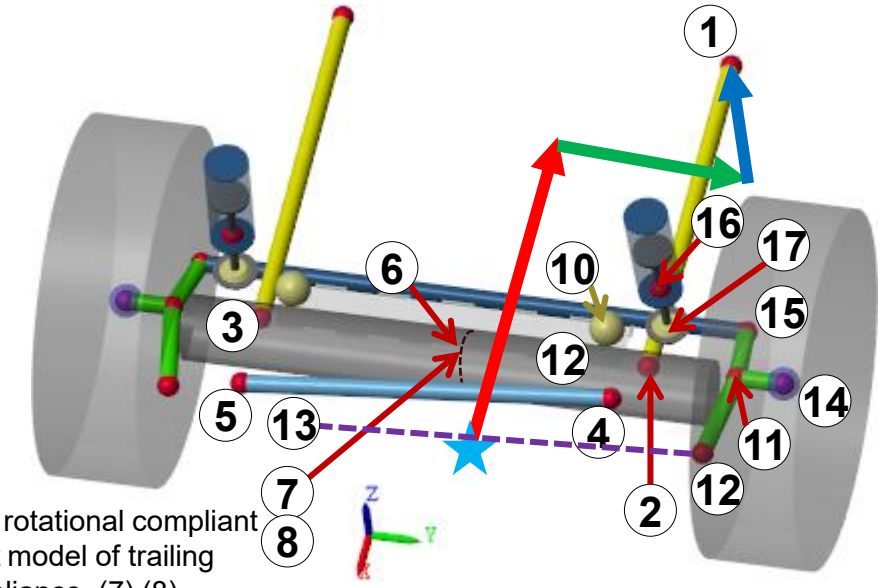
Two Trailing Arm with Panhard Rod

```
Vehicle.Chassis.SuspA1.AxleTA2PR.
        TrailingArm.sBody          (1)
                   .sAxleL         (2)
                   .sAxleR         (3)
        .PanhardRod.sBody          (4)
                   .sAxle          (5)
              Axle.sCG             (6)
                   .K              (7)
                   .d              (8)
          Bumpstop.sAxle           (9)
                   .sBody         (10)
           Upright.sAxleL         (11)
                   .sAxleR
                   .sDragLinkL    (12)
                   .sDragLinkR    (13)
                   .sWheelCentreL(14)
                   .sWheelCentreR
    SteeringLink.sOutboardL       (15)
                   .sOutboardR
              Shock.sTopL          (16)
                   .sTopR
                   .sBottomL      (17)
                   .sBottomR
```

Note: Panhard Rod suspension is asymmetrical.
Our model makes the following assumptions:

1. Any mirrored hardpoint attached to the **axle** may need to be **asymmetrical** (only at one point in its vertical travel will it be symmetrical). These points will need to have a left and right value.

2. Any mirrored hardpoint attached to the **chassis** will be **symmetrical**

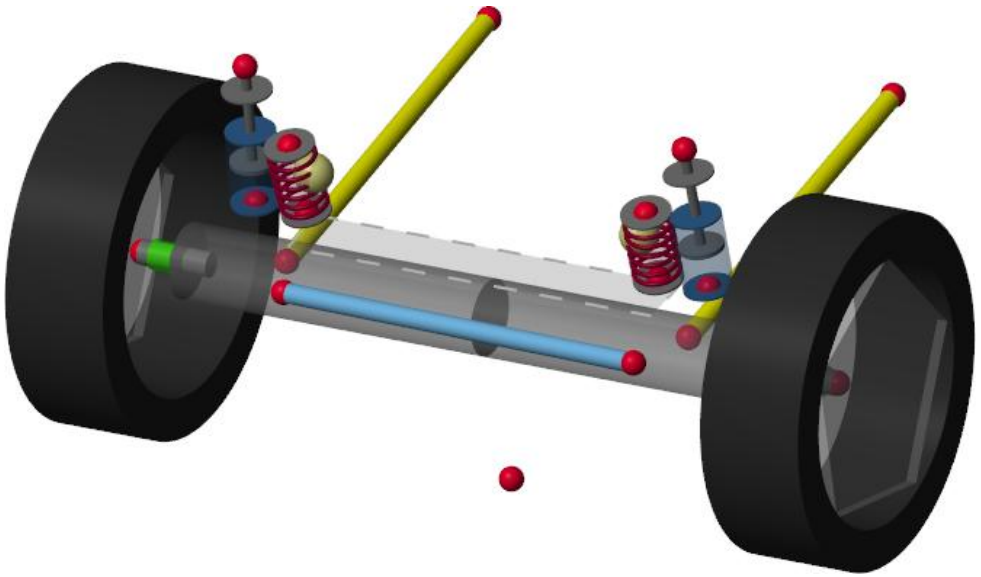


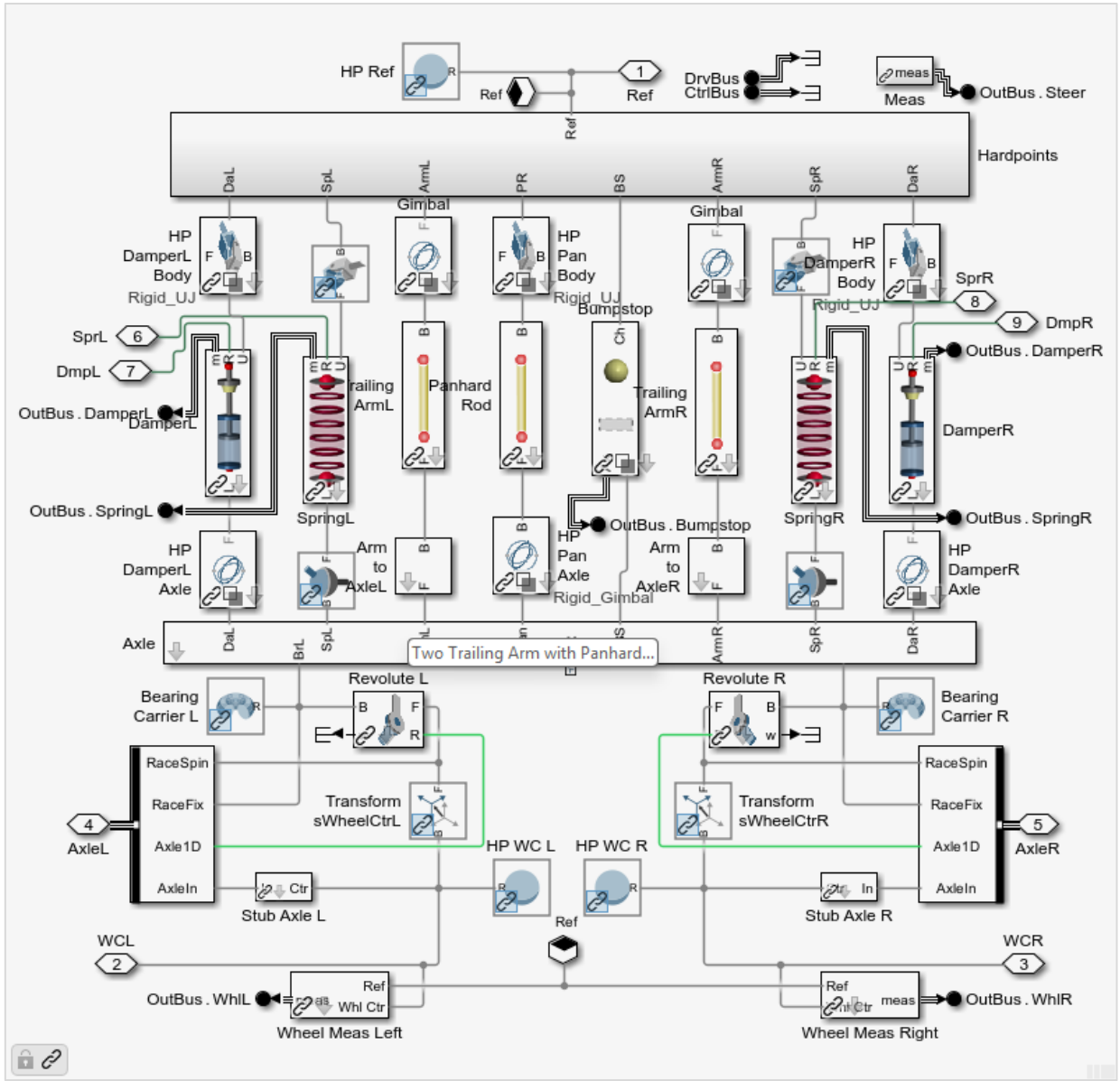
Axle is split with a rotational compliant element. Abstract model of trailing arm bushing compliance. (7) (8)

★ Reference Frame

Two Trailing Arm with Panhard Rod, No Steering

- Separate spring and damper connections





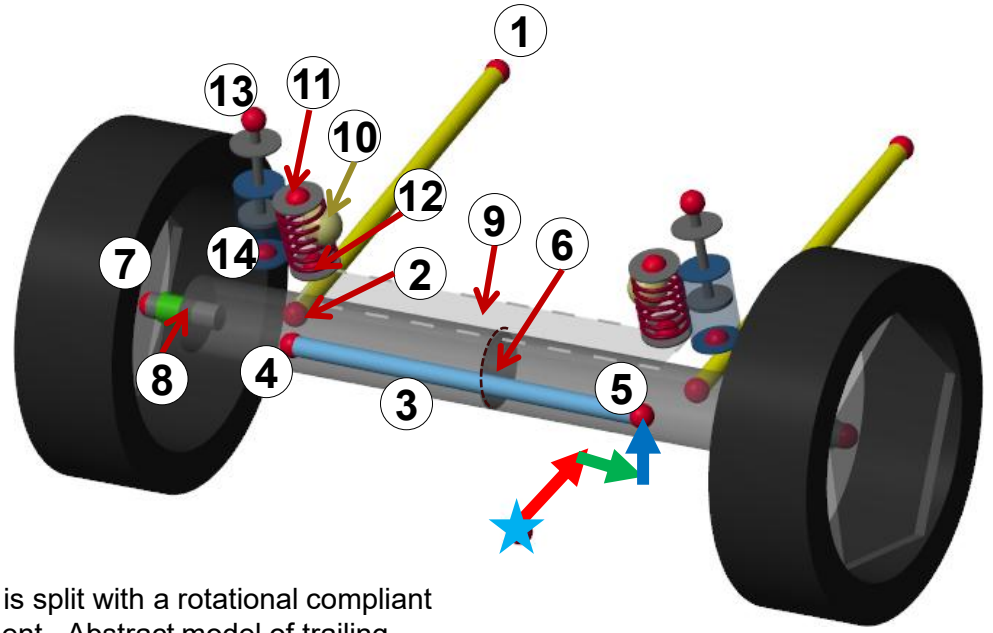
Two Trailing Arm with Panhard Rod, No Steering

```
Vehicle.Chassis.SuspA2.AxleTA2PR.
        TrailingArm.sBody          (1)
                   .sAxleL         (2)
                   .sAxleR         (3)
        .PanhardRod.sBody          (4)
                   .sAxle          (5)
              Axle.sCG             (6)
                   .sWheelCentreL  (7)
                   .sWheelCentreR
                   .sOutboardL     (8)
                   .sOutboardR
          Bumpstop.sAxle           (9)
                   .sBody         (10)
            Spring.sTopL          (11)
                   .sTopR
                   .sBottomL      (12)
                   .sBottomR
            Damper.sTopL          (13)
                   .sTopR
                   .sBottomL      (14)
                   .sBottomR
```

Note: Panhard Rod suspension is asymmetrical.
Our model makes the following assumptions:

1. Any mirrored hardpoint attached to the **axle** may need to be **asymmetrical** (only at one point in its vertical travel will it be symmetrical). These points will need to have a left and right value.

2. Any mirrored hardpoint attached to the **chassis** will be **symmetrical**

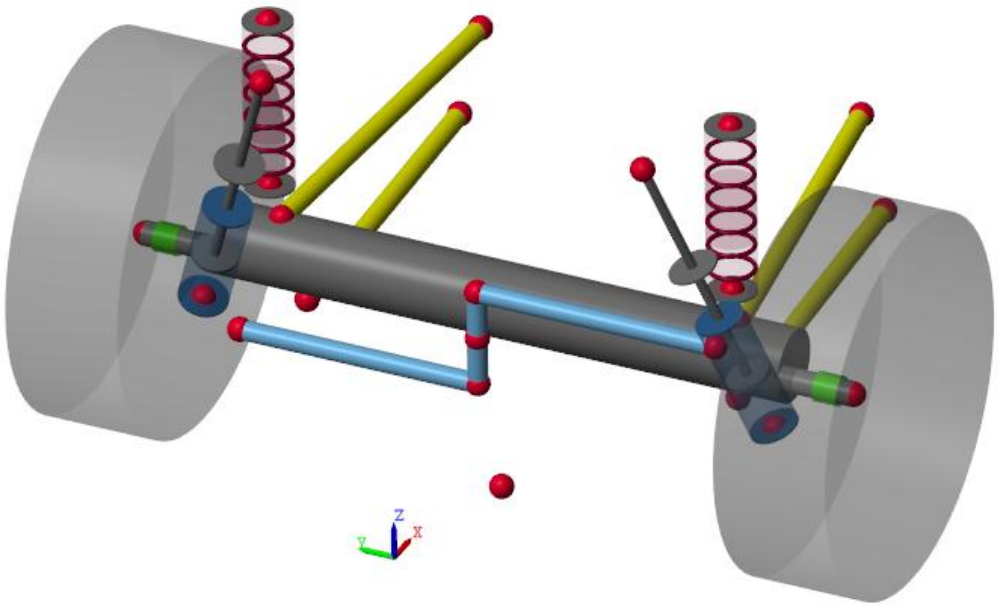


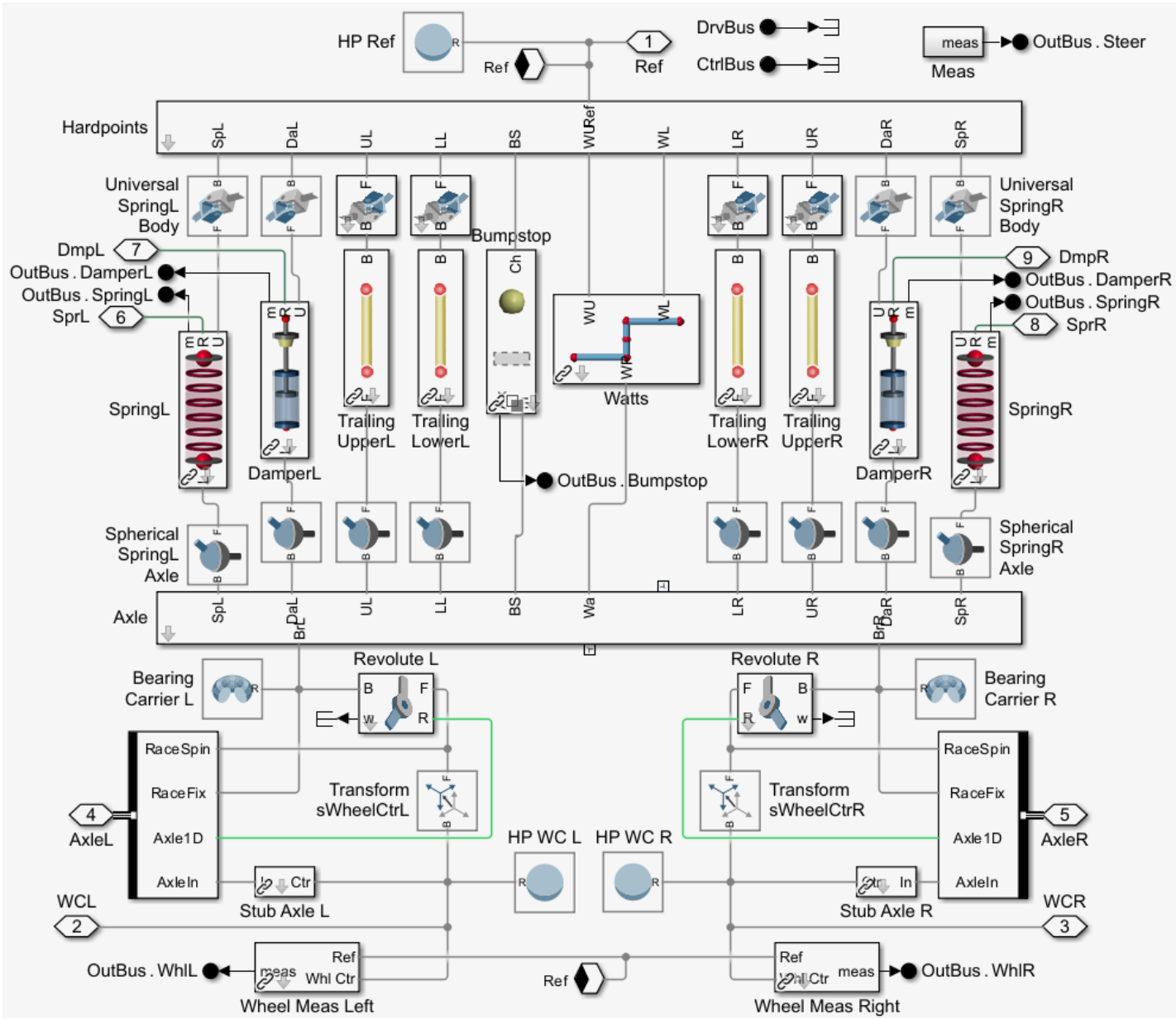
Axle is split with a rotational compliant element. Abstract model of trailing arm bushing compliance. (7) (8)

★ Reference Frame

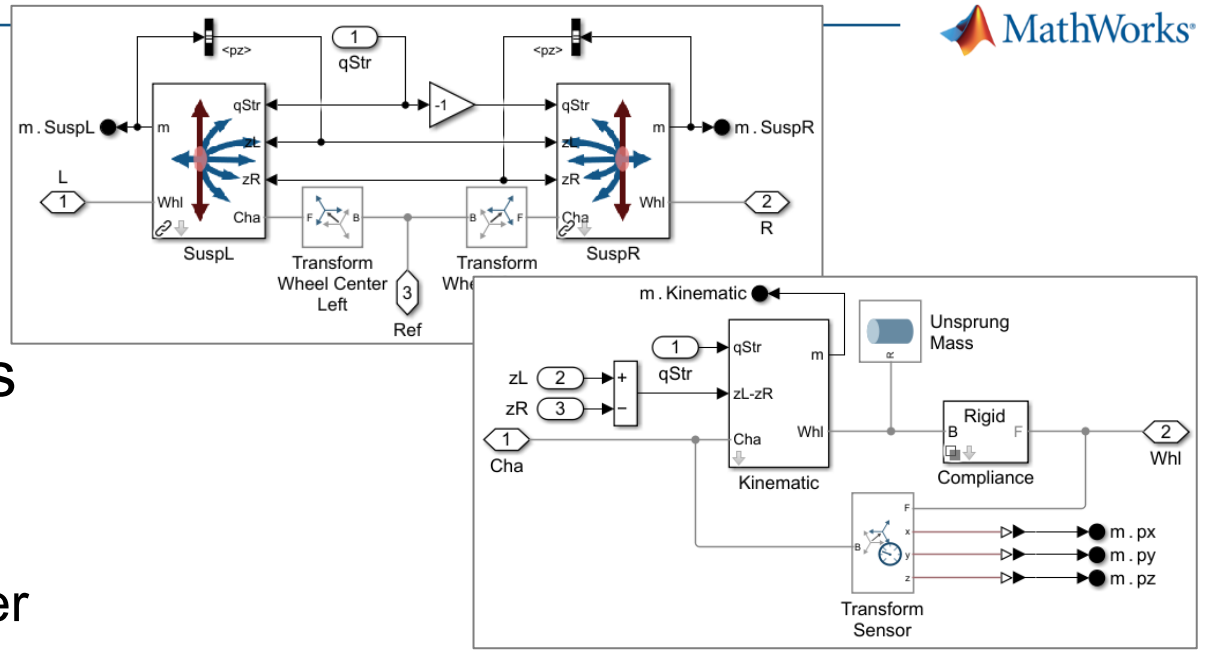
Four Trailing Arm with Watts Link

- Rigid axle with Watts Link to manage lateral movement of chassis with respect to axle





Lookup Table Suspension



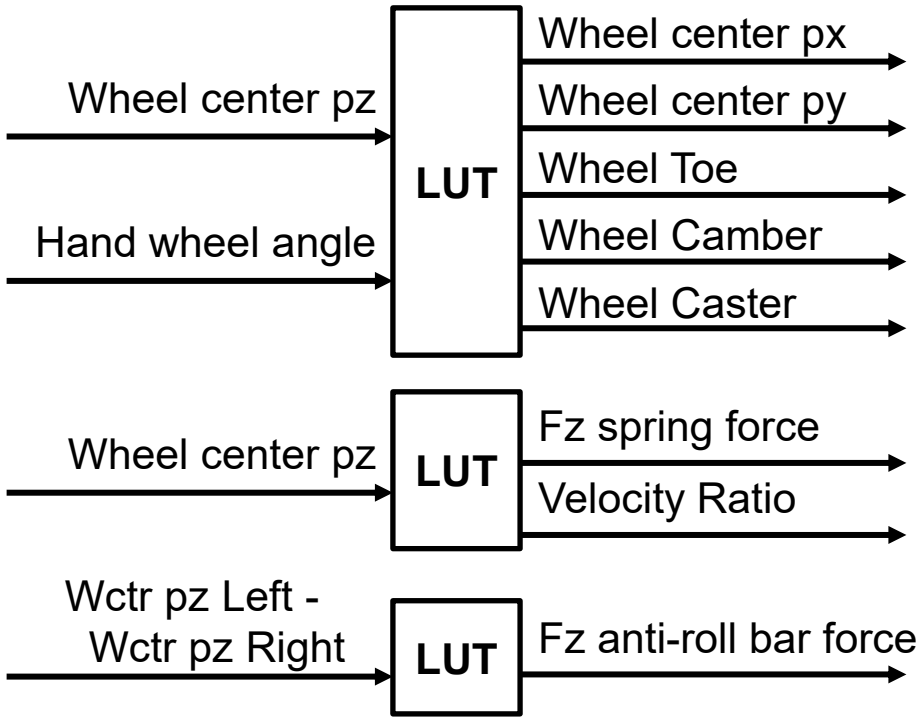
- Kinematics defined using lookup tables
  - Wheel lateral and longitudinal position and orientation depend wheel height
  - Optional 6-DOF compliance at wheel center
- Lookup tables derived from KnC tests
  - Exercise multi-link suspension
  - Record wheel position and orientation
  - Process into lookup tables
  - Includes spring stiffness and damping
- See script to try workflow

Simscape™ Vehicle Templates

Testing Examples

1. Serial Regenerative Braking, 2 Motor: Run, Edit Code, Results
2. Torque Vectoring: Run, Edit Code, Results
3. Four-Wheel Steering: Run, Edit Code, Results

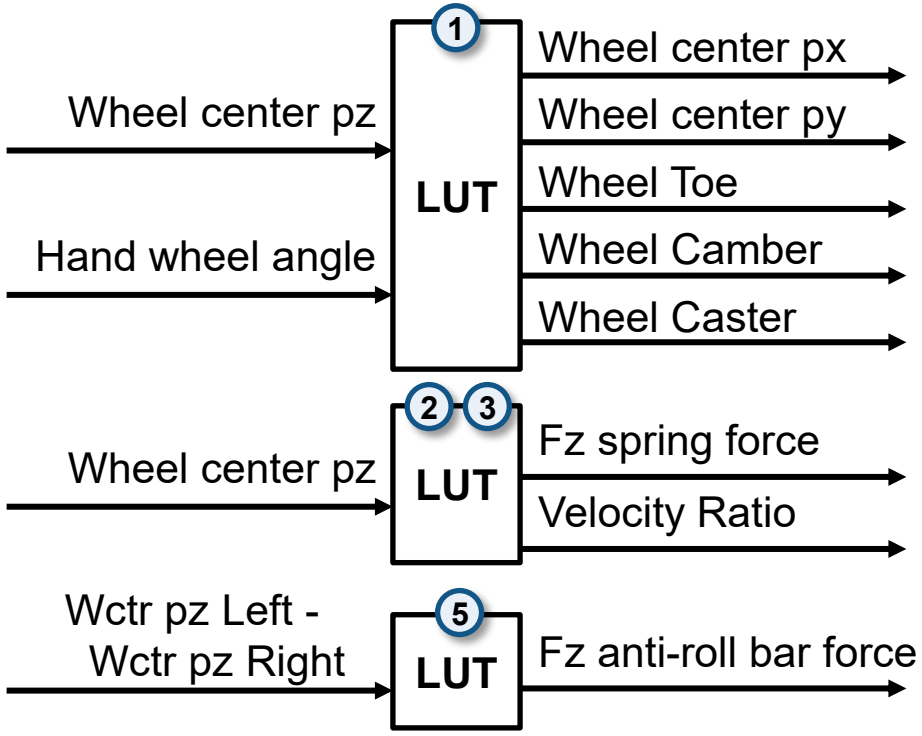
7. Driving in Reverse: Run, Edit Code, Results
8. Hydropneumatic Suspension: Run, Edit Code, Results
9. Generate GGV Diagram: Run, Edit Code, Results
10. Lookup-Table Suspension: Run, Edit Code, Results



Lookup Table Suspension

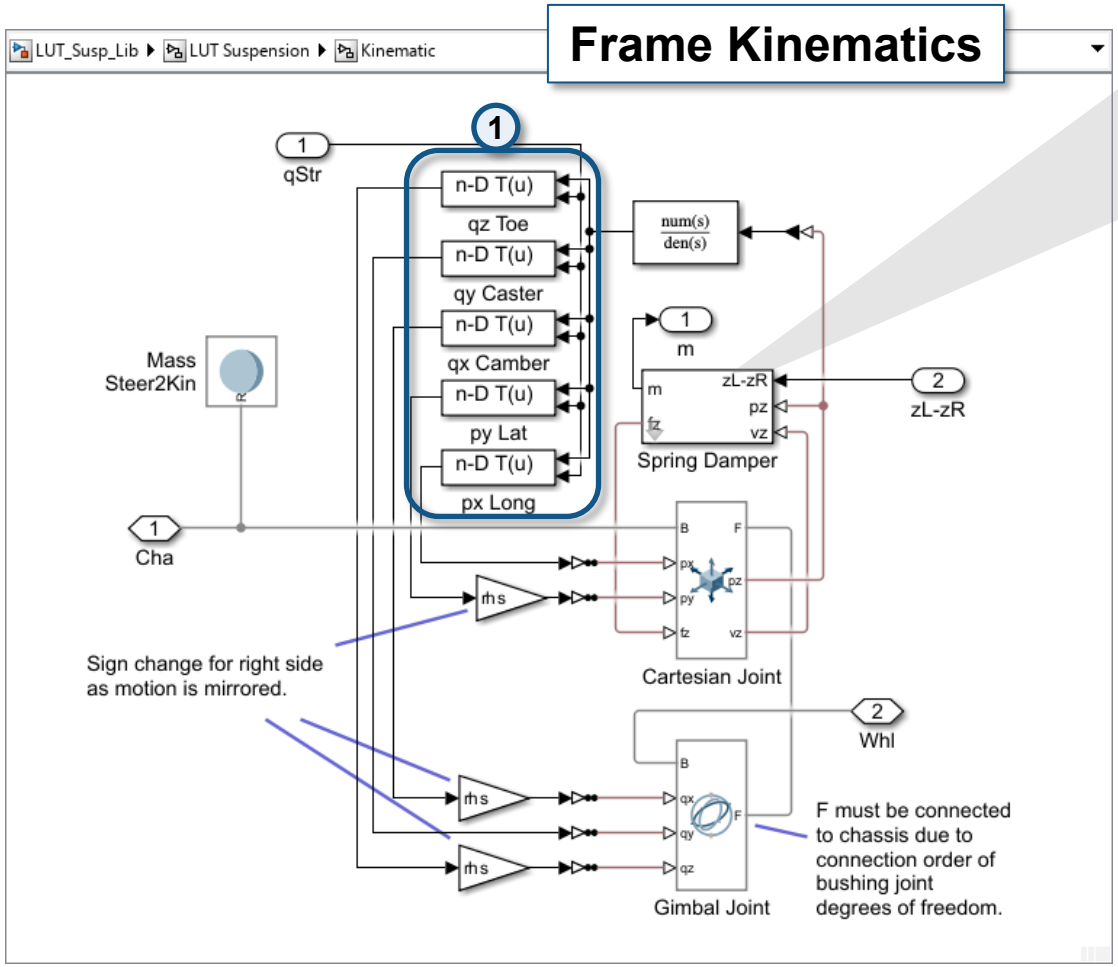
Table/Value	Sweep	Conditions	Measure
1. Kinematics (5x)	Wheel center height and steering angle	none	Wheel center frame position and orientation
2. Spring Force	Wheel center height	Steering = 0, damper and anti-roll bar off	Vertical force to actuate wheel center
3. Velocity Ratio	Wheel center height at constant speed	Steering = 0	Damper compression speed
4. Damping Coefficient	none	none	none
5. Anti-roll bar force	Wheel center height left and right out of phase	damper & anti roll bar off	Vertical force to actuate wheel center

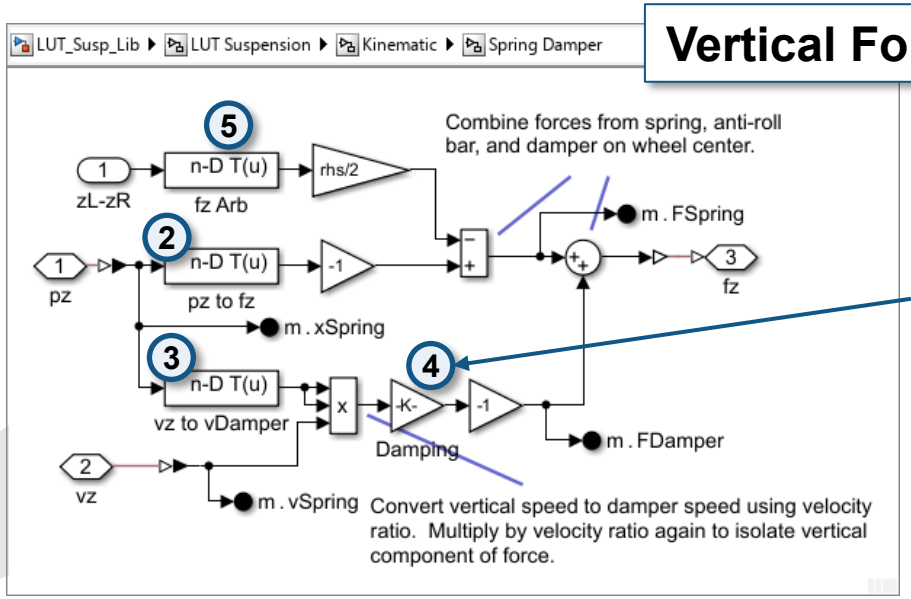




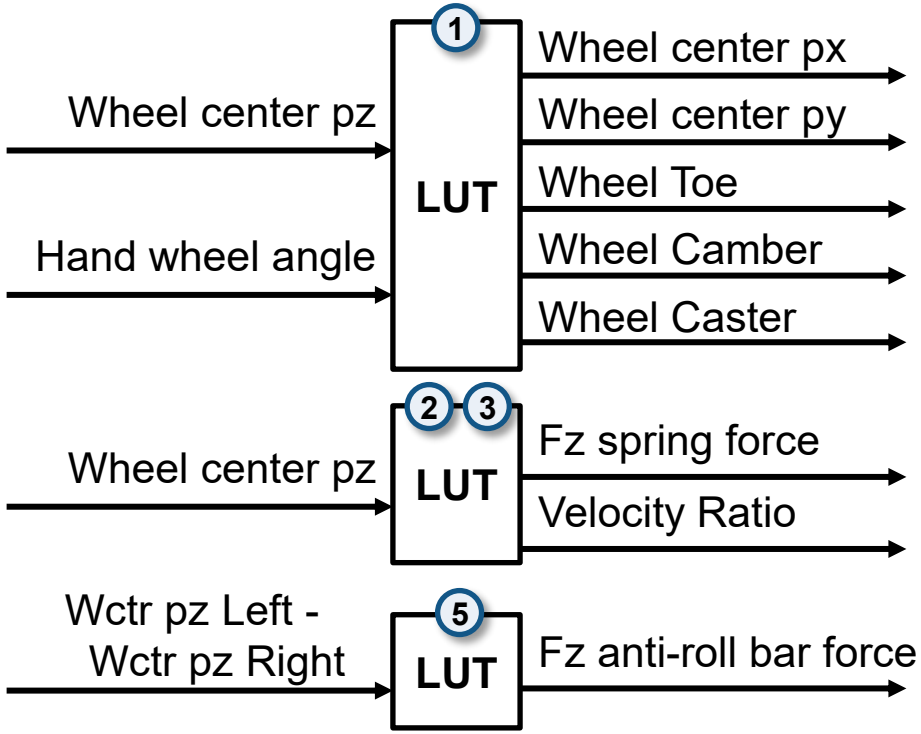
Lookup Table Suspension

- Implementation in Simulink
  Forces applied to multibody joints





Linear damping force implemented here. If different law used in original model, substitute that here.



Simscape Vehicle Templates
Springs and Dampers

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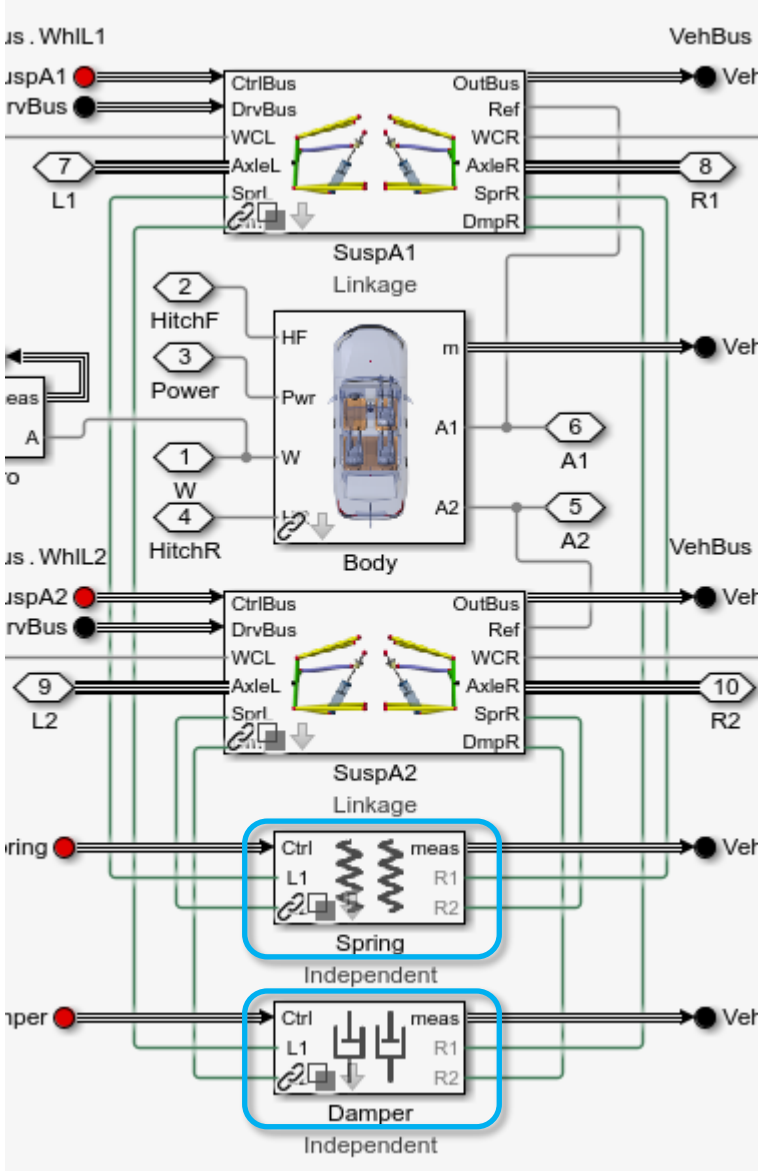
Springs and Dampers

- Interface
- Spring Force Laws
- Damper Force Laws

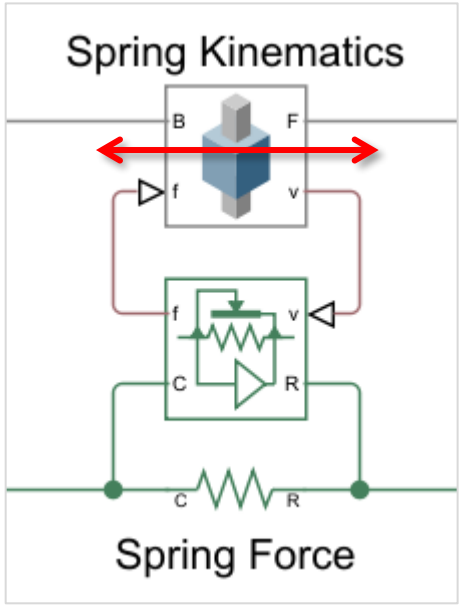
Springs and Dampers Interface

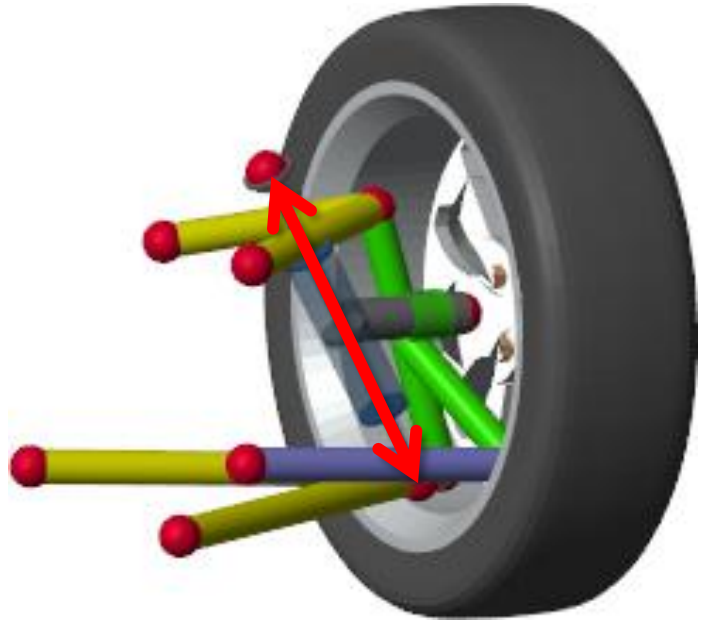
- Force laws for springs and dampers are modelled in distinct subsystems to permit many design variants
  - Fidelity (mechanical, hydraulic, lookup table)
  - Type (active, passive)
  - Connections (independent, left & right connected)
  - Asymmetry (Panhard rod, component wear)

- Permits integrated spring-damper force laws
  - Use no force law for one subsystem



Interface to 3D Model





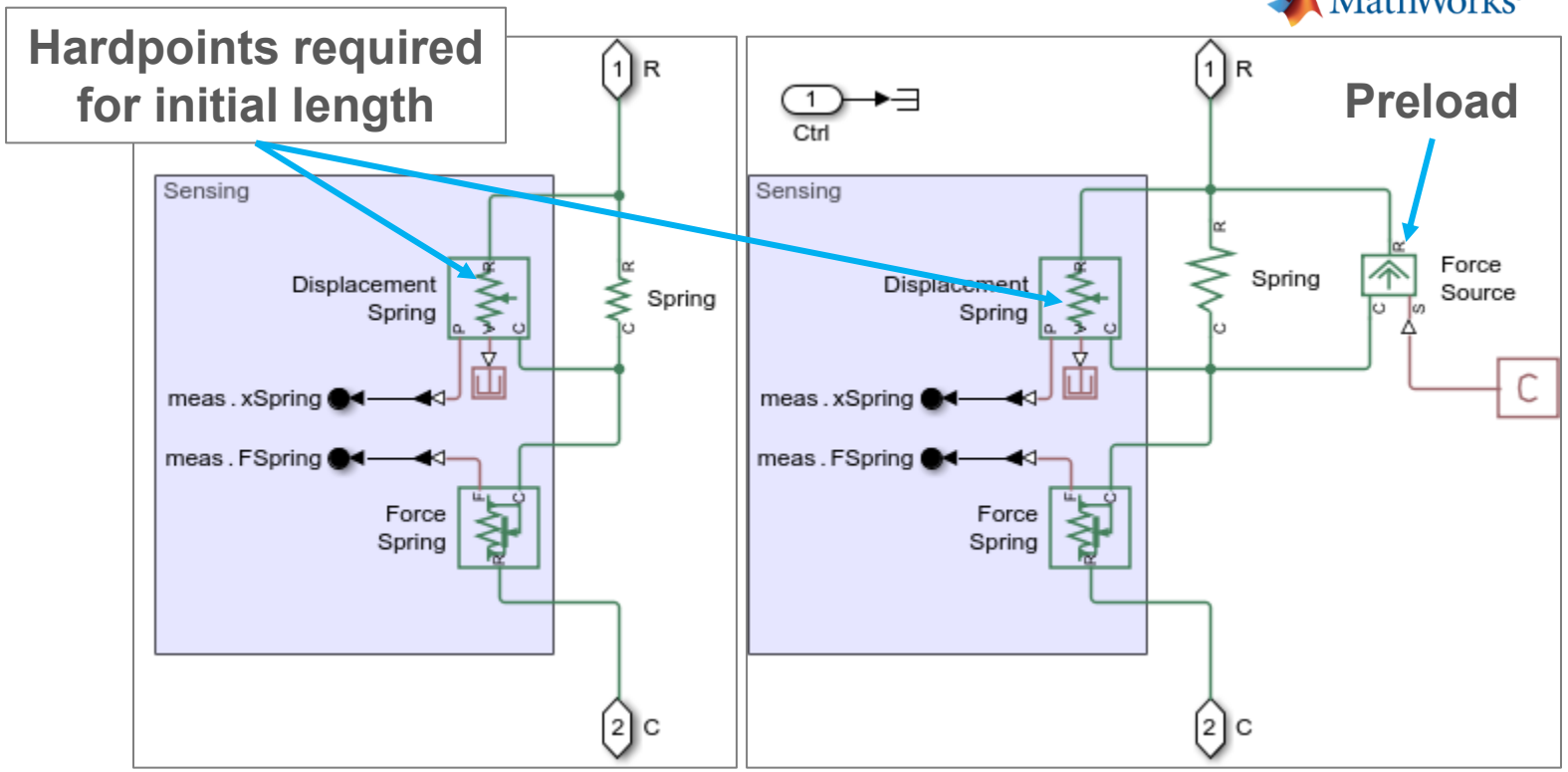
- Multibody model handles kinematic relationship between spring and damper endpoints and linkage
- 1D model calculates resulting force which is applied in multibody model
- Both models require hardpoint definitions
  - Assigned in separate places within the `Vehicle` structure
  - Must be consistent between 1D and 3D models
  - Spring and damper hardpoints can be distinct

```
>> Vehicle.Chassis.SuspA1.Linkage.Shock.sTop.Value
ans =   -0.0027    0.6200    0.6500
>> Vehicle.Chassis.Spring.Axle1.sTop.Value
ans =   -0.0027    0.6200    0.6500

>> Vehicle.Chassis.SuspA1.Linkage.Shock.sBottom.Value
ans =    0.0552    0.8500    0.1900
>> Vehicle.Chassis.Spring.Axle1.sBottom.Value
ans =    0.0552    0.8500    0.1900
```

Same for damper

Spring Force Laws

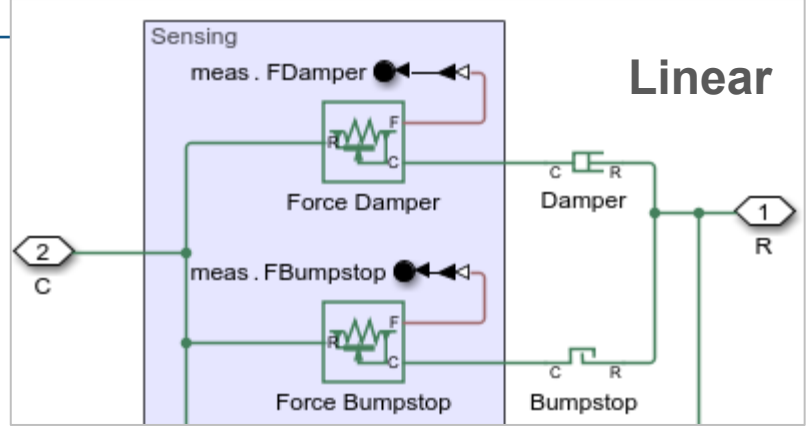


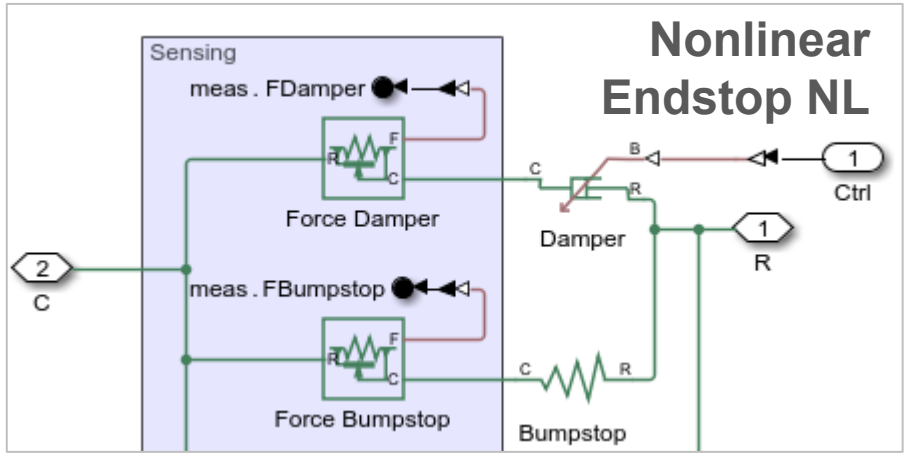
- Options
  - Linear
  - Nonlinear (Lookup Table)
  - Air-Oil Piston
- Preload
  - Linear: Initial compression or force
  - Nonlinear: Must be force
    Compression vs. force curve must go through [0,0]

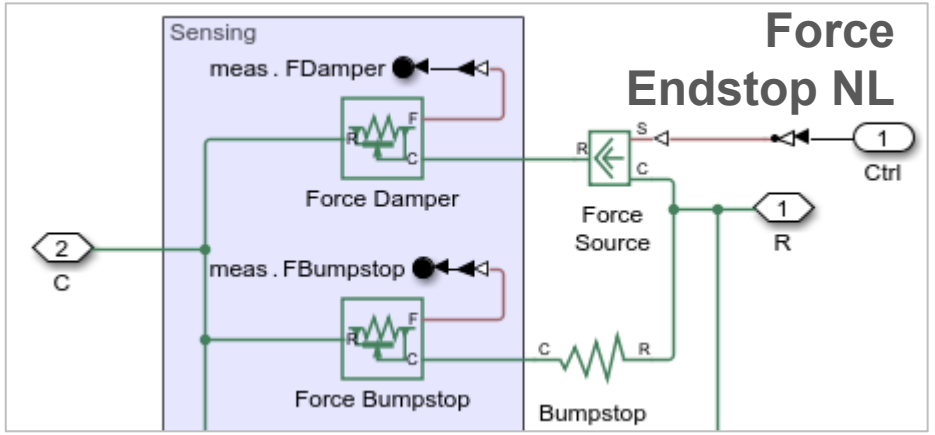
Damper Force Laws

- Damper model includes endstops (jounce and rebound)
- Options include linear, lookup table, and signal driven force laws

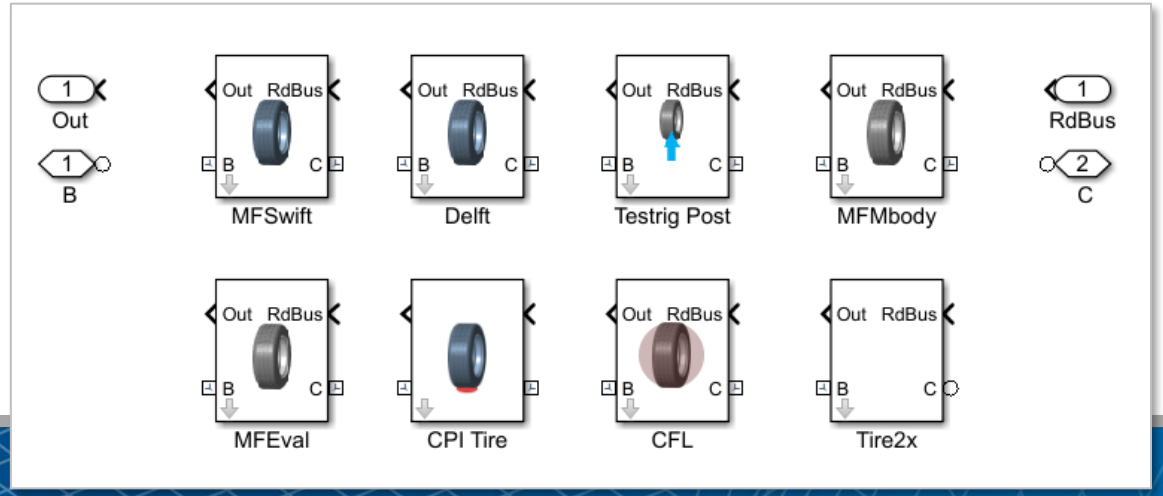
#	Variant	Force Law	Endstop
1	Linear	Linear	Hardstop
2	Nonlinear	Lookup Table	Hardstop
3	Nonlinear Endstop NL	Lookup Table	Lookup Table
4	Variable	Variable Damper	Hardstop
5	Variable Endstop NL	Variable Damper	Lookup Table
6	Force Endstop NL	Input Signal	Lookup Table

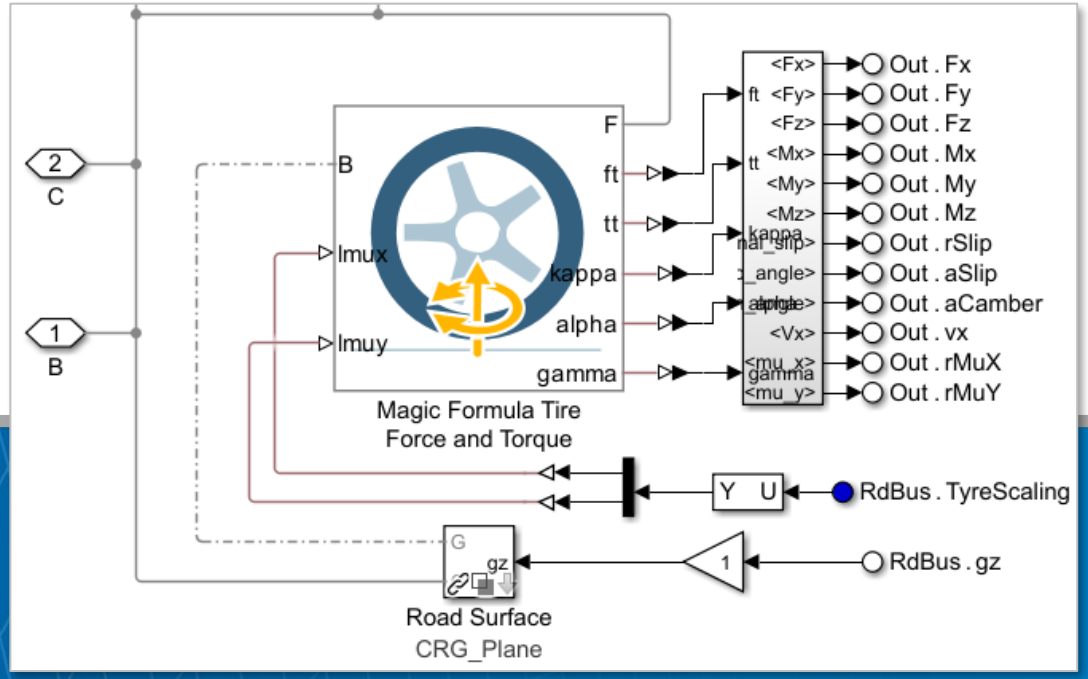






Simscape Vehicle Templates
Tires

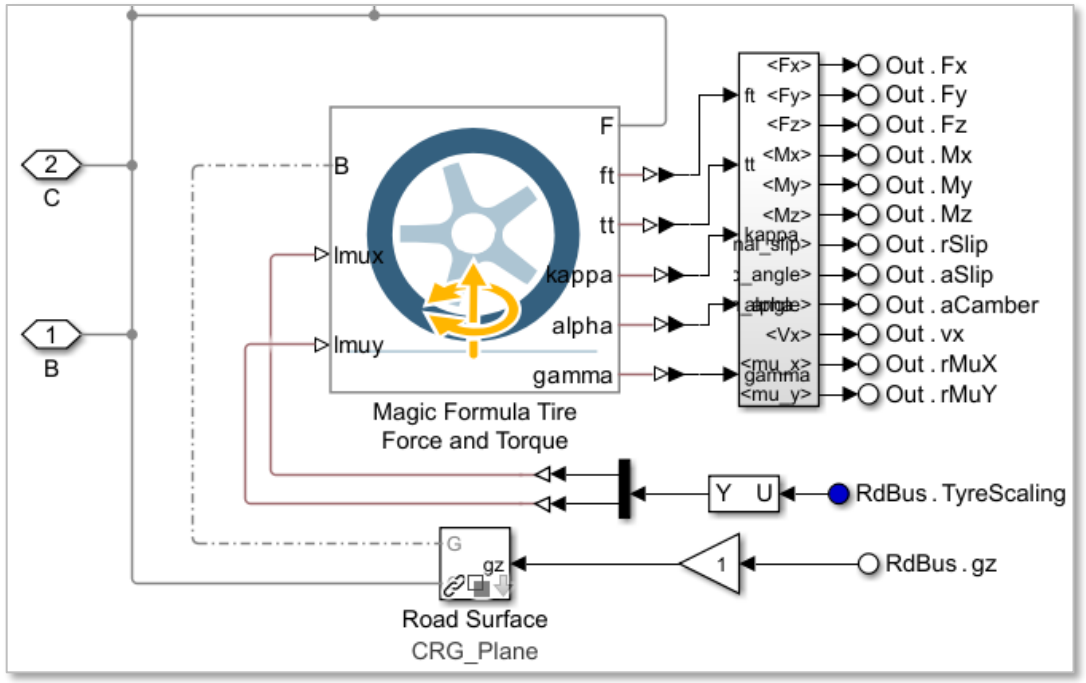




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Tires

- Interface to Suspension
- Interface to Road
- Parameterization
- Measurements



Tire Models Integrated with Simscape Vehicle Templates

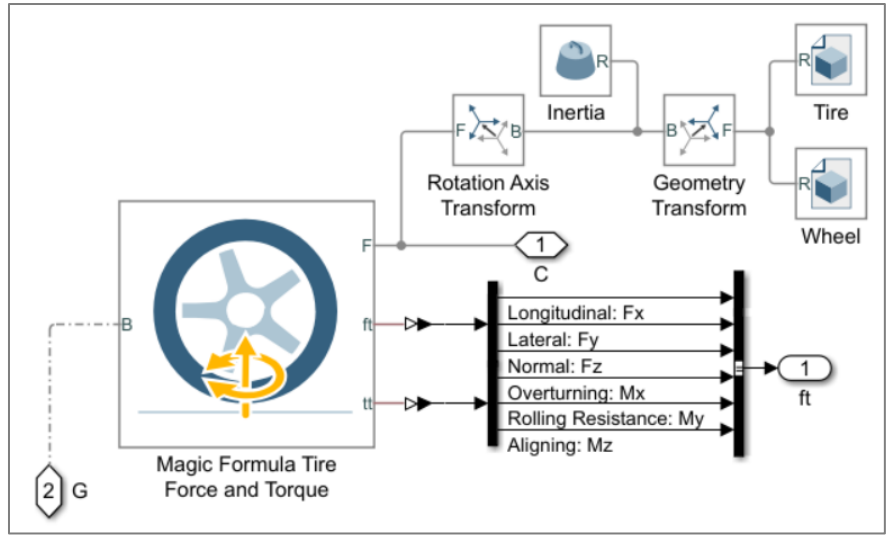
- We have directly integrated a few models
  - Any tire model can be integrated if you use the interfaces we provide

	Simscape	Simcenter	MFEval
Author	MathWorks	Siemens	Community
Category	Pacejka	Pacejka	Pacejka
Interface	Multibody	Signals	Signals
Flat Road	Yes	Yes	Yes
CRG Road	Yes*	Yes	No
Road Height Input	Yes	Yes	Yes
Combined Slip	Yes	Yes	Yes
Turn Slip	Yes	Yes	Yes
Transients	Exploring	Yes	Yes
Rigid Ring	Exploring	Yes	No

*See Use CRG with Multibody Tire

Tire Models Integrated with Simscape Vehicle Templates

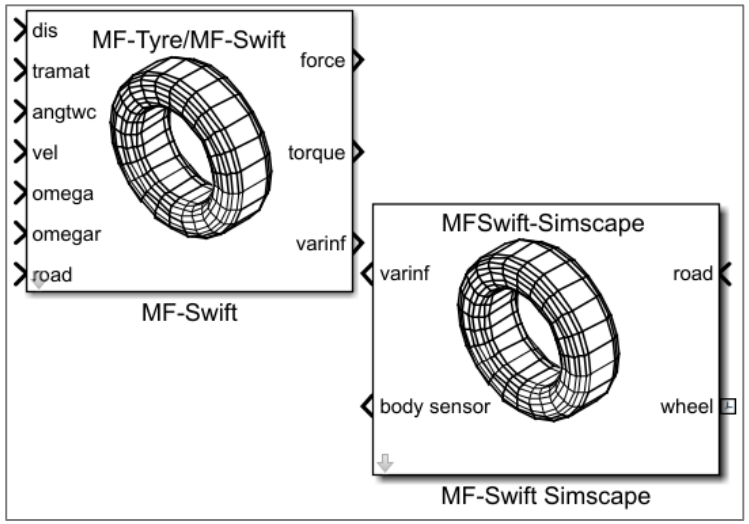
- Magic Formula Tire External Force and Torque
  - Part of Simscape Multibody
  - R2021b and higher



- MFEval
  - All releases

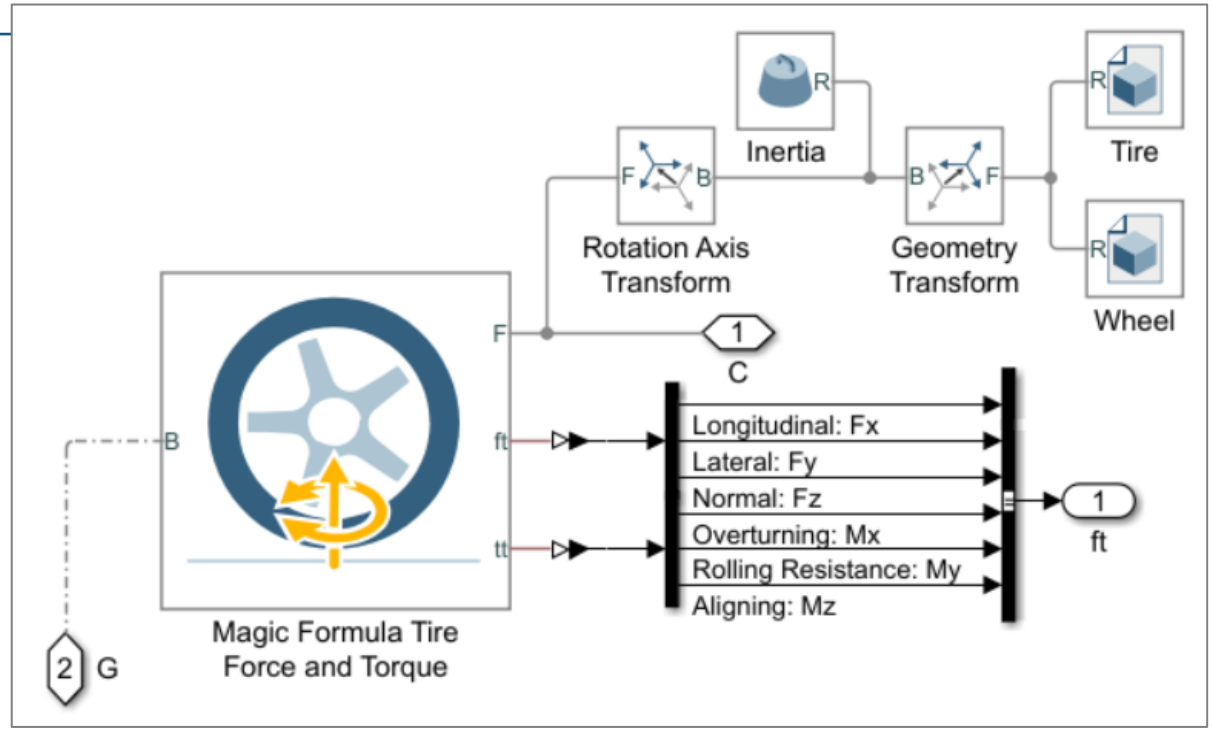


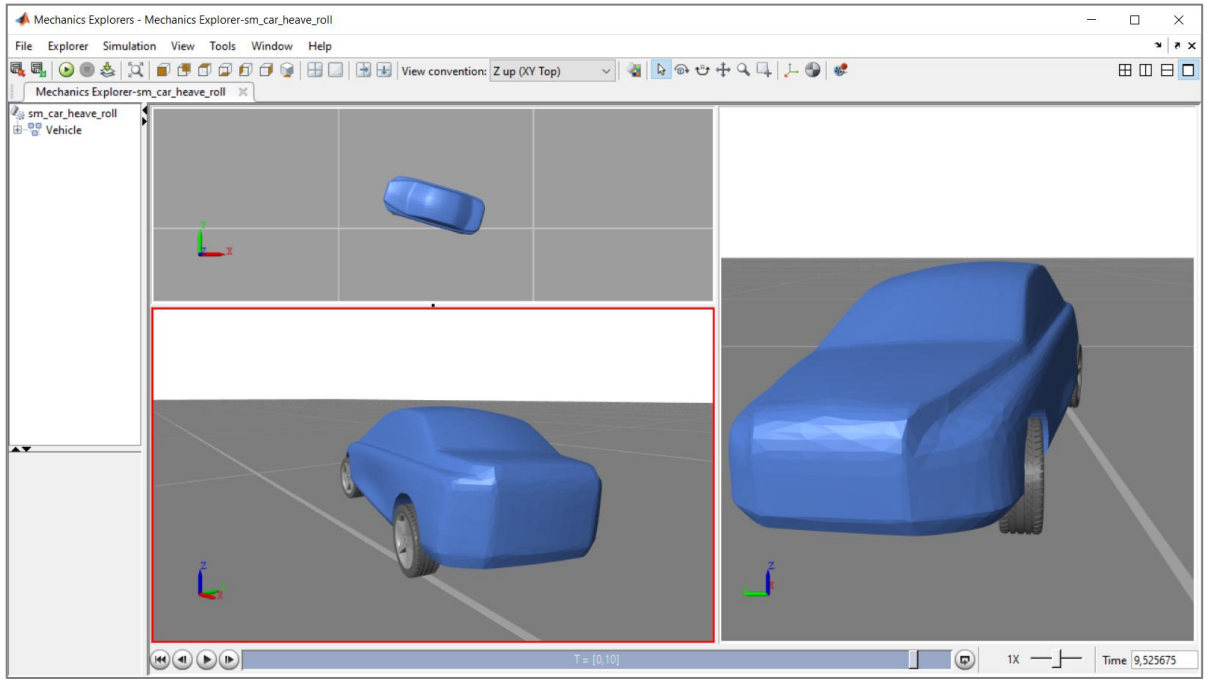
- MF-Swift/MF-Tyre from Siemens
  - All releases
  - Formerly known as Delft Tyre



Simscape Multibody
Magic Formula Tire

- Model the tire-road interaction force using the Magic Formula
  - Read tire parameters from a TIR file.
  - Sense contact force and torque between the tire and the road
  - Steady-state tire force and torque for the combined slip
- Multibody interface
  - Great numerical performance
  - Will enable detailed surface definition in future releases (flat surfaces in R2021b)



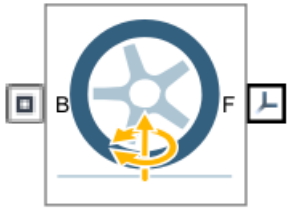



```
>> sm_car_heave_roll
```

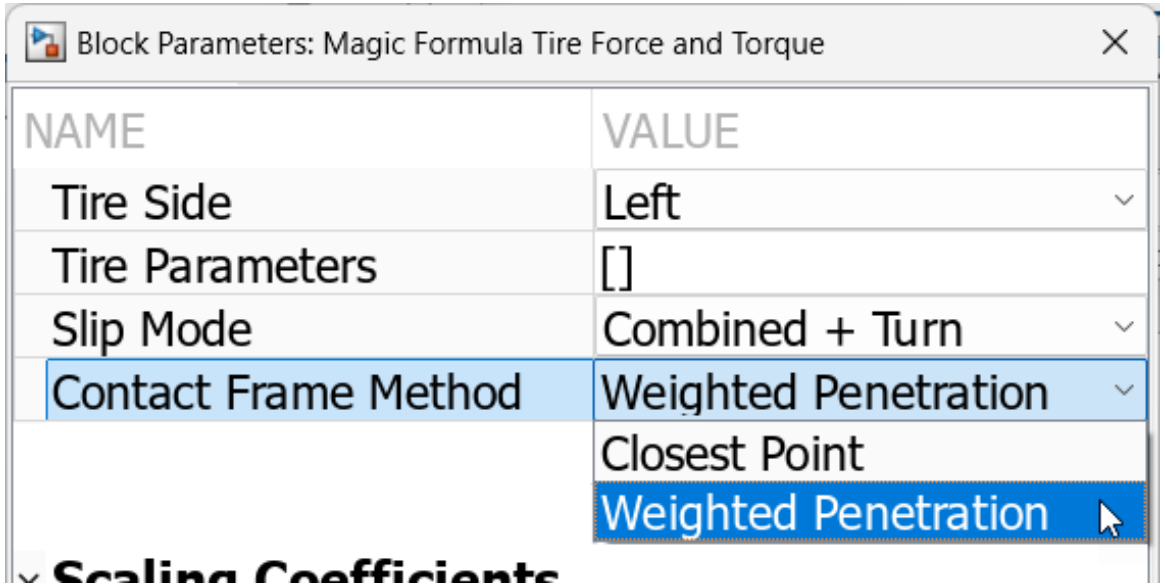
Simscape Multibody

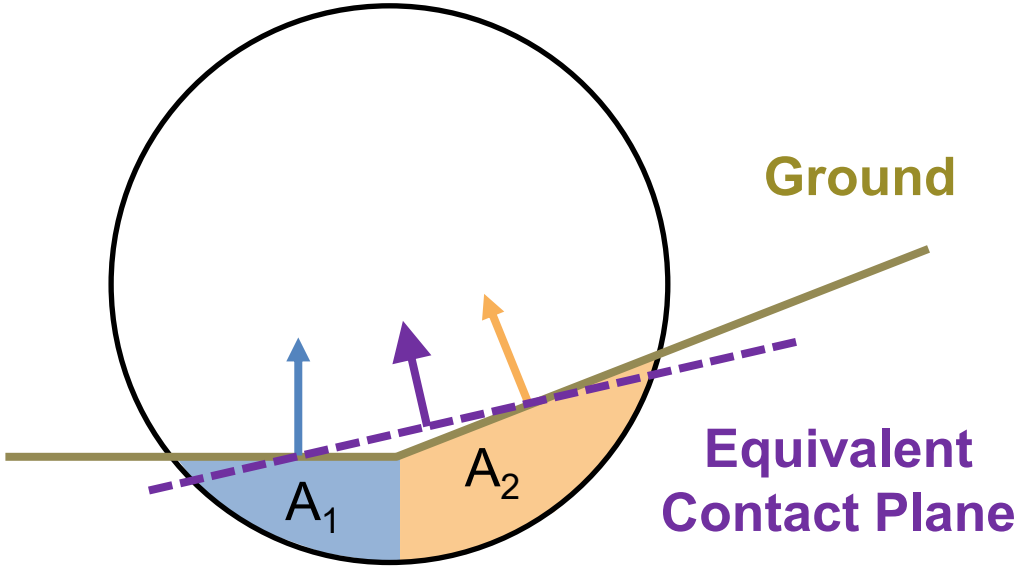
Tire Contact Weighted Penetration Method

- Approximate multi-point contact between tire and ground
  - Efficient modeling of rough road conditions
  - Compute tire radial deflection



Magic Formula Tire Force and Torque





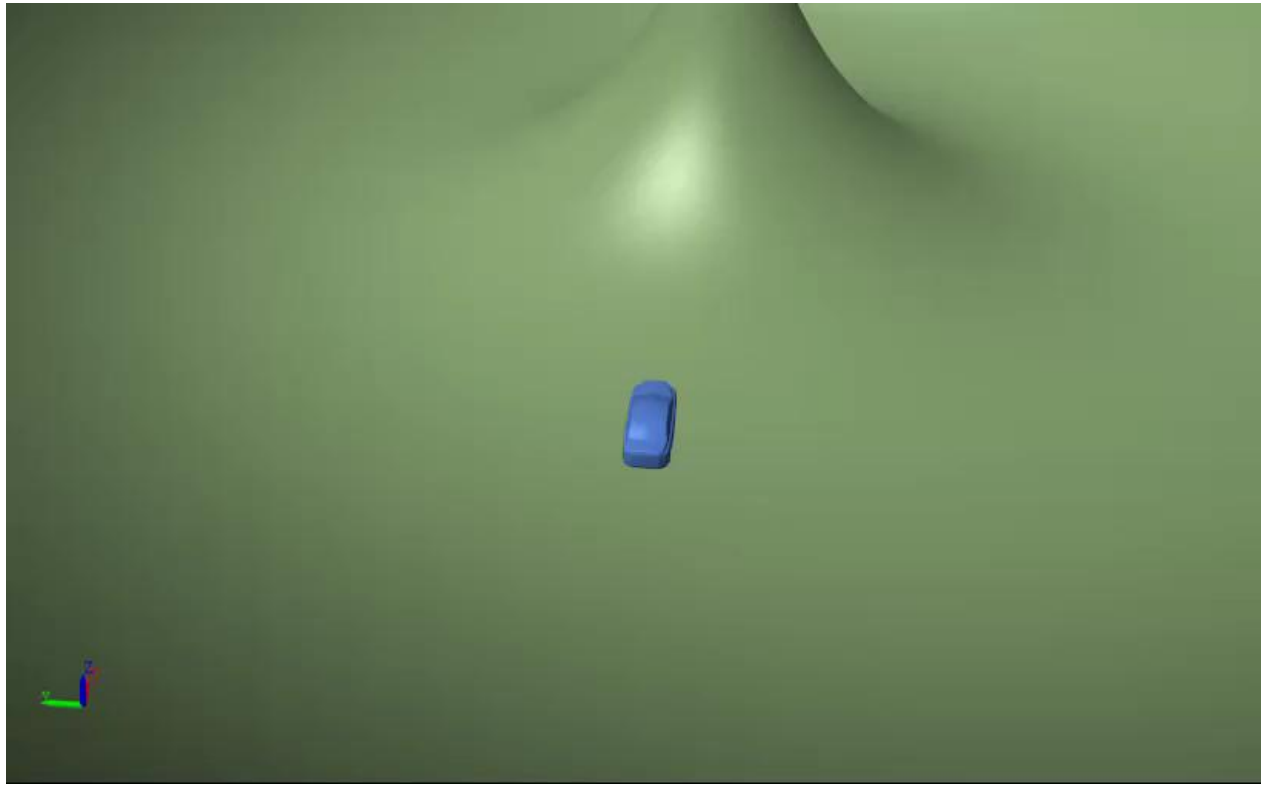
Simscape Multibody

Tires on Uneven Surface

- Model vehicle dynamics on uneven surfaces
  - Use the Grid Surface block with Magic Formula Tire Force and Torque block
- Model variations in surface friction
  - Vary friction scaling coefficients during simulation
- Sense location and orientation of contact frame

Stress Test of Simscape Multibody Tire Model

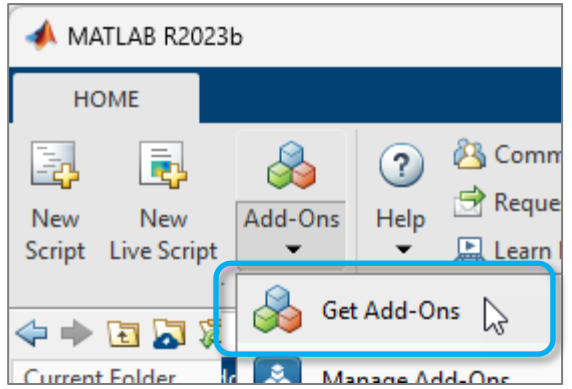
14 Degree-of-Freedom vehicle model
10x faster than real time

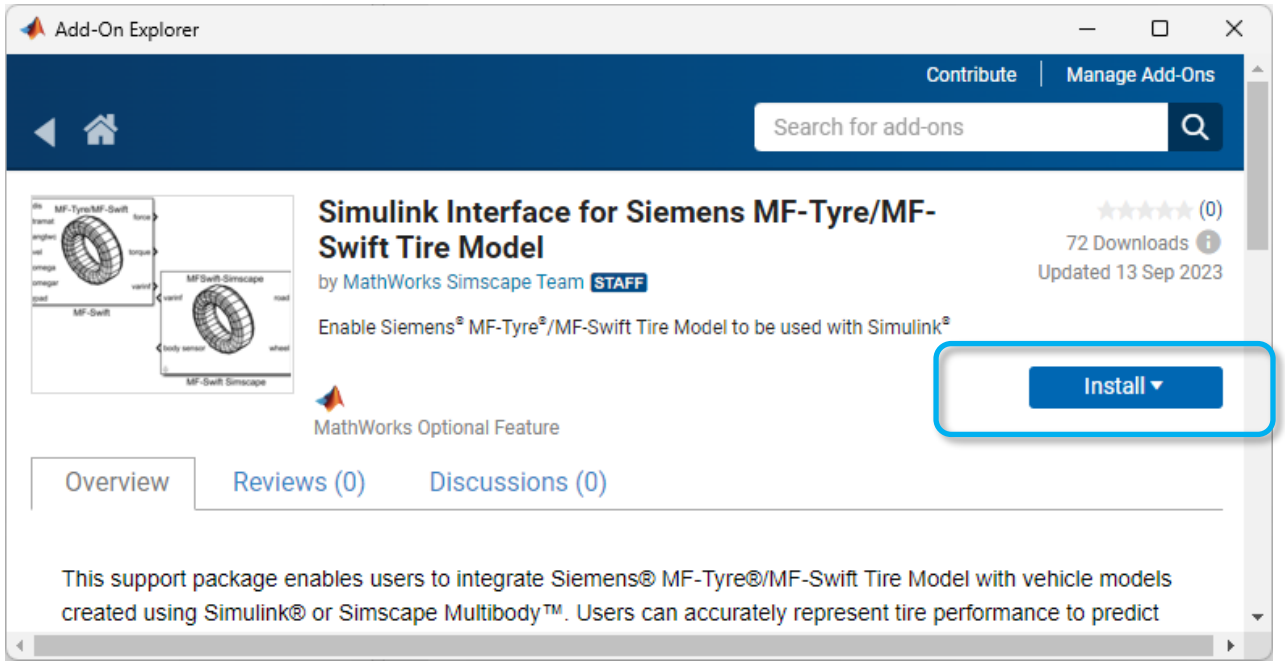


Simscape Multibody

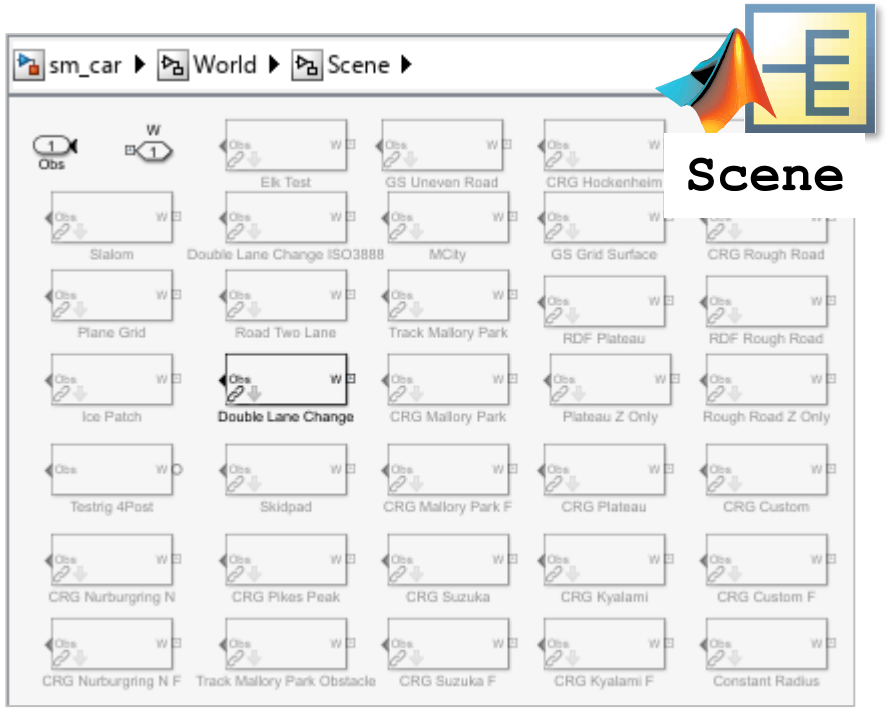
Simulink Interface for Siemens MF-Tyre/MF-Swift Tire Model

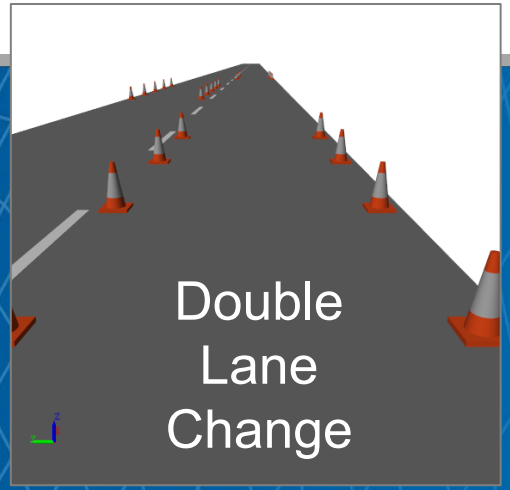
- Download MF-Tyre/MF-Swift Tire model directly from Add-on Explorer
  - Includes examples with Simscape Multibody, Simulink, and MATLAB
  - Available from R2021b forward

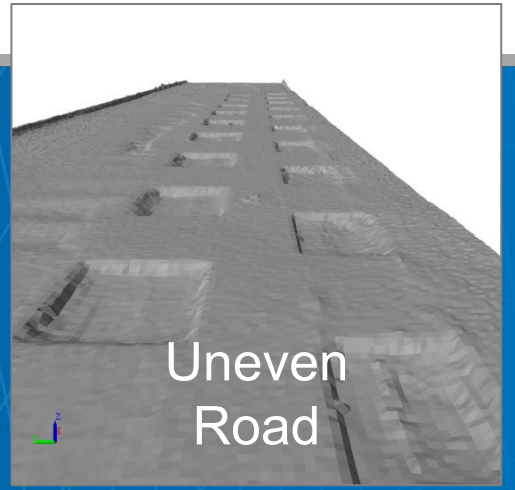


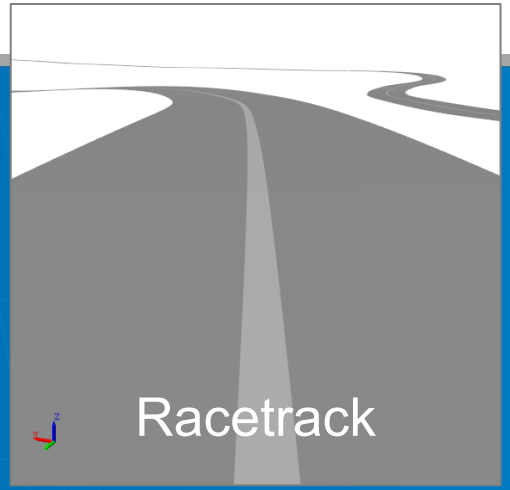


Simscape Vehicle Templates
Scene and Terrain





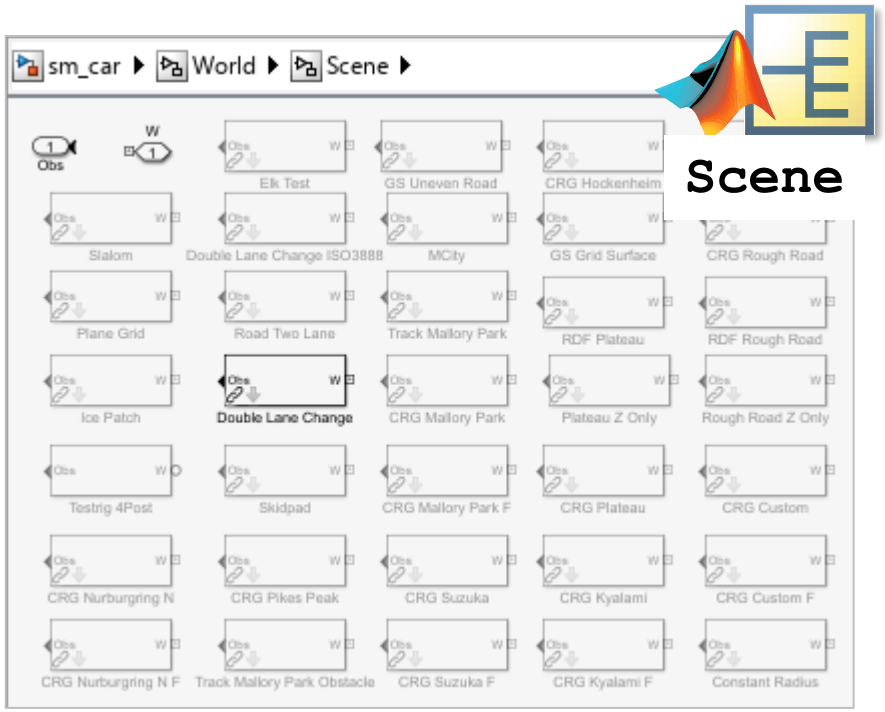




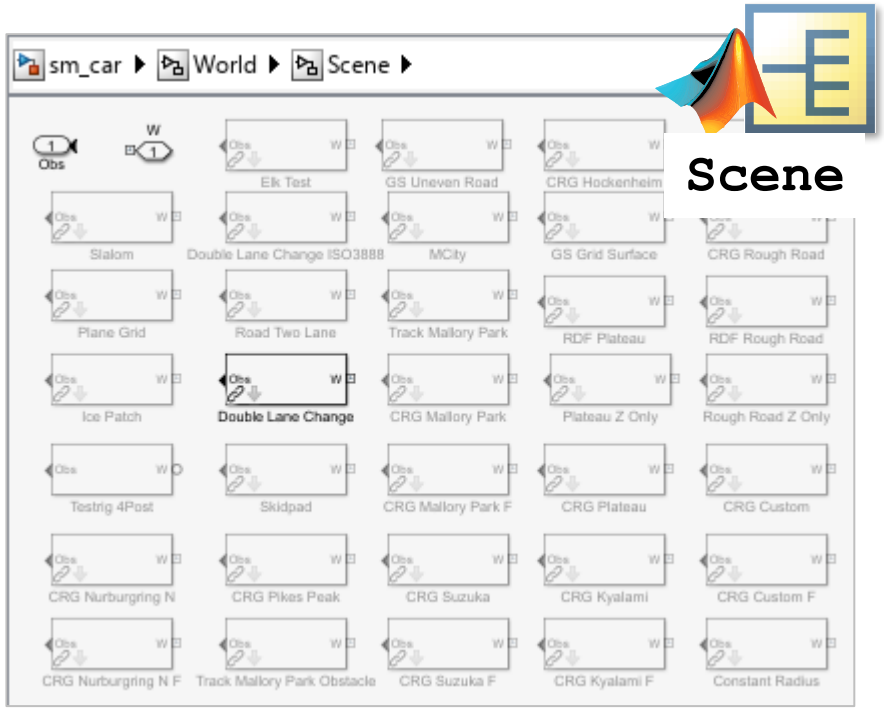
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Scene and Terrain

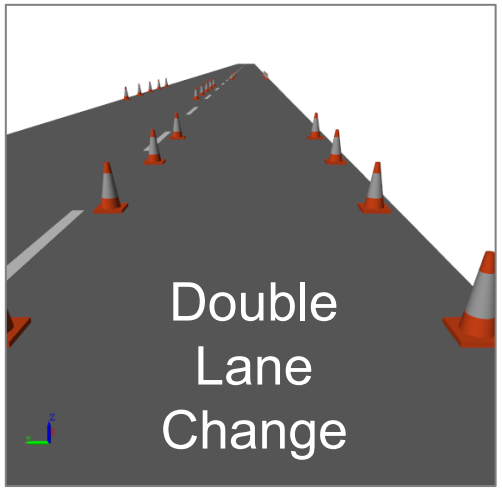
- Scene Definition
- Terrain: CRG
- Terrain: Grid Surface
- Terrain: Height Defined by Signal
- Terrain: Coefficient of Friction
- Scene Triggered Driver Overrides

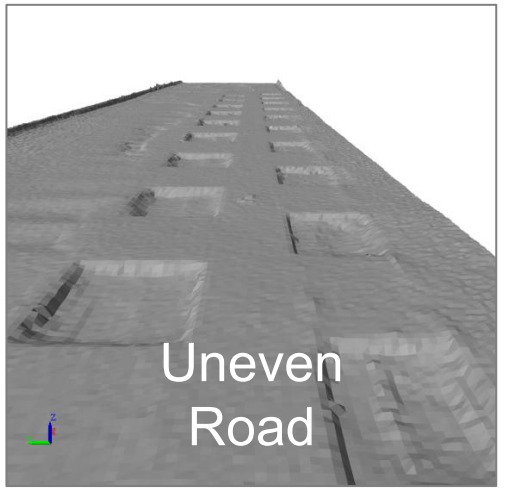


Scene Definition



- Provide visual context for maneuver
  - Road, lines, cones
  - Bumps, inclines
  - Parameterized using MATLAB
- Road surface for tire not defined here
  - Change in tire model affects how surface must be defined
- Elements in Scene can influence maneuver
  - Stoplights can trigger vehicle to pause



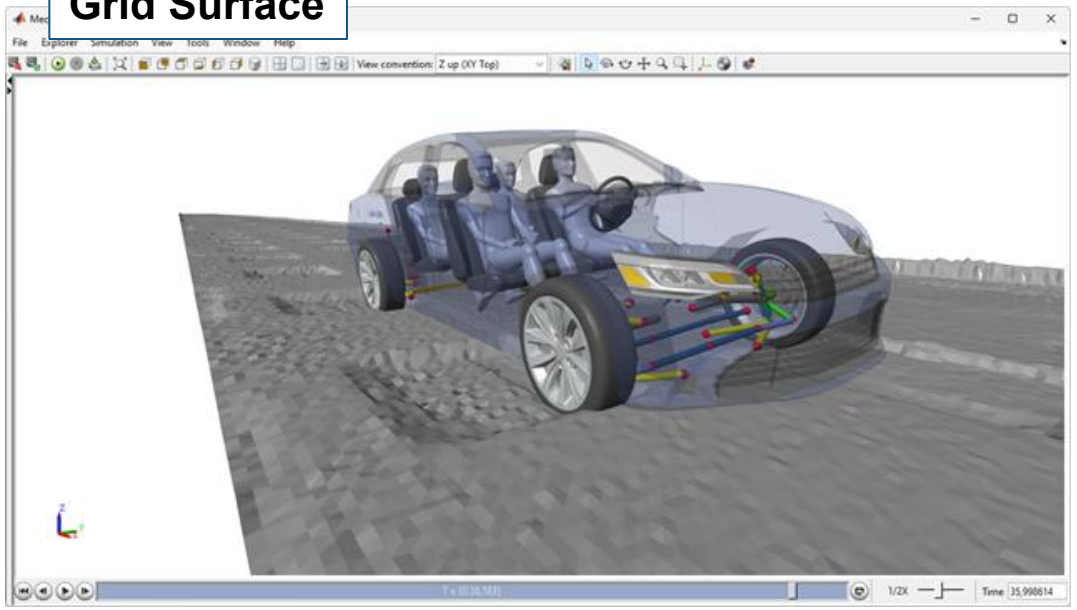




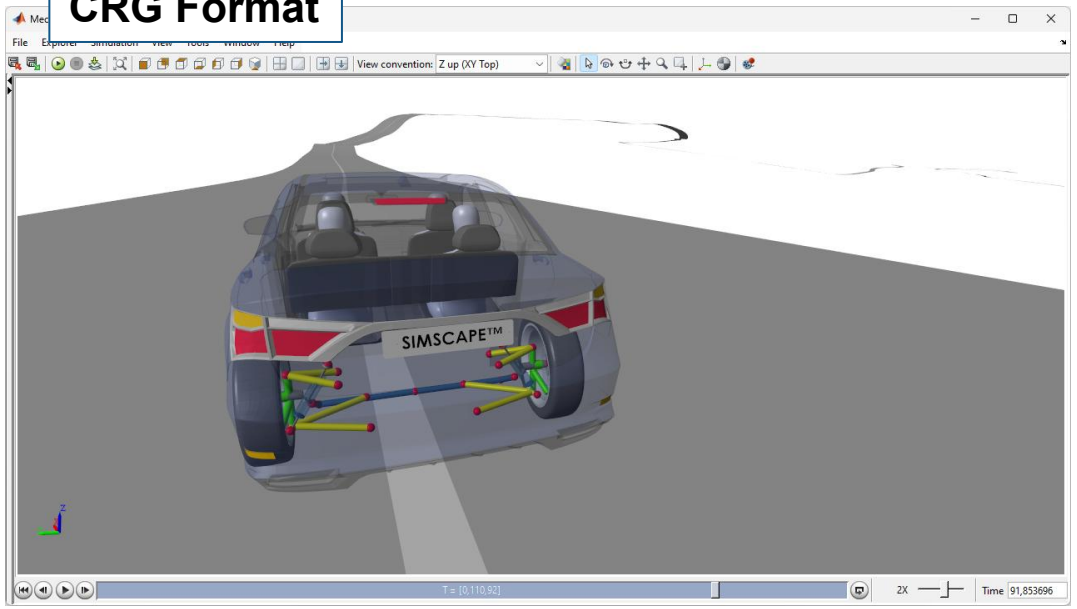
Terrain: Road Surface

- Road surfaces can be specified using
  - Grid Surface (x-y grid and heights)
  - Curved Regular Grid (CRG) format
  - Height input as signal
- Simscape Vehicle Templates uses tire models that support
  - Flat road surfaces
  - Varying coefficient of friction
  - Uneven road surfaces
- Additional tire models can be integrated
  - Look at subsystems for tires
  - Copy/paste subsystem, integrate tire with same interface

Grid Surface



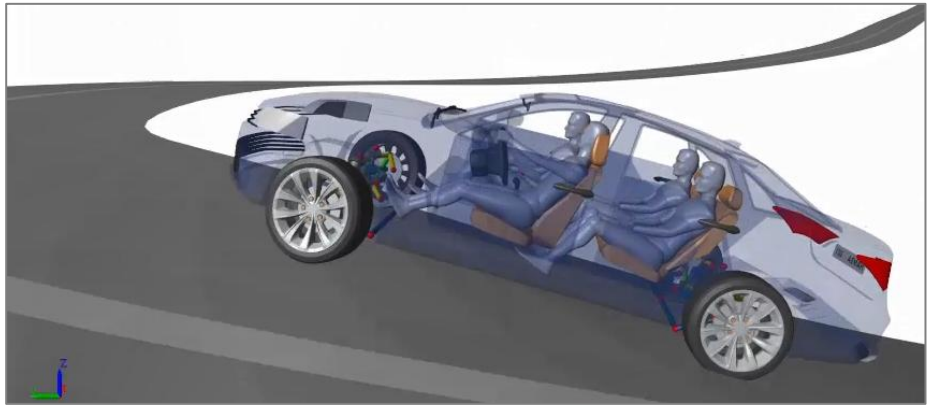
CRG Format

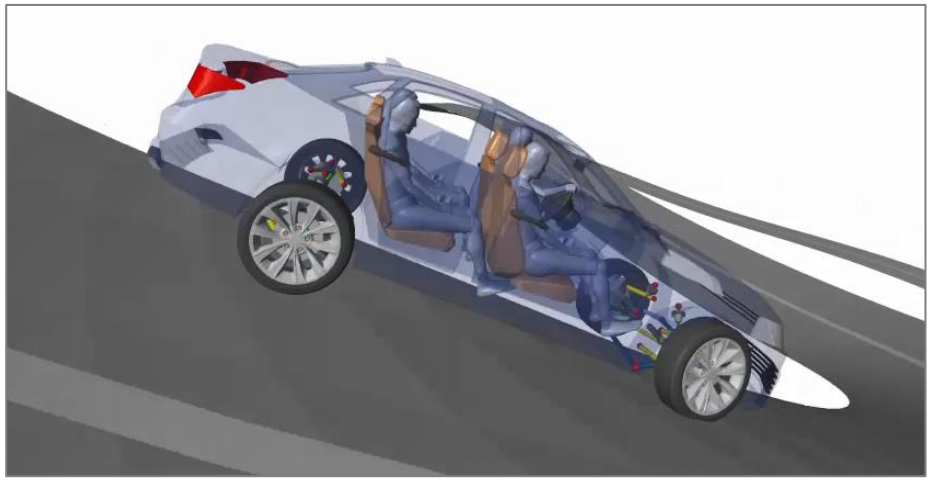


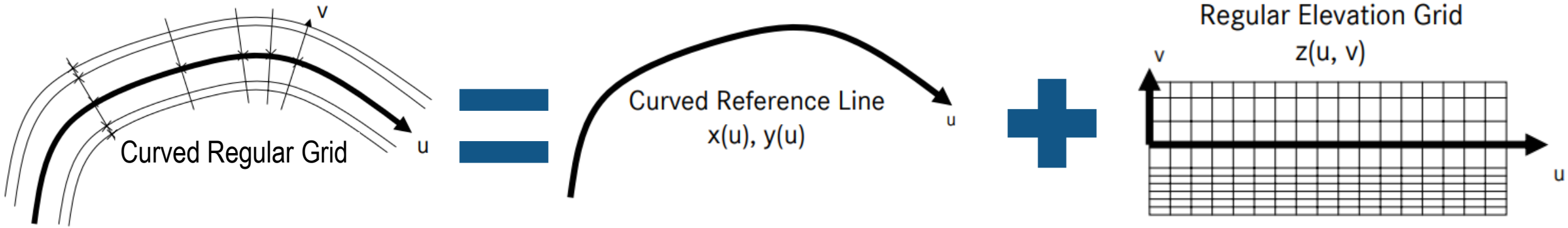
Terrain: CRG

Curved Regular Grid

- Frequent request from customers: 3D driving surface
  - Commercial vehicles (construction, agriculture, forestry)
  - Passenger vehicles
- MF-Swift and Delft-Tyre both support roads defined using CRG format





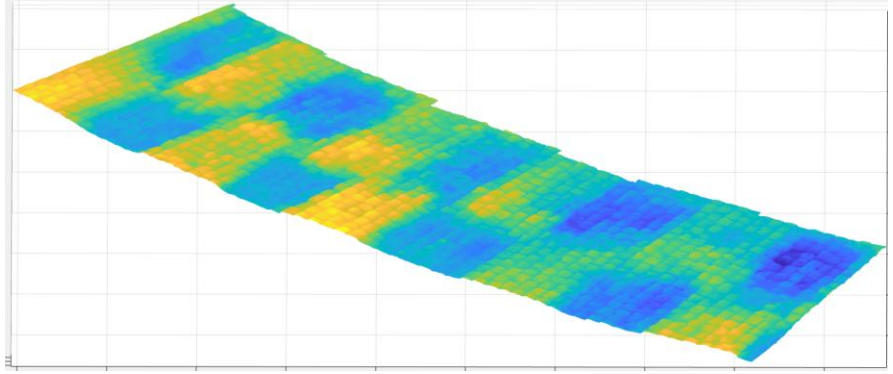


Terrain: CRG

Curved Regular Grid

- Frequent request from customers: 3D driving surface
  - Commercial vehicles (construction, agriculture, forestry)
  - Passenger vehicles
- MF-Swift and Delft-Tyre both support roads defined using CRG format
- Several famous roads are now part of Simscape Vehicle Templates
  - Includes examples of how to generate CRG file and driver profile based on road centerline

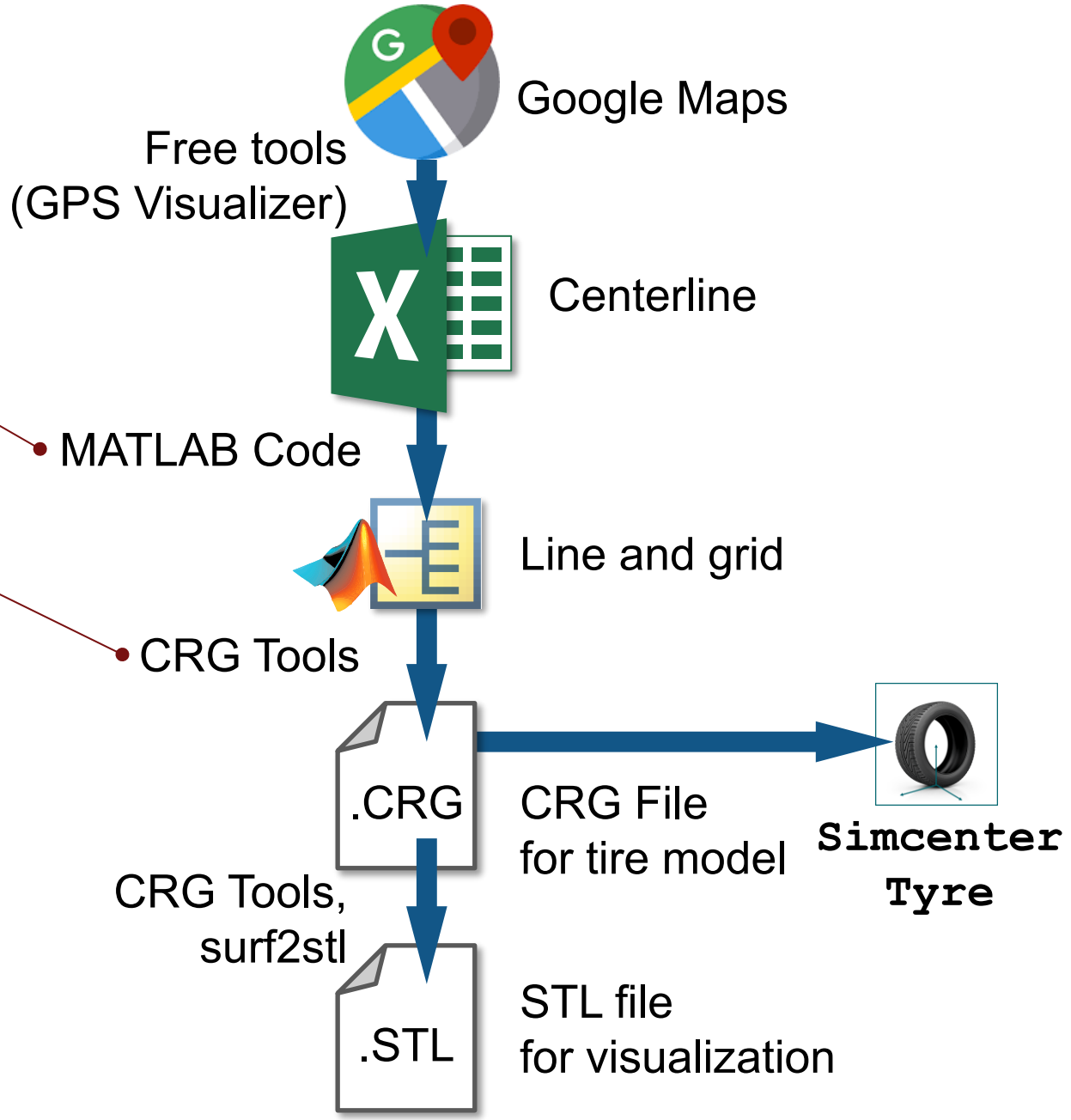
CRG Road XYZ map





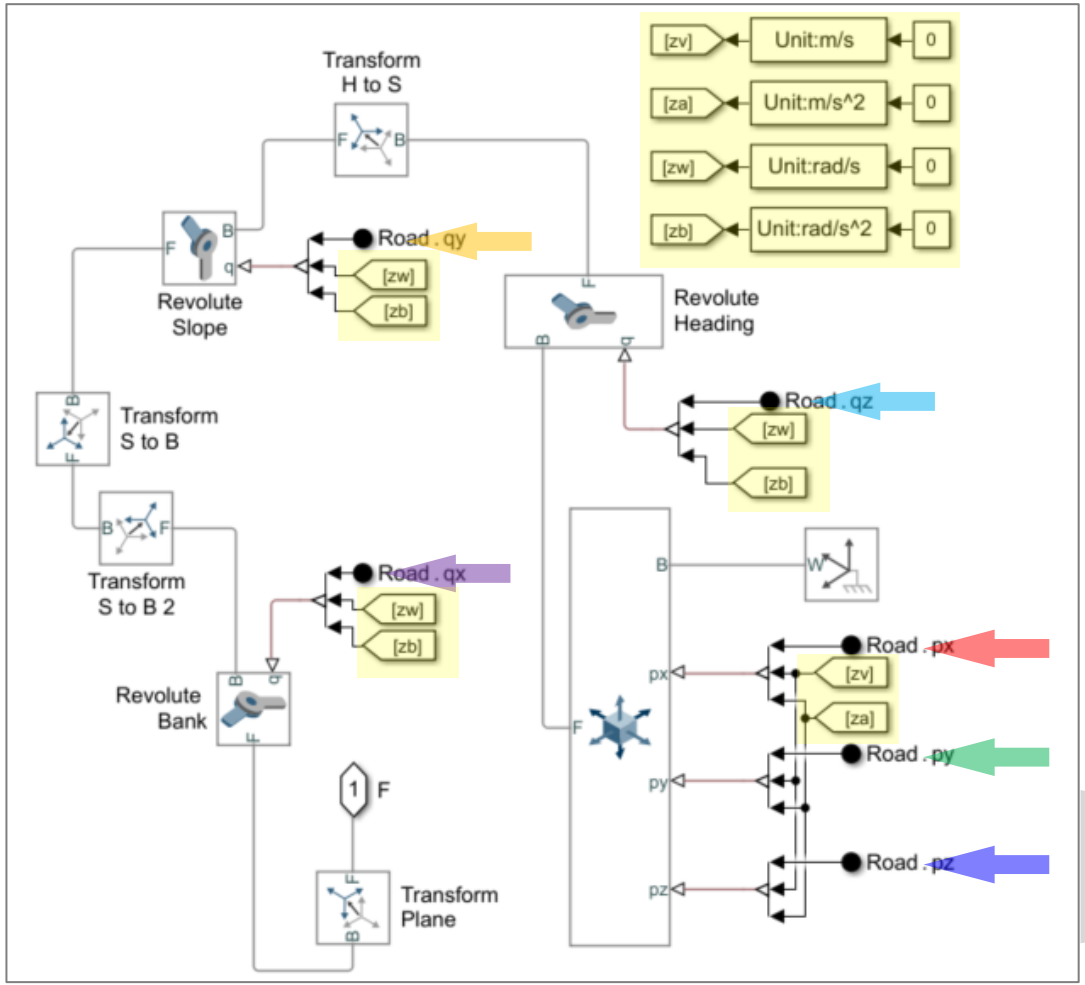
Terrain: CRG

- Current process converts GPS data to CRG and STL
  - Smoothing and interpolating necessary
  - For racetracks, tricky to get road start and finish to line up
- Also use CRG file to create trajectory for driver
  - Input: road curvature
  - Output: target speed and vehicle yaw vs. distance along path

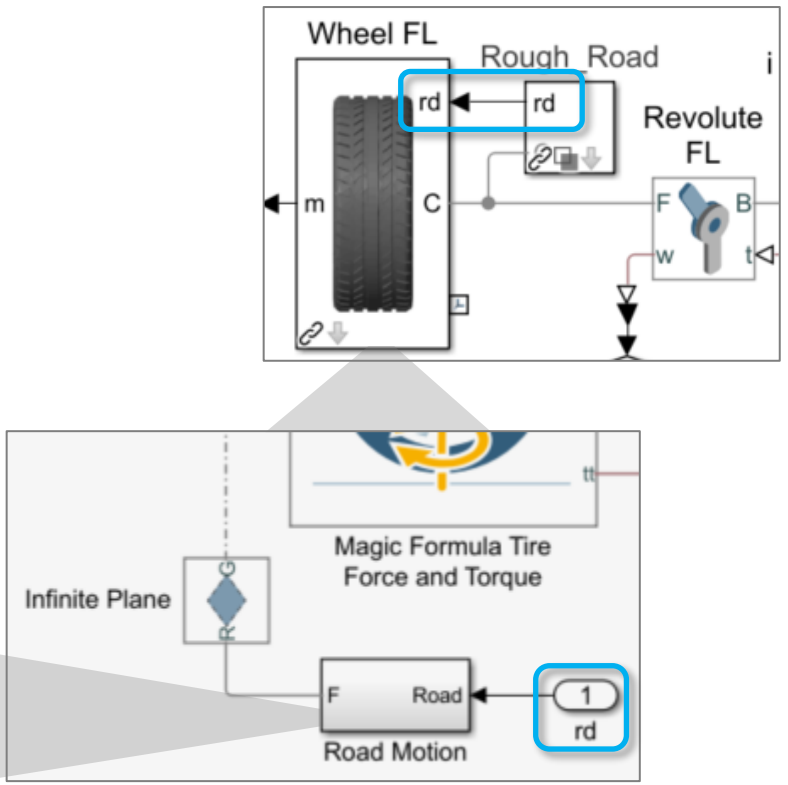


Use CRG with Multibody Tire

- Set x-y-z position, slope, bank, and heading angles of Infinite Plane, speed and accel = 0

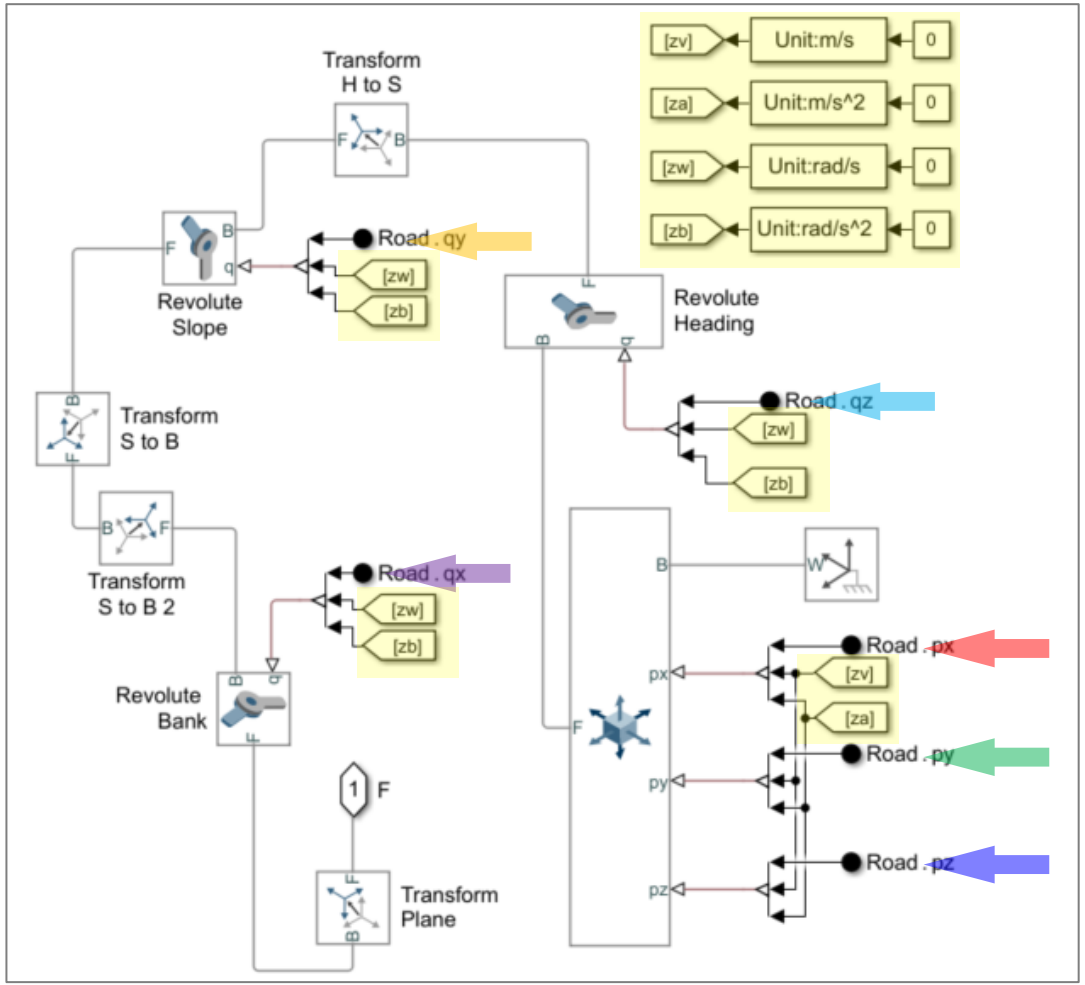


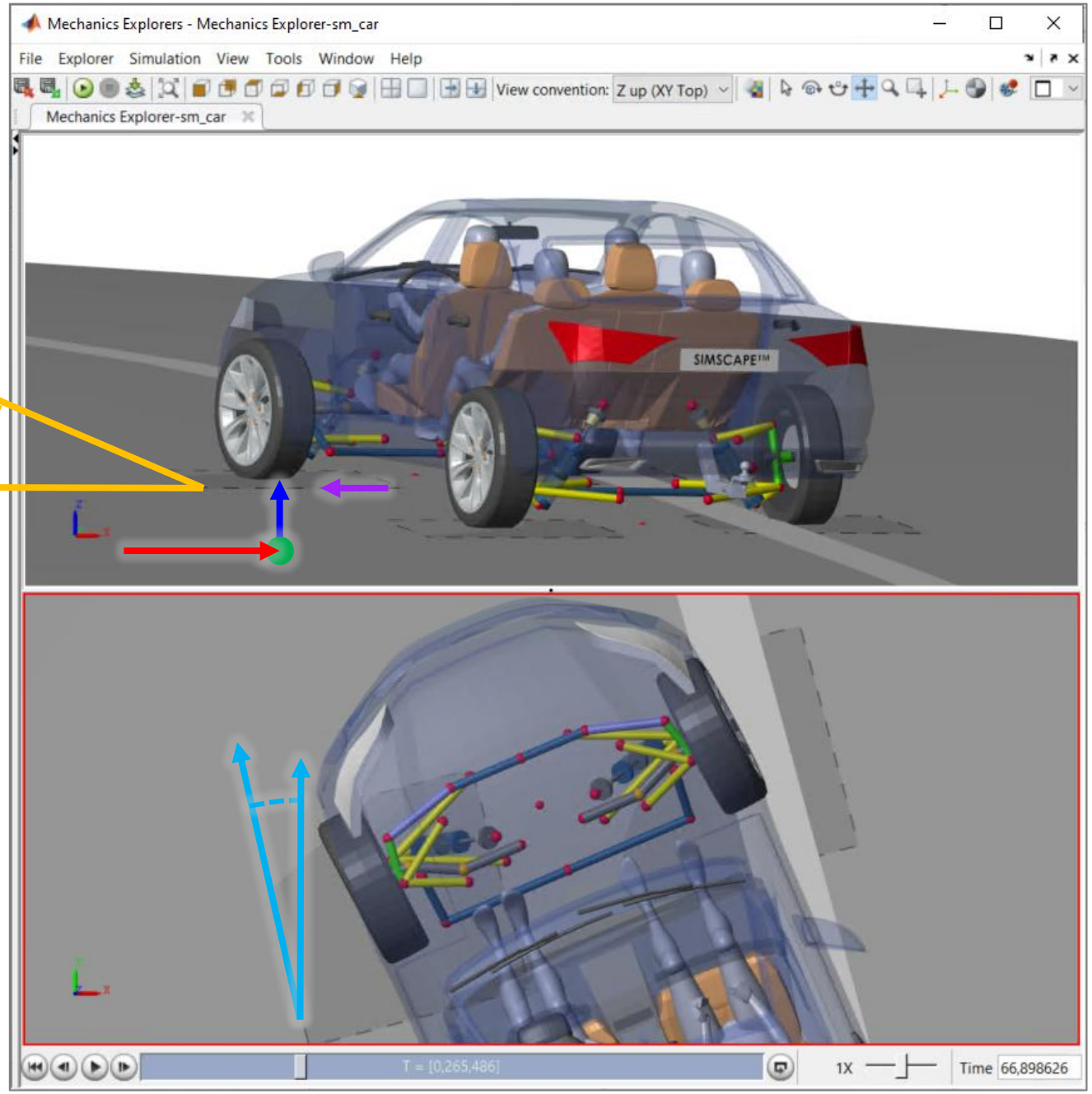
- Determine positions and angles based on position of wheel center
  - Search lookup table or query **c**urved-**r**egular **g**rid (CRG)



Use CRG with Multibody Tire

- Set x-y-z position, slope, bank, and heading angles of Infinite Plane; speed and accel = 0

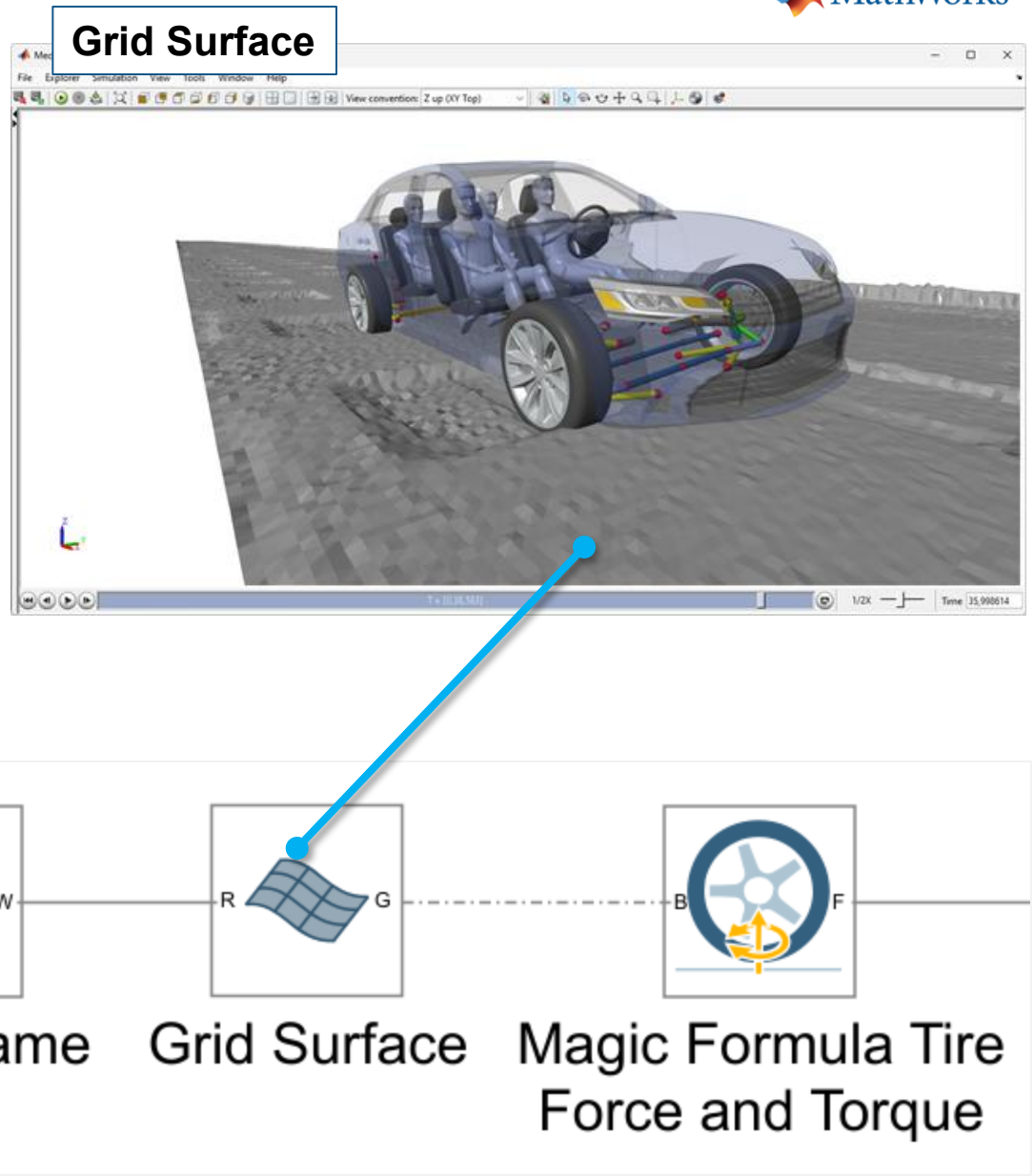




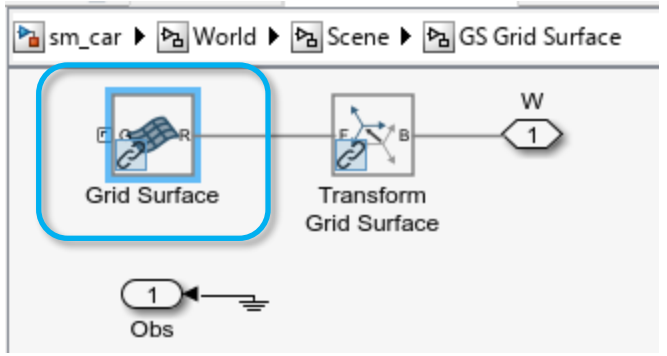
Terrain: Grid Surface

- Road can be defined as a grid
  - X vector, Y vector, Z matrix of heights
  - X and Y can be unevenly spaced
  - Define and refine in MATLAB

- Connect directly to Magic Formula Tire block
  - Tire accounts for intersection with multiple portions of grid surface
  - Weighted penetration is approximation for multi-point contact



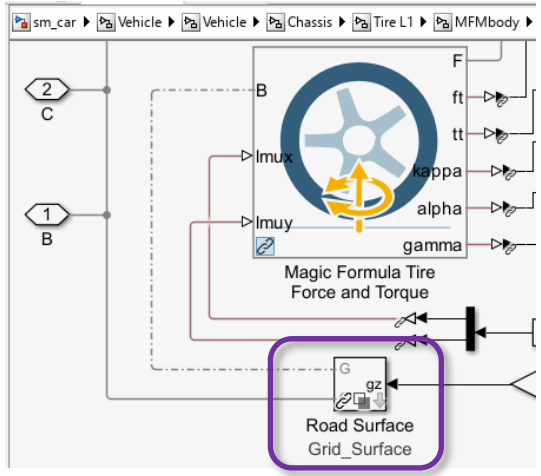
Terrain: Grid Surface



- Selecting Scene GS Grid Surface will use the Grid Surface block to visualize the surface

- Each Magic Formula Tire will be connected to a Grid Surface block if roadFile.Value field contains the string '.stl'
  - STL file does not need to exist

- These blocks all use the parameterization in Scene.GS_Grid_Surface
  - Only the Grid Surface in "Scene" is visualized
  - Copy your own data into Scene.GS_Grid_Surface for custom surfaces

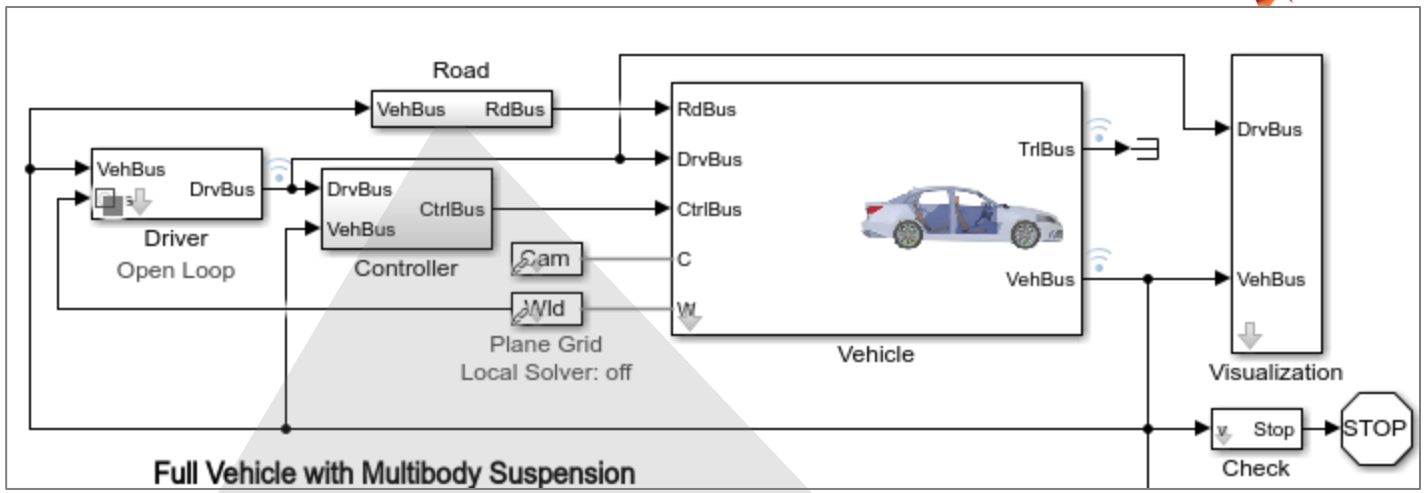


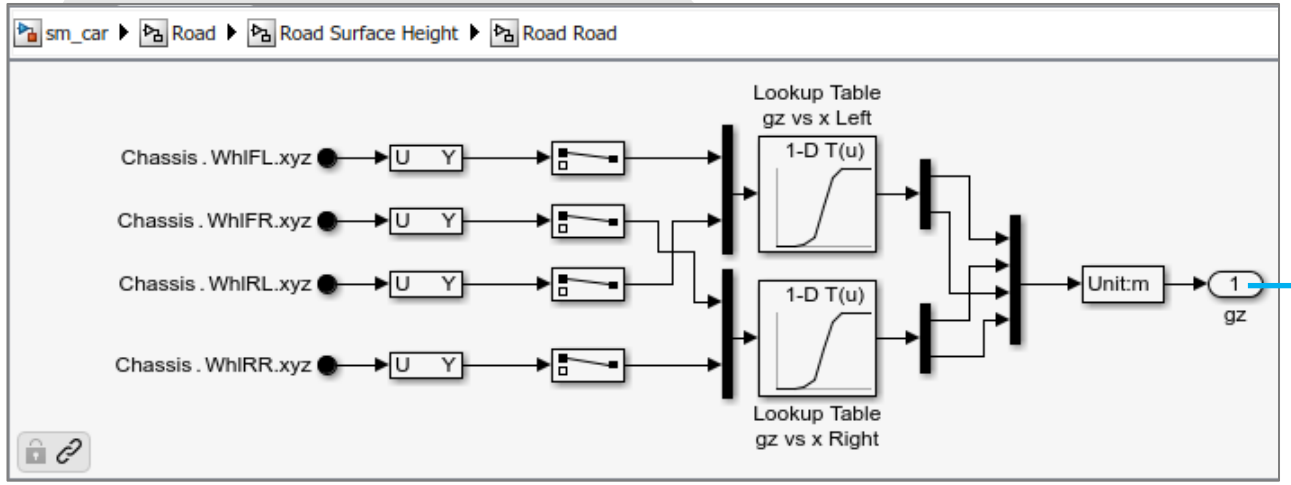
Once per wheel

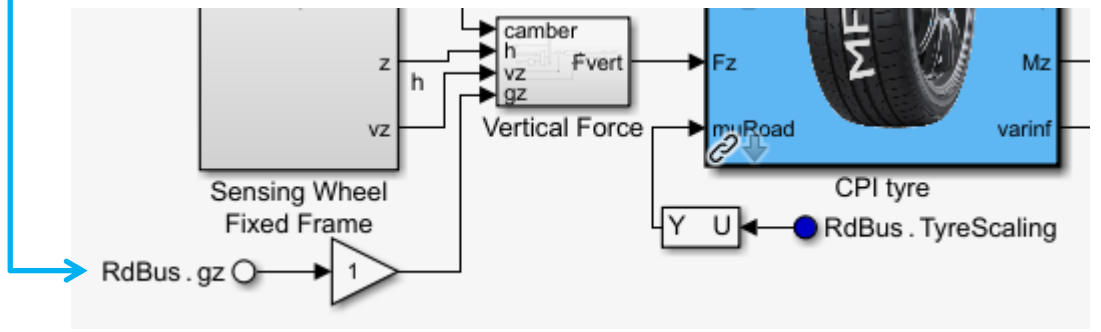
```
>> Vehicle.Chassis.TireA1.roadFile.Value
ans =
    'GS_Grid_Surface.stl'
```

Road Height From Signal MFeval, Multibody tire

- Vary road height with wheel center position
  - Inject bumps into suspension
  - Does model slope of road
- Separate variants for test case
  - Same lookup table to all wheels
    - Car can pitch but not roll
  - Separate tables for left and right side
    - Car can pitch and roll



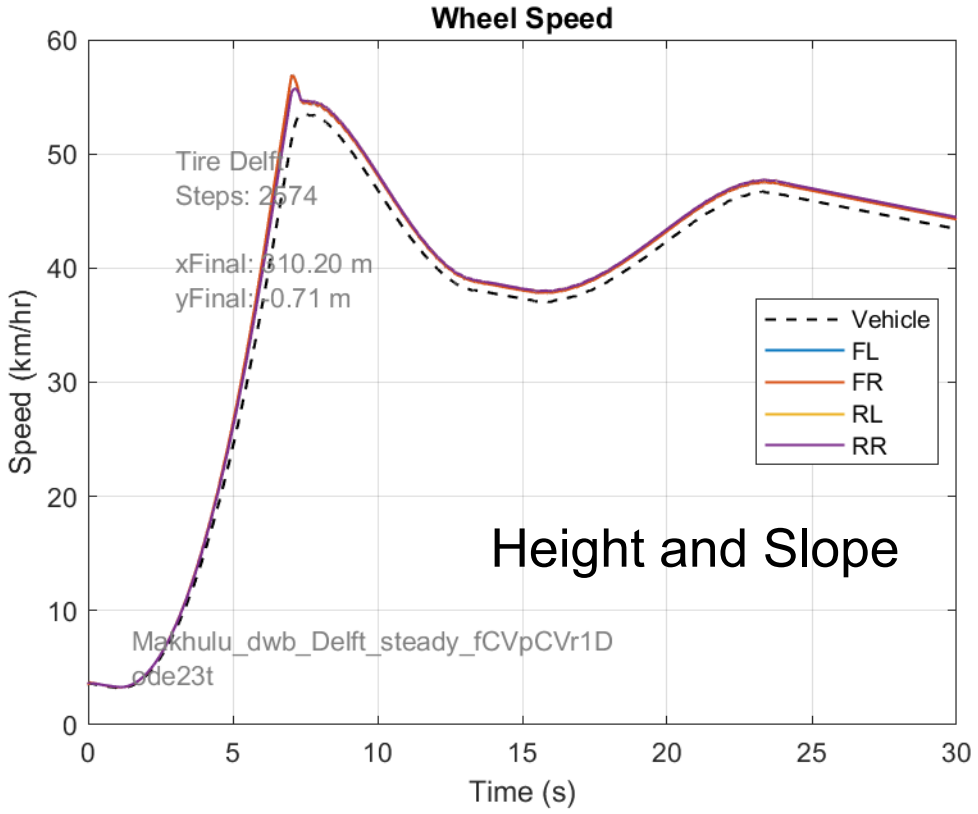


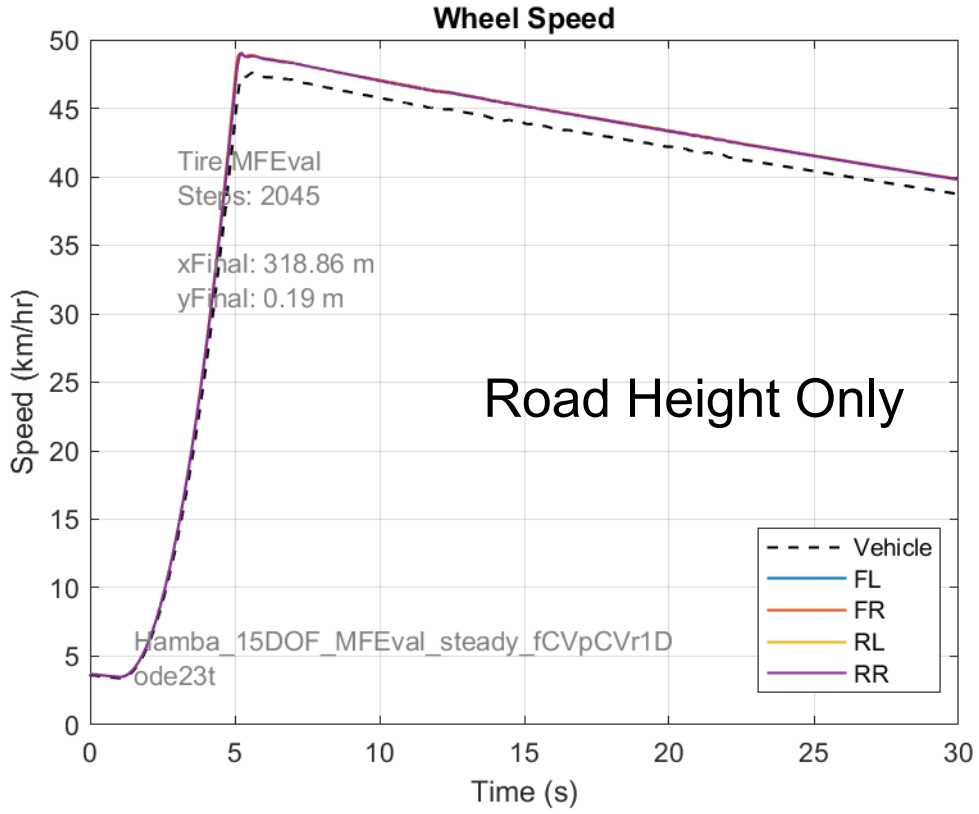


Effect of Ignoring Road Slope

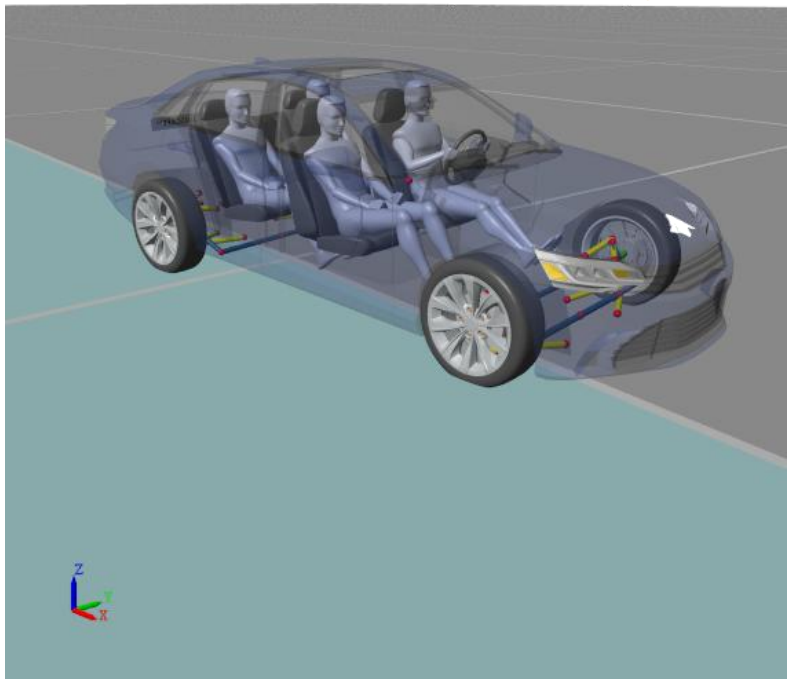
- If only the tire model only knows road height, it takes no additional effort to climb a hill
  - CRG format specifies slope and banking





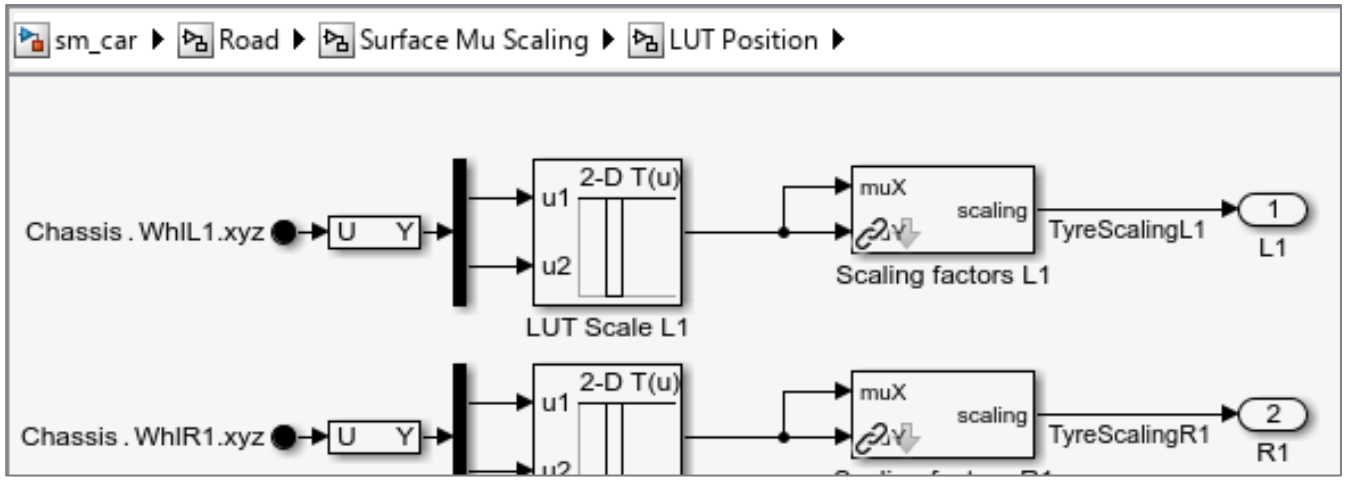


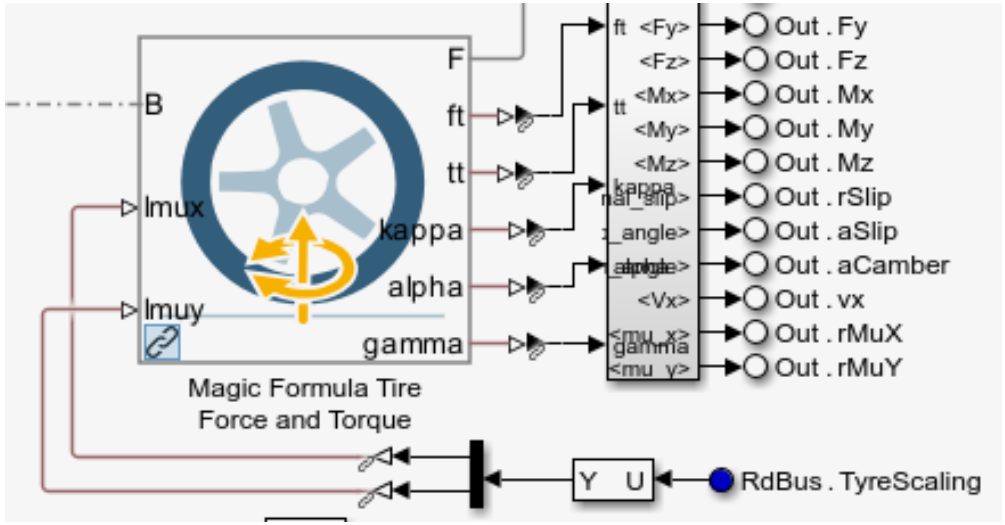
Terrain: Scale Coefficient of Friction



- Set scaling factor for friction coefficient based on wheel center x-y location
  - 2D lookup table (x, y) yields scaling factor
  - Pass to tire as an input during simulation
- Supported by Simscape Magic Formula Tire, MF-Swift, and MFeval

```
sm_car_config_maneuver('sm_car','Ice Patch');
```



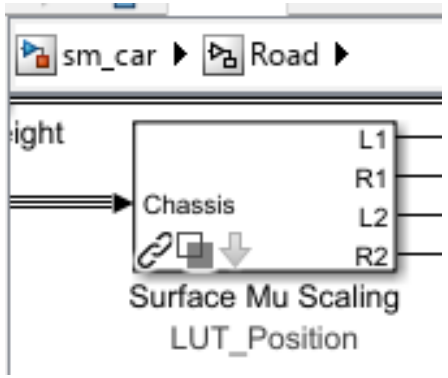


Road Surface Friction

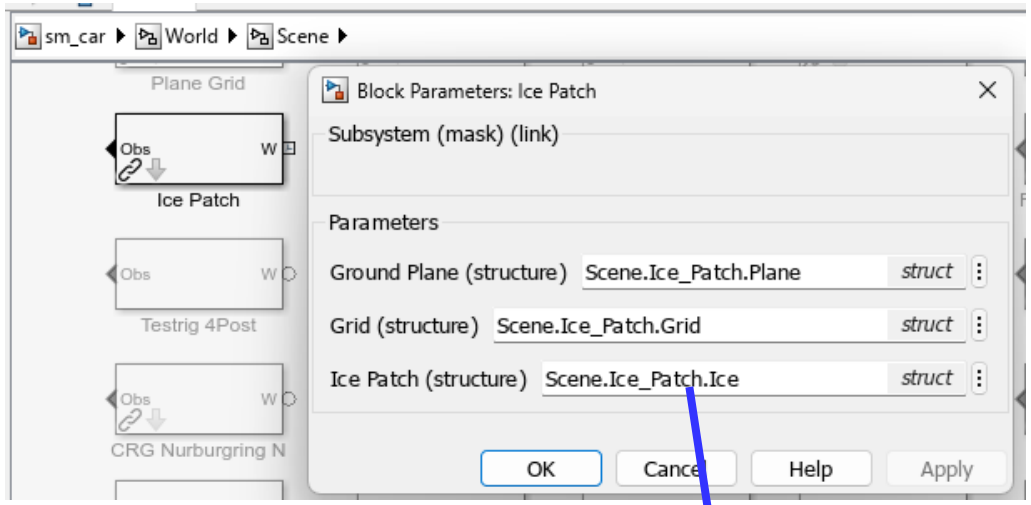
- Scaling coefficients for longitudinal and lateral friction can passed as signals to some tire models

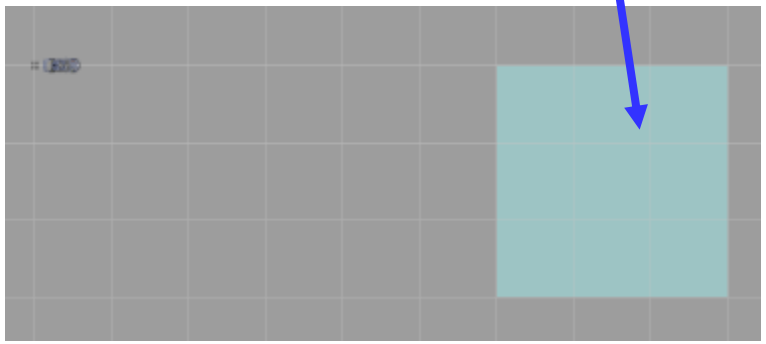
- Scaling parameters can be
  - Constant for entire surface
  - Defined using a 2-D lookup table (x position, y position)

- Ensure your scene has consistent parameters for any surface visualization



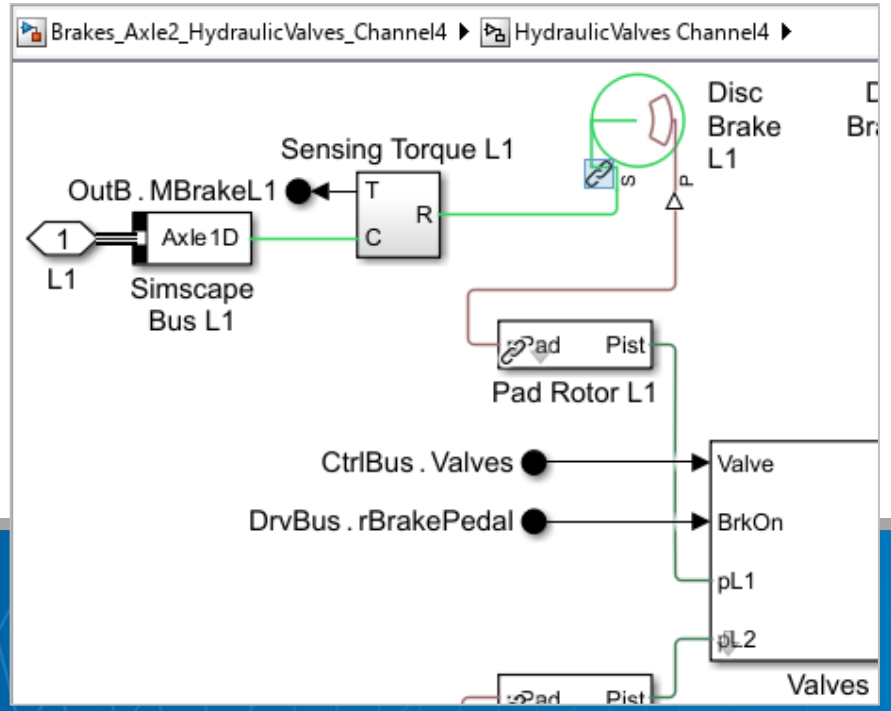

```
>> Scene.Mu_Scaling
ans =
  struct with fields:
          x: [0    59.9   60   90 90.1 200]
          y: [-60 -30.1 -30 -0.0100 0 60]
      scale: [6×6 double]
```





```
sm_car_config_maneuver('sm_car','Ice Patch');
```

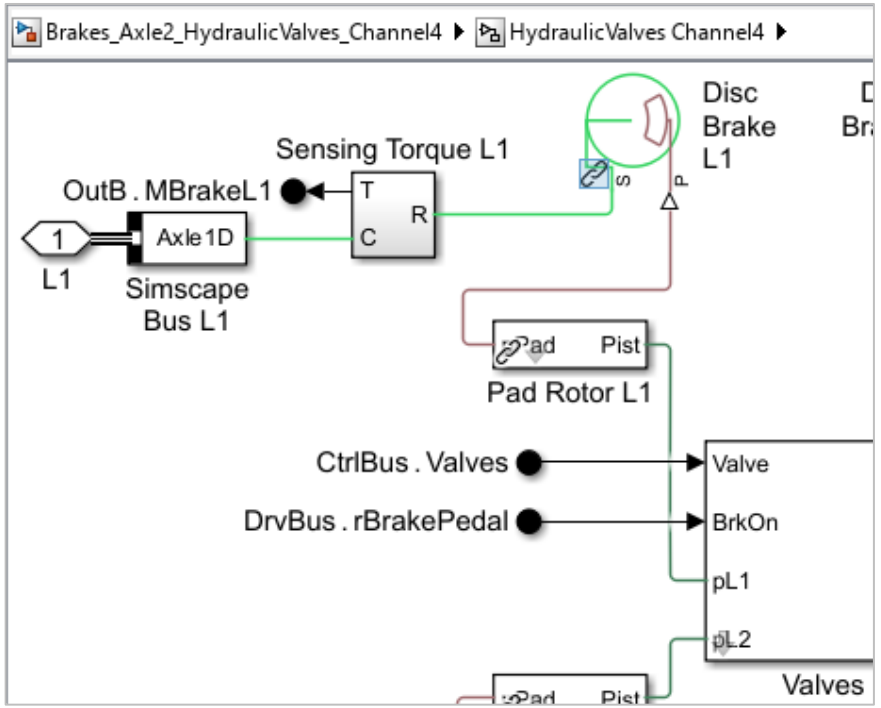
Simscape Vehicle Templates
Brakes



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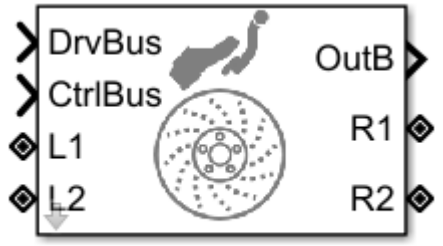
Brakes

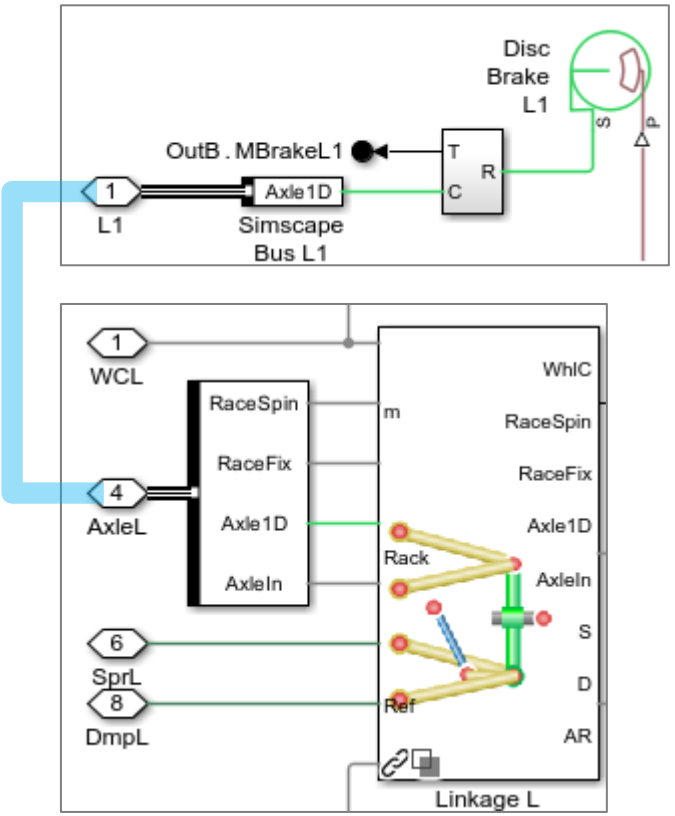
- Interface
- Abstract



Brakes Interface

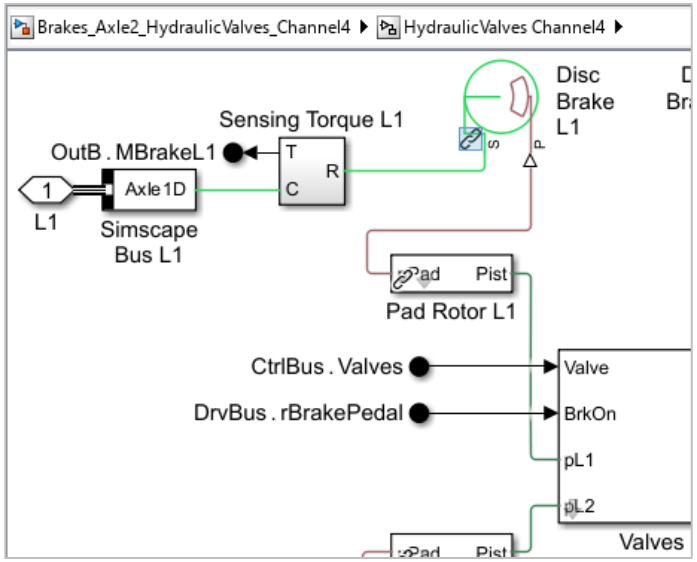
- Brake models connect to the same interface as the driveline, tire, and upright
  - Simscape bus with 1D and 3D connections
- Mechanical 1D model is sufficient for applying braking torque to shaft
  - 1D model is numerically very efficient
  - Enables connections to other domains
- Mechanical 3D connection is needed for reaction torque on upright
  - Not yet implemented

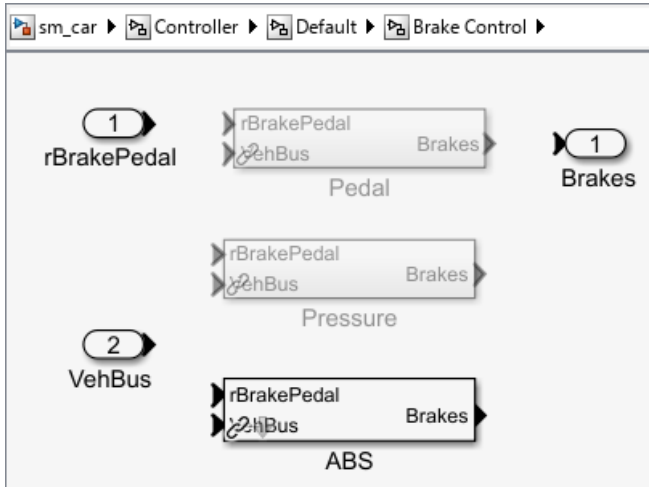




Brake Variants

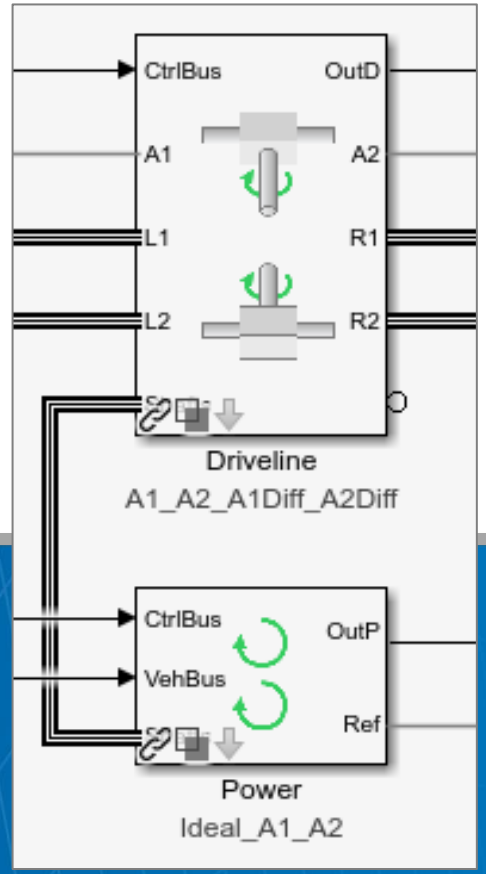
- Variants enable different methods of controlling brake pressure
  - Behavioral, physics-based
- Requires compatible controller
  - Pressure, valve control





#	Variant	Control Input	Pressure	Mechanical
1	Pedal Abstract Disc Disc	Pedal Ratio	Driver Action	Disc Brakes
2	Pressure Abstract Disc Disc	Pressure	Control System	Disc Brakes
3	Hydraulic Valves Channel4	Valve Commands	Hydraulic System	Disc Brakes
4	None	n/a	n/a	n/a

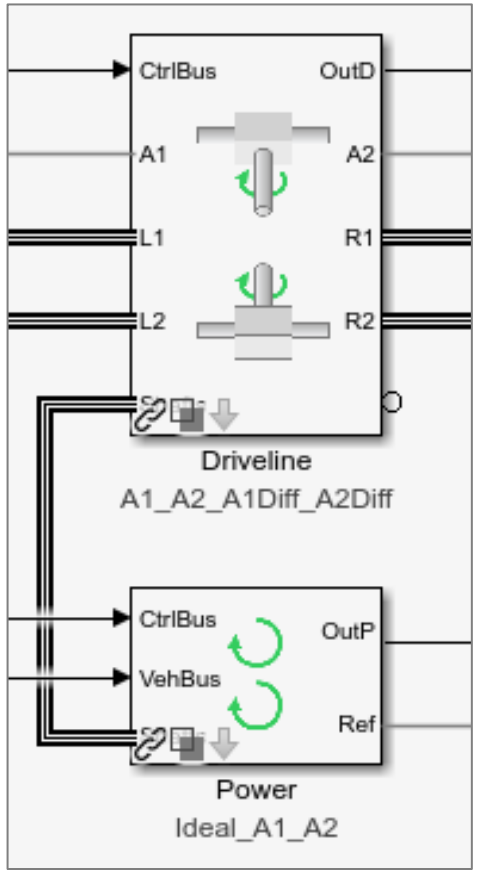
Simscape Vehicle Templates
Powertrain



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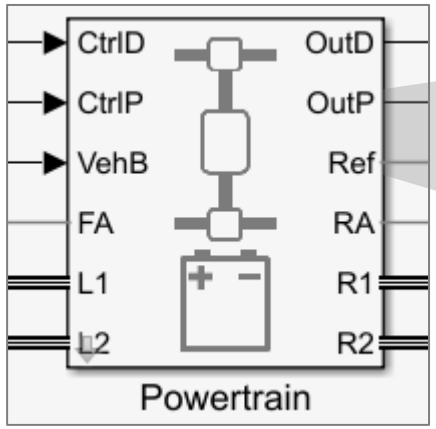
Powertrain

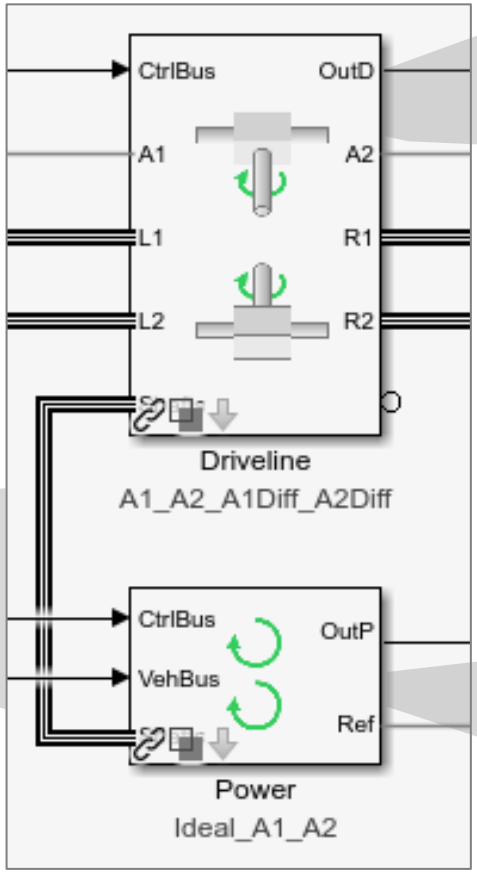
- Interface
- Power
- Driveline

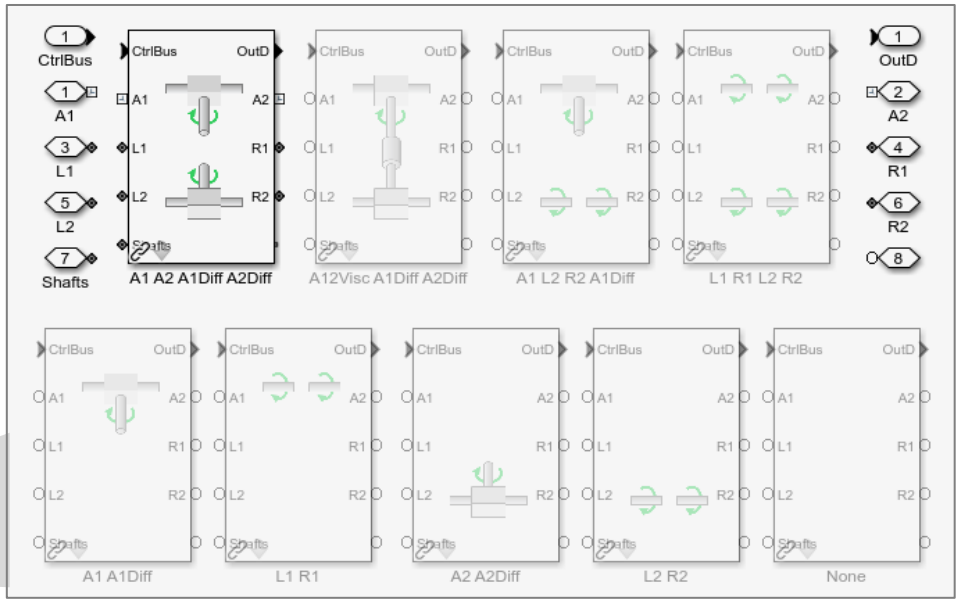


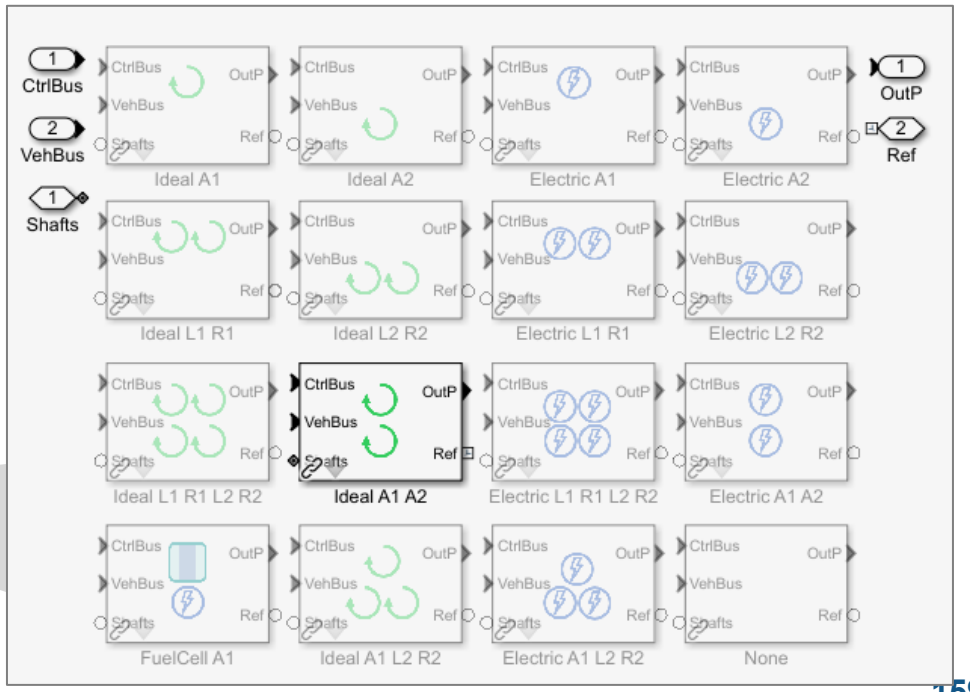
Powertrain Combines Power Sources and Mechanical Drivetrain

- Connect 1, 2, 3, or 4 driven shafts to differentials or directly to wheels
  - Ideal motors
  - Electric motors with torque limits
  - Optional cooling system



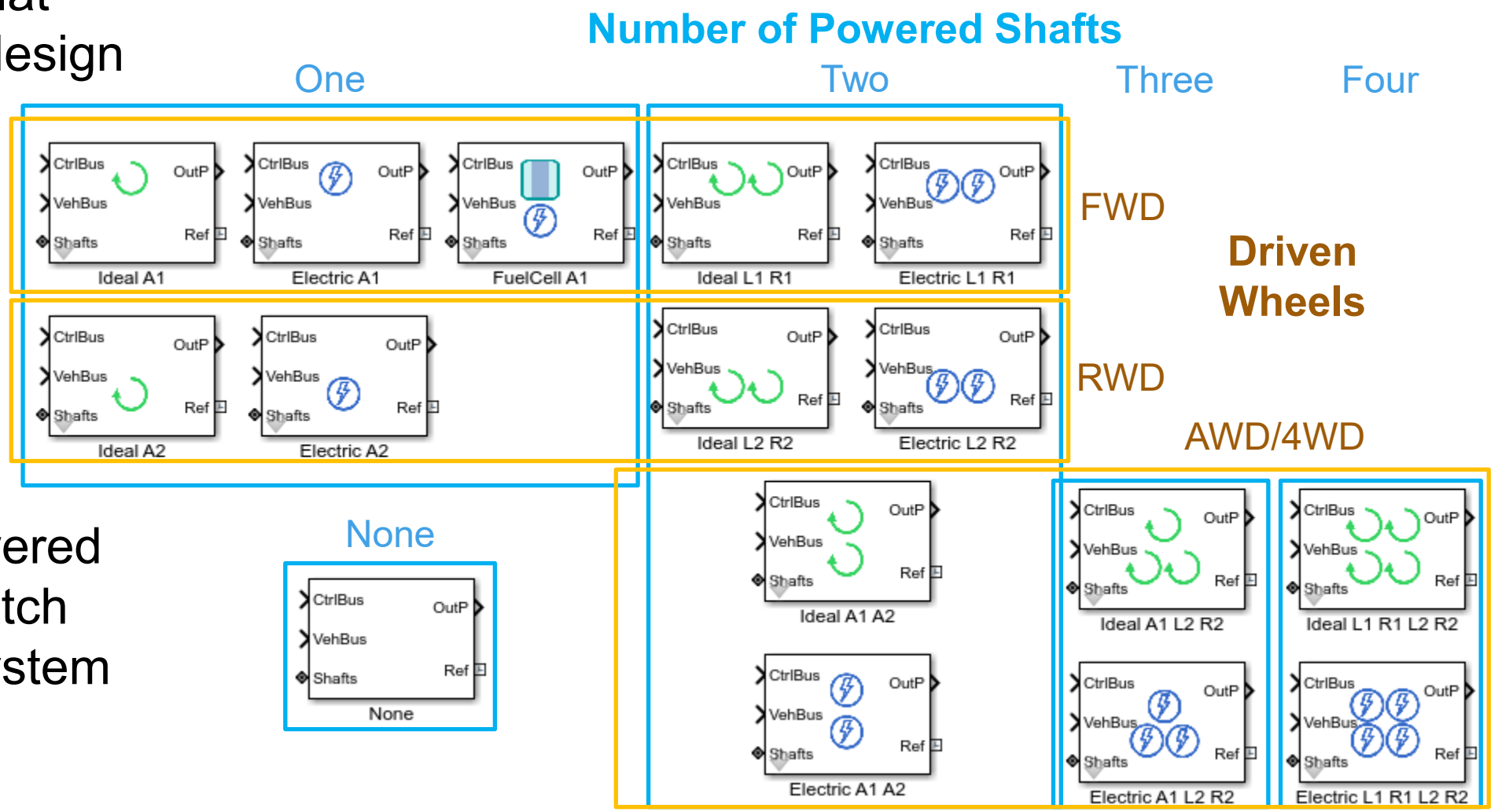






Power Supports AWD/4WD, FWD, RWD, and No Driven Wheels

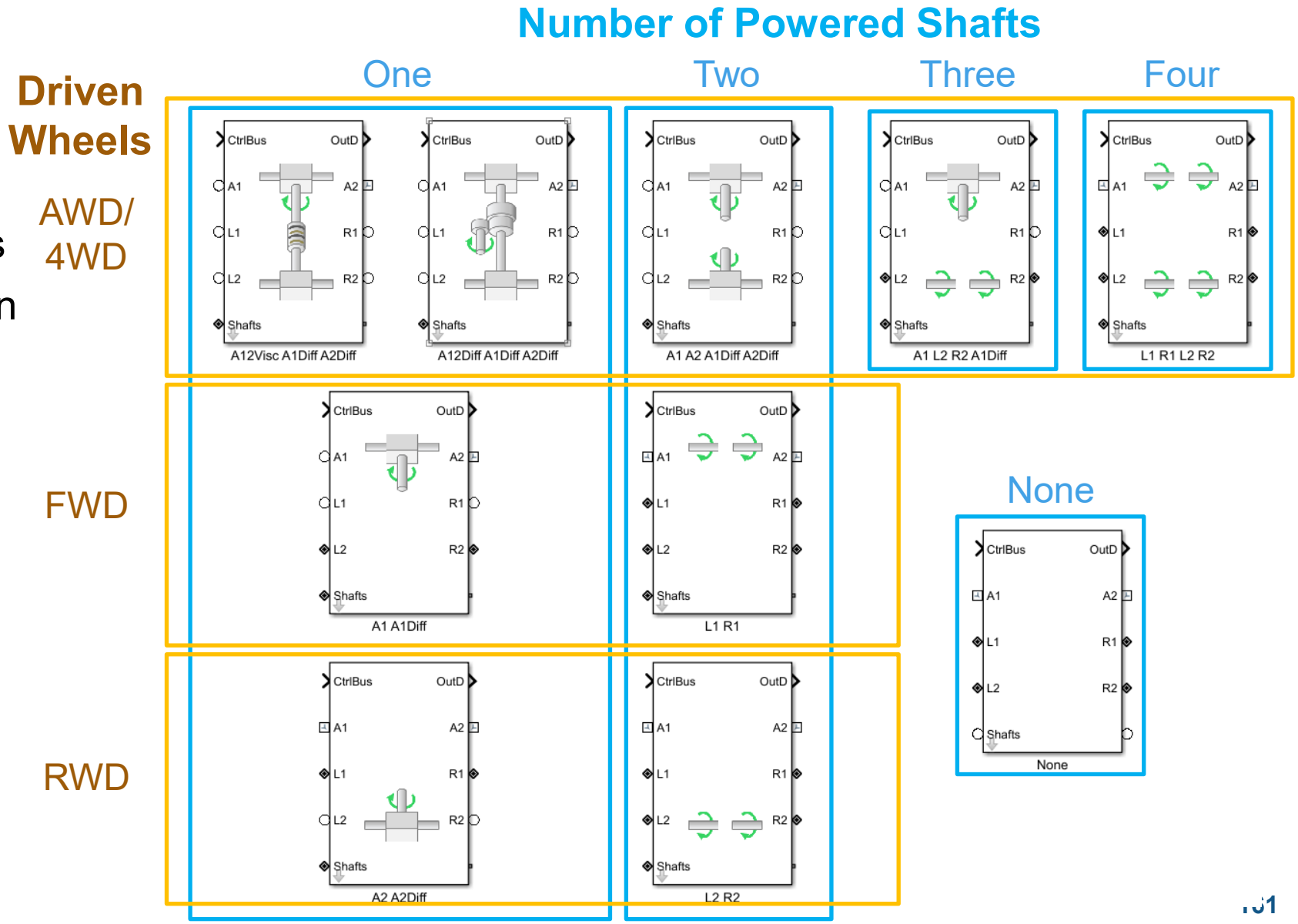
- Select power that matches your design
  - Number of driven shafts
  - Which axles are driven



- Number of powered shafts must match Driveline subsystem

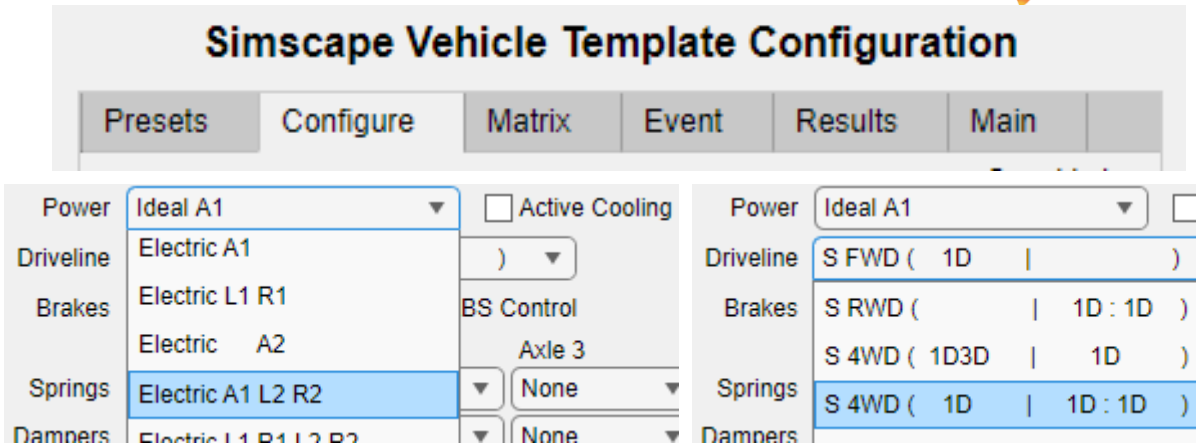
Driveline Supports AWD/4WD, FWD, RWD, and No Driven Wheels

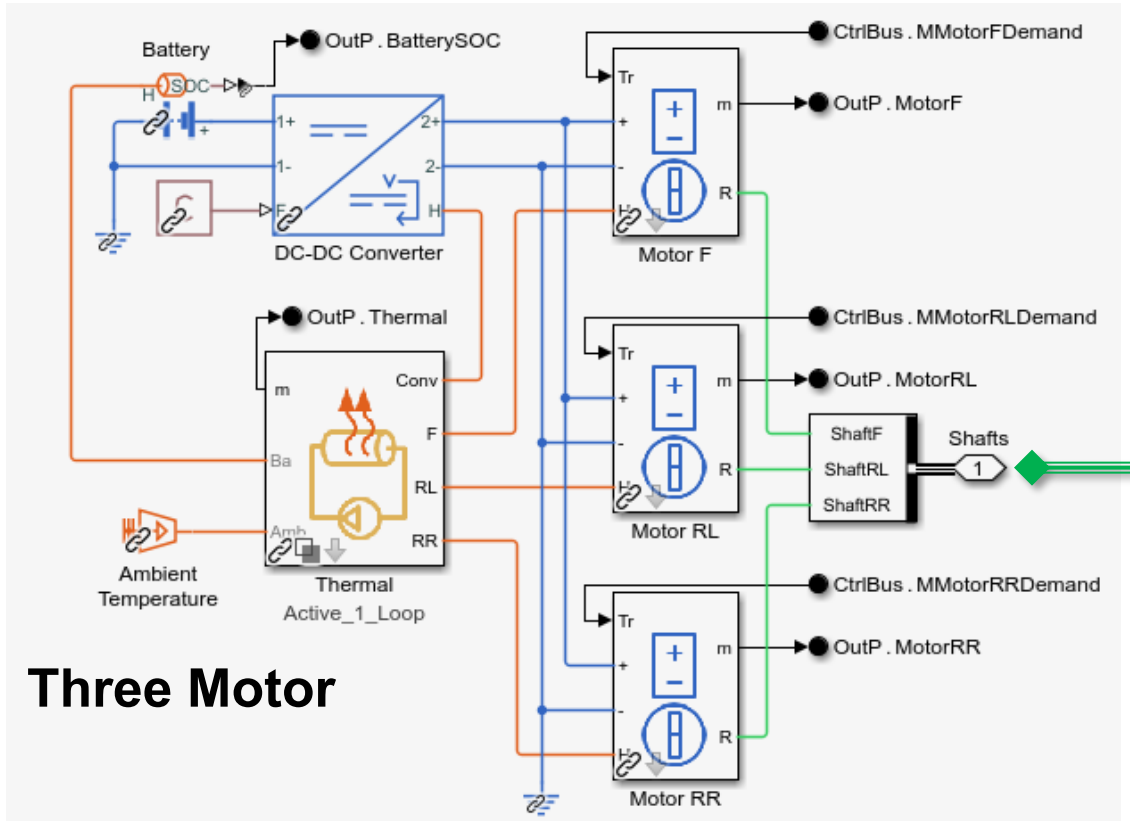
- Select drivetrain that matches your design
  - Number of driven shafts
  - Which wheels are driven
- Number of powered shafts must match power subsystem

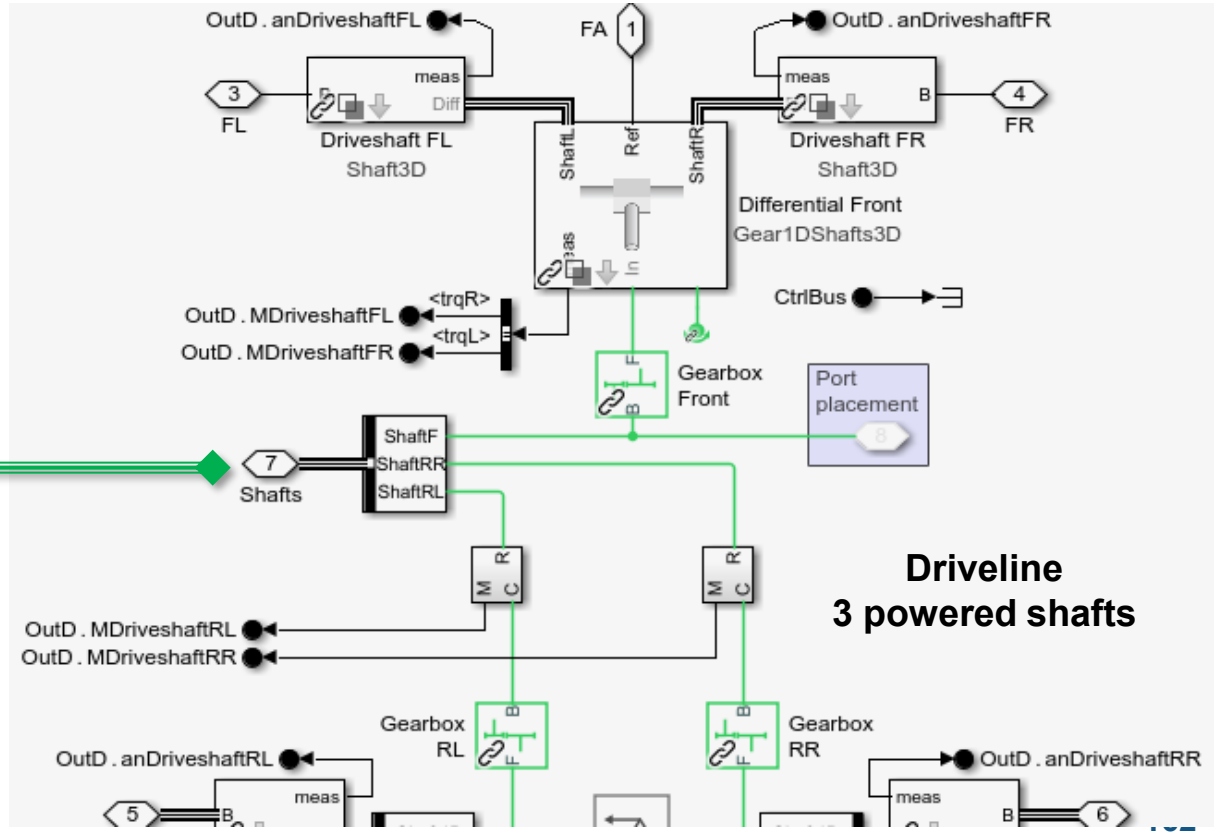


Electric Powertrain

- Example electric vehicle powertrain
  - Three motor (one front, two rear)

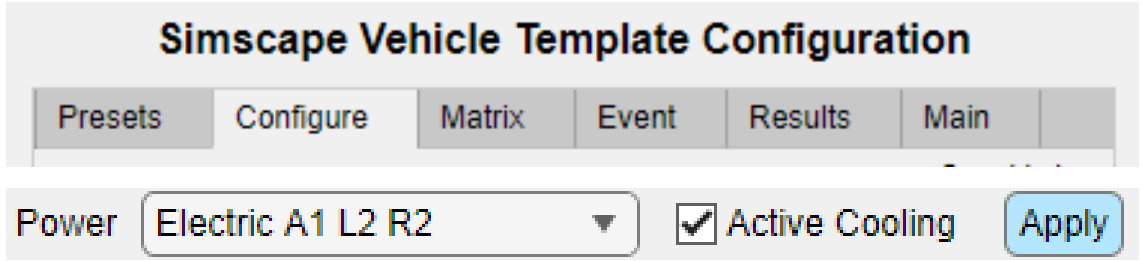


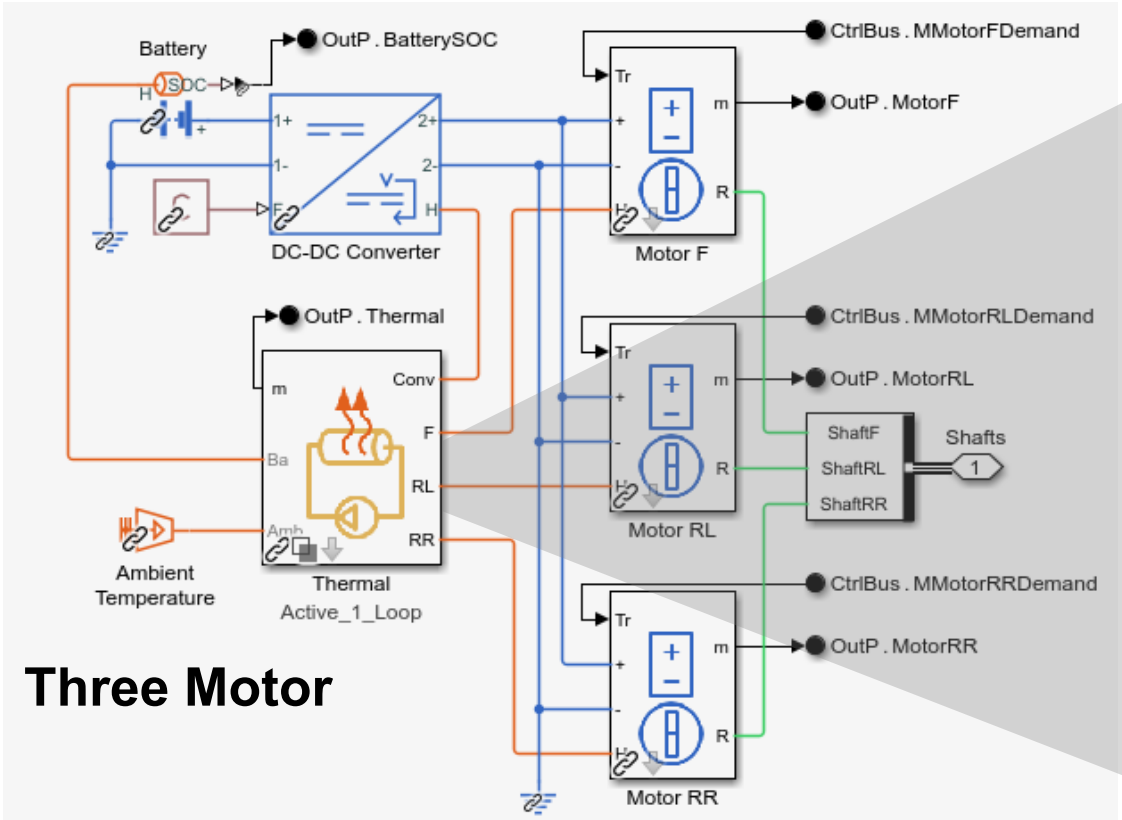




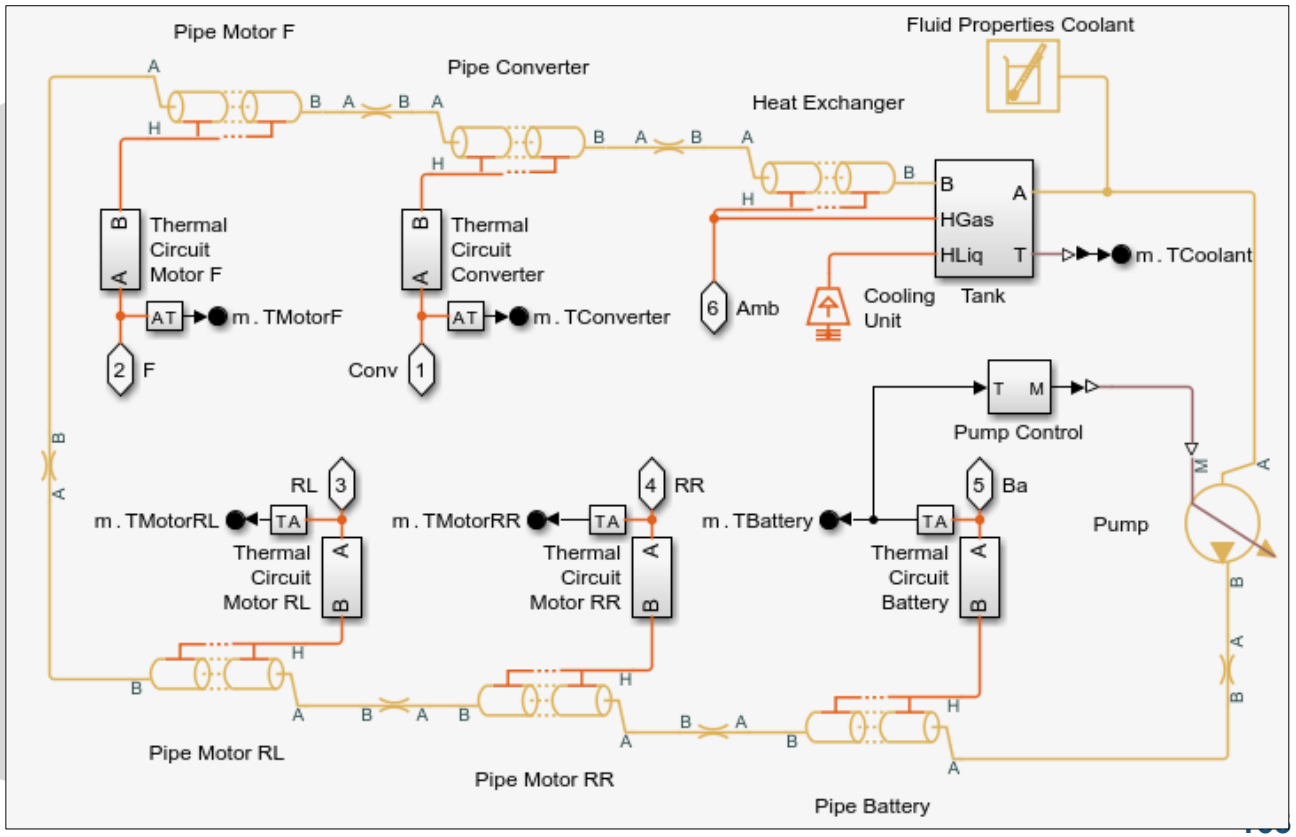
Cooling System for Motors, Battery, and Inverter

- Single cooling loop to extract heat
- For all motor variants



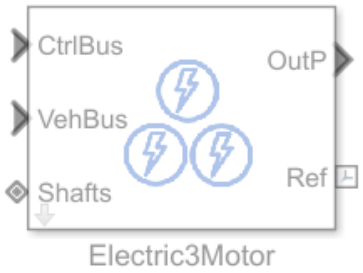


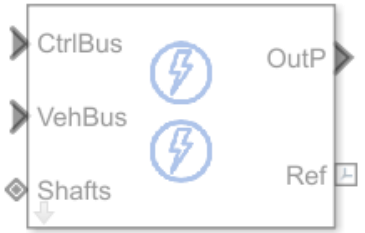
Three Motor

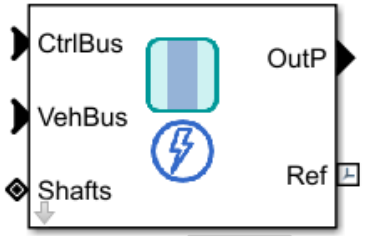


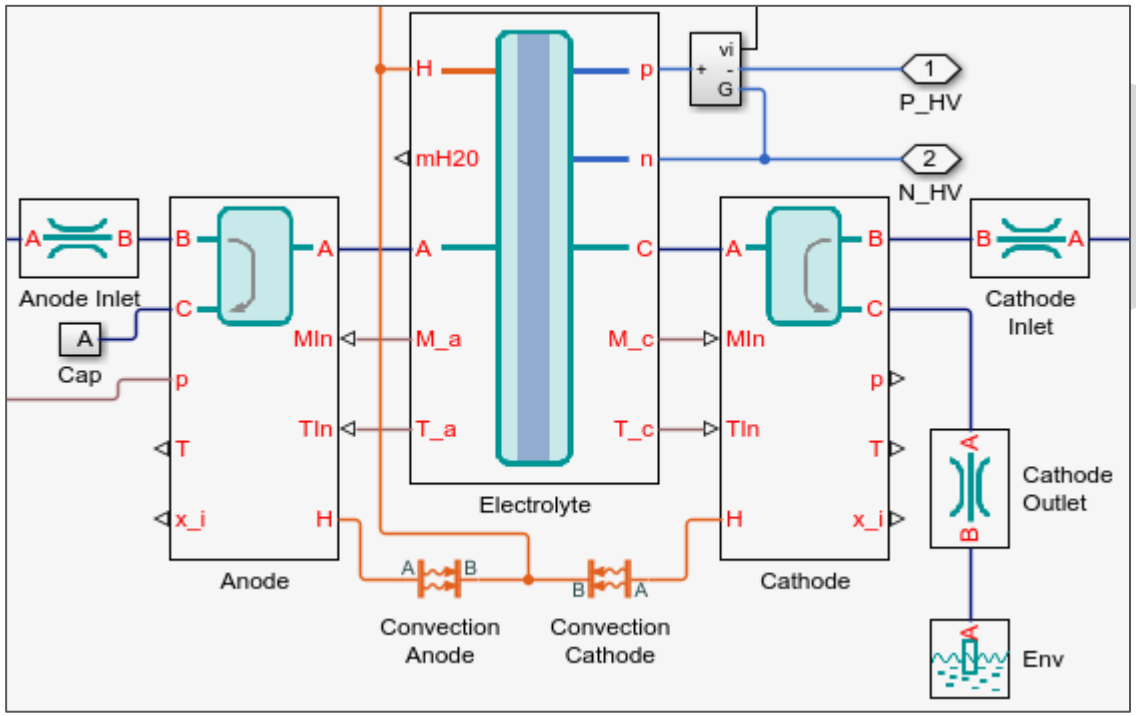
Fuel Cell Powertrain Variant

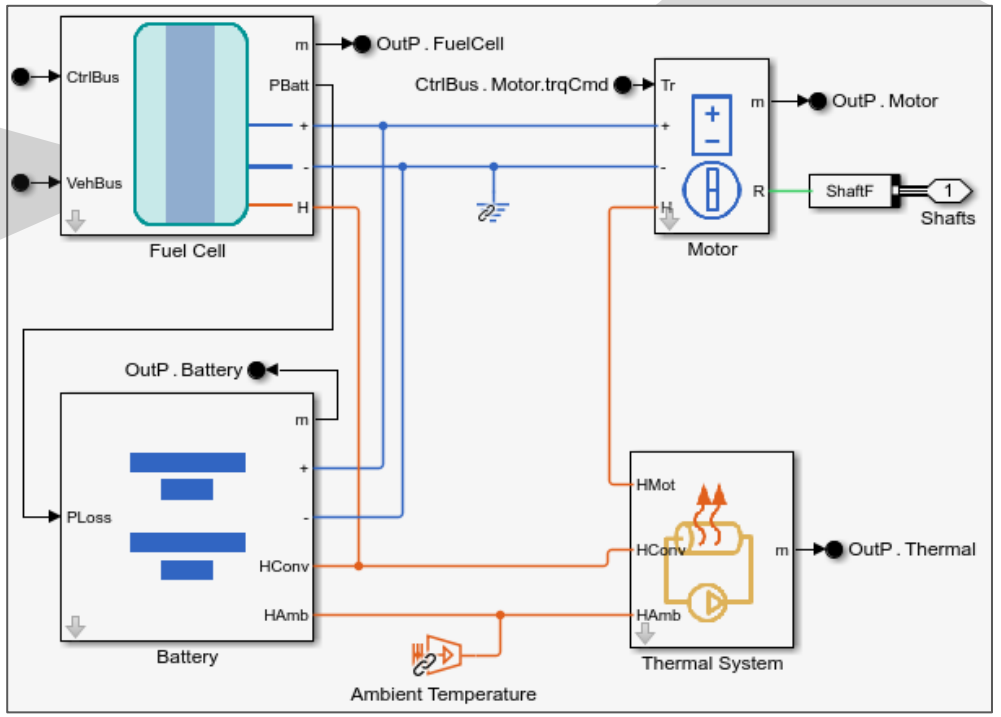
- Vehicle propelled by a single motor powered by a fuel cell and a battery





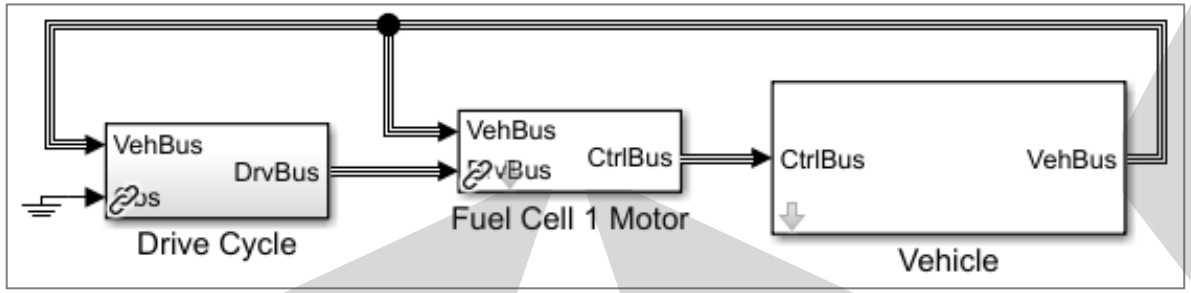


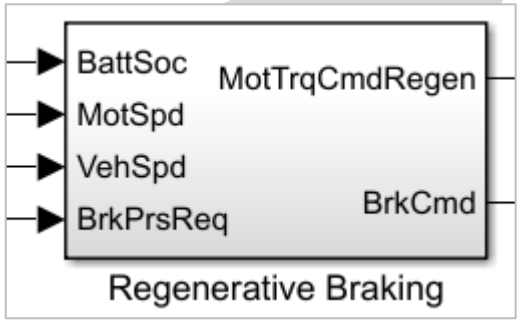


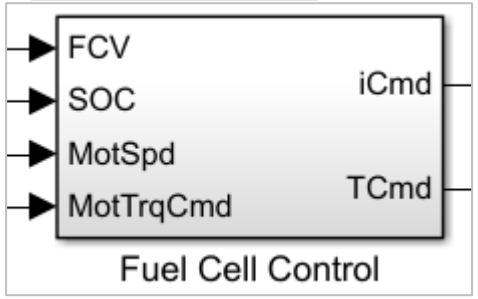


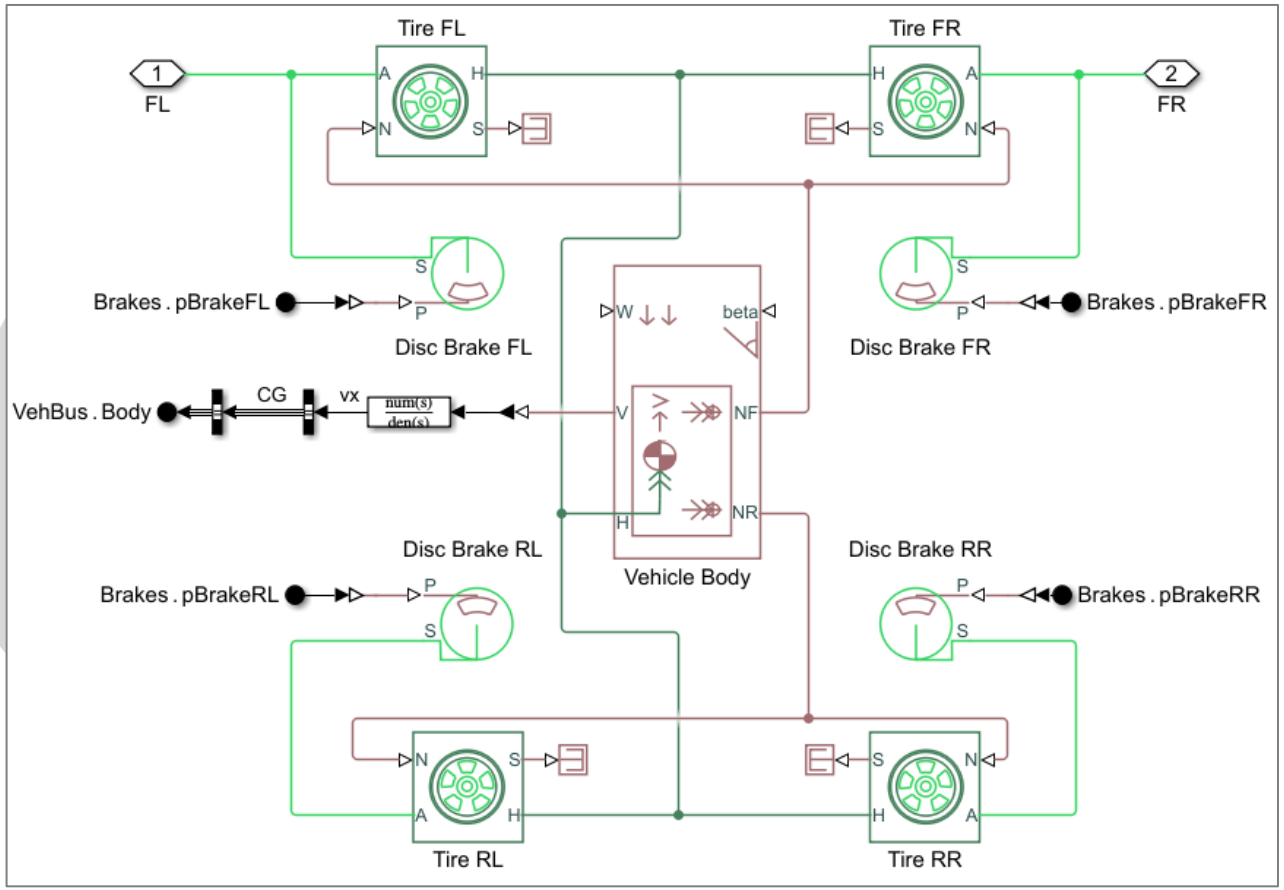
Fuel Cell Powertrain Variant

- Vehicle propelled by a single motor powered by a fuel cell and a battery
- Test with multibody or 1D driveline with powertrain control algorithms



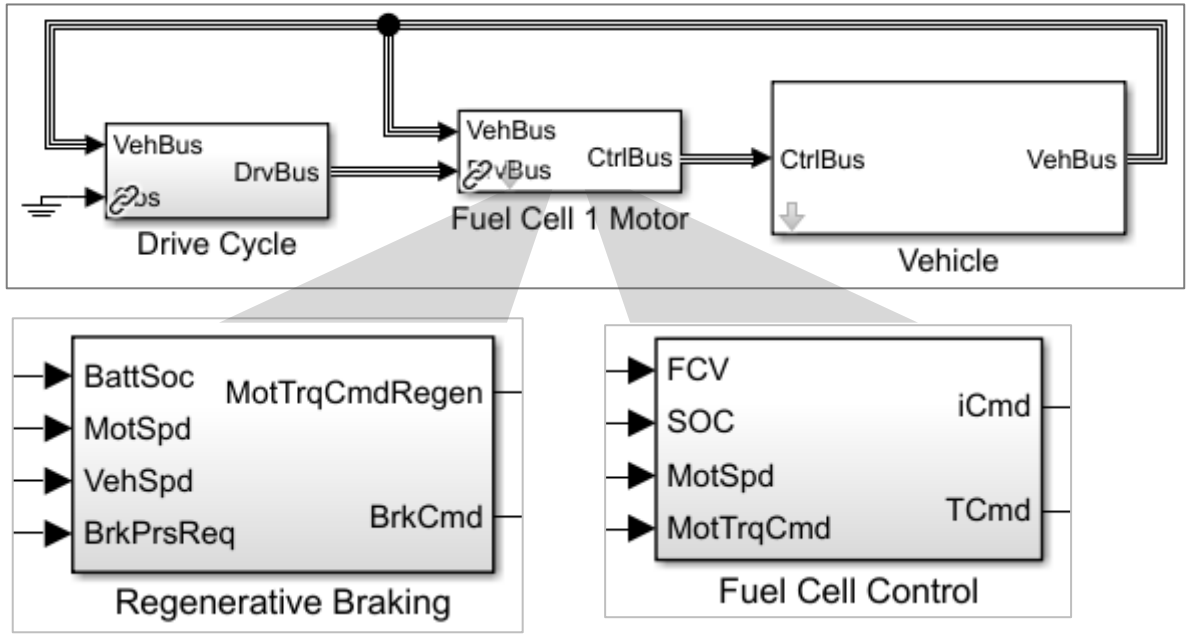


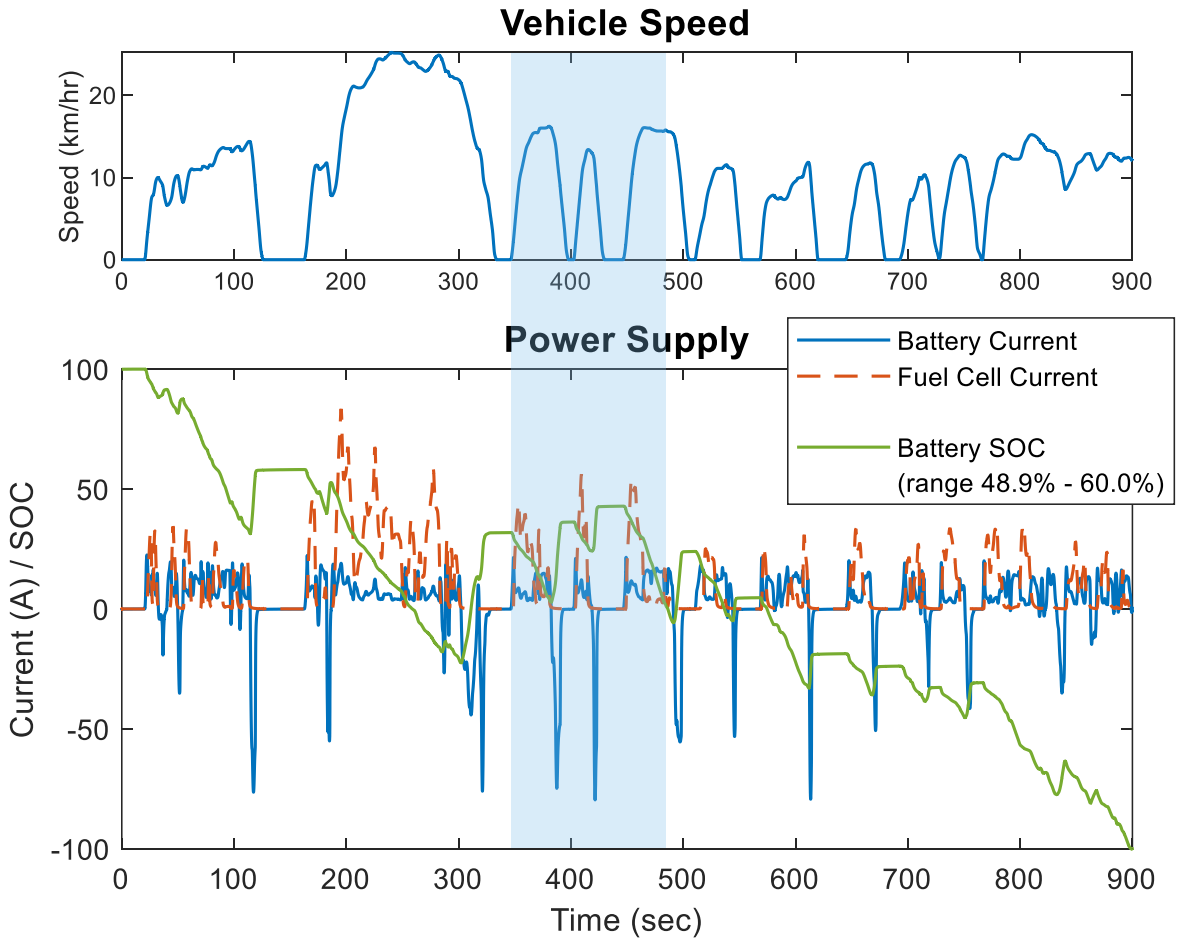




Fuel Cell Powertrain Variant

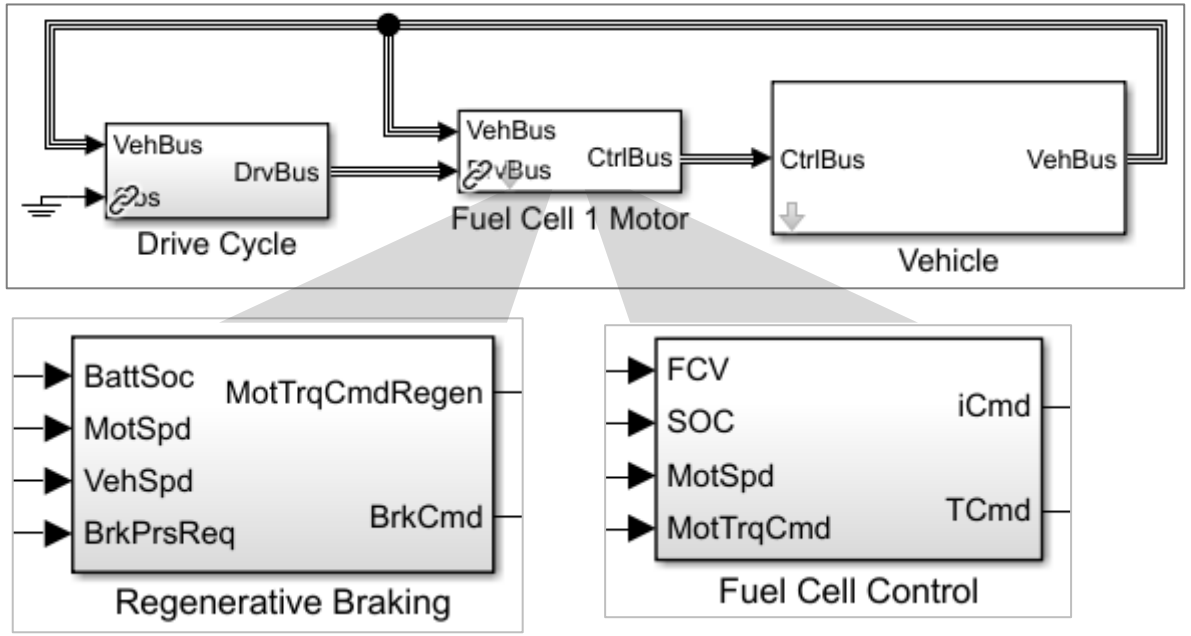
- Vehicle propelled by a single motor powered by a fuel cell and a battery
- Test with multibody or 1D driveline with powertrain control algorithms





Fuel Cell Powertrain Variant

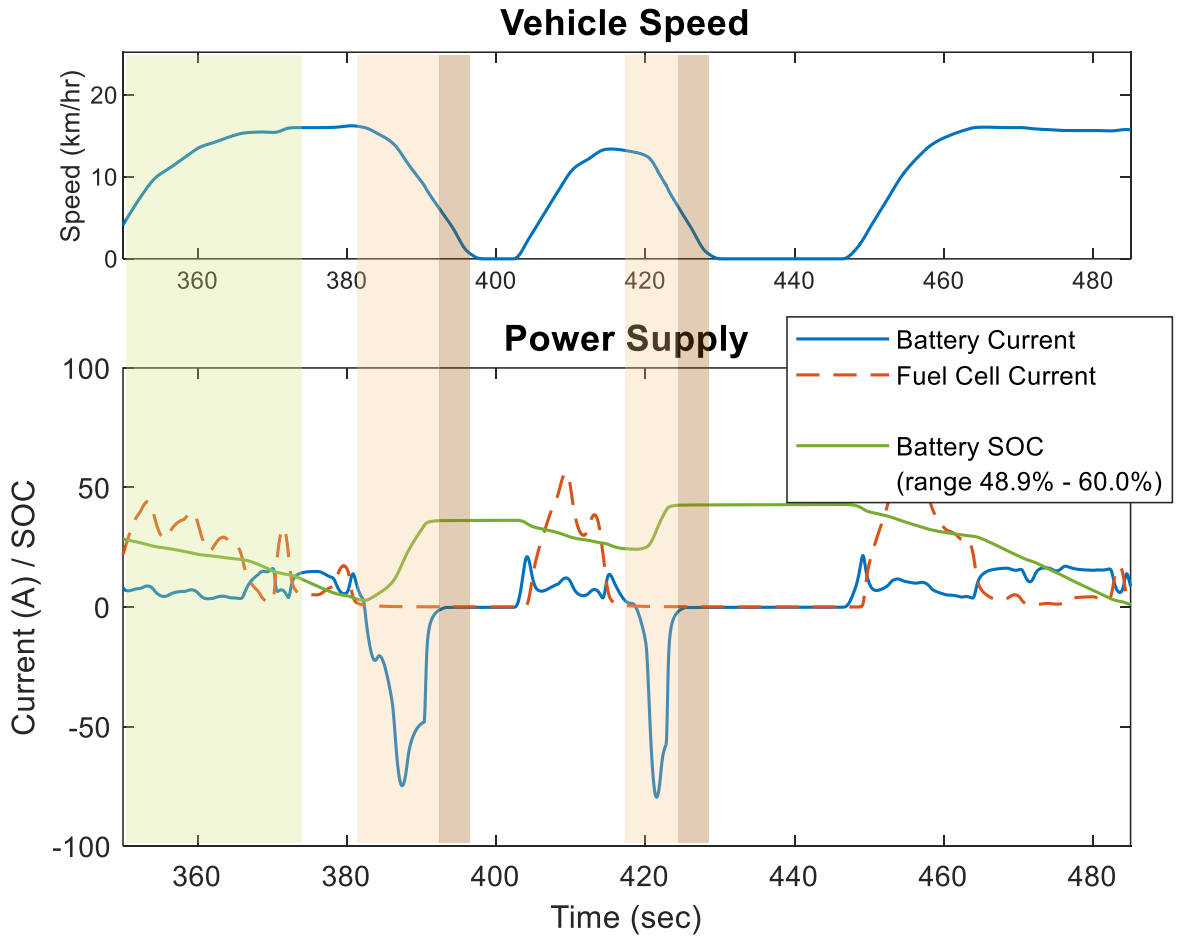
- Vehicle propelled by a single motor powered by a fuel cell and a battery
- Test with multibody or 1D driveline with powertrain control algorithms



Acceleration
SOC Drops
Power Output

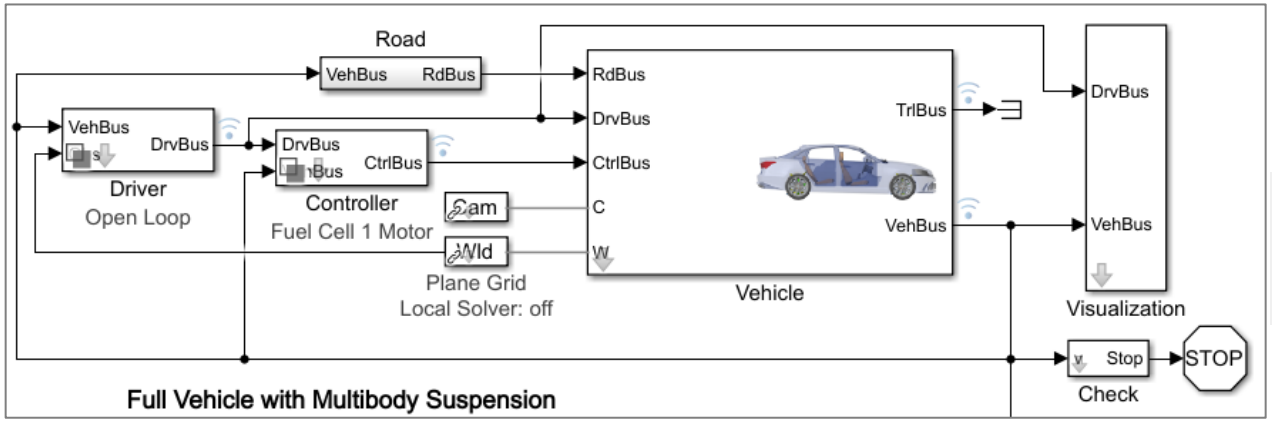
Deceleration
SOC Increases
Power to Battery

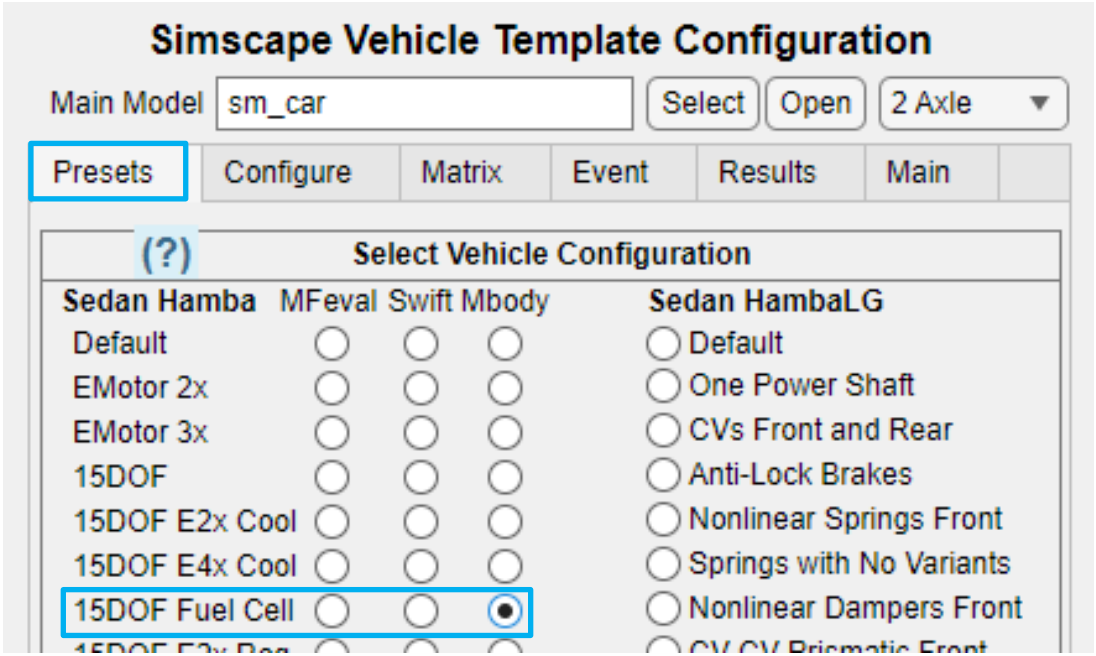
Braking
Mechanical
Only

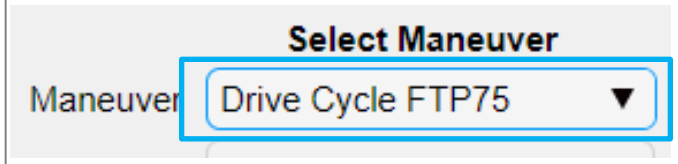


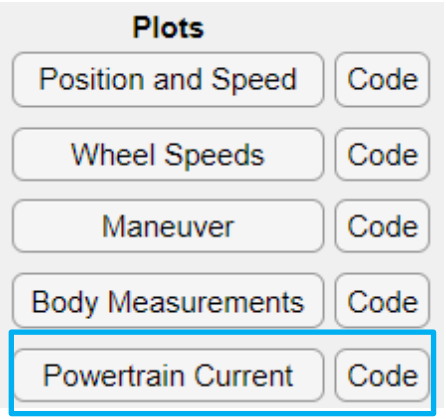
Fuel Cell Powertrain Variant

- Configuring the vehicle
  - Select preset to configure the powertrain, brakes, and controller



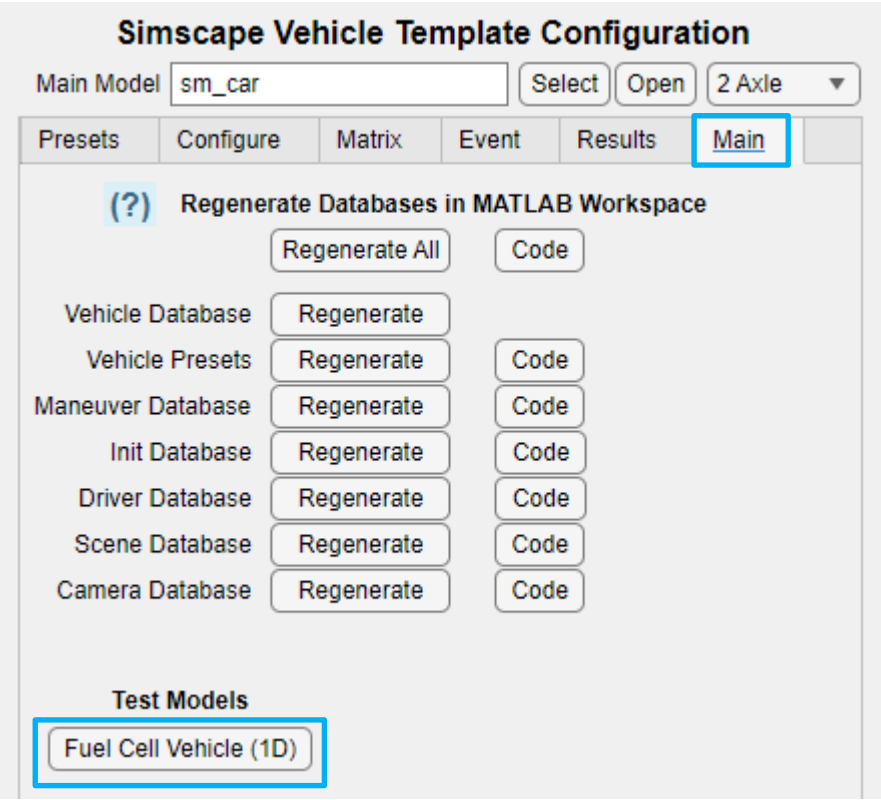


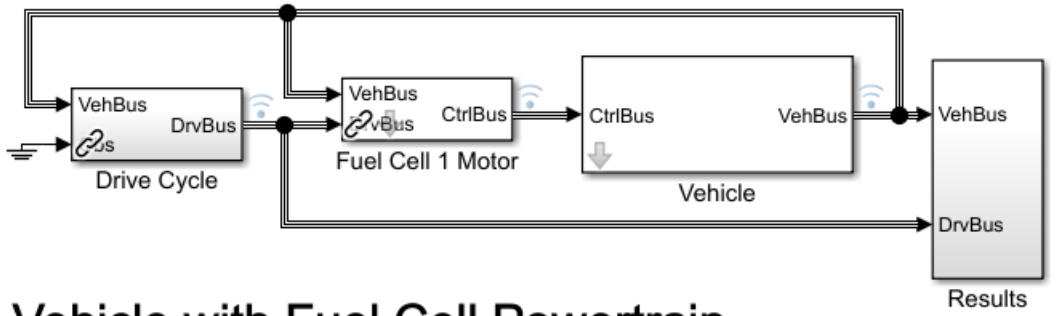




Fuel Cell Powertrain Variant

- Configuring the vehicle
  - Open 1D driveline model from Main tab





Vehicle with Fuel Cell Powertrain

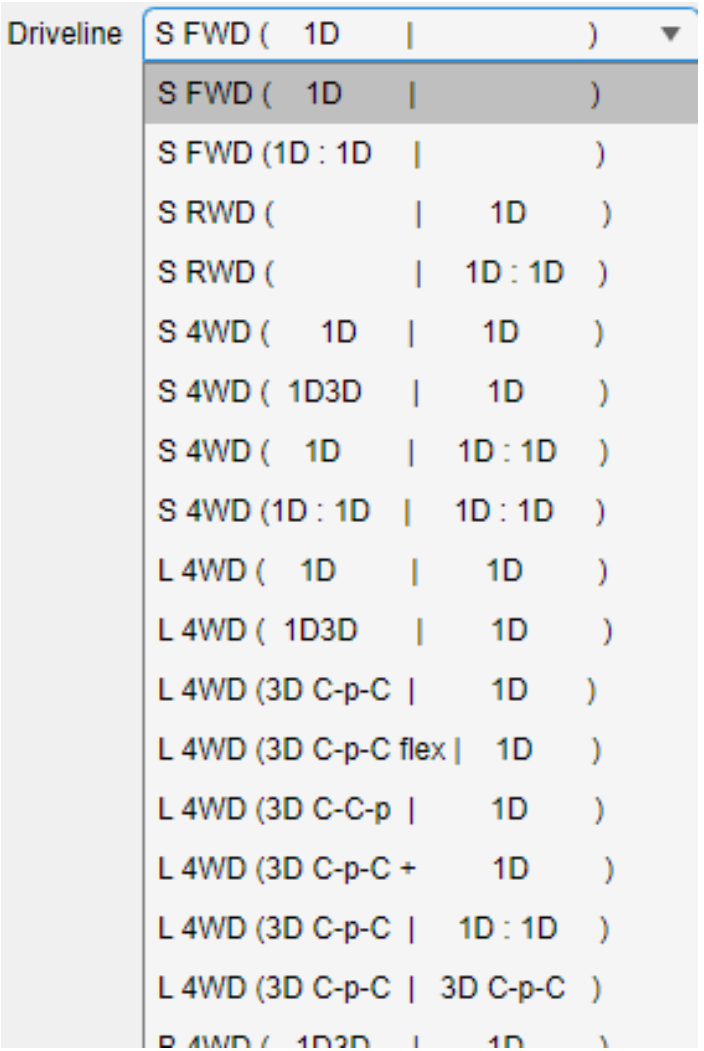
Load preset: Fuel Cell 1 Motor
Plot results (code)

Click hyperlink to set up test

Run model

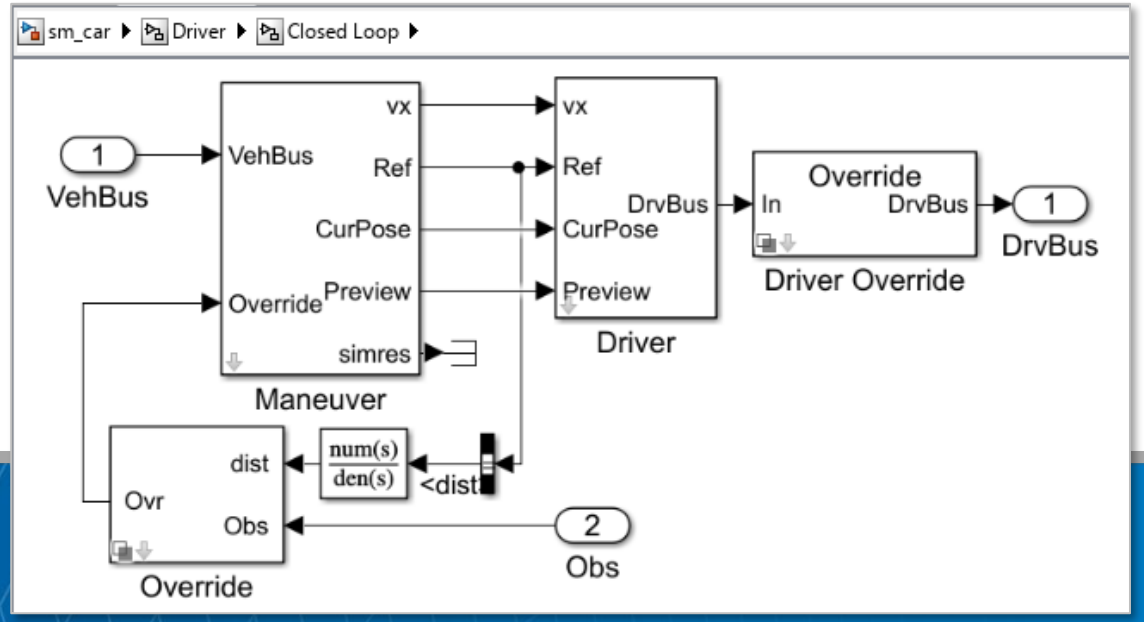
Click hyperlink to generate plot

Drivetrain Options



- Select number of powered shafts, to which element they are connected, and type of shaft model
- Code for options
  - S / L / B = vehicle instance
  - (<front axle options> | <rear axle options>)
  - (<front left> : <front right> | <rear left> : <rear right>)
  - (<front axle options> + <rear axle options>)
    Front and rear axle shafts are connected
  - 1D / 1D3D / 3D / … = fidelity of shaft model
    1D = only inertia about shaft axis
    1D3D = 3D inertia present but does not rotate
    3D = 3D rotating shaft
    CV-p-CV = Constant Velocity – Prismatic – Constant Velocity constraints
    flex = flexibility about shaft axis

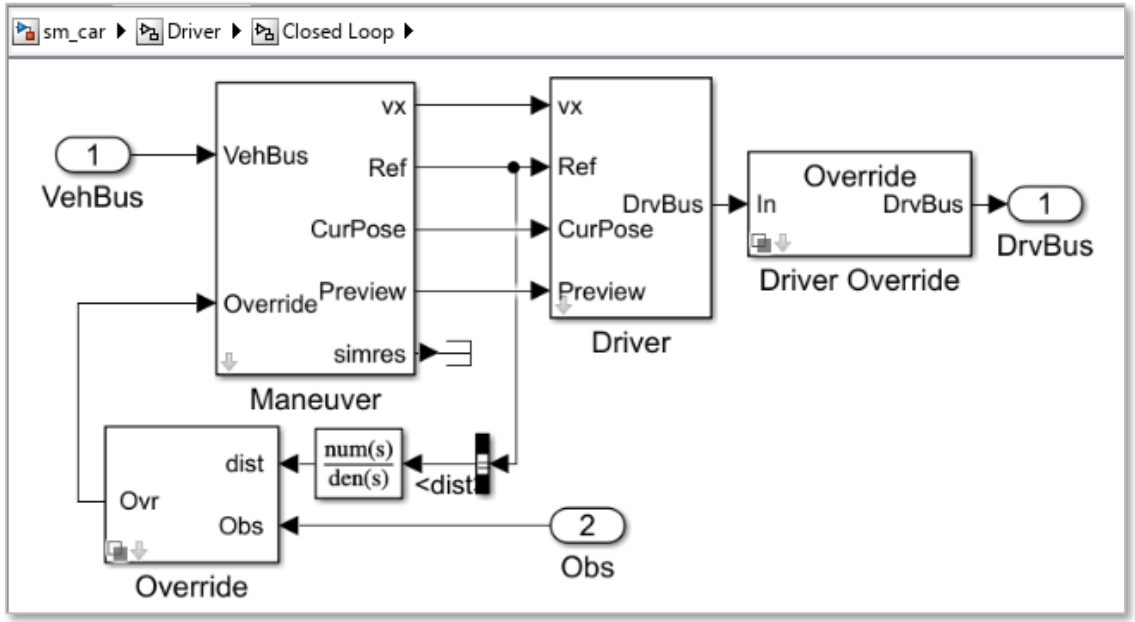
Simscape Vehicle Templates
Closed-Loop Driver Model



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Closed-Loop Driver Model

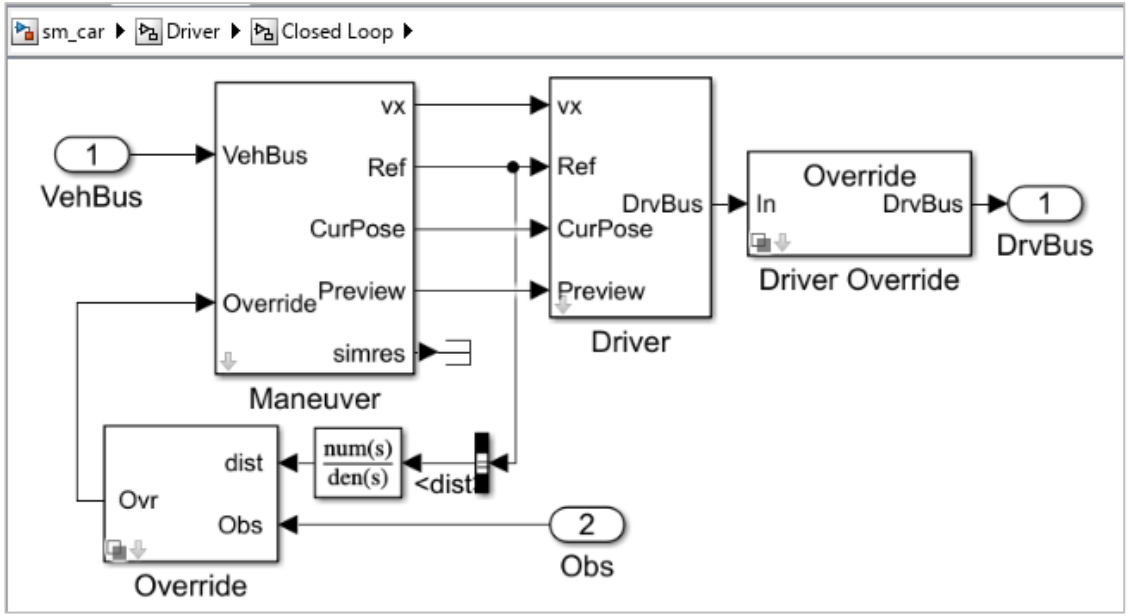
- Interface
- Longitudinal Driver
- Lateral Driver
- Driver Overrides

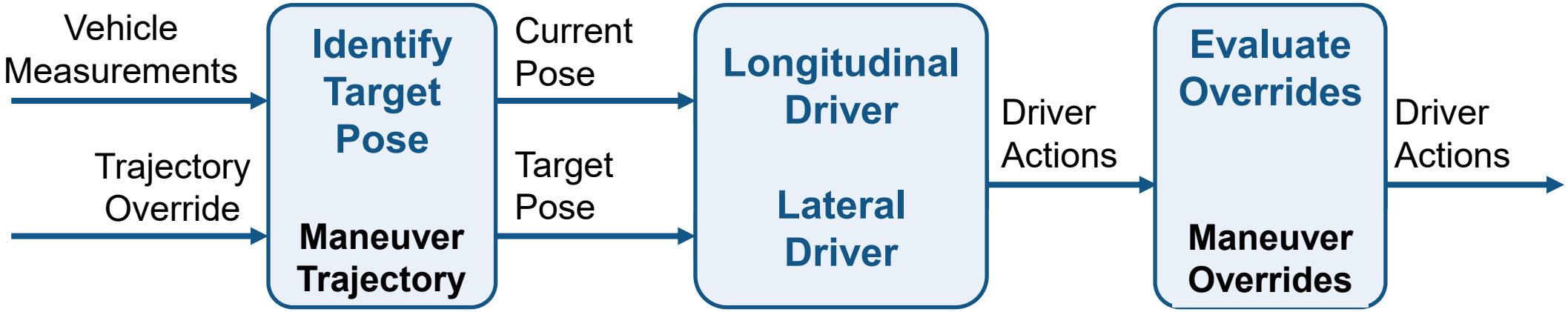


Closed-Loop Driver Model

Determine driver actions (steering, pedals) to follow trajectory

- Inputs
  - Vehicle measurements
  - Trajectory override triggers from Scene
- Trajectory
  - Target pose based on distance
  - Maneuver overrides (closed-loop to open-loop driver)

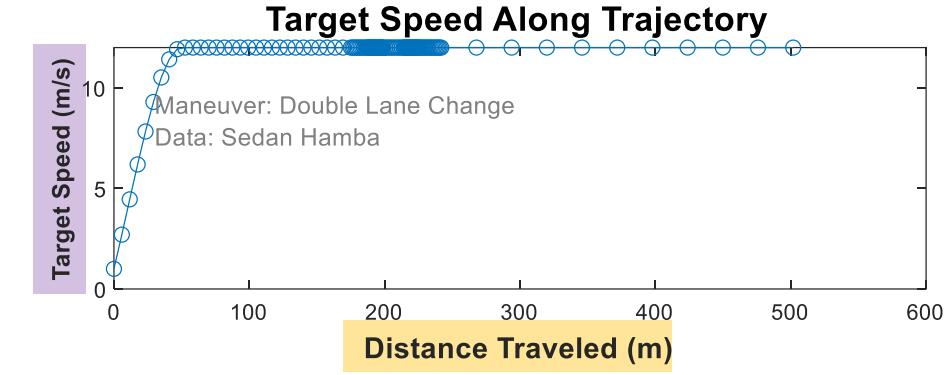


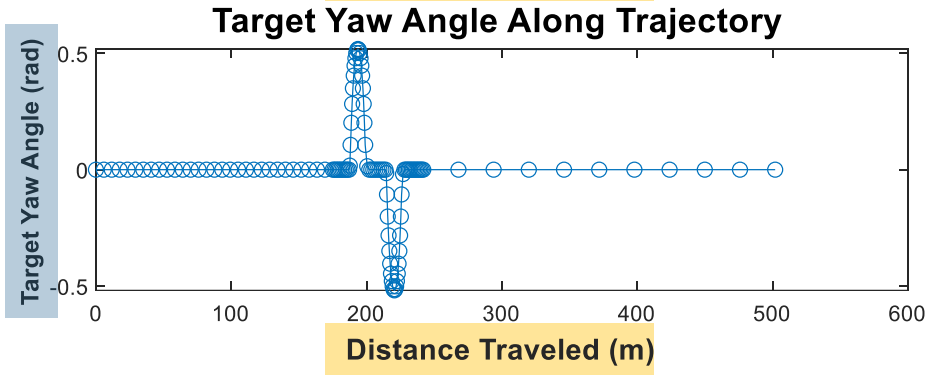


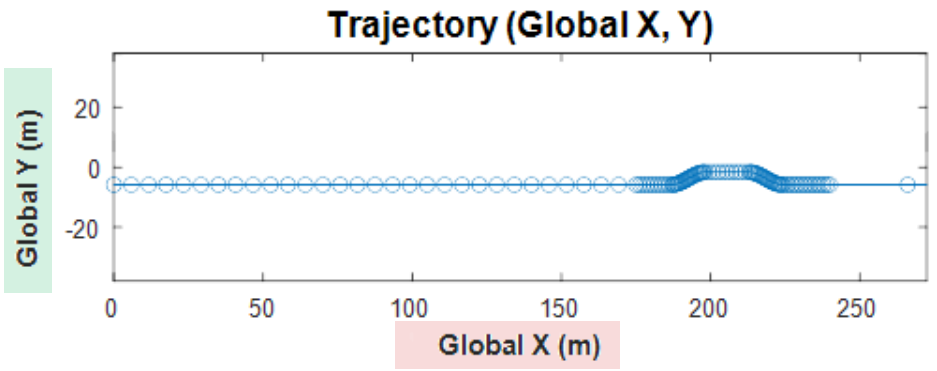
Trajectory Definition

- Closed Loop Test: **Driver Commands**
  - Target pose (x, y, z position, yaw angle) vs. distance along path (`xTrajectory`)
  - Driver model compares target pose to current vehicle pose and **determines driver actions**

```
Maneuver.Trajectory
                x: [1×1 struct]
                y: [1×1 struct]
                z: [1×1 struct]
      xTrajectory: [1×1 struct]
               vx: [1×1 struct]
             aYaw: [1×1 struct]
```

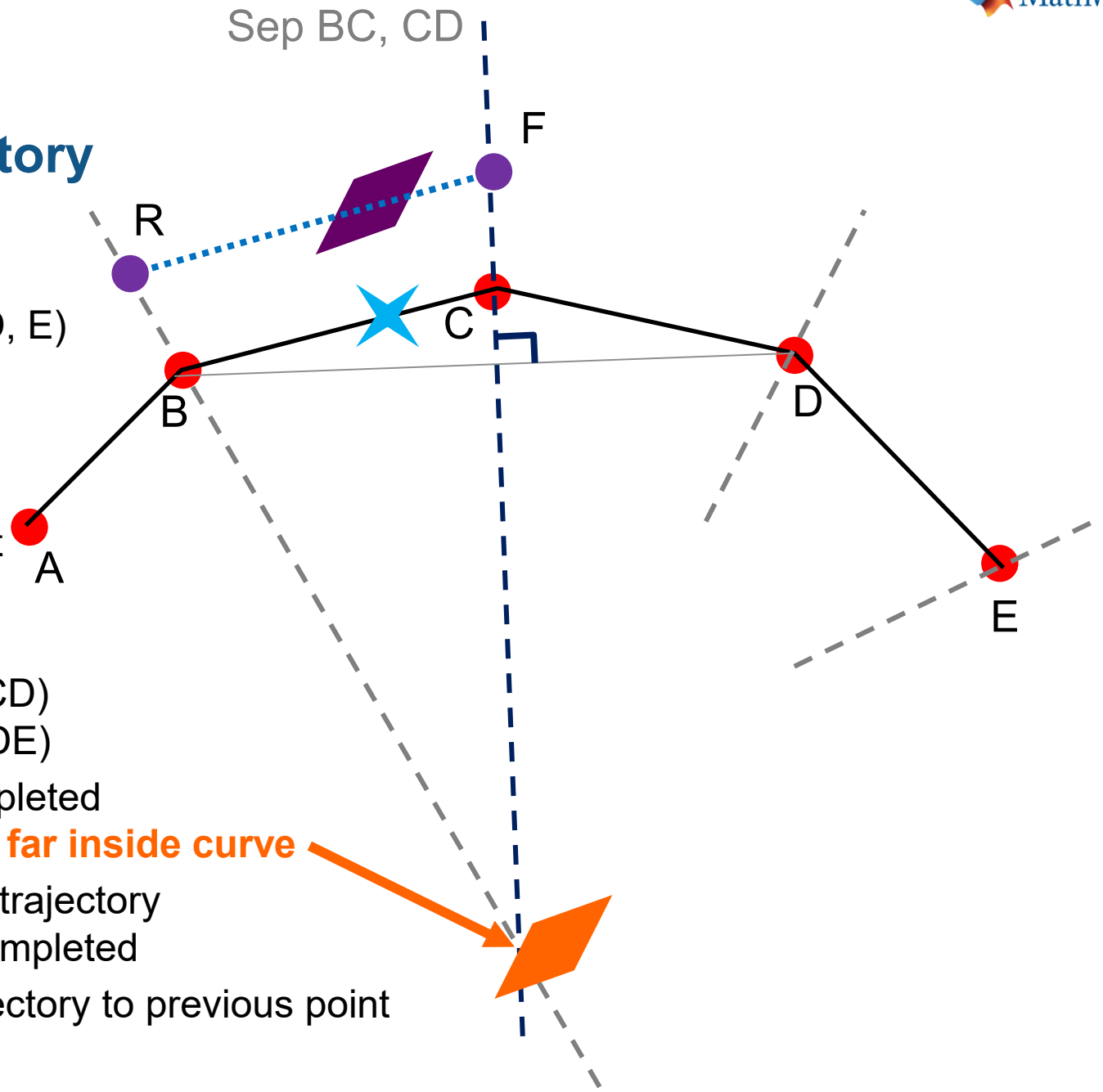




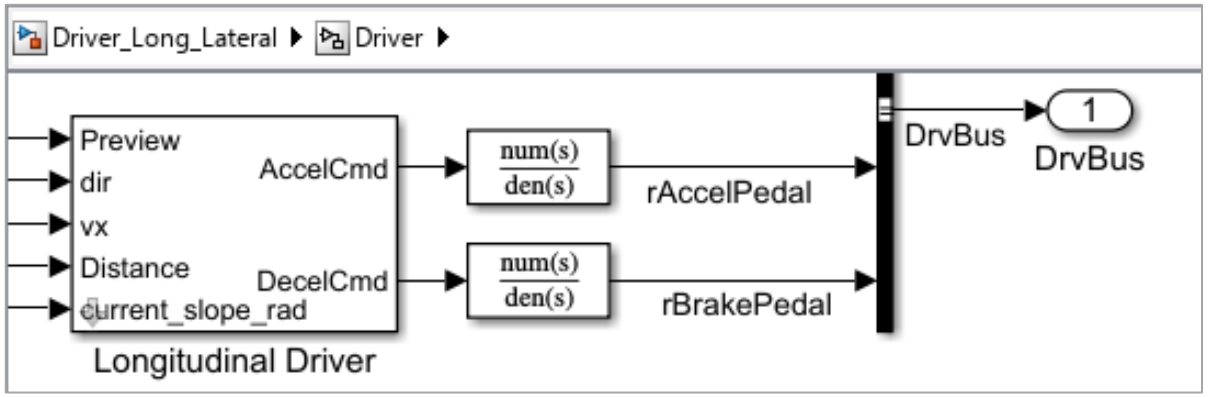


Path Following
Finding Distance Along Trajectory

1. Find point closest to **car position** (pt C)
2. Find 2 points ahead and behind (A, B, D, E)
3. Find orientation of line separating segments BC, CD
4. Is car on BC side or CD side?
5. Find line parallel to BC (or CD) segment upon which car lies (FR)
6. Find intersection of line FR with line separating AB & BC (or BC & CD) and line separating BC & CD (or CD & DE)
7. Find percentage of line FR car has completed
   Limit percentage to 100% in case car **is far inside curve**
8. Use percentage to determine portion of trajectory along current segment that has been completed
9. Add that distance to distance along trajectory to previous point to **calculate distance along trajectory**



Longitudinal Driver

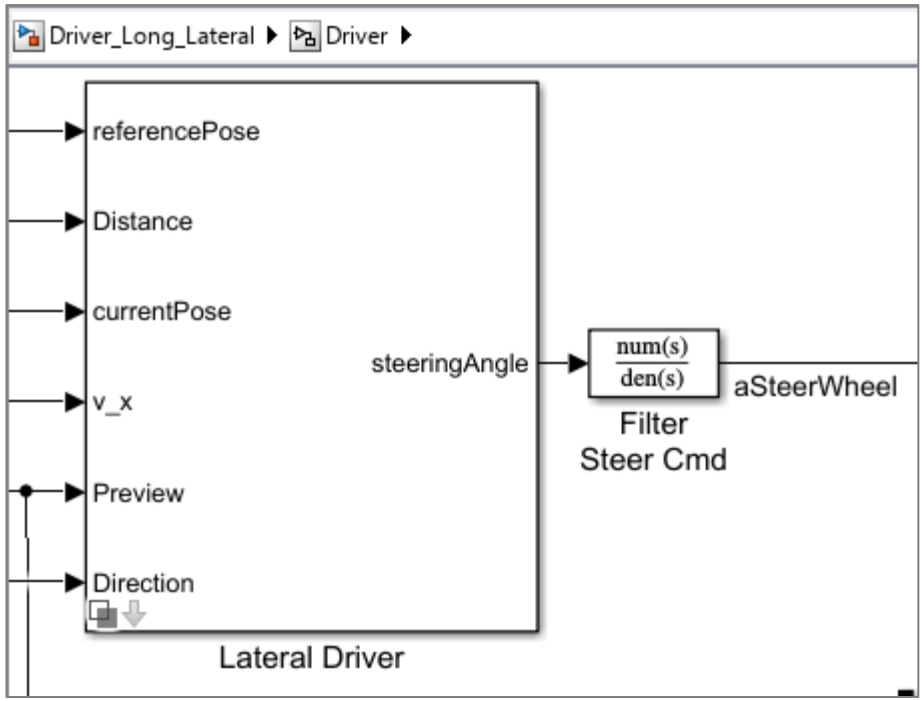


- PI controller with feedforward for velocity tracking
- Compares current speed to target speed
  - Target speed is from a point further along trajectory than the current closest point
  - Preview distance varies with speed
    Shorter at slow speeds, longer at high speeds
- Works when driving forward and reverse
- Fully customizeable

Lateral Driver

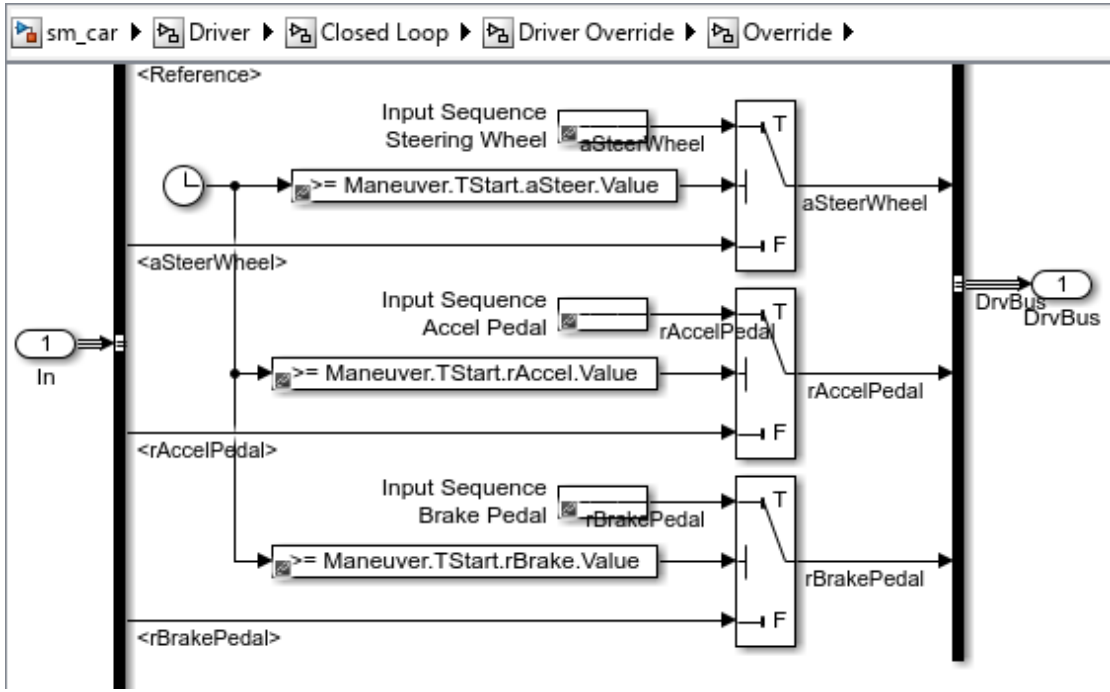
- Two options
  - Stanley (Hoffman, 2007, “Automobile Trajectory Tracking for Off-Road Driving
  - Pure Pursuit (Craig, 1992, “Implementation of the Pure Pursuit Path Tracking Algorithm”)

- Compares heading to trajectory to determine steering wheel angle
  - Pure Pursuit relies on geometric angle between current heading and look-ahead vector
  - Stanley sums heading error and nonlinear feedback term of cross-track error
- Both work in forward and reverse and are fully customizable

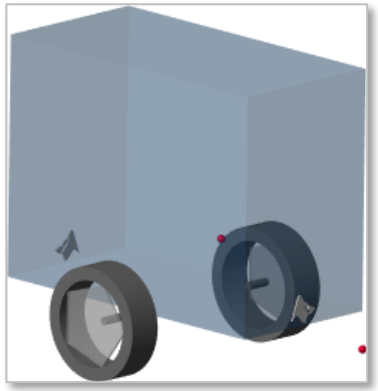


Maneuver Overrides

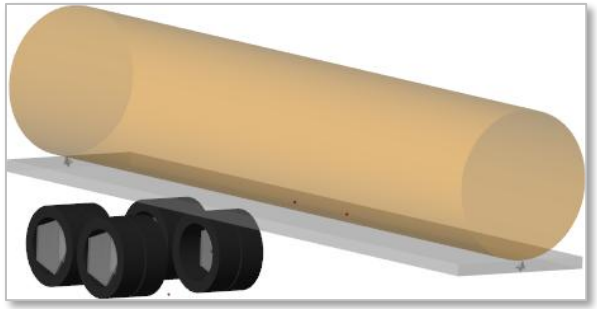
- Maneuvers can include a closed-loop and an open-loop trajectory
  Example: Coastdown
  - Phase 1: Closed-loop speed and steering to reach target speed along straight line
  - Phase 2: Open-loop pedal (no brake, no accel), closed-loop steering
- Override switches from
  closed-loop driver action (maintain speed)
  to open-loop driver action (pedal travel = 0)
- Overrides currently based on time,
  could be based on distance
  or other criteria



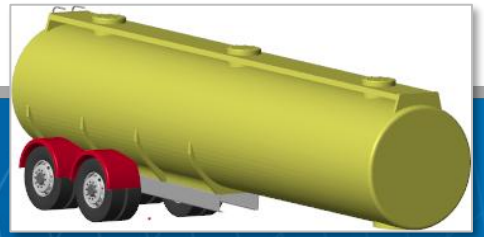
Simscape Vehicle Templates
Trailers

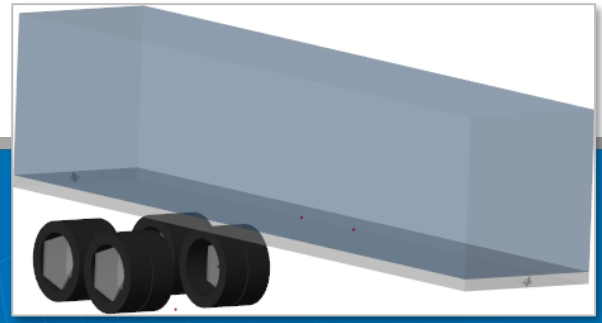












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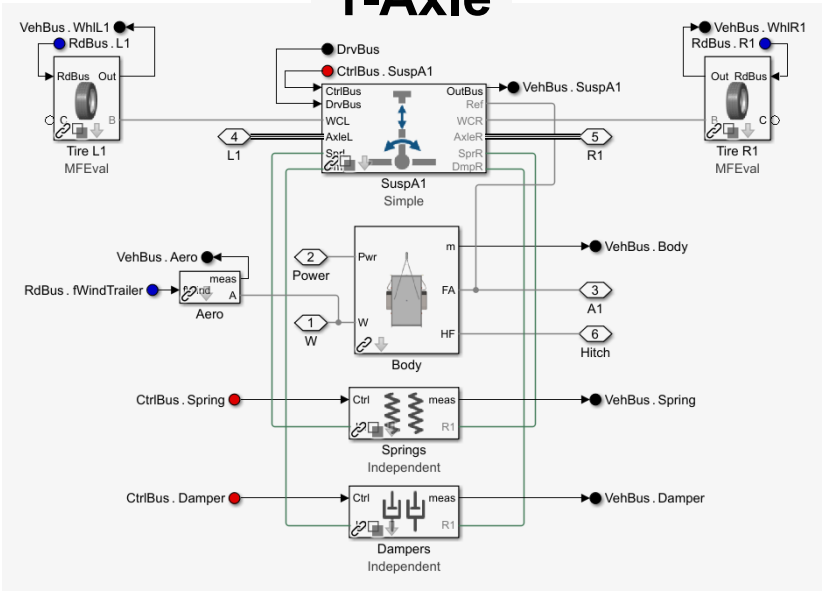
Trailers

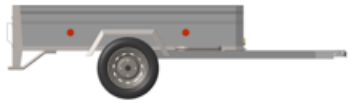
- Overview
- One-Axle Trailers
- Two-Axle Trailers

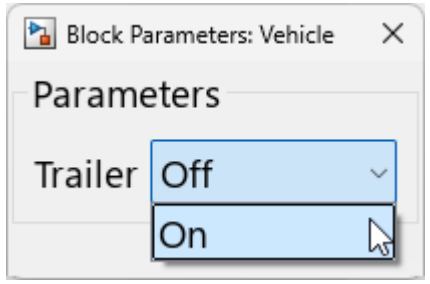
Trailers Overview

- Trailers have the same structure as the vehicles
  - Use same libraries
  - Two-axle trailer is <u>same</u> as two-axle vehicle
  - Are optional in model

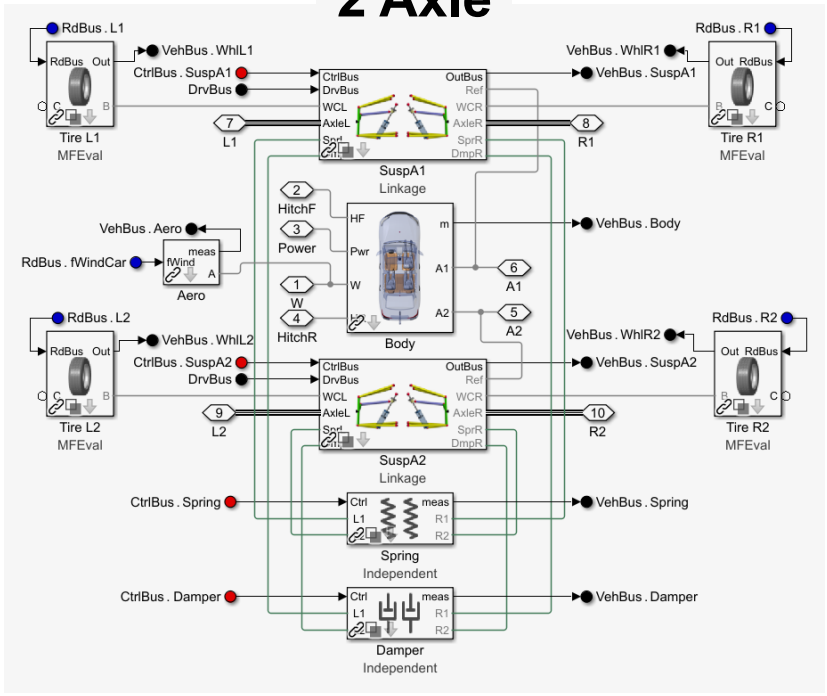
1-Axle

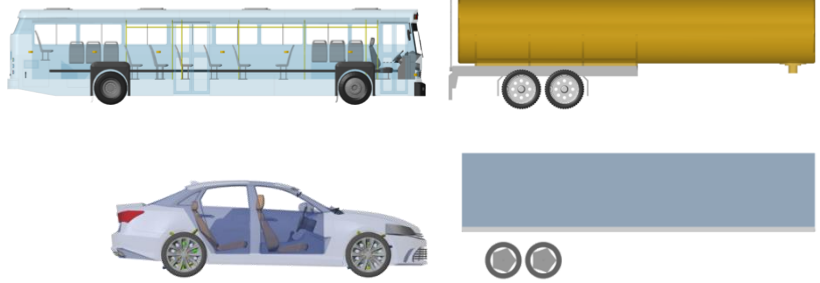






2 Axle

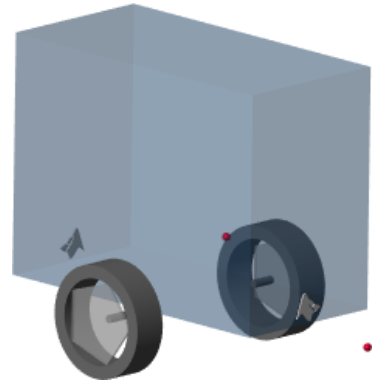




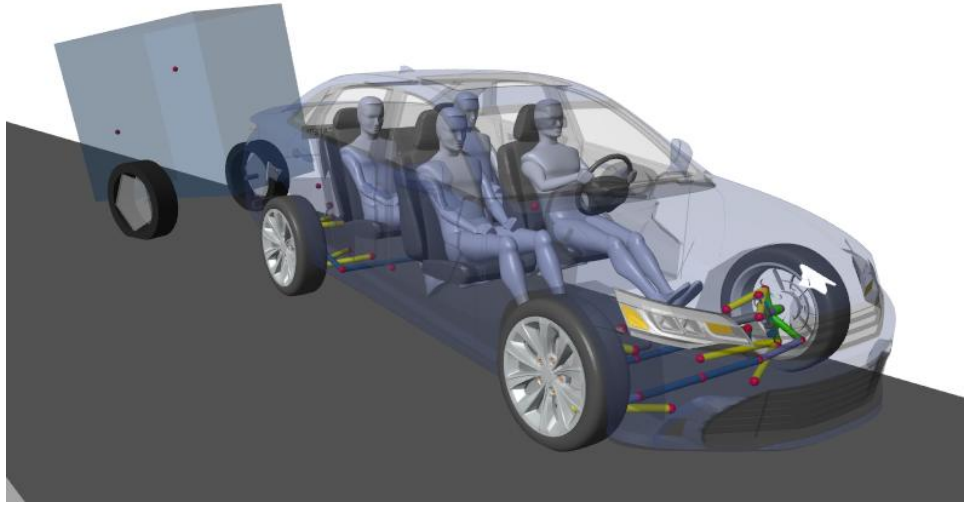
- Parameters: `Trailer`, `Init_Trailer`

One-Axle Trailers

- Parametric and CAD geometry
- Parameterized locations for CG, aerodynamic drag, and wind disturbance
- Test stability, braking, and powertrain



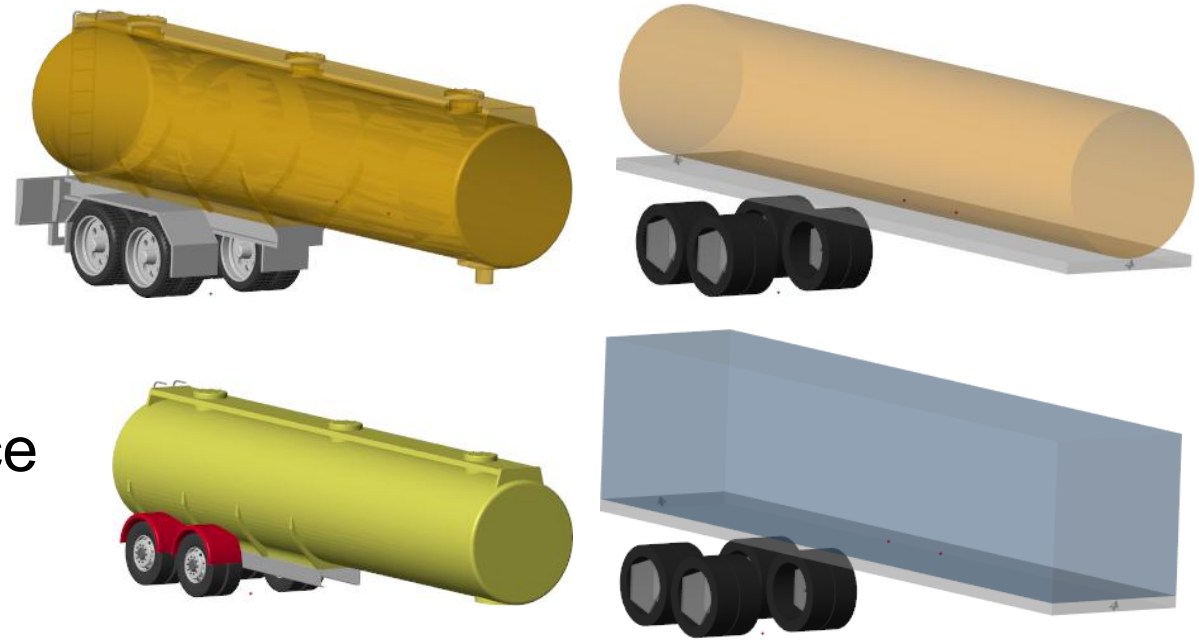


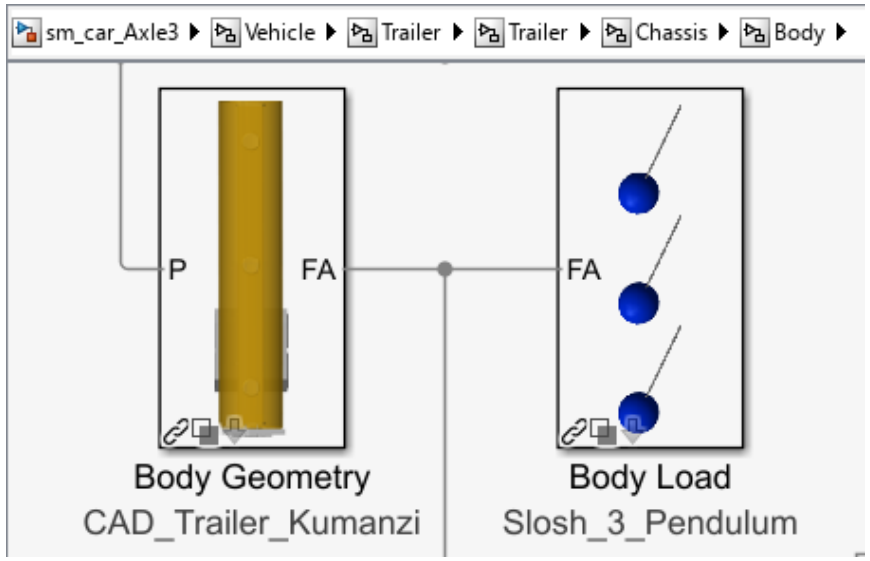


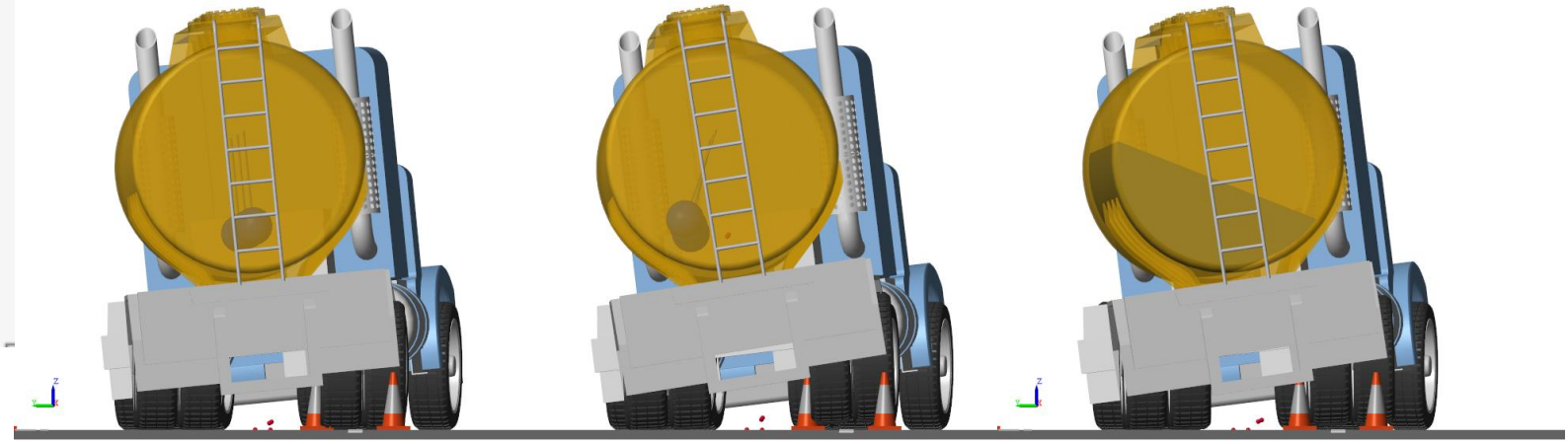
```
sm_car_config_maneuver('sm_car','Trailer Disturbance');
```

Two-Axle Trailers

- Parametric and CAD geometry
- Parameterized locations for CG, aerodynamic drag, and wind disturbance
- Customizable slosh models







Simscape Vehicle Templates Workflows

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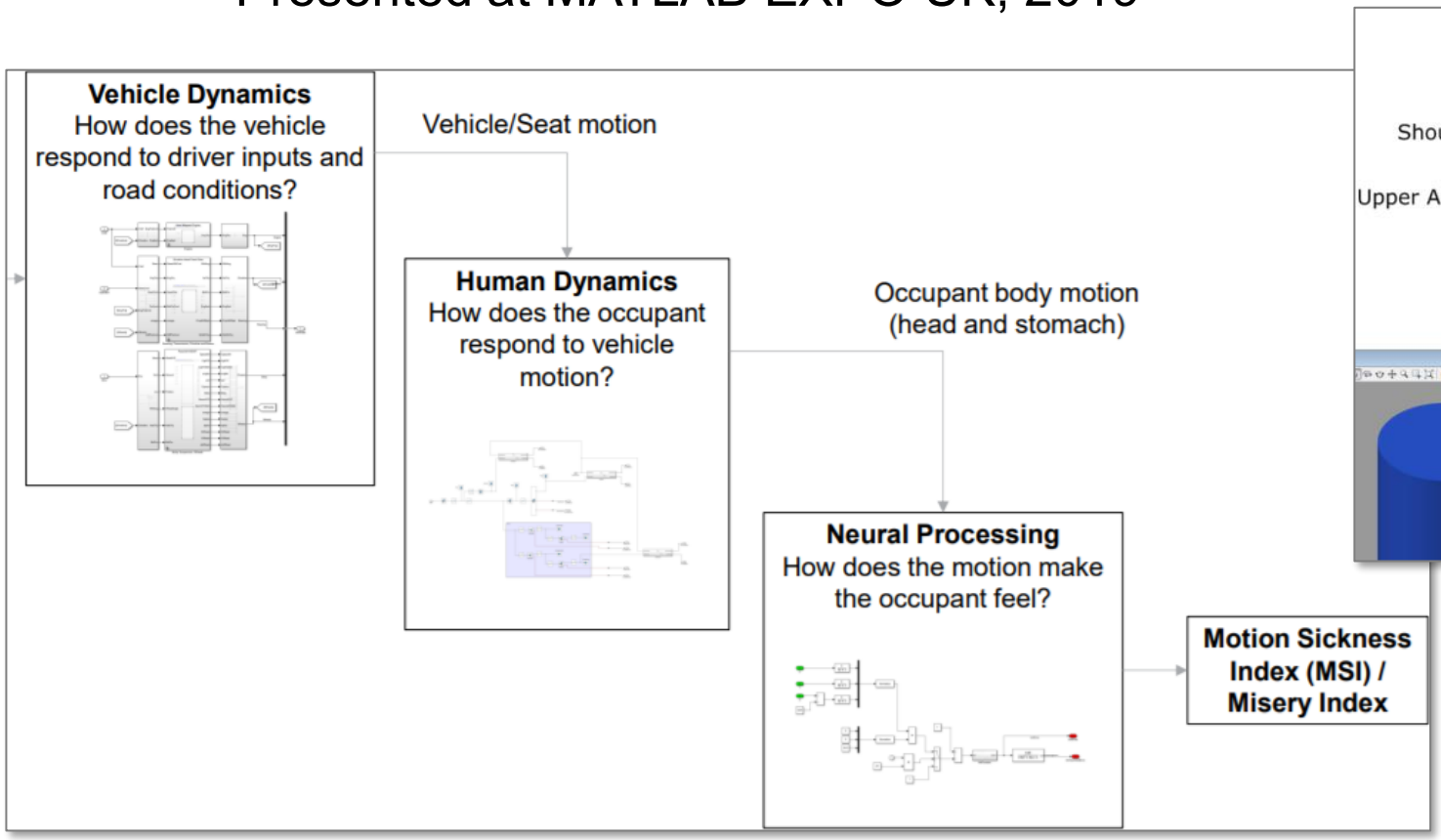
Workflows

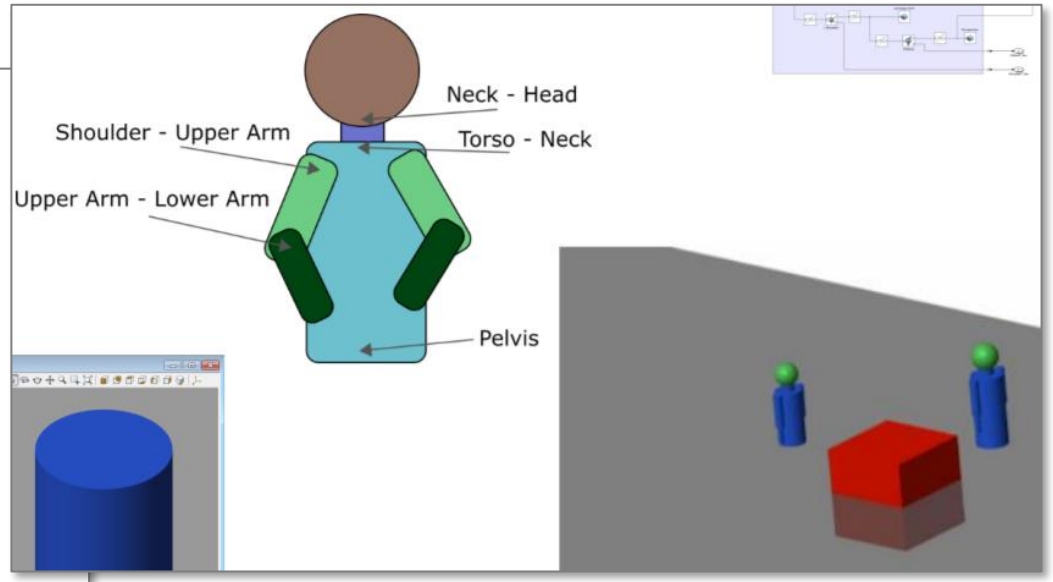
- ADAS
- Real-Time Simulation
- Optimization

Simscape™ Vehicle Templates

Use Case: Evaluate ADAS for Passenger Comfort

- True differentiator for autonomous vehicles may be passenger comfort
  - Ricardo uses simulation to predict “Motion Sickness“
  - Presented at MATLAB EXPO UK, 2019

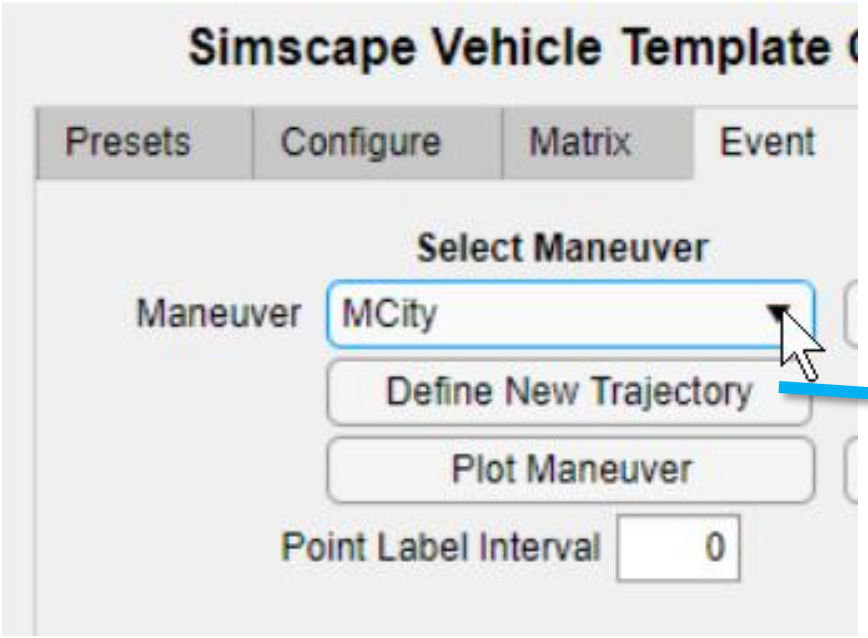


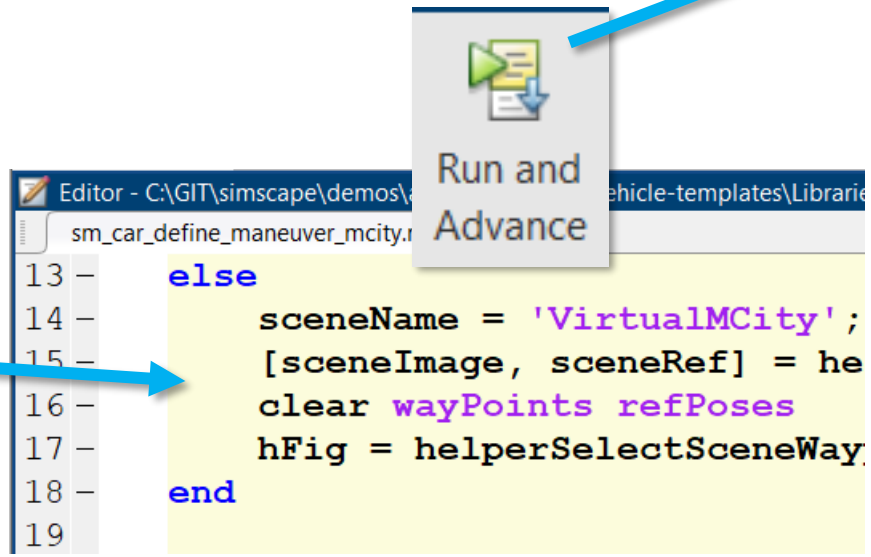


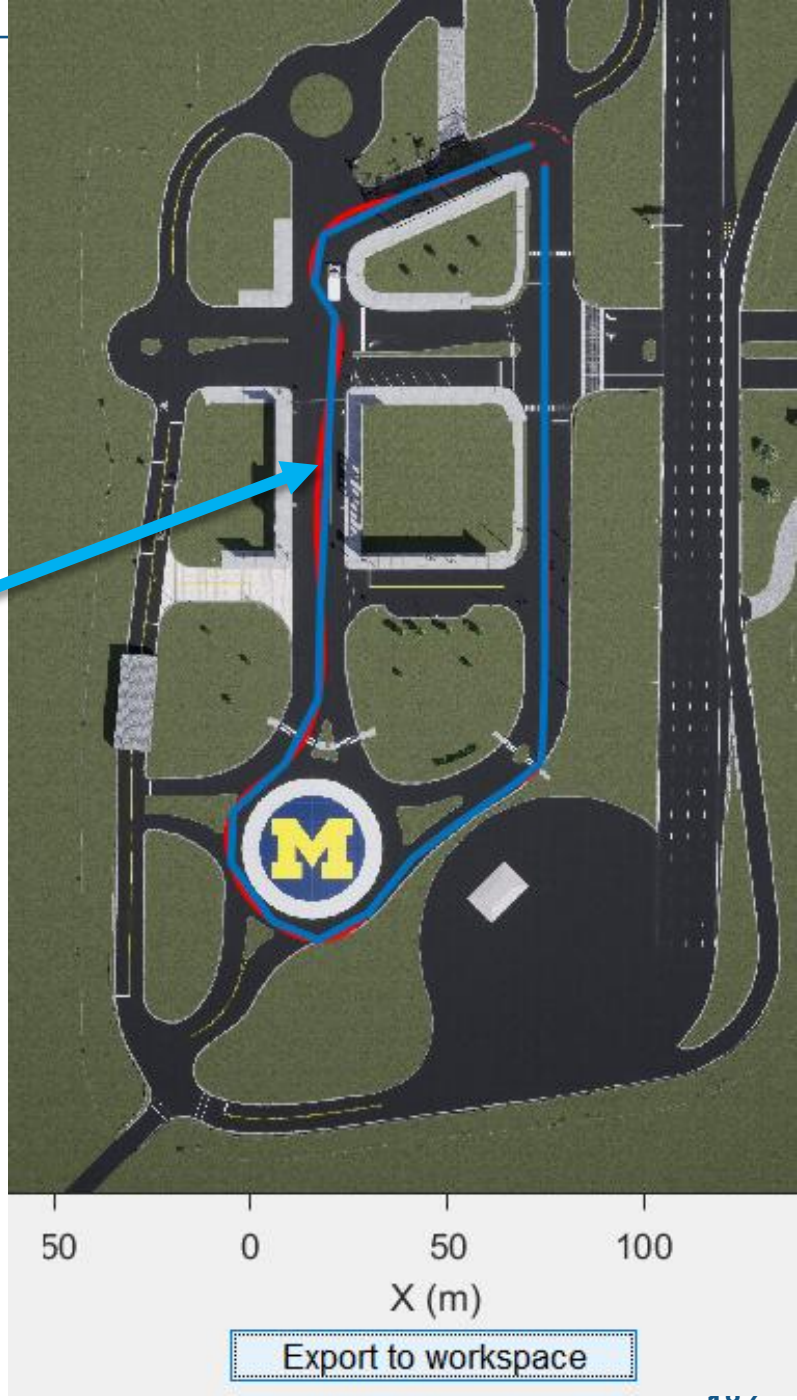
https://www.matlabexpo.com/uk/2019/proceedings.html

Define Route Graphically in MCity Maneuver

- Based on Automated Driving Toolbox example
  - Visualize Automated Parking Valet Using 3D Simulation



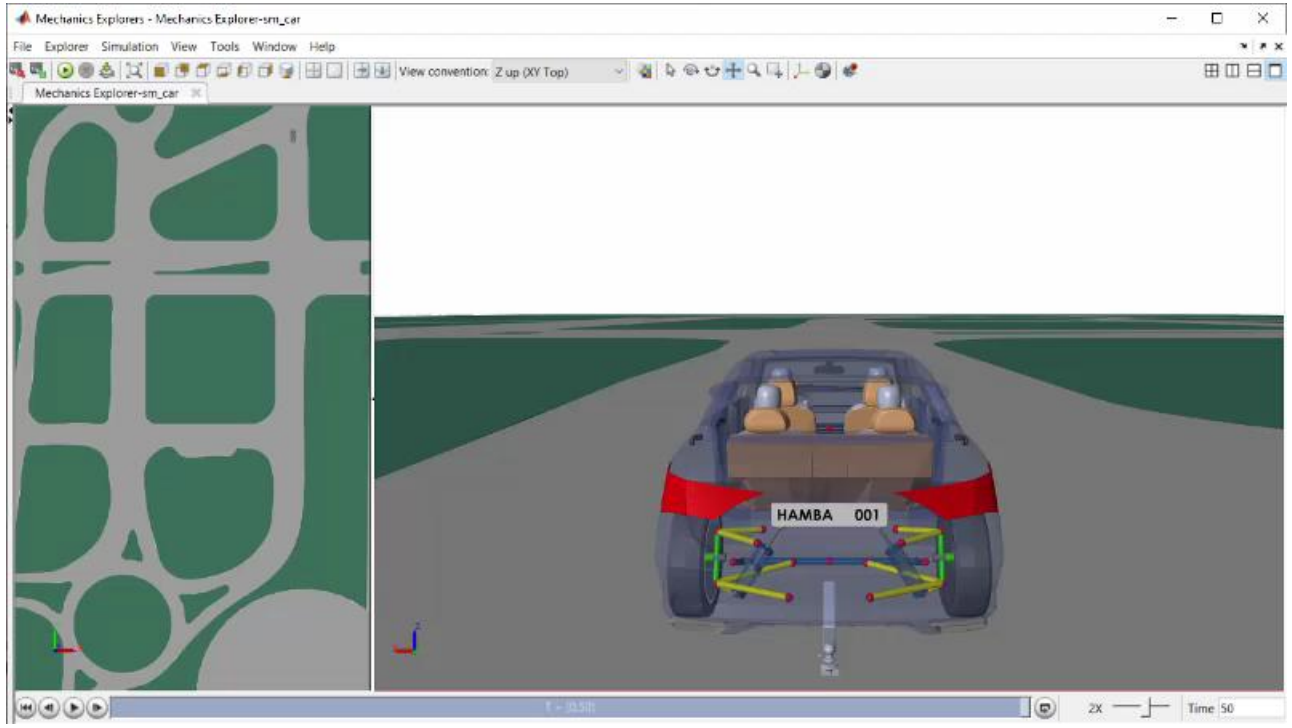




Simscape Vehicle Templates
Examples for ADAS Development

- Example for route planning based on Automated Driving Toolbox example [Visualize Automated Parking Valet Using 3D Simulation]

Animation in Simscape



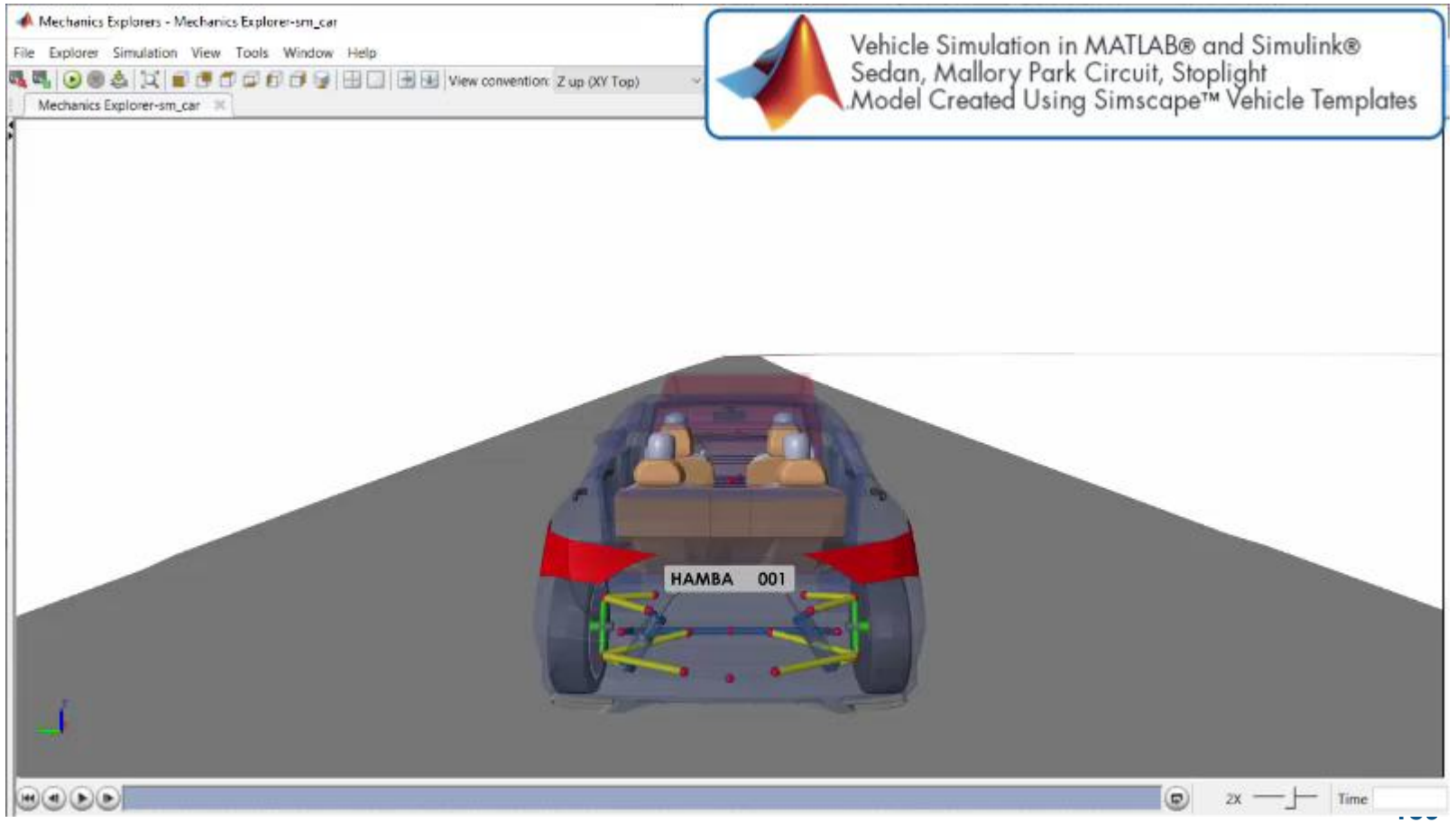
Animation in Unreal



Simscape Vehicle Templates
Examples for ADAS Development

- Example of driver reacting to environment: Stoplight

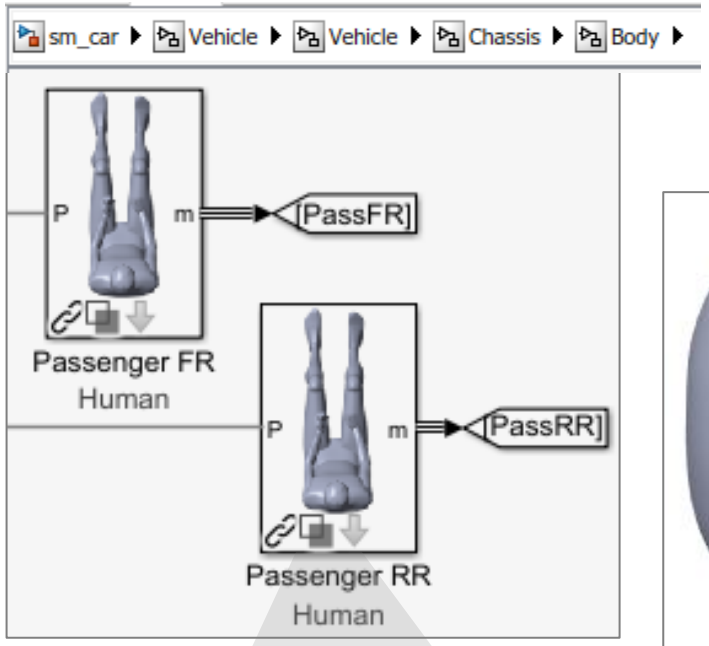
- Driver considers location of vehicle and state of stoplight
  - If in front of stoplight and stoplight is yellow, apply brakes.
  - Remain stopped until light turns green

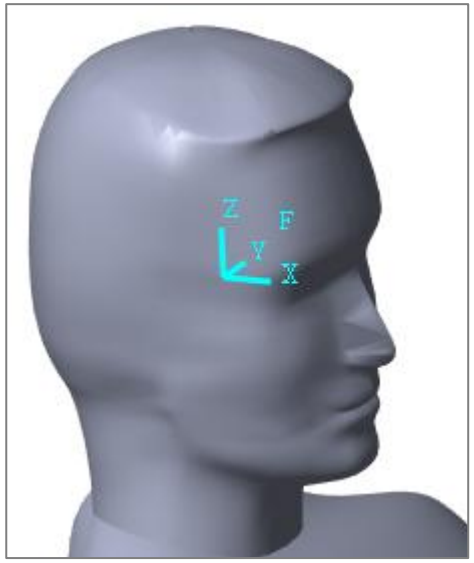


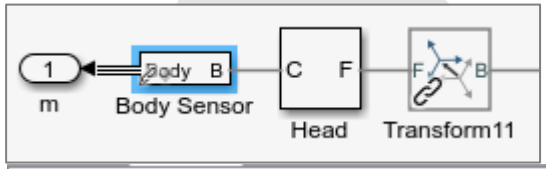
Sensor Added to Passengers and Driver

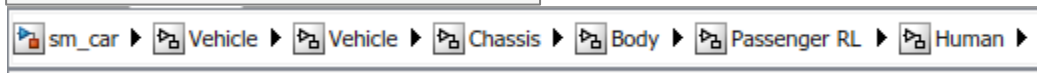
- Measures position, speed, acceleration, pitch, and roll using frame on human as reference frame





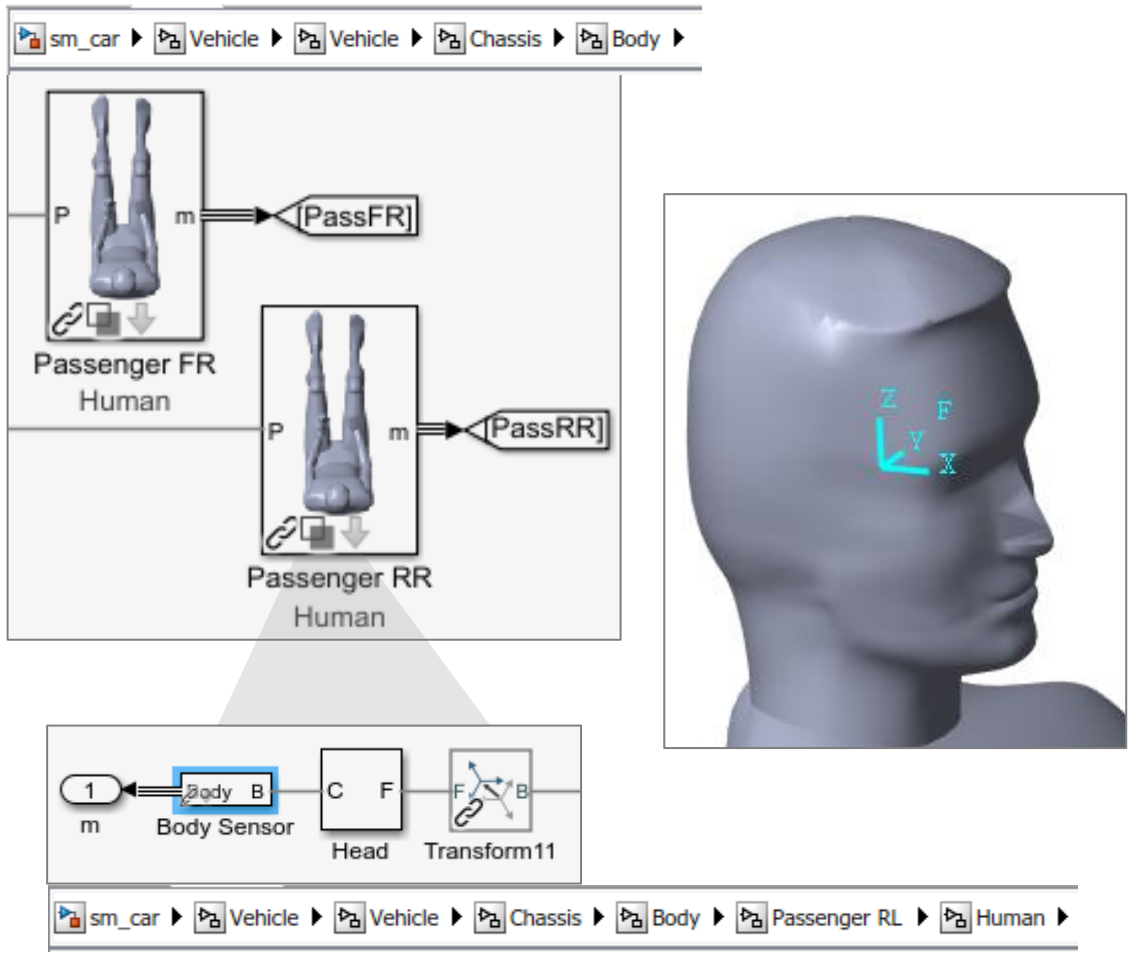






Sensor Added to Passengers and Driver

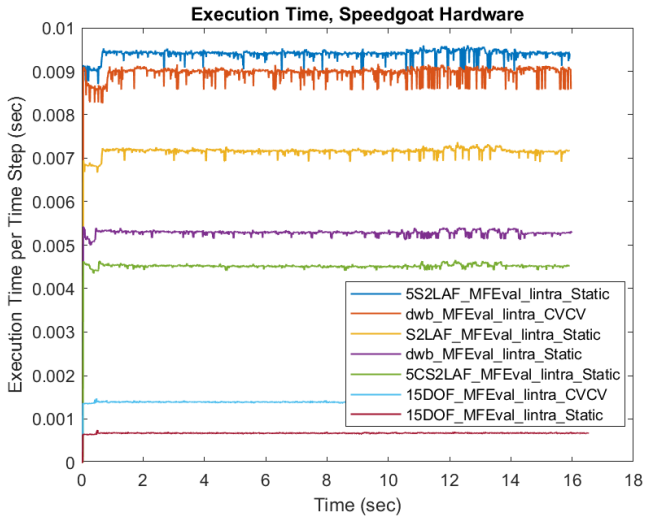
- Measures position, speed, acceleration, pitch, and roll using frame on human as reference frame
  - Passenger measurements are in Body hierarchy
    - VehBus.Chassis.Body.Passengers.FL.aPitch
  - Driver measurements are within Steer hierarchy
    - VehBus.Chassis.SuspF.Steer.Driver.aPitch

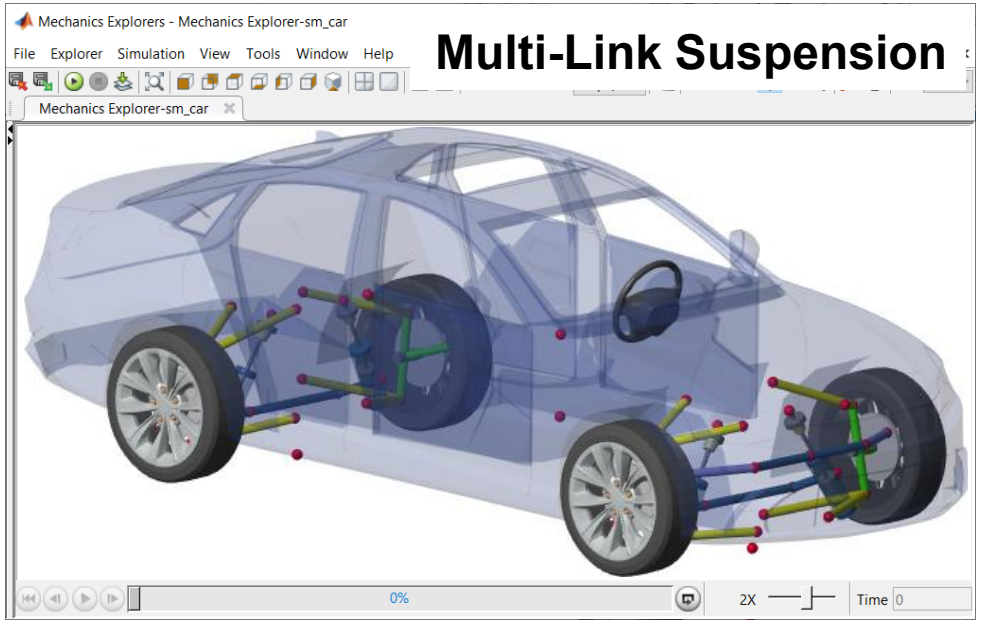


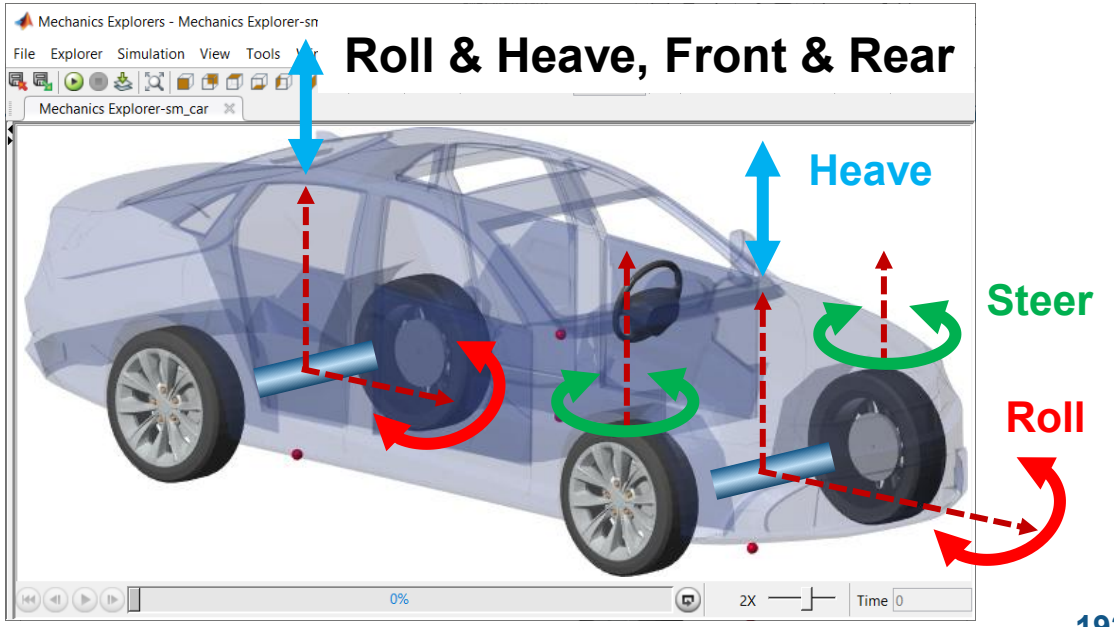
Real-Time Simulation

- Can define abstract models as well
  - Example: Suspension with roll and heave degrees of freedom on front and rear axles

Suspension	State Vector Size	Execution Time
Multi-link	155	9.5 ms
Roll & Heave	41	0.8 ms







Real-Time Simulation

- Can define abstract models as well
  - Example: Suspension with roll and heave degrees of freedom on front and rear axles

Code Execution Profiling Report for sm_car

The code execution profiling report provides metrics based on data collected from real-time simulation. Execution times are calculated from data recorded by instrumentation probes added to the generated code. See Code Execution Profiling for more information.

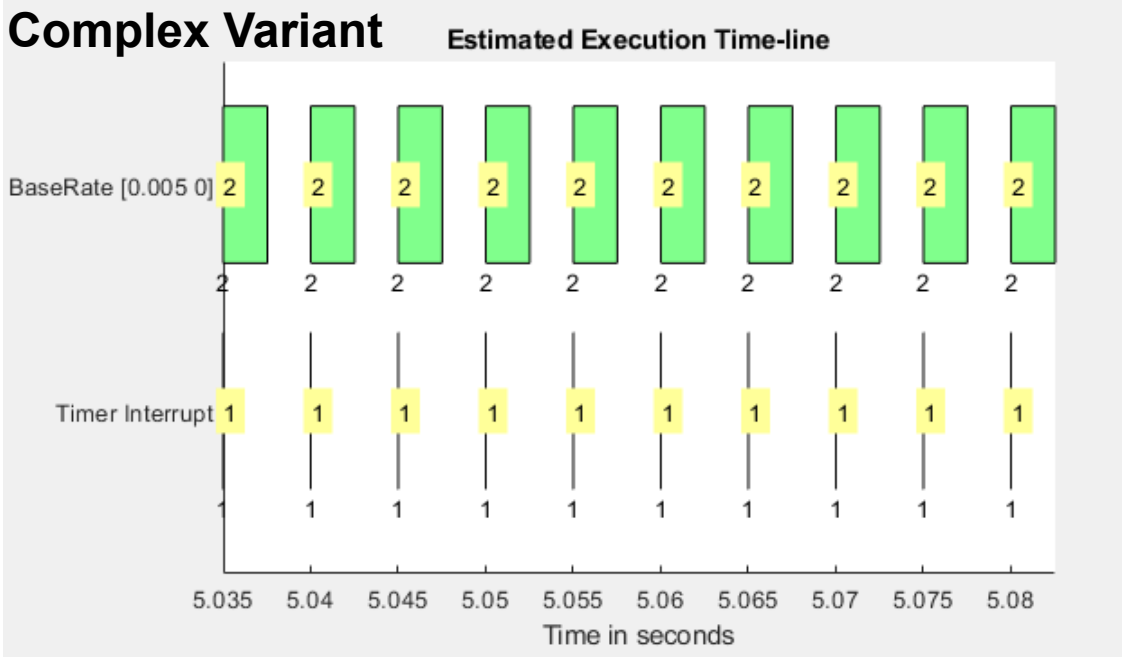
1. Summary

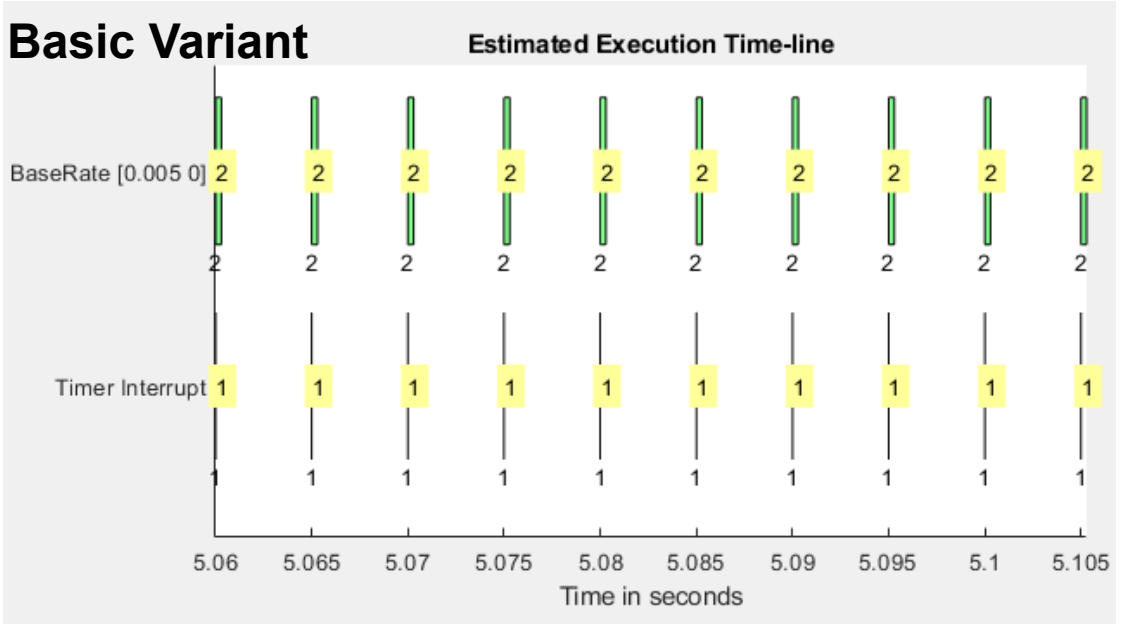
Total time	2517676274
Unit of time	ns
Command	report(, 'Units', 'seconds', 'ScaleFactor', '1e-09', 'NumericFormat', '%0.0f');
Timer frequency (ticks per second)	1e+09
Profiling data created	30-Jun-2020 15:41:58

2. Profiled Sections of Code

Section	Maximum Turnaround Time in ns	Average Turnaround Time in ns	Maximum Execution Time in ns	Average Execution Time in ns	Calls
Timer Interrupt	1770	496	1770	496	1001
BaseRate [0.005 0]	2636703	2514665	2636703	2514665	1001

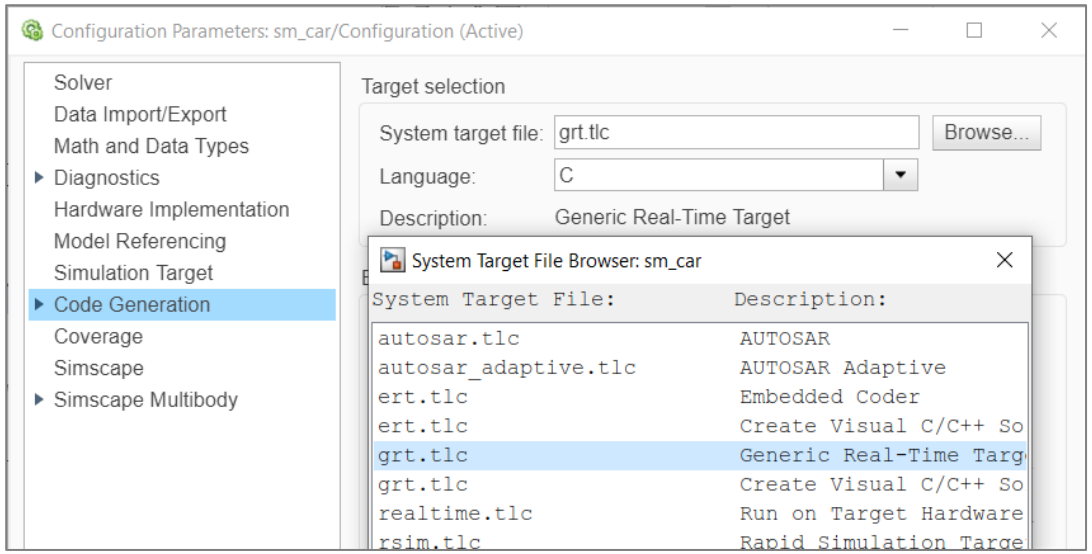
Profile Reports on Speedgoat Hardware

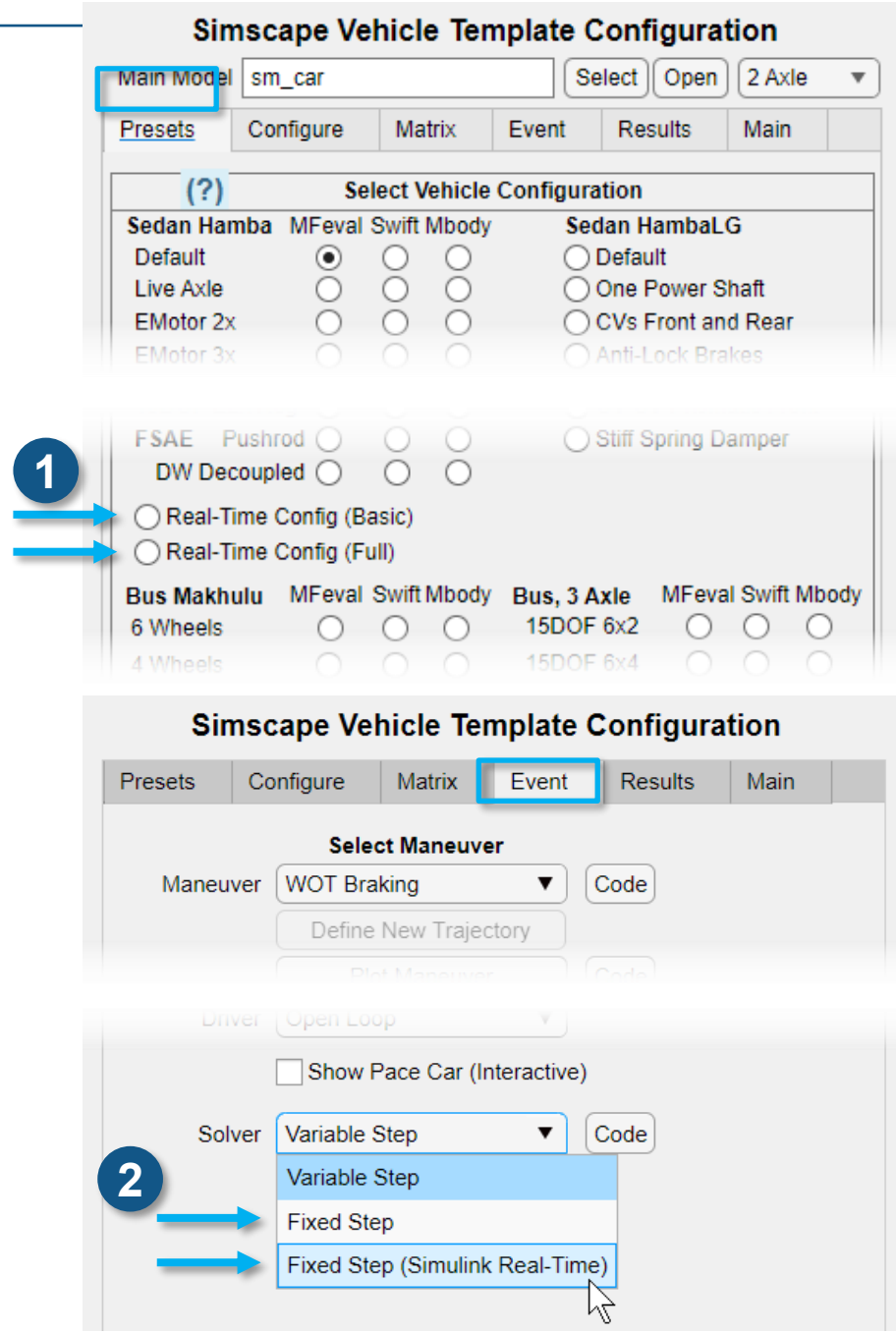




Real-Time Simulation

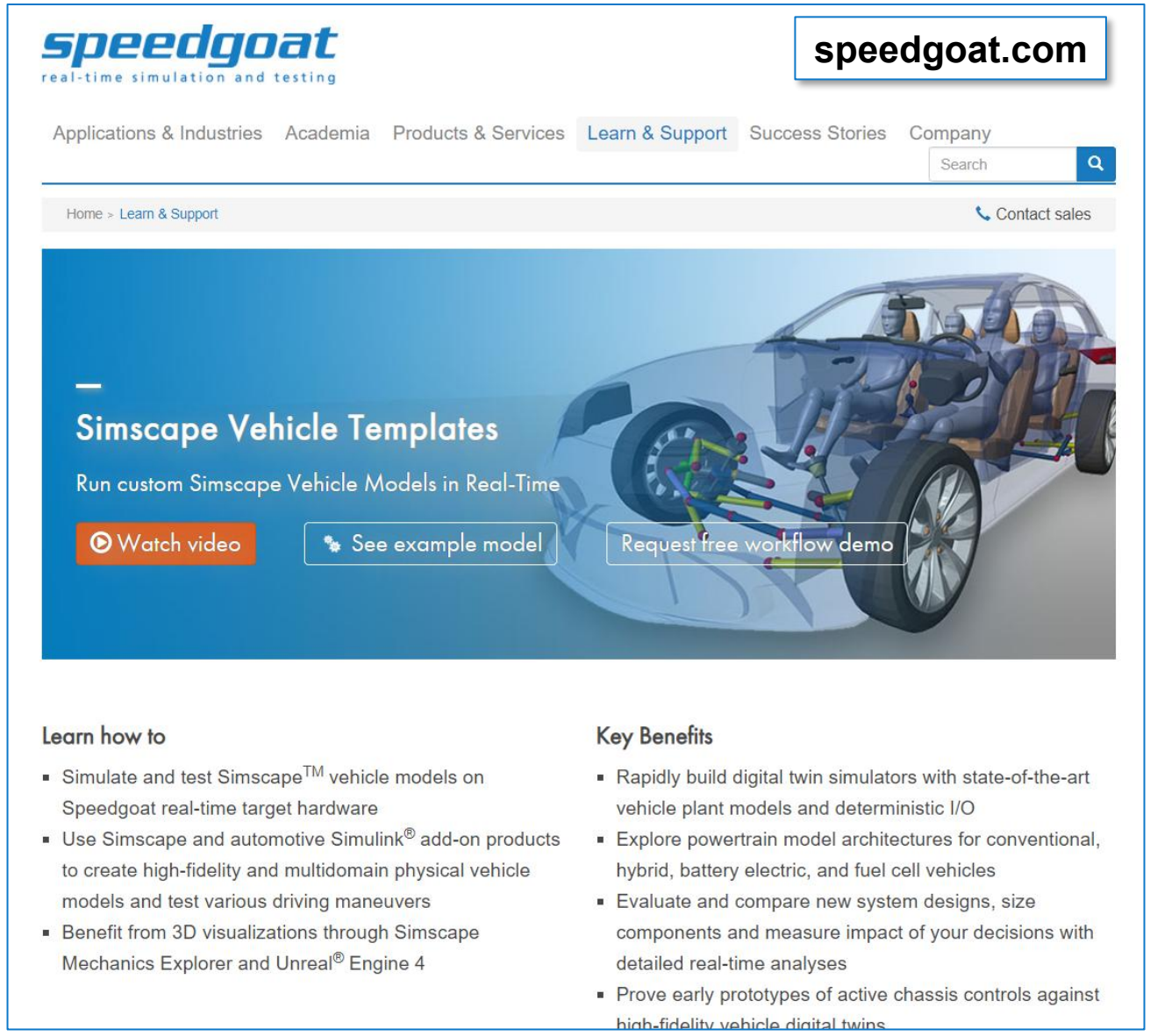
- Configure model for real-time execution
  1. Select appropriate preset
  2. Configure solver for fixed-step simulation.
  3. Select code generation target
     (dropdown for Simulink Real-Time,
     use Solver Configuration Panel for others)





Work with Speedgoat to use Simscape in Real-Time Testing

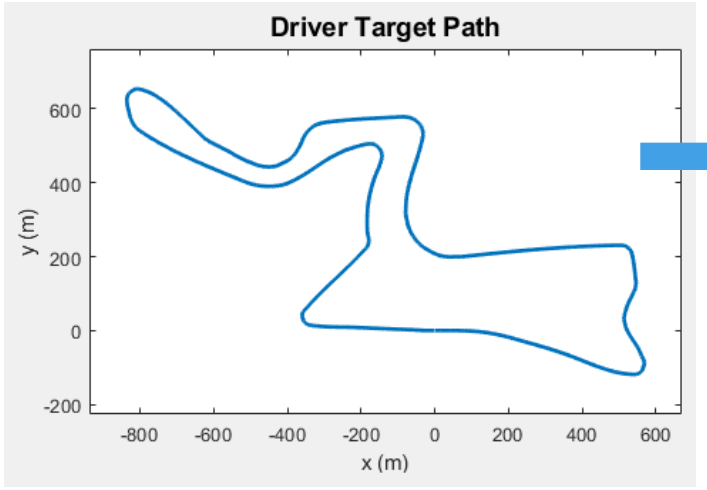
- Videos and explanations demonstrating use of Simscape in real-time simulation
  - HIL testing
  - Driver-in-the-Loop tests
- Hardware and model can be tailored to use case
  - Chassis controls
  - Powertrain design

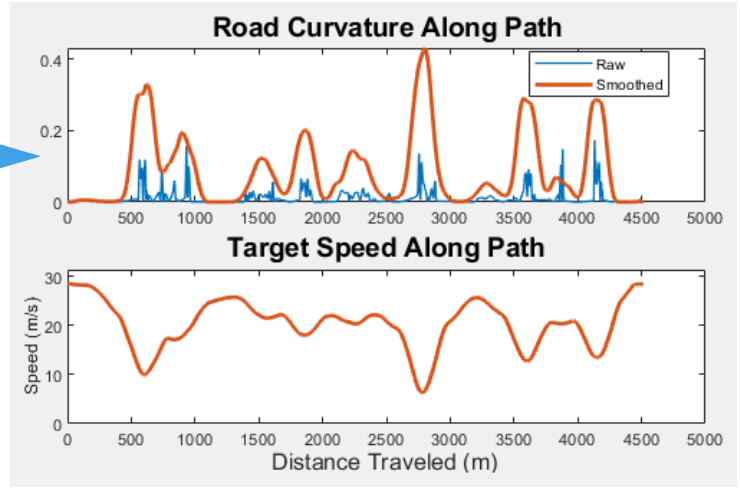


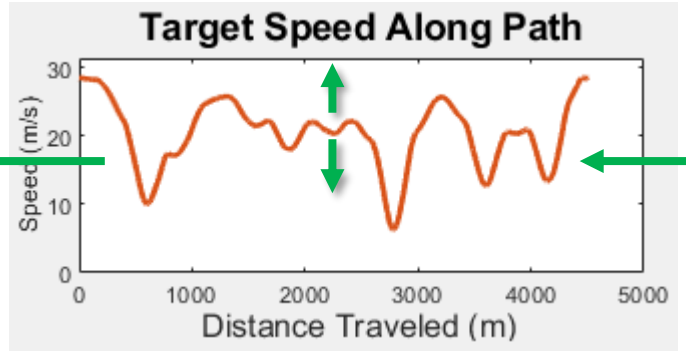
https://www.speedgoat.com/.../simscape-vehicle-templates

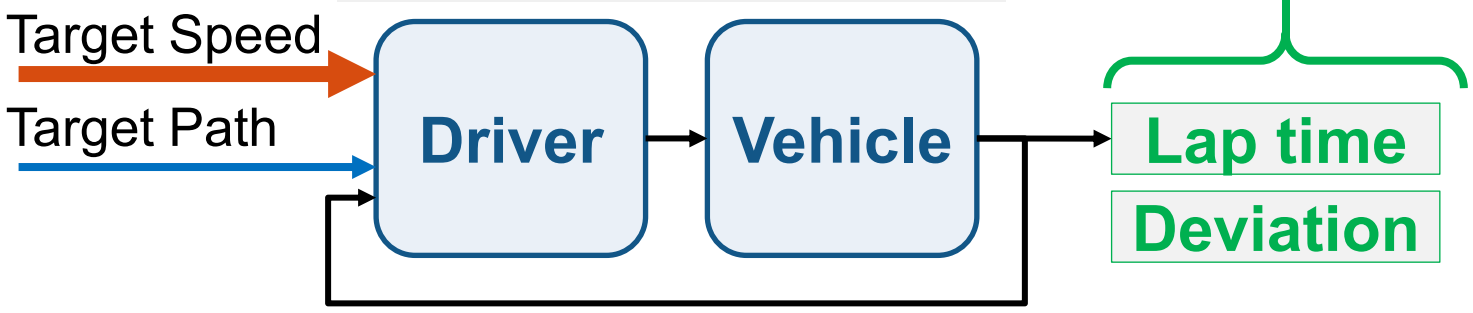
Optimization Example: Use Simulation to Optimize Lap Time

- Simple example:
  Adjust target speed along path to minimize lap time
  - Assume fixed target path
  - Calculate road curvature
  - Convert curvature to target speed (not acceleration – easier to set up)
- Requires:
  - Measuring lap time
  - Parameters to adjust target speed along path
  - Measuring lateral deviation from target path



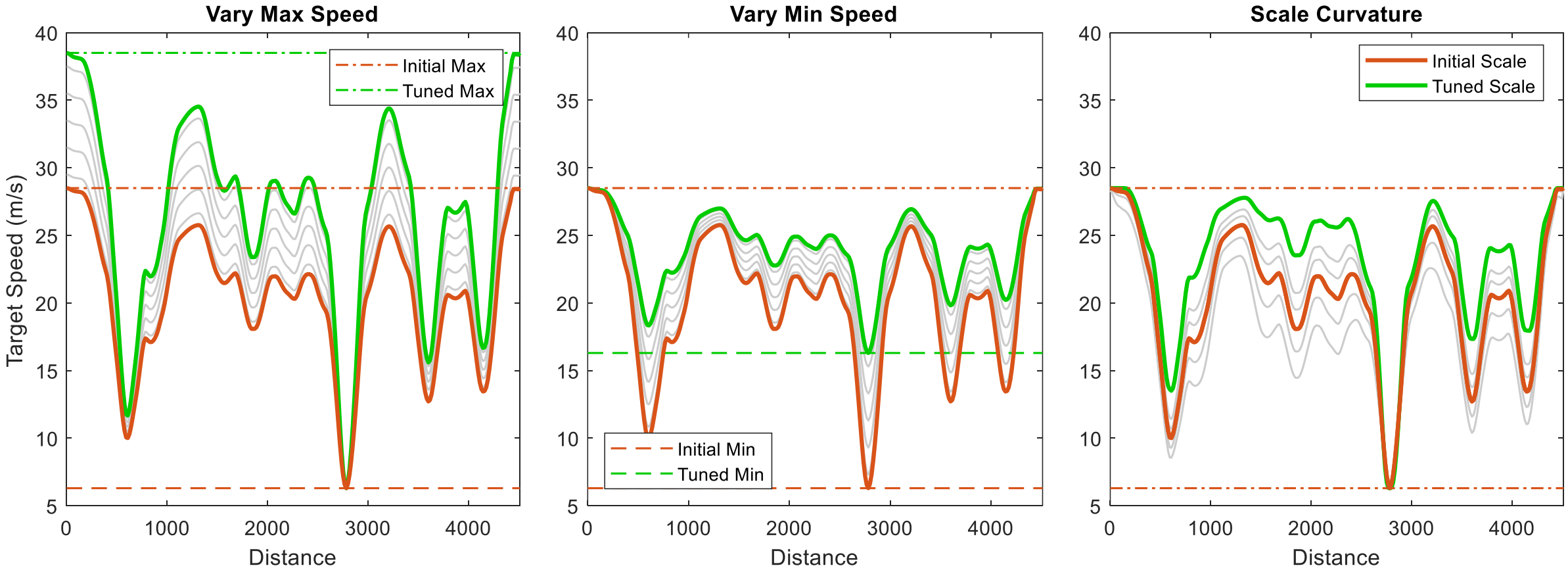






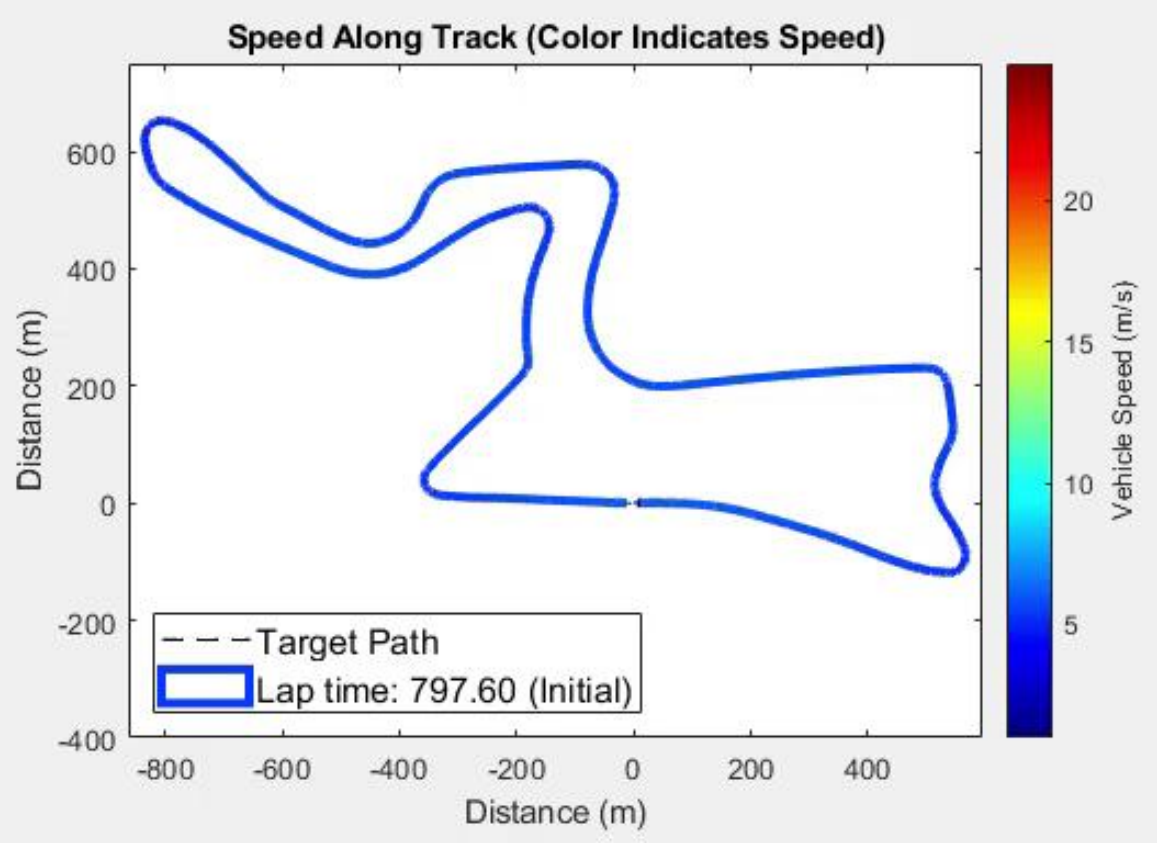
Simple Method for Adjusting Target Speed

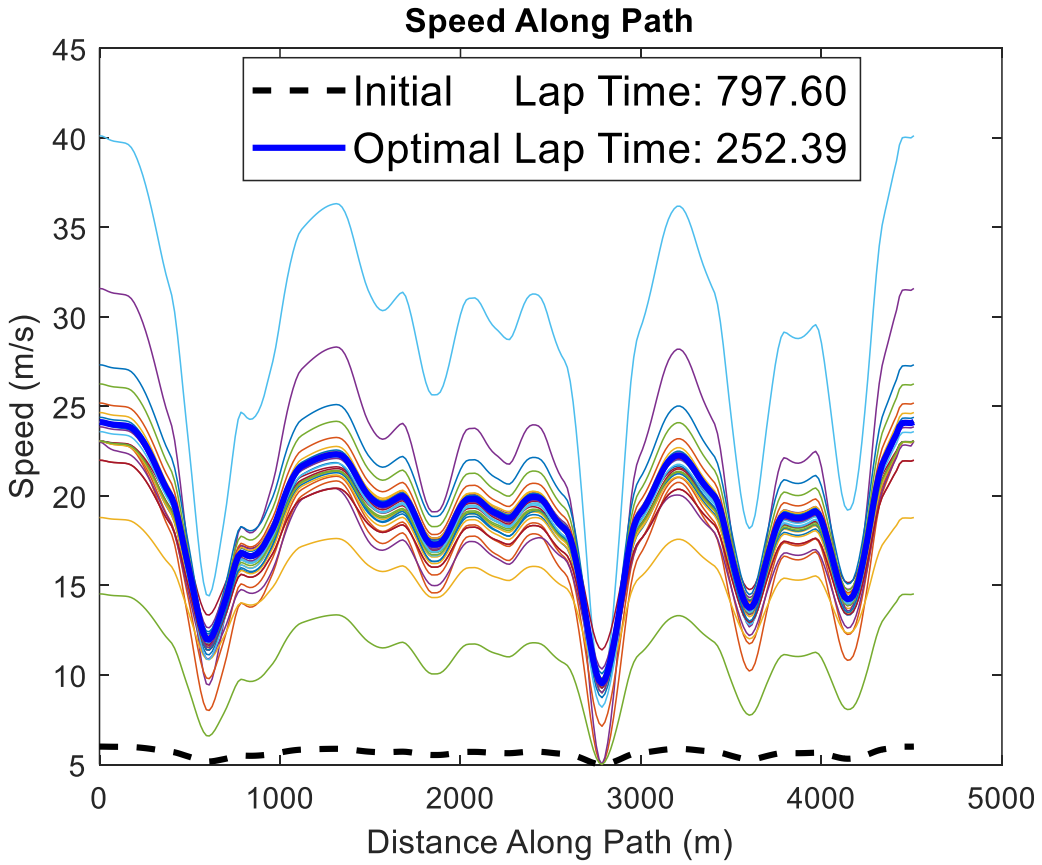
- Three parameters: Maximum Speed, Minimum Speed, Scaling Factor
  - Applied to default trajectory derived from path curvature



Plausible Results Produced From Challenging Problem

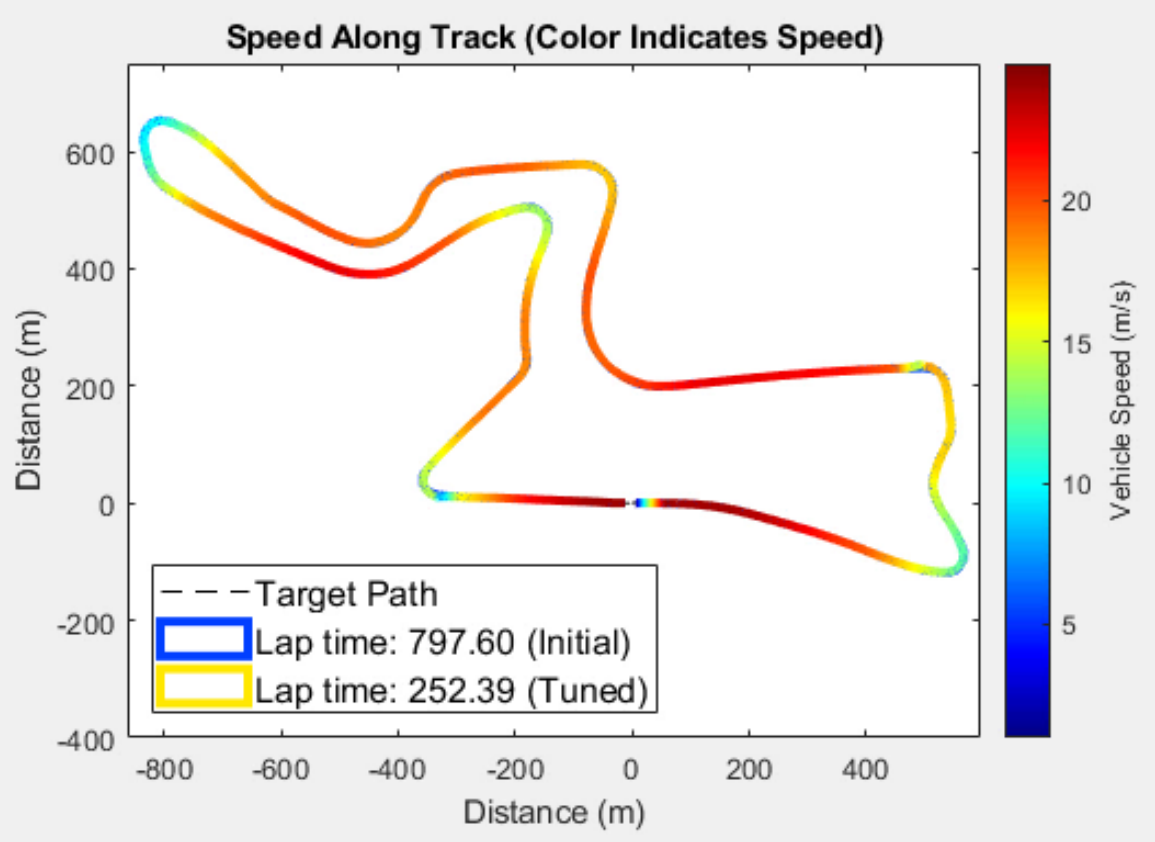
- Better results than tuning by hand
- Managed to keep car on the track (most of the time)

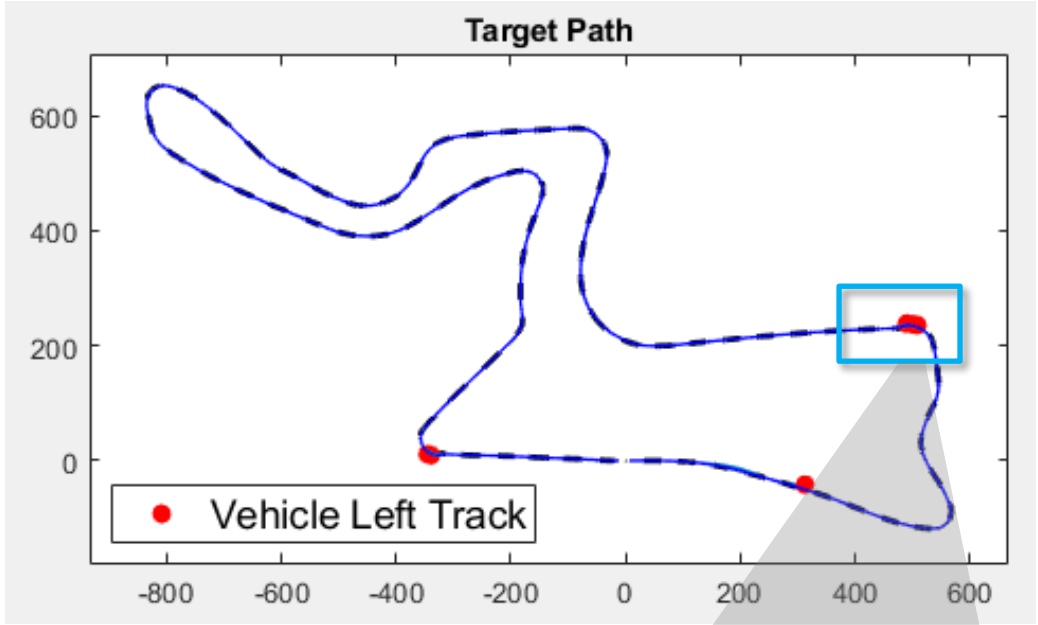


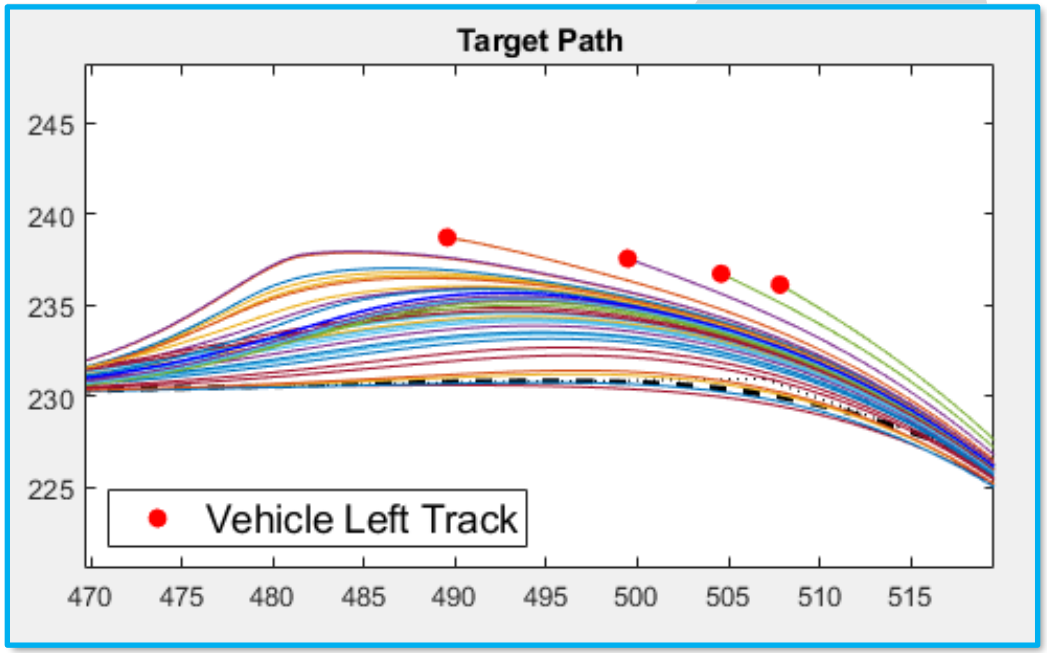


Plausible Results Produced From Challenging Problem

- Better results than tuning by hand
- Managed to keep car on the track



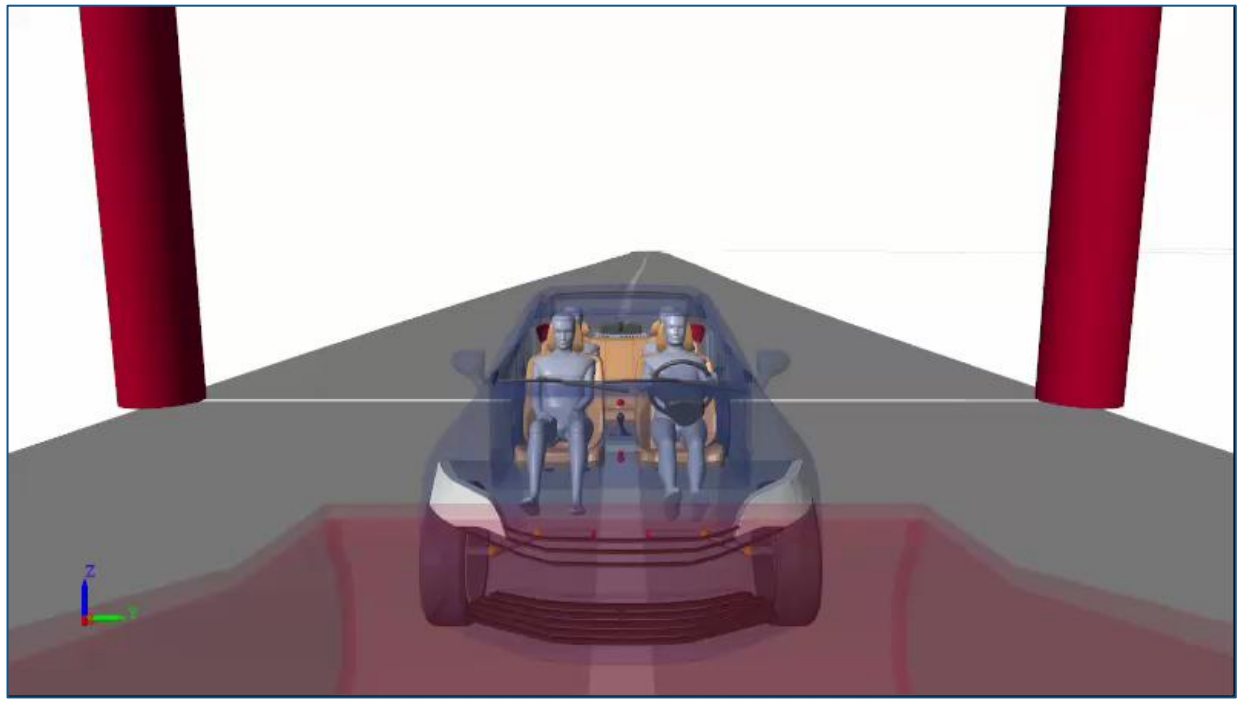




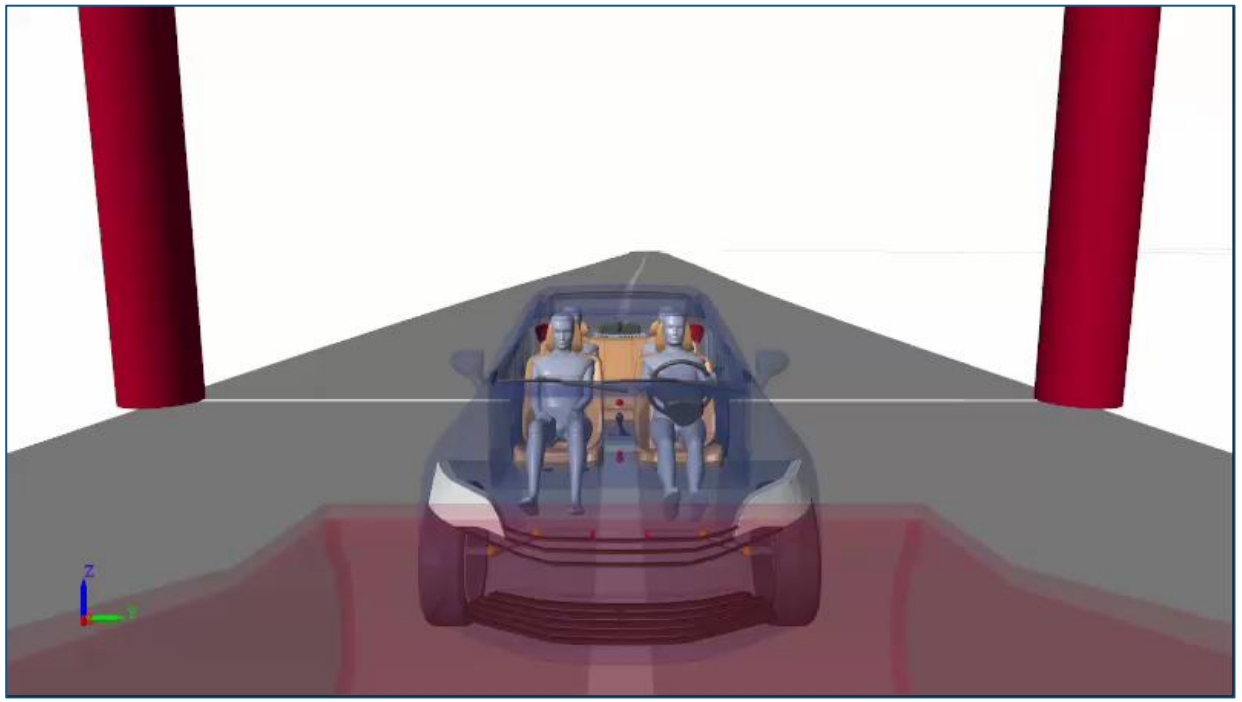
Plausible Results Produced From Challenging Problem

- Better results than tuning by hand
- Managed to keep car on the track (most of the time)

Failed Run

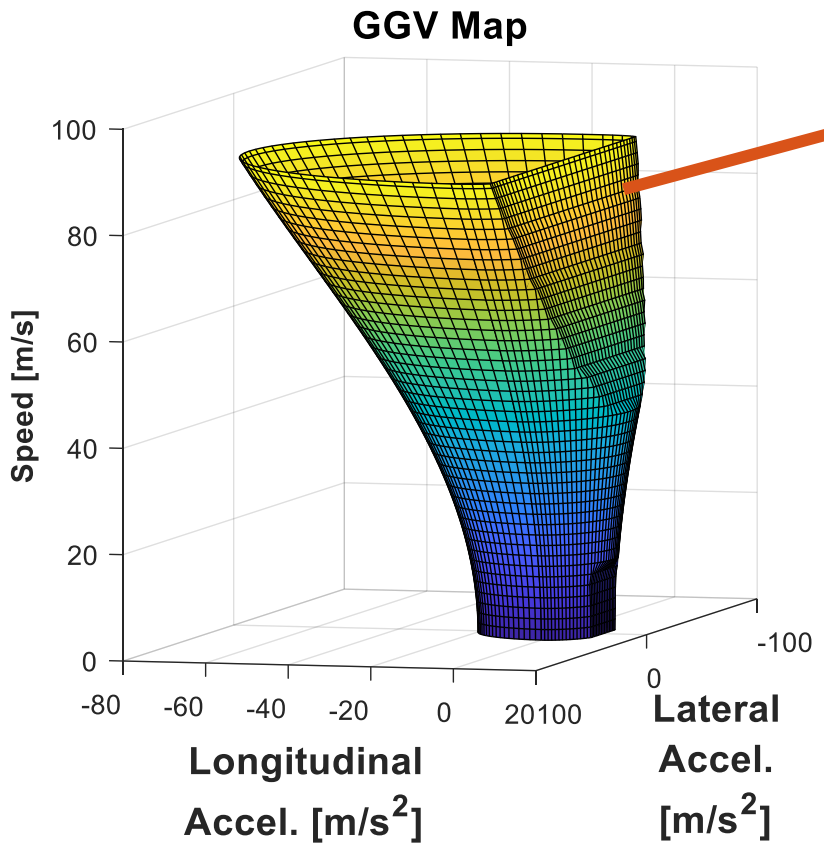


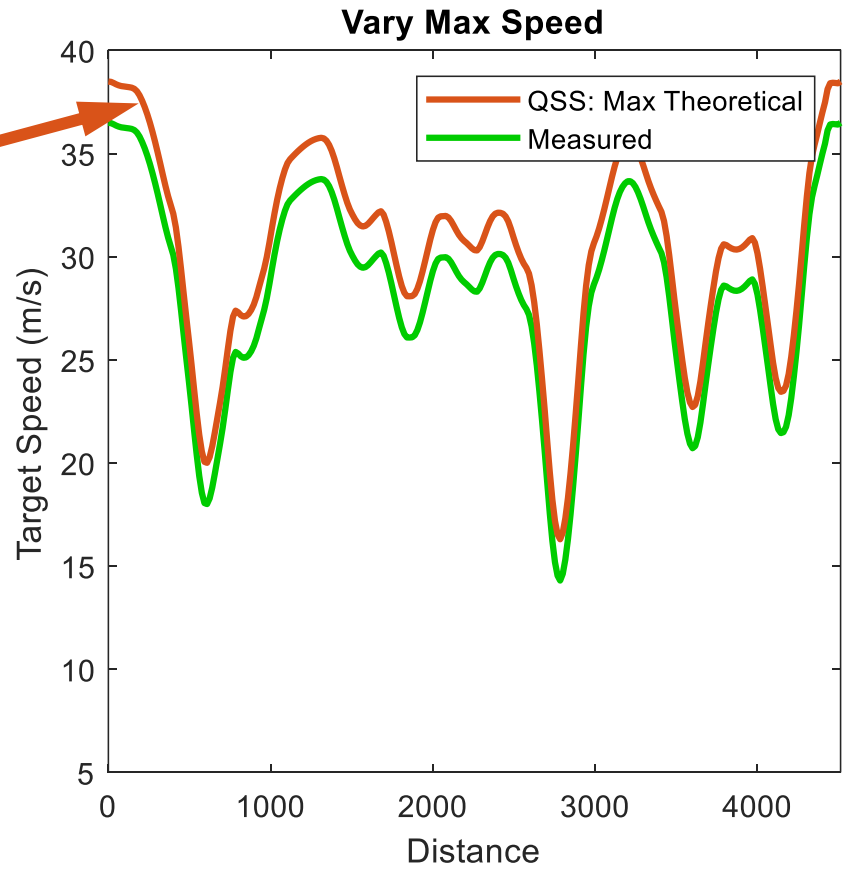
Successful Run



Common Method for Calculating Minimum Lap Time Quasi-Steady State (QSS)

- Assume vehicle achieves maximum acceleration possible in any direction at any speed





"Measured" – does not include QSS assumptions

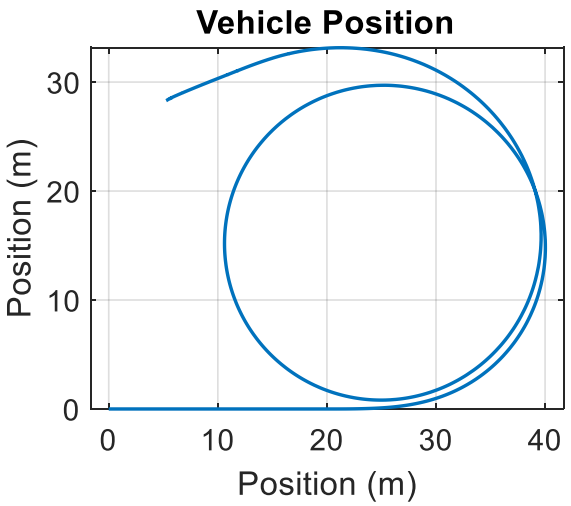
- Transitions not at maximum acceleration
- Dynamic effects (dampers)
- Driver behavior

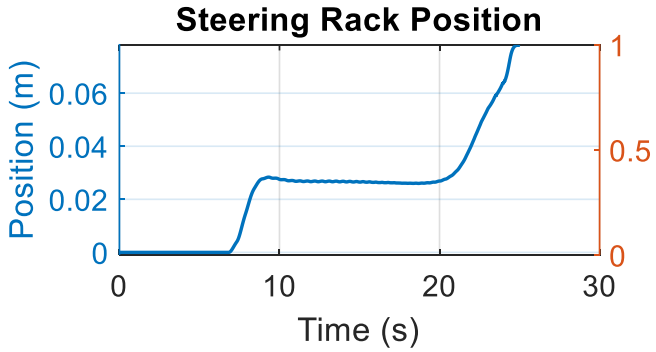
Many papers on methods to increase accuracy of QSS using hybrid approaches (including multibody models)

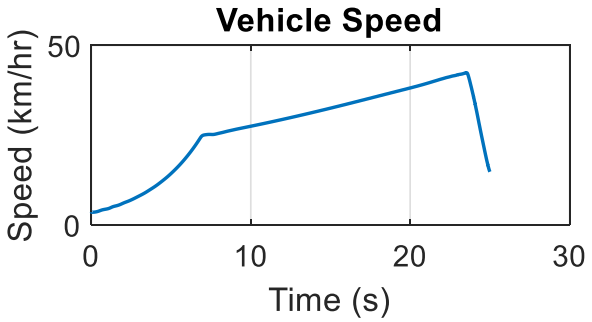
Simscape Vehicle Templates could be used to refine GGV Map

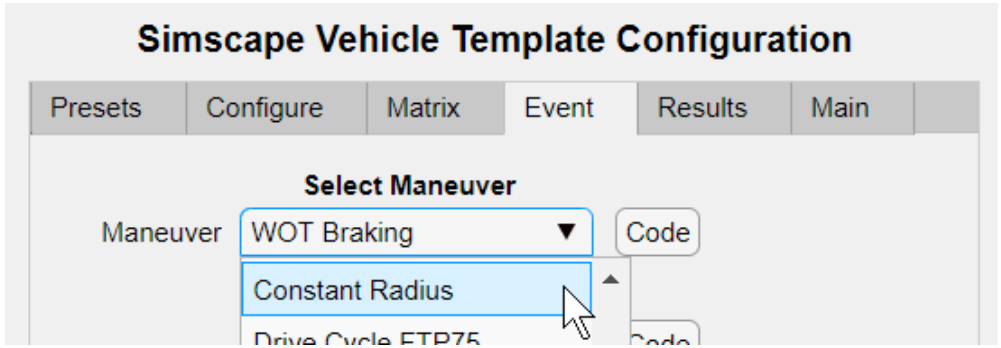
Constant Radius Maneuver

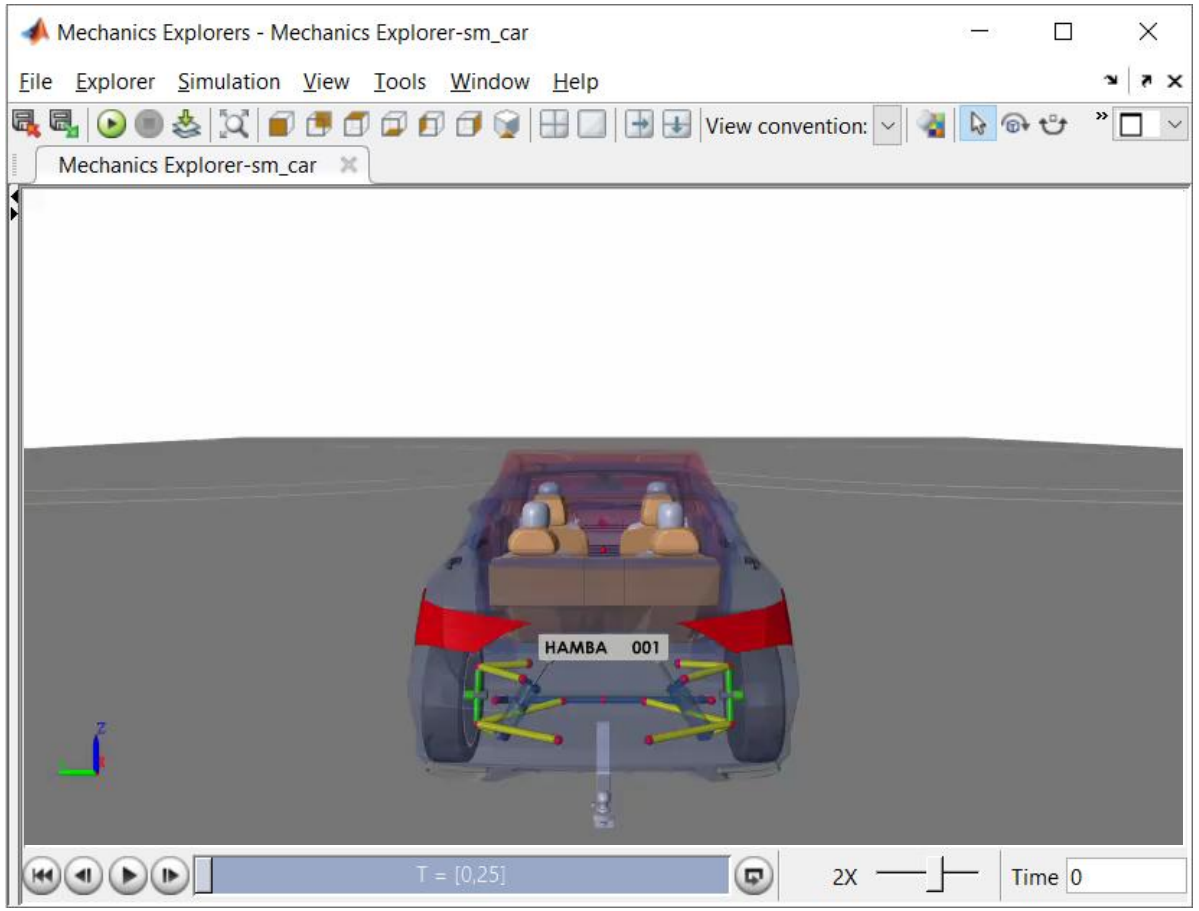
- Driver attempts to follow a circular path as the vehicle speed increases
- Used to test suspension and find maximum lateral acceleration











Simscape Vehicle Templates
Managing Customization



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Managing Customization

- Custom Parameters
- Custom Templates
- Version Control



Customizations

- Confining customizations to simply parameters makes it easy to use latest version
- Adding custom variants requires careful maintenance if you wish to use latest version
- Version control is critical for such a complex model

